3GPP TSG RAN WG1 Meeting #120bis
Wuhan, China, April 07th – 11th, 2025

R1-250xxxx

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Document for: **Comments**

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3GPP TSG RAN WG1 Meeting #120bis R1-250xxxx
Wuhan, China, April 07th – 11th, 2025

Main facts summary

The 3GPP TSG WG RAN1#120 meeting, hosted by ETSI, was held at the INTERCONTINENTAL ATHENAEUM hotel, Athens, Greece.
The meeting started at 9:00 on Monday 17th February and finished at 14:59 (local time) on Friday 21st February 2025.

The number of attending delegates, having confirmed their participation through the electronic check-in application, was **666.**
Note that the number of registered (F2F) was 800, and the number of registered (online participation) was 292.

As RAN1#120 was a Face-to-Face meeting with one-way remote participation, GoToWebinar (GTW) conference calls were set-up for the full duration of the meeting and to the 3 online sessions only.

The schedule of the RAN1#120 Online Sessions is detailed in Annex A-2.

The list of action points that require RAN1 post meeting follow-up is listed in Annex F (end of document).

The number of contributions for this meeting was 1639.

Note: The amount of documents includes those discussed during the email discussion session post meeting.

	Contribution is agreed/approved
	Contribution is not pursued (no consensus)
	Further discussion is required – come-back needed
	Working assumption
	Specific action needed from MCC
	Contribution is for email discussion/approval

1 Opening of the meeting

Mr Younsun Kim (RAN1 Chair) welcomed the participants of the RAN WG1 #120 meeting and opened the meeting at 09:00.

Mr Aris Papasakellariou from Samsung on behalf of ETSI and the hosting companies/sponsors, welcomed the delegates, and detailed the domestic arrangements for the full week.

1.1 Call for IPR

The attention of the delegates to the meeting of this Technical Specification Group was drawn to the fact ***that 3GPP Individual Members have the obligation*** *under the IPR Policies of their respective Organizational Partners to* ***inform their respective*** *Organizational Partners* ***of Essential IPRs they become aware of****.*
The delegates take note that they are hereby invited:

- *to investigate whether their organization or any other organization owns IPRs which were, or were likely to become Essential in respect of the work of 3GPP.*
- *to notify their respective Organizational Partners of all potential IPRs, e.g., for ETSI, by means of the IPR Information Statement and the Licensing declaration forms (e.g. see the ETSI IPR forms https://www.etsi.org/intellectual-property-rights/ipr).*

More information is available via the "Legal" page on the web site: https://www.3gpp.org/about-3gpp/legal-matters .

1.2 Competition law statement

The Chair also drew Member's attention to the fact that 3GPP activities are subject to antitrust and competition laws and that compliance with said laws is therefore required of any participant of this WG meeting including the Chair and Vice Chairs. In case of question, please contact your legal counsel.
The present meeting will be conducted with strict impartiality and in the interests of 3GPP.
Furthermore, the Chair reminded Members that timely submission of work items/contributions in advance of WG meetings is important to allow for full and fair consideration of such matters.

1.3 Network usage conditions

The PCG has laid down the following network usage conditions:

Users shall not use the network to engage in illegal activities. This includes activities such as copyright violation, hacking, espionage or any other activity that may be prohibited by local laws.

Users shall not engage in non-work related activities that consume excessive bandwidth or cause significant degradation of the performance of the network.
Since the **network is a shared resource**, users should exercise some basic etiquette when using the 3GPP network at a meeting. It is understood that high bandwidth applications such as downloading large files or video streaming might be required for business purposes, but delegates should be strongly discouraged in performing these activities for personal use. Downloading a movie or doing something in an interactive environment for personal use essentially wastes bandwidth that others need to make the meeting effective. The meeting Chair should remind end users that the network is a shared resource; the more one user grabs, the less there is for another. Email and its attachments already take up significant bandwidth (certain email programs are not very bandwidth efficient). In case of need the chair can ask the delegates to restrict IT usage to things that are essential for the meeting itself.

1. DON'T place your WiFi device in ad-hoc mode
2. DON'T set up a personal hotspot in the meeting room
3. DO try 802.11a if your WiFi device supports it
4. DON'T manually allocate an IP address
5. DON'T be a bandwidth hog by streaming video, playing online games, or downloading huge files
6. DON'T use packet probing software which clogs the local network (e.g., packet sniffers or port scanners)

1.4 Check-in for Registered Delegates

The attention of the delegates to this meeting was drawn to the fact that it is not permitted to check in other delegates on their behalf. In the event of technical difficulties preventing check-in, delegates should present themselves in person to the Secretary.

1.5 Recording of RAN1 Meeting

Recording of the GoToWebinar sessions of the present meeting is strictly prohibited. No individual or entity - including the speakers and/or the authors - may electronically record any portion of the meeting without prior written consent of the Chair and all the RAN1 meeting participants.

2 Approval of Agenda

R1-2500138 **Draft Agenda of RAN1#120 meeting** **RAN1 Chair** (rev of R1-2500001)
Decision: No comments, the agenda is approved.
RRC parameters: Sorour (Ericsson) to send an email and clarify the handling of Rel-19 RRC parameters for this meeting.
Schedule of the week: As a follow-up to Iridium's query, swap the order on Tuesday evening session in order to handle IoT NTN TDD mode after the uplink sub-agendas (9.11).

In addition to the agenda, the following additional guidance is provided.
R1-2500004 **RAN1#120 Meeting Timelines, Scope, Process** **RAN1 Chair, ETSI MCC**
Decision: The document is noted for information.

3 Highlights from RAN plenary

R1-2500002 **Highlights from RAN#106** **RAN1 Chair**
Decision: The document is noted for information.

4 Approval of Minutes from previous meeting

R1-2500003 **Report of RAN1#119 meeting** **ETSI MCC**
The document was presented by Patrick Merias - MCC (ETSI Mobile Competence Center) and provides the report from RAN1#119 meeting in Orlando.
Decision: No comments, the report is approved.

5 Incoming Liaison Statements

Rel-17 RedCap, SDT, CovEnh
R1-2500007 **LS response on waveform determination for PUSCH scheduled by fallbackRAR UL grant and MsgA PUSCH** **RAN2, CATT**
Decision: RAN2 response for RAN1 LS in R1-2409301. No further action necessary from RAN1.

Rel-18 Coverage Enh
R1-2500008 **Reply LS on the time period for R18 preamble repetition** **RAN2, Huawei**
Relevant tdocs:
R1-2500328 Discussion on time period determination for PRACH repetition vivo
R1-2500819 Discussion on the time period determination for multiple PRACH transmission Samsung
R1-2501066 Discussion on the time period for R18 preamble repetition Huawei, HiSilicon
R1-2501102 Discussion on time period determination for PRACH transmission with preamble repetitions China Telecom
R1-2501265 Discussion on time period for Rel-18 preamble repetition ZTE Corporation, Sanechips
Decision: RAN2 response for RAN1 LS in R1-2409262. To be discussed in agenda item 8.1 (Coverage Enh) - Frank (Huawei).

Rel-18 MIMO
R1-2500012 **LS on UL 8Tx** **RAN2, Samsung**
Relevant tdocs:
R1-2500195 Discussion on LS from RAN2 on UL 8Tx CATT
R1-2500196 Draft reply LS on UL 8Tx CATT
R1-2500248 Draft reply LS to RAN2 on UL 8Tx ZTE Corporation, Sanechips
R1-2500321 Draft reply LS on UL 8Tx vivo
R1-2500428 Discussion on LS on UL 8Tx OPPO
R1-2500540 Discussion of RAN2 LS on 8Tx UL Transmission Ericsson
R1-2500564 Draft Reply LS on UL 8Tx Google
R1-2500705 Draft Reply LS on UL 8Tx Lenovo

R1-2500826 Draft reply LS on UL 8Tx Samsung
R1-2501137 Discussion on response to RAN2 LS on UL 8Tx Qualcomm Incorporated
R1-2501187 Discussion on LS reply for UL 8Tx NTT DOCOMO, INC.
R1-2501239 Discussion on LS to RAN1 on UL 8Tx ASUSTeK
R1-2501327 Discussion on UL 8Tx Huawei, HiSilicon
Decision: RAN2 is requesting RAN1 input to finalize the MAC specification support of UL 8TX. RAN1 response is necessary and to be handled in agenda item 8.1 (MIMO) - Sa (Samsung).

Rel-18 MIMO
R1-2500013 LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP RAN2, CATT
Relevant tdocs:
R1-2500193 Discussion on LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP CATT
R1-2500194 Draft reply LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP CATT
R1-2500322 Draft reply LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP vivo
R1-2500429 Discussions on LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP OPPO
R1-2500538 Draft reply LS to RAN2 on differentiation of sDCI mTRP, mDCI mTRP and sTRP ZTE Corporation, Sanechips
R1-2500599 Discussion on RAN2 LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP NEC
R1-2500706 Draft Reply LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP Lenovo
R1-2500763 Draft reply LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP Apple
R1-2500827 Draft reply LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP Samsung
R1-2500895 Draft LS reply on differentiation of sDCI mTRP, mDCI mTRP and sTRP Nokia
R1-2501282 Discussion on LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP Ericsson
R1-2501332 Discussion on differentiation of sDCI mTRP, mDCI mTRP and sTRP Huawei, HiSilicon
Decision: RAN2 is requesting RAN1 input on simultaneous unified TCI state(s) update for a list of cells. RAN1 response is necessary and to be handled in agenda item 8.1 (MIMO) - Jiayi (CATT)

Rel-18 Mobility Enh
R1-2500010 LS on LTM UE capabilities for cross-band operation RAN2, Qualcomm
Relevant tdocs:
R1-2500197 Draft reply LS on LTM UE capabilities for cross-band operation CATT
R1-2500247 Draft reply LS to RAN2 on LTM UE capabilities for cross-band operation ZTE Corporation, Sanechips
R1-2500320 Draft reply LS on LTM UE capabilities for cross-band operation vivo
R1-2500416 Discussion on LS on LTM UE capabilities for cross-band operation Nokia
R1-2500431 Discussion on LS on LTM UE capabilities for cross-band operation OPPO
R1-2500704 Draft Reply LS on LTM UE capabilities for cross-band operation Lenovo
R1-2500764 Draft reply LS on LTM UE capabilities for cross-band operation Apple
R1-2500822 Draft reply LS on LTM UE capabilities for cross-band operation Samsung
R1-2501063 Discussion on RAN2 LS on LTM UE capabilities for cross-band operation Huawei, HiSilicon
R1-2501140 Discussion on RAN2 LS on LTM UE capabilities for cross-band operation Qualcomm Incorporated
R1-2501246 Discussion on RAN2 LS on LTM UE capabilities for cross-band operation Ericsson
Decision: RAN2 is requesting RAN1 input to address potential confusion on configuration based on per band LTM UE capabilities for cross-band LTM switch operation. RAN1 response is necessary and to be handled in agenda item 8.2.

Rel-18 Positioning
R1-2500014 LS on co-existence of SL-CA and SL PRS transmission/reception for Dedicated SL-PRS resource pool RAN2, vivo
Relevant tdocs:
R1-2500191 Discussion on co-existence of SL-CA and SL PRS transmission/reception in dedicated SL-PRS resource pool CATT, CICTCI
R1-2500192 Draft reply LS on co-existence of SL-CA and SL PRS transmission/reception in dedicated SL-PRS resource pool CATT, CICTCI
R1-2500327 Draft reply LS on co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool vivo
R1-2500412 Draft Reply LS on co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool Intel Corporation
R1-2500433 Discussion on LS about co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool OPPO
R1-2500434 Draft reply LS to RAN2 on co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool OPPO

R1-2500578 Draft reply LS on co-existence of SL-CA and SL PRS ZTE Corporation, Sanechips
R1-2500820 Discussion on RAN2 LS about co-existence of SL-CA and SL PRS transmission/reception in dedicated SL-PRS resource pool Samsung
R1-2501069 Discussion on coexistence of SL CA and SL PRS in dedicated RP Huawei, HiSilicon
R1-2501300 Discussion on LS on co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool Ericsson
R1-2501301 Draft reply LS on co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool Ericsson
R1-2501352 Discussion of RAN2 LS on co-existence of SL-CA and SL PRS transmission/reception for Dedicated SL-PRS resource pool Nokia
Decision: RAN2 is requesting RAN1 input on the restriction of SL-PRS transmission/reception in a dedicated SL-PRS resource pool in a band (pre-)configured with SL CA. RAN1 response is necessary and to be handled in agenda item 8.1 (Positioning) - Yuan (vivo).

Rel-18 NR-NTN
R1-2500006 LS on soft satellite switch with re-sync RAN2, Huawei
Relevant tdocs:
R1-2500273 Discussion on the LS on soft satellite switch with re-sync CMCC
R1-2500324 Draft reply LS on soft satellite switch with resync vivo
R1-2500378 Discussion on the LS on soft satellite switch with re-sync ZTE Corporation, Sanechips
R1-2500435 Discussion on soft satellite switch with re-sync OPPO
R1-2500436 Draft reply LS on soft satellite switch with re-sync OPPO
R1-2500760 Discussion on RAN2 LS on Soft Satellite Switch with Re-sync Apple
R1-2500761 Draft Reply LS to RAN2 on Soft Satellite Switch with Re-sync Apple
R1-2500825 Discussion on RAN2 LS about soft satellite switch with re-sync Samsung
R1-2500990 Discussion on RAN2 LS on soft satellite switch with re-sync Ericsson
R1-2500996 Discussion of LS response related to soft satellite switch with re-sync Nokia
R1-2501038 Discussion on reply LS on soft satellite switch with re-sync MediaTek Inc.
R1-2501070 Discussion on soft satellite switch with re-sync Huawei, HiSilicon
R1-2501139 Discussion on RAN2 LS on soft satellite switch with re-synch Qualcomm Incorporated
R1-2501324 Draft reply LS on soft satellite switch with re-sync Huawei, HiSilicon
Decision: RAN2 is requesting RAN1 input on possible introduction of a temporarily shifted SSB pattern during [t-ServiceStart, t-Service] for soft satellite switching with re-sync. RAN1 response is necessary and to be handled in agenda item 8.1 (NTN) - Xiaolei (Huawei).

Rel-18 NR-NTN
R1-2500023 Reply LS to RAN1 and RAN2 on FR2-NTN inclusion to specifications RAN4, vivo
Decision: RAN4 response to RAN1 LS in R1-2407406. No further action necessary from RAN1.

Rel-19 NTN
R1-2500009 LS on PWS support in RRC_CONNECTED for NB-IoT NTN RAN2, InterDigital
Relevant tdocs:
R1-2500326 Draft reply LS on PWS support in RRC_CONNECTED vivo
R1-2500379 Discussion on the LS on PWS support in RRC_CONNECTED for NB-IoT NTN ZTE Corporation, Sanechips
R1-2500448 Discussion on PWS support in RRC_CONNECTED for NB-IoT NTN OPPO
R1-2500449 Draft reply LS on PWS support in RRC_CONNECTED for NB-IoT NTN OPPO
R1-2500510 Discussion on LS from RAN2 on PWS reception in RRC_CONNECTED Nokia, Nokia Shanghai Bell
R1-2500676 Discussion LS on PWS for NB-IoT in RRC connected Ericsson
R1-2500758 Discussion on LS reply on PWS support in RRC_CONNECTED for NB-IoT NTN CATT
R1-2500824 Discussion on RAN2 LS about PWS support in RRC_CONNECTED for NB-IoT NTN Samsung
R1-2501047 Draft Reply for the LS on PWS support in RRC_CONNECTED for NB-IoT NTN InterDigital, Inc.
R1-2501064 Discussion on PWS support in RRC_CONNECTED for NB-IoT NTN Huawei, HiSilicon
R1-2501138 Discussion on response to RAN2 LS on PWS support in RRC_CONNECTED for NB-IoT NTN Qualcomm Incorporated
R1-2501316 Draft reply LS on PWS support in RRC_CONNECTED for NB-IoT NTN Huawei, HiSilicon
Decision: RAN2 is requesting RAN1 input on the support for reception of PWS (ETWS and CMAS) notifications in RRC_CONNECTED for NB-IoT from RAN1 perspective. RAN1 response is necessary and to be handled in agenda item 9.11.4 - Umer (InterDigital).

Rel-19 NTN
R1-2500018 **LS on Ku band numerology** **RAN4, Eutelsat**
Relevant tdocs:
R1-2500325 Draft reply LS on Ku band numerology vivo
R1-2500387 Draft reply LS on Ku band numerology THALES
R1-2500675 Discussion LS on Ku band numerology Ericsson
R1-2500759 Discussion on LS reply on Ku band numerology CATT
R1-2500762 Draft Reply LS to RAN4 on Ku Band Numerology Apple
R1-2500823 Discussion on RAN4 LS about Ku band numerology Samsung
R1-2500986 Draft Reply LS on Ku band numerology Eutelsat Group
R1-2500997 Discussion of LS response related to Ku band numerology Nokia
R1-2501054 Discussion on reply to LS on Ku band numerology Sharp
R1-2501068 Discussion on the possibility to mix FR1 numerology and FR2 numerology for Ku band Huawei, HiSilicon
Decision: RAN4 is requesting RAN1 input on whether there is any issue if different numerologies are used for different Ku bands. RAN1 response is necessary and to be handled in agenda item 9.11 - Keith (Eutelsat).

Rel-19 AI/ML
R1-2500388 **LS on LMF-based AI/ML Positioning for Case 2b** **SA2, vivo**
R1-2500389 **LS on LMF-based AI/ML Positioning for case 3b** **SA2, Ericsson**
Relevant tdocs:
R1-2500695 Draft reply LS on LMF-based AI/ML Positioning for Case 2b vivo
R1-2500571 Discussion on reply LS on LMF-based AI/ML Positioning for Case 2b CATT,CICTCI
R1-2500572 Draft reply LS on LMF-based AI/ML Positioning for Case 3b CATT, CICTCI
R1-2500613 Draft LS reply on LMF-based positioning for case 2b NEC
R1-2500614 Draft LS reply on LMF-based positioning for case 3b NEC
R1-2500616 Discussion on LS on LMF-based AI/ML Positioning for case 3b OPPO
R1-2500617 Discussion on LS on LMF-based AI/ML Positioning for Case 2b OPPO
R1-2500618 [Draft] Reply LS on LMF-based AI/ML Positioning for Case 2b ZTE Corporation, Sanechips
R1-2500619 [Draft] Reply LS on LMF-based AI/ML Positioning for Case 3b ZTE Corporation, Sanechips
R1-2500634 Discussion on SA2 LSs on LMF-based Positioning for Cases 2b and 3b Lenovo
R1-2500695 Draft reply LS on LMF-based AI/ML Positioning for Case 2b vivo
R1-2500696 Draft reply LS on LMF-based AI/ML Positioning for Case 3b vivo
R1-2500708 Discussion for SA2 LS on LMF-based AI/ML Positioning for Case 2b InterDigital, Inc.
R1-2500745 Discussion for SA2 LS on LMF-based AI/ML Positioning for Case 3b InterDigital, Inc.
R1-2501302 Discussion for SA2 LS on LMF-based AI/ML Positioning for Case 2b Ericsson
R1-2501303 Discussion for SA2 LS on LMF-based AI/ML Positioning for Case 3b Ericsson
R1-2501304 Draft reply LS on LMF-based AI/ML Positioning for Case 2b Ericsson
R1-2501305 Draft reply LS on LMF-based AI/ML Positioning for Case 3b Ericsson
R1-2501306 Draft LS reply on LMF-based AI/ML Positioning for Case 2b Nokia
R1-2501308 Draft LS reply on LMF-based AI/ML Positioning for Case 3b Nokia
R1-2501330 Discussion on the LS reply to SA2 on AI/ML Positioning of Case 2b Huawei, HiSilicon
R1-2501331 Discussion on the LS reply to SA2 on AI/ML Positioning of Case 3b Huawei, HiSilicon
Decision:SA2 (in R1-2500388) is requesting RAN1 input on current progress and future work plans on data types and procedures defined for LMF-based AI/ML Positioning case 2b. RAN1 response is necessary.
SA2 (in R1-2500389) is requesting RAN1 input on current progress and future work plans on data types defined for case 3b. RAN1 response is necessary.
To be handled in agenda item 9.1.2 - Huaming (vivo).

Rel-19 NES
R1-2500020 **Reply LS on timeline for On-demand SSB operation on SCell** **RAN4, Ericsson**
Relevant tdocs:
R1-2500948 Discussion on RAN4 LS on timeline for On-demand SSB operation on SCell LG Electronics
R1-2500975 Discussion on RAN4 Reply LS on Timeline for On-demand SSB Operation on Scell Nokia, Nokia Shanghai Bell
Decision: RAN4 response to RAN1 LS in R1-2407565. To be taken into account in agenda item 9.5.

Rel-19 Duplex Evolution

R1-2500019 LS on CSI-RS measurement with SBFD operation RAN4, MediaTek
Relevant tdocs:
R1-2500142 Draft reply LS on CSI-RS measurement with SBFD operation ZTE Corporation, Sanechips
R1-2500158 Draft reply LS on CSI-RS measurement with SBFD operation Spreadtrum, UNISOC
R1-2500190 Draft reply LS on CSI-RS measurement with SBFD operation CATT
R1-2500272 Discussion on RAN4 LS on CSI-RS measurement with SBFD operation CMCC
R1-2500323 Draft reply LS on CSI-RS measurement with SBFD operation vivo
R1-2500458 Discussion on RAN4 LS for CSI-RS measurement with SBFD operation OPPO
R1-2500589 Draft reply to LS on CSI-RS based measurements for SBFD NEC
R1-2500668 Discussion on RAN4 LS on CSI-RS measurement with SBFD operation MediaTek Inc.
R1-2500709 Draft reply LS on CSI-RS measurement with SBFD operation Xiaomi
R1-2500752 Discussion on RAN4 LS on CSI-RS measurement with SBFD operation Ericsson
R1-2500753 [Draft] Reply LS on CSI-RS measurement with SBFD operation Ericsson
R1-2500821 Discussion on LS on CSI-RS measurement with SBFD operation Samsung
R1-2501006 Discussion on RAN4 LS on CSI-RS measurement with SBFD Nokia, Nokia Shanghai Bell
R1-2501136 Draft reply to RAN4 LS on CSI-RS measurement with SBFD operation Qualcomm Incorporated
R1-2501188 Discussion on LS on CSI-RS measurement with SBFD operation NTT DOCOMO, INC.
R1-2501329 Discussion on RAN4 LS on CSI-RS measurement with SBFD operation Huawei, HiSilicon
Decision: RAN4 is requesting clarification from RAN1 on CSI-RS measurement with SBFD operation to progress with their RRM requirement. RAN1 response is necessary and to be handled in agenda item 9.3 - Mohamed (MediaTek)

Rel-19 LP-WUS
R1-2500005 Reply LS to R2-2407921/ R1-2407559 RAN2, Vodafone
Decision: RAN2 response to RAN1 LS in R1-2407559. To be taken into account in agenda item 9.6.

Rel-19 Mobility Enh
R1-2500011 RAN2 agreements on Inter-CU LTM and Conditional LTM RAN2, Lenovo
Decision: RAN2's update on their progress on inter-CU LTM and conditional LTM. To be taken into account in agenda item 9.9.

Rel-19 Mobility Enh
R1-2500027 Reply LS for LS on the support of semi-persistent CSI-RS resource for LTM candidate cells RAN3, CATT
Decision: RAN3 response to RAN1 LS in R1-2409283. RAN3 confirmed it is feasible to specify the signaling for coordination between serving cell and candidate cell(s) on the transmission of semi-persistent CSI-RS(s). To be taken into account in agenda item 9.9.

Rel-19 XR
R1-2500021 Response LS on gaps/restrictions that are caused by RRM measurements RAN4, Qualcomm
Relevant tdocs:
R1-2501342 Discussion on RAN4 LS response on skipping RRM measurement gaps/restrictions Ericsson
Decision: RAN4 response to RAN1 LS in R1-2407561. RAN4 confirmed that Alt 1-1 is feasible from RAN4 perspective with X=5ms as baseline, and X=3ms with optional UE capability. To be taken into account in agenda item 9.10.
Ericsson commented that sending a reply and confirm RAN1 working assumption will be ok.

Rel-19 ISAC
R1-2501370 Availability of Group Reports from ETSI ISAC ISG on use cases and channel models ETSI ISG ISAC, xiaomi, Mitsubishi, ZTE
Decision: ETSI is sharing their reports on ISAC channel modelling approaches, RCS modelling, micro-Doppler modelling, etc. To be taken into account in agenda 9.7.

RAN1 was cc-ed in the following incoming LSs:
R1-2500015 Reply LS to SA5 on AIML data collection RAN2, Nokia
R1-2500016 Reply LS to SA2 on AIML data collection RAN2, InterDigital
R1-2500017 LS on inter-gNB RACH-less HO in NTN RAN2, Nokia

R1-2500022 Reply LS on *carrierBandwidth* configuration for less-than-5MHz carriers RAN4, ZTE Corporation
R1-2500024 LS on *AdditionalSpectrumEmission* in NR SL Pre-configuration RAN4, LG Electronics
R1-2500025 LS on simultaneous operation between GNSS and NR NTN RAN4, Huawei
R1-2500026 LS on UE capability for FDD-FDD inter-band CA simultaneous Rx/Tx RAN4, CATT
R1-2500028 Reply LS on RAN3 vs RAN2 Basketball Match RAN3, ZTE
R1-2500029 Reply on AIML data collection SA5, Nokia
R1-2500030 Reply LS on AIML data collection SA5, Samsung
R1-2500031 Reply LS on AIML data collection SA2, InterDigital
R1-2500032 LS on SSB position restrictions for less-than-5MHz Scells RAN4, Qualcomm

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Incoming LS received during the week:
R1-2501569 Reply LS on the STx2P UE Features RAN2, ZTE
Decision: RAN2 asks RAN1 to confirm whether the combination of the cross-carrier scheduling and MDCI MTRP PDSCH/PUSCH operation is supported in R18. If not supported, RAN1 to answer the following 2 questions:
Q1: Are the FGs 40-6-3a-1 and 40-6-3b-2 also applicable for FR1?
Q2: For both FGs 40-6-3a-1 and 40-6-3b-2 does a UE capability value need to be signaled for each SCS in a given Component Carrier?
To be handled in agenda item 8.2.

6 Pre-Rel-18 E-UTRA Maintenance

R1-2501012 Draft Rel-15 eMTC parameter correction for flexible starting PRB ZTE Corporation, Sanechips
Decision: With or without these changes, nothing is broken in Rel-15 spec. Not considered as an essential change, draft CR is not pursued.

7 Pre-Rel-18 NR Maintenance

[120-Pre-R18-NR] – RAN1 Chair
To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

DSS
R1-2500143 Draft CR on DCI size alignment for UL grant ZTE Corporation, Sanechips
Decision: Comeback on Tuesday. To be moderated by Shuaihua (ZTE).
From Tuesday session
R1-2501488 FL summary of DCI size alignment for UL grant Moderator (ZTE)
Keep the current text as it is in the spec. Draft CR is not pursued.

MIMO
R1-2500594 Draft CR on PTRS in TS 38.211 NEC
R1-2501434 Summary on CR on PTRS in TS 38.211 Moderator (NEC)
Agreement
The TP in the conclusion section of R1-2501434 is agreed for Rel-18 as alignment CR on TS38.211.

R1-2500329 Draft CR on QCL assumption of AP CSI-RS vivo
R1-2500330 Draft mirror CR on QCL assumption of AP CSI-RS (Rel-18 mirror) vivo
Decision: Comeback on Tuesday. To be moderated by Rakesh (vivo).
From Tuesday session
R1-2501491 Summary of discussion on QCL assumption Moderator (vivo)
Agreement
TP#2 in R1-2501491 is agreed as alignment CR for Rel-17 TS38.214.

NR Extension to 71GHz
R1-2501268 Correction to slot aggregation configuration with multi-PUSCH Nokia
Decision: Comeback on Tuesday. To be moderated by Karri (Nokia).

R1-2501514 [120-Pre-R18-NR] Summary for slot aggregation configuration with multi-PUSCH Moderator (Nokia)
From Tuesday session
Agreement
Fix the specification to enable the operation according to the RAN1#107-e agreement by clarifying that the *pusch-AggregationFactor* is not configured with *pusch-TimeDomainAllocationListForMultiPUSCH* (i.e. Rel-16 behavior), if *extendedK2* (i.e. Rel-17 operation) is not configured. This would need to be captured in the first statement in TS 38.214:

> If a UE is configured with higher layer parameter *pusch-TimeDomainAllocationListForMultiPUSCH* but not configured with *extendedK2* in *pusch-TimeDomainAllocationListForMultiPUSCH*, the UE does not expect to be configured with *pusch-AggregationFactor*.

And propagate the same change also to the TS 38.331 RRC parameter field description:

> ***pusch-TimeDomainAllocationListForMultiPUSCH***
> Configuration of the time domain resource allocation (TDRA) table for multiple PUSCH (see TS 38.214 [19], clause 6.1.2). The network configures at most 64 rows in this TDRA table in *PUSCH-TimeDomainResourceAllocationList-r16* configured by this field. This field is not configured simultaneously with *pusch-AggregationFactor* if *extendedK2* is not configured. The network does not configure the *pusch-TimeDomainAllocationListForMultiPUSCH-r16* simultaneously with the *pusch-TimeDomainAllocationListDCI-0-1-r16*. The network does not configure the *pusch-TimeDomainAllocationListForMultiPUSCH-r16* simultaneously with the *numberOfSlotsTBoMS-r17*.

Send an LS to RAN2 to lift the restriction on RRC specification. Check draft CR and LS on Thursday.

R1-2501527 Draft CR on Slot aggregation configuration with multi-PUSCH Moderator (Nokia)
From Thursday session
Agreement
The TP in R1-2501527 is agreed for Rel-17. Final CRs in:
R1-2501579 CR on Slot aggregation configuration with multi-PUSCH Moderator (Nokia), Ericsson
(Rel-17, 38.214, NR_ext_to_71GHz-Core, CR0656, Cat F) is agreed.
R1-2501580 CR on Slot aggregation configuration with multi-PUSCH Moderator (Nokia), Ericsson
(Rel-18, 38.214, NR_ext_to_71GHz-Core, CR0657, Cat A) is agreed.

R1-2501526 Draft LS on Slot aggregation configuration with multi-PUSCH Moderator (Nokia)
Decision: The draft LS is endorsed. Final LS is approved in R1-2501593.

Coverage Enhancement
R1-2500757 Draft CR to 38.214 for alignment of parameter name pusch-FrequencyHoppingInterval Ericsson
Parameter name alignment between 38.331 and 38.214.
Agreement
The TP in R1-2500757 is agreed for Rel-17 as alignment CR on TS38.214.

UE Power Savings
R1-2501035 On DCI 1_0/0_0 monitoring with Rel-17 SSSG switching MediaTek Inc.
Decision: Comeback on Tuesday. To be moderated by James (MediaTek).

R1-2501036 On SSSG reset with RRC reconfiguration MediaTek Inc.
Decision: Comeback on Tuesday. To be moderated by James (MediaTek).

R1-2501037 On R17 SSSG reset with BWP switch triggered by RA procedure MediaTek Inc.
Decision: Comeback on Tuesday. To be moderated by James (MediaTek).

R1-2501383 Moderator summary 1 of SSSG switching on R17 UE power saving Moderator (MediaTek)
From Tuesday session
Conclusion
It is common understanding in RAN1 that when the fields *PDCCH-Config* and *searchSpaceGroupIdList*/*searchSpaceGroupIdList-r17* are both absent in a RRC reconfiguration which does not indicate BWP change, UE does not reset to SSSG 0.

Final summary in R1-2501384.

New RAT, TEI
R1-2500765 **UCI multiplexing procedure within BWP switch delay Apple**
Decision: Comeback on Tuesday. To be moderated by Ali (Apple).
R1-2501515 **FL summary #1 of UCI multiplexing procedure within BWP switch delay** **Moderator (Apple)**
From Tuesday session
Proposal for Conclusion: It is RAN1 understanding that within BWP switch delay,
- UE first applies multiplexing between a grant on the cell with BWP switch, and another grant on another cell. No spec change is needed.
- UE expects the indicated PUCCH resource for P-CSI in both old and new BWPs to occupy the same set of symbols

Comeback on Thursday.
Thursday decision: No conclusion could be achieved.

R1-2501326 **Discussion on Type0-PDCCH CSS set monitoring** **Huawei, HiSilicon**
Decision: Comeback on Tuesday. To be moderated by Xinghua (Huawei).
Tuesday decision: Further discussion has concluded that there is no need to clarify the UE behavior related to Type0-PDCCH CSS monitoring. Not pursued.

R1-2501267 **Initialization of PRACH preamble generation for restricted set type B** **Nokia**
Decision: Comeback on Tuesday. To be moderated by Karri (Nokia).
From Tuesday session:
No conclusion. More time is needed for the companies to check and understand the two interpretations. Draft CR is not pursued in this meeting.

Follow up discussion on soft buffer size determination for 1024QAM rank restricted UE from RAN1#119.
R1-2501266 Soft buffer size determination for 1024QAM rank restricted UE Nokia
R1-2501513 [120-Pre-R18-NR] Summary for soft buffer for 1024QAM Moderator (Nokia)
R1-2501567 **[120-Pre-R18-NR] Summary#2 for soft buffer for 1024QAM** **Moderator (Nokia)**
R1-2501568 **Correction on LBRM TBS determination for UE indicating 1024QAM rank restricted capability** **Moderator (Nokia)**
From Thursday session
Agreement
The TP in R1-2501568 is agreed for Rel-17. Final CRs in:
R1-2501581 **Correction on LBRM TBS determination for UE indicating 1024QAM rank restricted capability** **Moderator (Nokia), Ericsson, xiaomi**
(Rel-17, 38.212, NR_DL1024QAM_FR1-Core, CR0211, Cat F) is agreed.
R1-2501582 **Correction on LBRM TBS determination for UE indicating 1024QAM rank restricted capability** **Moderator (Nokia), Ericsson, xiaomi**
(Rel-18, 38.212, NR_DL1024QAM_FR1-Core, CR0212, Cat A) is agreed.

Follow up discussion on power control for 2-step RACH from RAN1#119.
R1-2501067 Discussion on p0-nominal Huawei, HiSilicon
R1-2501320 Draft CRs for p0-nominal Huawei, HiSilicon
R1-2501516 **Moderator summary on discussion of p0-nominal** **Moderator (Huawei)**
From Thursday session
Conclusion
For a cell configured with 2-step RACH feature:
- When UE dedicated RRC configuration is provided to a UE for the cell, all the three parameters P0-PUSCH-AlphaSet, p0-NominalWithoutGrant and p0-NominalWithGrant are expected to be configured.

Follow up discussion on skipping uplink transmission in case of BWP switching from RAN1#119
R1-2501321 Discussion on skipping uplink transmission in case of BWP switching Huawei, HiSilicon
R1-2501322 Correction on skipping uplink transmission in case of BWP switching Huawei, HiSilicon

R1-2501510 Summary #1 of discussion on skipping uplink transmission in case of BWP switching Moderator (Huawei)
R1-2501565 Summary #2 of discussion on skipping uplink transmission in case of BWP switching Moderator (Huawei)
From Thursday session
No consensus whether the issue could be resolved by implementation vs gNB implementation.

Final summary in R1-2501566.

8 Maintenance on Release 18

8.1 Maintenance on Rel-18 work items

[120-R18-Maintenance] – RAN1 Chair
To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

MIMO

R1-2500200 Discussion on missing RRC parameter for PRACH transmission in 2TA enhancement CATT
R1-2501422 Summary on RRC parameter for PRACH transmission in 2TA Moderator (CATT)
From Monday session
Agreement
- Introduce a new parameter to RACH-ConfigTwoTA-r18 for indicating the restricted sets of cyclic shift for PRACH associated with additional configured PCI.

Agreement
- Send an LS to RAN2 to ask for adding a new RRC parameter in RACH-ConfigTwoTA-r18 for indicating the restricted sets of cyclic shift for PRACH associated with additional configured PCI.

Comeback to check draft LS on Tuesday (Jiayi)
R1-2501476 Summary#2 on RRC parameter for PRACH transmission in 2TA Moderator (CATT)
Decision: The draft LS in section 2 of R1-2501476 is endorsed.
R1-2501540 LS on RRC parameter for PRACH transmission in 2TA RAN1, CATT
Decision: The LS is approved.

R1-2500536 Draft CR on transmission power reduction for STxMP in TS 38.213 ZTE Corporation, Sanechips
R1-2500537 Discussion on transmission power reduction for STxMP in TS 38.213 ZTE Corporation, Sanechips
Decision: Comeback on Tuesday. To be moderated by Yang (ZTE).
R1-2501462 Summary of discussion on transmission power reduction for STxMP in TS 38.213 Moderator (ZTE Corporation)
Decision: Comeback on Thursday.
R1-2501597 Summary#2 of discussion on transmission power reduction for STxMP in TS 38.213 Moderator (ZTE Corporation)
From Thursday session
No consensus - different views from companies requiring more time to double check the discrepancy between RAN1 and RAN4 on this issue.

R1-2500831 Draft CR on timeline for STxMP Samsung
R1-2501444 Summary of discussion on the timeline issue for STxMP Moderator (Samsung)
Decision: Comeback on Tuesday. To be moderated by Sa (Samsung).
R1-2501489 Summary #2 of discussion on the timeline issue for STxMP Moderator (Samsung)
From Tuesday session
Agreement
The following TP is agreed for 38.214 section 6.4.
----- Start of TP -----
If the first uplink symbol in the PUSCH allocation for a transport block, including the DM-RS, as defined by the slot offset K_2 and K_{offset}, if configured, and the start S and length L of the PUSCH allocation indicated by '*Time domain resource assignment*' of the scheduling DCI and including the effect of the timing advance, is no earlier than at symbol L_2, where L_2 is defined as the next uplink symbol with its CP starting $T_{proc,2}$ =

$max\left((N_2 + d_{2,1} + d_2 + d_3)(2048 + 144) \cdot \kappa 2^{-\mu} \cdot T_C + T_{ext} + T_{switch}, d_{2,2}\right)$ after the end of the reception of the last symbol of the PDCCH carrying the DCI scheduling the PUSCH, then the UE shall transmit the transport block. When the PDCCH reception includes two PDCCH candidates from two respective search space sets, as described in clause 10.1 of [6, TS 38.213], for the purpose of determining the last symbol of the PDCCH carrying the DCI scheduling the PUSCH, the PDCCH candidate that ends later in time is used.

....

- d_3 is set as reported by UE if the UE is configured with *sTx-2Panel* and the PUSCH ~~is~~ associated with a value of *coresetPoolIndex* is fully/partially overlapping in time domain and is fully/partially/non-overlapping in frequency domain with another PUSCH that is associated with a different value of *coresetPoolIndex* on a same serving cell; otherwise $d_3 = 0$.

----- End of TP -----

Final CR is in:

R1-2501564 **CR on timeline for STxMP** **Moderator (Samsung), ZTE, Huawei, HiSilicon**
(Rel-18, 38.214, NR_MIMO_evo_DL_UL-Core, CR0655, Cat F) is agreed.

R1-2501189 **Draft CR on Rel.18 DMRS ports indication by DCI format 0_3/1_3/4_2** **NTT DOCOMO, INC., Huawei, HiSilicon**
From Monday session
Agreement
TP in R1-2501189 is agreed for alignment CR on Rel-18 TS38.212.

R1-2501346 **Discussion on Rel.18 DMRS ports indication by DCI format 0_3/1_3/4_2** **Huawei, HiSilicon**
Decision: Comeback on Tuesday. To be moderated by Zhening (Huawei).
R1-2501419 **Summary on Rel.18 DMRS ports indication by DCI format 0_3/1_3/4_2** **Moderator (NTT DOCOMO)**
Comeback on Thursday.
R1-2501557 **Summary#2 on Rel.18 DMRS ports indication by DCI format 0_3/1_3/4_2** **Moderator (NTT DOCOMO)**
From Thursday session
Agreement
Introduce a UE capability to support the simultaneous configuration of Rel-18 DMRS ports and DCI format 1_3.

- UE capability reporting granularity is per band.
- Note: It is RAN1 understanding that no other UE capability for simultaneous configuration of Rel.18 DMRS and Rel.18 DCI format 1_3 function need to be introduced.

MIMO (Alignment/editorial issues)

R1-2500430 **Draft CR on RRC paramters for uplink multi-panel transmission** **OPPO**
From Monday session
Agreement
TP is agreed for alignment CR on Rel-18 3TS8.214.

R1-2500535 **Draft CR on the formula of further enhanced further enhanced Type II port selection codebook for CJTZTE Corporation, Sanechips**
From Monday session
Agreement
TP is agreed for alignment CR on Rel-18 3TS8.214.

R1-2500596 **Draft CR on CSI parameters in TS38.214** **NEC**
From Monday session
Agreement
TP is agreed for alignment CR on Rel-18 3TS8.214.

R1-2500597 **Draft CR on priority formula for predicted CSI in TS38.214** **NEC**
From Monday session
Agreement
TP is agreed for alignment CR on Rel-18 3TS8.214.

R1-2500813 Draft CR on RRC parameter alignment of Rel-18 MIMO enhancements in TS 38.214 ZTE Corporation, Sanechips
Decision: Comeback on Tuesday.
R1-2501461 Draft CR on RRC parameter alignment of Rel-18 MIMO enhancements in TS 38.214 Moderator (ZTE Corporation), Sanechips
From Tuesday session
Agreement
TP in R1-2501461 is agreed for alignment CR on Rel-18 38.214.

R1-2500832 Draft CR on Rel-18 Type-II codebook refinement for high/medium velocities and the TRS-based TDCP reporting Samsung
Decision: Comeback on Tuesday.
R1-2501480 Draft CR on Rel-18 Type-II codebook refinement for high/medium velocities and the TRS-based TDCP reporting Samsung
From Tuesday session
Agreement
TP in R1-2501480 is agreed for alignment CR on Rel-18 38.214.

R1-2500941 Correction on downlink DM-RS resource mapping Nokia
Decision: Comeback on Tuesday.
From Tuesday session
Agreement
TP in R1-2500941 is agreed for alignment CR on Rel-18 38.211.

R1-2501243 Draft CR on RRC name alignment of Rel-18 MIMO enhancement in TS 38.213 ZTE Corporation, Sanechips
From Monday session
Agreement
TP in R1-2501243 is agreed for alignment CR on Rel-18 38.213.

R1-2501328 Correction on notation of number of TRPs for CJT codebooks Huawei, HiSilicon
Decision: Not essential. Draft CR is not pursued.

From AI5
R1-2500012 LS on UL 8Tx RAN2, Samsung
Decision: RAN2 is requesting RAN1 input to finalize the MAC specification support of UL 8TX. RAN1 response is necessary - Sa (Samsung).
R1-2501490 Summary of reply LS on 8Tx Moderator (Samsung)
R1-2501563 Summary#2 of reply LS on 8Tx Moderator (Samsung)
From Thursday session
Agreement
- Proposal 1: Regarding Item 1 of the incoming LS R1-2500012, RAN1 is fine with RAN2's assumption, with the understanding that it is RAN2 internal assumption to implement RAN2 specifications on UL 8Tx. There is no RAN1 specification change needed based on the RAN2 assumption.
- Proposal 2: Regarding Item 2 of the incoming LS R1-2500012, for UL 8Tx 2TB transmission scheduled by one DCI, for the case where MAC entity only generates a single TB due to the uplink skipping, this case was not considered in previous RAN1 meetings and the current RAN1 specifications do not define UE behaviours to support this case.

Capture the above in a response LS to RAN2.
R1-2501618 [Draft] Reply LS on UL 8Tx Moderator (Samsung)
Friday decision: The draft LS is endorsed. Final LS is approved in R1-2501636.

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R1-2500013 LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP RAN2, CATT
Decision: RAN2 is requesting RAN1 input on simultaneous unified TCI state(s) update for a list of cells. RAN1 response is necessary - Jiayi (CATT)
R1-2501477 Summary on LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP Moderator (CATT)
From Tuesday session

Agreement
The following is agreed as response to RAN2 LS in R1-2500013.
- The answer to question 1A is NO

Comeback on Thursday for the responses to the remaining questions.
R1-2501538 Summary#2 on LS on differentiation of sDCI mTRP and sTRP Moderator (CATT)
From Thursday session
No RAN1 consensus on differentiation the cells operated as sTRP vs sDCI mTRP.
Send an LS to RAN2 to inform them of the above response.
R1-2501611 Reply LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP RAN1, CATT
Friday decision: The LS is approved.

Network Energy Savings

R1-2500543 Draft CR on PMI prediction during cell DTX Google
R1-2500544 Discussion on PMI prediction during cell DTX Google
Decision: Comeback on Tuesday. To be moderated by Yushu (Google).
R1-2501497 Summary on PMI prediction during cell DTX Moderator (Google)
From Tuesday session
Draft CR in R1-2500543 is not pursued.

R1-2500828 Correction on the zero padding description for CSI part 1 Samsung
R1-2500829 Discussion on the zero padding for CSI part 1 Samsung
From Monday session
Agreement
TP in R1-2500828 is agreed in principle. Companies to check draft CR in R1-2501454.
R1-2501454 Draft Final CR on the zero padding for CSI part 1 on NES Moderator (Samsung)
From Thursday session
R1-2501559 Correction on the zero padding for CSI part 1 on NES Moderator (Samsung)
(Rel-18, 38.214, Netw_Energy_NR-Core, CR0654, Cat F) is agreed.

Rel-18 maintenance on Mobility Enhancement, Multi-Carrier Enhancement, Coverage discussed in RAN1 session chaired by Xiaodong.

Mobility Enhancement

R1-2501543 Session notes for 8.1 (Maintenance on Rel-18 work items - Mobility Enhancement) Ad-Hoc Chair (CMCC)
Friday decision: The session notes are endorsed and contents reflected below.

R1-2500332 Discussion on L1-RSRP reporting for unmeasured LTM L1-RSRP resources vivo
R1-2500333 Draft CR on L1-RSRP reporting for unmeasured LTM L1-RSRP resources vivo
R1-2500814 Discussion on misalignment of aperiodic CSI reporting configuration in TS 38.214 and TS 38.331 for LTM ZTE Corporation, Sanechips

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R1-2500334 Draft CR on the PRACH retransmission indicator field included in the PDCCH order vivo
R1-2501492 Summary of discussion on PRACH retransmission indicator field included in the PDCCH order Moderator (vivo)
From Tuesday session
Agreement
Adopt the following TP to Section 7.3.1.2.1, TS38.212 in principle.
----- Start of TP -----
----------Omitted Text----------
For PRACH retransmission indicator - 0 or 1 bit
1bit if the UE is configured with higher layer parameter *EarlyUL-SyncConfig*. This field indicates initial transmission or retransmission of PRACH according to Table 7.3.1.2.1-3 if the cell indicated by Cell indicator field is a candidate cell, and this field is reserved if ~~the cell indicated by~~ the value of Cell indicator field is ~~a serving cell but not a candidate cell~~ zero.
----------Omitted Text----------
----- End of TP -----
R1-2501504 Draft CR on the PRACH retransmission indicator field included in the PDCCH order Moderator (vivo), Nokia
Agreement

Draft CR in R1-2501504 is endorsed in principle. Final CR (Rel-18, 38.212, NR_Mob_enh2-Core, CR0213, Cat F) is agreed in R1-2501637.

R1-2501314 Draft CR on LTM PRACH and serving cell UL transmission in a same band MediaTek, Apple, NTT DOCOMO, Ericsson, Nokia, ZTE, Huawei, Vivo, Google
R1-2501450 Summary on LTM PRACH and serving cell UL transmission Moderator (MediaTek)
R1-2501558 Draft CR on LTM PRACH and serving cell UL transmission in a same band Moderator (MediaTek)
From Wednesday session
Agreement
The Draft CR in R1-2501558 is endorsed in principle. Final CR in:
R1-2501575 CR on LTM PRACH and serving cell UL transmission in a same band Moderator (MediaTek), Apple, NTT DOCOMO, Ericsson, Nokia, ZTE, Huawei, Vivo, Google, Fujitsu, Samsung
(Rel-18, 38.213, NR_Mob_enh2-Core, CR0695, Cat F) is agreed.

R1-2501578 Final summary on LTM PRACH and serving cell UL transmission Moderator (MediaTek)

Multi-Carrier Enhancement

R1-2501544 Session notes for 8.1 (Maintenance on Rel-18 work items – Multi-Carrier) Ad-Hoc Chair (CMCC)
Friday decision: The session notes are endorsed and contents reflected below.

R1-2501065 Maintenance of Rel-18 Multicarrier Enhancements Huawei, HiSilicon
R1-2501554 Corrections on determination of DCI format 0_3 for unpaired spectrum operation Huawei, HiSilicon (rev of R1-2501317)
R1-2501318 Correction on SCell dormancy indication case 2 for DCI format 1_3 Huawei, HiSilicon

R1-2501532 Corrections on determination of DCI format 0_3 in case of dormancy Huawei, HiSilicon
From Wednesday session
Agreement
The Draft CR in R1-2501532 is endorsed in principle. Final CR in:
R1-2501572 Corrections on determination of DCI format 0_3 in case of dormancy Huawei, HiSilicon, Ericsson, Fujitsu, Nokia, Samsung, ZTE
(Rel-18, 38.212, NR_MC_enh-Core, CR0210, Cat F) is agreed.

Coverage Enhancement

R1-2501542 Session notes for 8.1 (Maintenance on Rel-18 work items - Coverage enhancements) Ad-Hoc Chair (CMCC)
Friday decision: The session notes are endorsed and contents reflected below.

R1-2500331 Draft CR on time period determination for PRACH repetition vivo
R1-2501319 Draft CR on the time period for R18 preamble repetition Huawei, HiSilicon
R1-2501423 Summary#1 of discussion on the time period for R18 preamble repetition Moderator (Huawei)
From Wednesday session
R1-2501533 Draft CR on the time period for R18 preamble repetition Moderator (Huawei), vivo, ZTE
Agreement
The Draft CR in R1-2501533 to Section 8.1 R18 TS38.213 is endorsed as alignment CR.

Rel-18 maintenance on NR Extension to 71GHz, IoT-NTN/NR-NTN, Positioning, Sidelink discussed in RAN1 session chaired by David.

R1-2501545 Session notes for 8.1 Maintenance on Rel-18 work items – Positioning, Sidelink, IoT NTN, NR NTN, NR extension to 71 GHz Ad-Hoc Chair (Huawei)
Friday decision: The session notes are endorsed and contents reflected below.

NR Extension to 71GHz

R1-2500335 Correction on transition time of UL BWP change triggered by DCI format 0_1 scheduling multiple PUSCHs vivo

From Monday session
Agreement
The draft CR in R1-2500335 is endorsed. Final CR in:
R1-2501501 Correction on transition time of UL BWP change triggered by DCI format 0_1 scheduling multiple PUSCHs vivo, ASUSTek
(Rel-18, 38.213, TEI18, NR_ext_to_71GHz-Core, CR0693, Cat F) is agreed.
Note: This Cat F CR is a correction of previous Cat A CR#0661 in Tdoc R1-2407440.

IoT-NTN/NR-NTN

R1-2501234 Correction on RACH-less handover for NTN ASUSTeK
Decision: Comeback on Tuesday, to be moderated by Denny (ASUSTeK)
R1-2501507 Summary of RACH-less HO for NTN Moderator (ASUSTeK)
From Wednesday session
Agreement
The TP in section 4 of R1-2501507 is endorsed. Final CR in:
R1-2501596 Correction on RACH-less handover for NTN Moderator(ASUSTeK), Huawei, Hisilicon
(Rel-18, 38.213, NR_NTN_enh-Core, CR0696, Cat F) is agreed.

R1-2500336 Discussion on E-UTRAN measurement in IoT-NTN vivo
R1-2500511 Discussion on E-UTRAN measurement in IoT NTN Nokia, Nokia Shanghai Bell
R1-2501421 Summary of discussion on E-UTRAN measurement in IoT NTN Moderator (vivo)
Presented in Monday session.

R1-2501486 Summary#2 of discussion on E-UTRAN measurement in IoT NTN Moderator (vivo)
From Tuesday session
Agreement
Send an LS to ask RAN4 the following:

- Whether any of the following measurement quantities, described in clause 5.2 of TS 36.214, are applicable for NB-IoT-based NTN and/or eMTC-based NTN:
  - DL RS TX power
  - Received Interference Power
  - Thermal Noise Power
- For the measurement quantitie(s) that are applicable, if any, whether reference point(s) need to be defined.

Comeback for draft LS on Wednesday.
R1-2501511 Draft LS on measurement quantities for IoT NTN Moderator (vivo)
Decision: The draft LS is endorsed. Final LS is approved in R1-2501512.

From AI5
R1-2500006 LS on soft satellite switch with re-sync RAN2, Huawei
Decision: RAN2 is requesting RAN1 input on possible introduction of a temporarily shifted SSB pattern during [t-ServiceStart, t-Service] for soft satellite switching with re-sync. RAN1 response is necessary - Xiaolei (Huawei).
R1-2501505 Summary of discussion on the LS reply of soft-satellite switching with re-sync Moderator (Huawei)
Presented in Tuesday session.

R1-2501541 Summary#2 of discussion on the LS reply of soft-satellite switching with re-sync Moderator (Huawei)
From Wednesday session
Agreement
The following RAN1 reply to RAN2 is endorsed.
RAN1 would like to ask RAN2 for further clarification on the question and assumptions below:

- Is RAN2 asking whether a Rel-18 UE (based on current Rel-18 RAN1 specifications) can maintain the pre-acquired DL synchronization started between [t-ServiceStart, t-Service] with target satellite if the SSB pattern (e.g., SSB offset) is changed (determined by ssb-TimeOffset) at t-Service, or whether RAN2 assumes that some RAN1 specification work may be needed?
- RAN1 assumes that the meaning of “the SSB pattern is changed” is that only the time offset is changed (may be determined by ssb-TimeOffset).
- RAN1 assumes that the SIB19 of the source satellite includes ephemeris information of the target satellite, and that such information will not get outdated by the time the satellite switch occurs.

R1-2501506 Draft reply LS on soft satellite switch with re-sync Moderator (Huawei)
Decision: The draft LS is endorsed. Final LS is approved in R1-2501606.

Positioning

R1-2501463 Summary#1 of PSCCH DMRS sequence generation in a dedicated SL PRS resource pool Moderator (CATT)
R1-2500199 Draft CR on PSCCH DMRS sequence generation in a dedicated SL PRS resource pool CATT, CICTCI
From Monday session
Agreement
The draft CR in R1-2500199 is endorsed. Final CR in:
R1-2501502 CR on PSCCH DMRS sequence generation in a dedicated SL PRS resource pool Moderator (CATT), CATT, CICTCI
(Rel-18, 38.211, NR_pos_enh2-Core, CR0148, Cat F) is agreed.

From AI5
R1-2500014 LS on co-existence of SL-CA and SL PRS transmission/reception for Dedicated SL-PRS resource pool RAN2, vivo
Decision: RAN2 is requesting RAN1 input on the restriction of SL-PRS transmission/reception in a dedicated SL-PRS resource pool in a band (pre-)configured with SL CA. RAN1 response is necessary - Yuan (vivo).
R1-2501448 Summary#1 on LS of co-existence of SL-CA and SL PRS transmission/reception in dedicated SL-PRS resource pool Moderator (vivo)
R1-2501449 [Draft] Reply LS on co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool Moderator (vivo)
From Monday session
Agreement
The draft reply LS in R1-2501449 is endorsed. Final LS is approved in R1-2501479**.**

Sidelink

R1-2500198 Discussion on the usage of two candidate starting symbols CATT, CICTCI
R1-2500432 Discussion on maintenance of 2 candidate starting symbols in SL-U OPPO
R1-2500581 Draft CR on the application condition of two candidate starting symbols in TS 38.213 ZTE Corporation, Sanechips
R1-2500830 Discussion on two candidate starting symbols for sidelink unlicensed Samsung
R1-2501438 FLS#1 on 2 candidate starting symbols in TS 38.213 Moderator (ZTE)
From Monday session
Conclusion (For Issue 1 in R1-2501438)
The specifications are aligned with the following, and no CR is necessary:
- *sl-StartSymbol* is only used for operation without shared spectrum channel access.
- *sl-StartingSymbolFirst* and/or *sl-StartingSymbolSecond* are only used for operation with shared spectrum channel access.

Conclusion (For Issue 2 in R1-2501438)
sl-StartingSymbolFirst will always be provided for operation with shared spectrum channel access, either as a default value or explicitly (pre)configured.

R1-2500580 Draft CR on power control of PSFCH in TS 38.213 ZTE Corporation, Sanechips
R1-2501439 FLS#1 on power control of PSFCH in TS 38.213 Moderator (ZTE)
From Monday session
Agreement
The TP under Alt1 in section 3 of R1-2501439 is endorsed. Final CR in:
R1-2501503 CR on power control of PSFCH in TS 38.213 Moderator (ZTE), ZTE Corporation, Sanechips, Samsung, vivo, OPPO
(Rel-18, 38.213, NR_SL_enh2-Core, CR0694, Cat F) is agreed.

R1-2500585 Draft CR on RRC parameter alignment in TS 38.213 NEC
R1-2501445 Moderator summary of RRC parameter alignment in TS 38.213 Moderator (NEC)
From Monday session

Agreement
The draft CR in R1-2500585 is endorsed as part of the Rel-18 alignment CR for TS 38.213.

R1-2500586 **Draft CR on RRC parameter alignment in TS 38.214 NEC**
R1-2501446 **Moderator summary of RRC parameter alignment in TS 38.214 Moderator (NEC)**
From Monday session
Agreement
The draft CR in R1-2500586 is endorsed as part of the Rel-18 alignment CR for TS 38.214.

R1-2500595 Draft CR on TAG for sidelink resource allocation mode 1 in TS 38.214 NEC
R1-2501447 **Summary on Draft CR on TAG for sidelink resource allocation mode 1 in TS 38.214 Moderator (NEC)**
Monday decision: Comeback on Tuesday, to be moderated by Peng (NEC)
R1-2501498 **Summary#2 on Draft CR on TAG for sidelink resource allocation mode 1 in TS 38.214 Moderator (NEC)**
From Tuesday session
Conclusion (confirmed on Wednesday)
From RAN1 perspective, for Rel-18 sidelink mode 1 operation, the serving cell configured with two TAGs was never discussed. RAN1 has not determined whether this can already be supported by RAN1 specifications.

8.2 Rel-18 UE Features

R1-2501387 **Session Notes of AI 8.2 Ad-Hoc Chair (NTT DOCOMO, INC.)**
Friday decision: The session notes are endorsed and contents reflected below.

[120-R18-UE_features] – Hiroki (NTT DOCOMO)
Email discussion on Rel-18 UE features
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500833 UE features for Rel-18 work items Samsung
R1-2501325 Rel-18 UE Features Huawei, HiSilicon
R1-2501385 **Summary of discussion on Rel-18 NR UE features Moderator (NTT DOCOMO, INC.)**

R1-2501386 **Summary#2 of discussion on Rel-18 NR UE features Moderator (NTT DOCOMO, INC.)**
Agreement:
Introduce following FG for UE 8Tx PUSCH processing capability for codebook.
- FG name
  - UE 8Tx PUSCH processing capability for codebook
- Component
  - 1. Require additional timeline to process two TBs for 8Tx PUSCH
- Prerequisite
  - 40-7-1 ~~or 40-7-2~~
- Consequence if not supported
  - UE should process two TBs within legacy timeline
- Type
  - Per FS
- Need of FR1/FR2 differentiation
  - n/a
- Note
  - candidate values:
  - UE reports candidate value independently for each SCS in unit of symbols
  - For 15kHz SCS: {1,2,4}
  - For 30kHz SCS: {1,2,4,8}
  - For 60kHz SCS: {2,4,8,16}
  - For 120kHz SCS: {4,8,16,32}
  - For 480kHz SCS: {16,32,64, 128}
  - For 960kHz SCS: {32,64,128, 256}
  - Note: SCS above is the minimum between SCS of the scheduling DCI and SCS of the scheduled PUSCH

    - Note that the proposed UE capability is reported by UE only when UE reports the Maximum number of PUSCH MIMO layers to be {5,6,7,8}
- Mandatory or optional
    - Optional with capability signaling

Agreement:
Introduce following FG for UE 8Tx PUSCH processing capability for non-codebook.
- FG name
    - UE 8Tx PUSCH processing capability for non-codebook
- Component
    - 1. Require additional timeline to process two TBs for 8Tx PUSCH
- Prerequisite
    - ~~40-7-1 or~~ 40-7-2
- Consequence if not supported
    - UE should process two TBs within legacy timeline
- Type
    - Per FS
- Need of FR1/FR2 differentiation
    - n/a
- Note
    - candidate values:
    - UE reports candidate value independently for each SCS in unit of symbols
    - For 15kHz SCS: {1,2,4}
    - For 30kHz SCS: {1,2,4,8}
    - For 60kHz SCS: {2,4,8,16}
    - For 120kHz SCS: {4,8,16,32}
    - For 480kHz SCS: {16,32,64, 128}
    - For 960kHz SCS: {32,64,128, 256}
    - Note: SCS above is the minimum between SCS of the scheduling DCI and SCS of the scheduled PUSCH
    - Note that the proposed UE capability is reported by UE only when UE reports the Maximum number of PUSCH MIMO layers to be {5,6,7,8}
- Mandatory or optional
    - Optional with capability signaling

Agreement:
Reply to RAN2 that following per-band capabilities refer to the source band.
- ue-TA-Measurement-r18
- ltm-BeamIndicationJointTCI-r18
- ltm-BeamIndicationSeparateTCI-r18
- rach-EarlyTA-Measurement-r18
- ltm-MAC-CE-JointTCI-r18
- ltm-MAC-CE-SeparateTCI-r18
- ta-IndicationCellSwitch-r18

R1-2501631 Summary#3 of discussion on Rel-18 NR UE features Moderator (NTT DOCOMO, INC.)

Agreement:
Introduce following FG for simultaneous Configuration of Rel-18 DL DMRS and DCI format 1_3.
- FG name
    - Simultaneous Configuration of Rel-18 DL DMRS and DCI format 1_3
- Component
    - Indicates whether the UE can be configured with both Rel-18 DL DMRS and DCI format 1_3
- Prerequisite
    - one of {49-1, 49-1b} and 40-4-1
- Consequence if not supported
    - Rel-18 DL DMRS and DCI format 1_3 should not be configured simultaneously regardless of on scheduling cell or on the scheduled cell(s).
- Type
    - Per band
- Note
    - None

- Mandatory or optional
  - Optional with capability signaling

Agreement:
Reply to RAN2 with following RAN1 answers.
- Answer to Q1: No, the FGs are only applicable to FR2. RAN1 has not agreed to support cross-carrier scheduling of MDCI based STxMP PUSCH.
- Answer to Q2: the capability value does not need to be reported for each of FR2 SCS in a given CC. UE reports candidate value for the applicable SCS in unit of symbols.
- Consequently, the FGs are updated as below.

40-6-3a-1	1. Require additional timeline to process multiple TBs for codebook multi-DCI based STx2P PUSCH+PUSCH for DG+DG Note. This FG can be applied for CG+DG also if UE can support those FG.	Per FSPC	FR2 only	candidate values: UE reports candidate value ~~independently~~ for the applicable ~~each~~ SCS in unit of symbols ~~For 15kHz SCS: {1,2,4}~~ ~~For 30kHz SCS: {1,2,4,8}~~ For 60kHz SCS: {2,4,8,16} For 120kHz SCS: {4,8,16,32} For 480kHz SCS: {16,32,64,128} For 960kHz SCS: {32,64,128,256} ~~Note: SCS above is the minimum between SCS of the scheduling DCI and SCS of the scheduled PUSCH.~~
40-6-3b-2	1. Require additional timeline to process multiple TBs for non-codebook multi-DCI based STx2P PUSCH+PUSCH for DG+DG Note. This FG can be applied for CG+DG also if UE can support those FG.	Per FSPC	FR2 only	candidate values: UE reports candidate value ~~independently~~ for the applicable ~~each~~ SCS in unit of symbols ~~For 15kHz SCS: {1,2,4}~~ ~~For 30kHz SCS: {1,2,4,8}~~ For 60kHz SCS: {2,4,8,16} For 120kHz SCS: {4,8,16,32} For 480kHz SCS: {16,32,64,128} For 960kHz SCS: {32,64,128,256} ~~Note: SCS above is the minimum between SCS of the scheduling DCI and SCS of the scheduled PUSCH.~~

From Friday session
R1-2501388 Updated RAN1 UE features list for Rel-18 NR after RAN1#120 Moderators (AT&T, NTT DOCOMO, INC.)
Decision: The updated list after RAN1#120 in R1-2501388 is agreed.

R1-2501389 Draft LS on Rel-18 RAN1 UE features lists for NR after RAN1#120 Moderators (AT&T, NTT DOCOMO, INC.)
Decision: The draft LS to RAN2 (cc: RAN4) on the updated RAN1 UE features list for NR after RAN1#120 is endorsed. Final LS is approved in R1-2501390.

9 Release 19

9.1 Artificial Intelligence (AI)/Machine Learning (ML) for NR Air Interface

Please refer to RP-243244 for detailed scope of the WI.
Rapporteur to provide initial input on higher layer signalling under agenda item 9.1. For input on higher layer signalling from any other source, please include it as part of your tdoc to relevant sub-agenda items.

R1-2501546 Session notes for 9.1 (AI/ML for NR Air Interface) Ad-Hoc Chair (CMCC)
Friday decision: The session notes are endorsed and contents reflected below.

[120-R19-AI/ML] – Taesang (Qualcomm)
Email discussion on Rel-19 AI/ML
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2501143 Rapporteur view on higher layer signalling of Rel-19 AI-ML for NR air interface Qualcomm Incorporated

9.1.1 Specification support for beam management

R1-2500050 Discussion on specification support for AI/ML-based beam management FUTUREWEI

R1-2500066	Discussion on AI/ML-based beam management	ZTE Corporation, Sanechips
R1-2500089	Discussion on AI/ML for beam management	Huawei, HiSilicon
R1-2500159	Discussion on AIML for beam management	Spreadtrum, UNISOC
R1-2500201	Discussion on AI/ML-based beam management	CATT
R1-2500254	Discussion on AI/ML for beam management	China Telecom
R1-2500274	Discussion on specification support for beam management	CMCC
R1-2500337	Specification support for beam management	vivo
R1-2500390	Specification Support for AI/ML for Beam Management	Kyocera
R1-2500391	AI/ML for beam management	Ericsson
R1-2500404	Specification support for beam management	Tejas Networks Limited
R1-2500465	On specification for AI/ML-based beam management	OPPO
R1-2500512	Discussion on AI/ML for beam management	Ofinno
R1-2500529	Discussion on AI/ML for beam management	InterDigital, Inc.
R1-2500545	AI/ML based Beam Management	Google
R1-2500555	Discussion on AIML beam management	TCL
R1-2500560	Discussion on specification support for beam management	Panasonic
R1-2500565	Discussions on AI/ML for beam management	LG Electronics
R1-2500590	Discussion on specification support for beam management	NEC
R1-2500635	AI/ML specification support for beam management	Lenovo
R1-2500642	Discussion on Specification Support for Beam Management	Sony
R1-2500669	Discussion on specification support for beam management	Ruijie Networks Co. Ltd
R1-2500686	Specification support for AI-enabled beam management	NVIDIA
R1-2500710	Discussion on AI/ML for beam management	Xiaomi
R1-2500766	Enhancements for AI/ML enabled beam management	Apple
R1-2500834	Discussion for supporting AI/ML based beam management	Samsung
R1-2500877	Specification support for beam management	KDDI Corporation
R1-2500900	Discussion on specification support for beam management	ETRI
R1-2500925	Discussion on specification support on AI/ML for beam management	Fujitsu
R1-2500962	Discussion on specification support for AI/ML beam management	Transsion Holdings
R1-2500970	AI/ML for Beam Management	Nokia
R1-2501013	Discussion on specification support for AIML-based beam management	MediaTek Inc.
R1-2501085	AI/ML for Beam Management	Meta
R1-2501103	Specification support for beam management	Intel Corporation
R1-2501104	Discussion on AI/ML for beam management	HONOR
R1-2501130	Discussions on specification support for beam management	Sharp
R1-2501144	Specification support for AI-ML-based beam management	Qualcomm Incorporated
R1-2501190	Discussion on AI/ML for beam management	NTT DOCOMO, INC.
R1-2501226	Discussion on AI/ML based beam management	KT Corp.
R1-2501235	Discussion on AIML based beam management	ASUSTeK
R1-2501262	Discussions on AI/ML for beam management	CAICT
R1-2501270	Specification support for beam management	Fraunhofer HHI, Fraunhofer IIS
R1-2501333	Specification support for AI/ML beam management	ITL

R1-2501440 FL summary #1 for AI/ML in beam management Moderator (Samsung)

From Monday session

Agreement

For report content of inference results for UE-sided model, where the largest RSRP value is quantized to a 7-bit value in the range [-140, -44] dBm with 1dB step size, and the differential RSRP is quantized to a 4-bit value with 2 dB step size.

Note: the model output is UE implementation and it doesn't have to be RSRP subject to dBm value.

Agreement

For report content of inference results for UE-sided model for BM-Case 1, the RSRP of predicted beam(s)in the report of inference results, is the predicted RSRP, where the predicted RSRP is based on AI/ML output.

Note: how to capture it in the spec is a separate discussion.

R1-2501441 FL summary #2 for AI/ML in beam management Moderator (Samsung)

From Tuesday session

Agreement

For UE-side AI/ML model inference and BM-Case2, for the quantization of a RSRP value of inference results in a report over multiple future time instances,

- the largest RSRP value based on prediction of all time instances is the reference RSRP, and differential RSRPs in the report are computed relative to the reference RSRP.
    - The time instance information of the beam with the largest RSRP are additionally indicated in the report.

R1-2501442 FL summary #3 for AI/ML in beam management Moderator (Samsung)
From Wednesday session
Agreement
For inference, for BM-Case 2 of UE-side model,
- The time gap between two consecutive future time instances is configured by RRC, and the number of future time instance(s) *N* is configured by RRC.
    - time gap is [10ms, 20ms, 40ms, 80ms, 160ms]
    - N = [1, 2, 4, 8]
- Reference time of the earliest time instance for the predicted results is based on the most recent occasion of the CSI-RS/SSB resource in Set B for measurement
    - Where the most recent occasion of the CSI-RS/SSB resource of set B is the latest CSI-RS/SSB occasion no later than the corresponding CSI reference resource of the corresponding inference report.

R1-2501443 FL summary #4 for AI/ML in beam management Moderator (Samsung)
Presented in Thursday session

R1-2501594 FL summary #5 for AI/ML in beam management Moderator (Samsung)
From Friday session
Agreement
For UE-sided model, for configuring the resource for data collection purpose, support
- *CSI-ReportConfig* can used for configuring the resources for data collection purpose without CSI report.
    - One *CSI-ResourceConfigId* is configured for Set A.
    - One *CSI-ResourceConfigId* is configured for Set B.
    - Note: UE performs measurement on all resources
    - One or two associated IDs can be configured in *CSI-ReportConfig*
        - When Set B is equal or a subset of set A (i.e., *NZP-CSI-RS-ResourceId*/*SSB-Index* in the resource set for Set B is within the *NZP-CSI-RS-ResourceId*/*SSB-Index* in the resource set for Set A), one associated ID is configured,
        - Otherwise, one associated ID is configured for Set A and another one associated ID is configured for Set B
- FFS: whether/how to support 'aperiodic' CSI RS

Note: This is not related to whether/how to support delivery/transmission of the collected data for training for UE-sided model.

Agreement
For UE-sided model, in *CSI-ReportConfig* for inference
- One or two associated IDs can be configured in *CSI-ReportConfig*
    - When Set B is equal or a subset of set A (i.e., *NZP-CSI-RS-ResourceId*/*SSB-Index* in the resource set for Set B is within the *NZP-CSI-RS-ResourceId*/*SSB-Index* in the resource set for Set A), one associated ID is configured,
    - Otherwise, one associated ID is configured for Set A and another one associated ID is configured for Set B
- FFS: At least BM-Case 1, the applicability for 'aperiodic' CSI RS

Final summary in R1-2501595.

9.1.2 Specification support for positioning accuracy enhancement

From AI5
R1-2500388 LS on LMF-based AI/ML Positioning for Case 2b SA2, vivo
R1-2500389 LS on LMF-based AI/ML Positioning for case 3b SA2, Ericsson

Decision:SA2 (in R1-2500388) is requesting RAN1 input on current progress and future work plans on data types and procedures defined for LMF-based AI/ML Positioning case 2b. RAN1 response is necessary.
SA2 (in R1-2500389) is requesting RAN1 input on current progress and future work plans on data types defined for case 3b. RAN1 response is necessary - Huaming (vivo).

R1-2501521 Summary#1 on reply LS to SA2 on LMF-based AI/ML Positioning for case 2b and 3b Moderator (vivo)

From Wednesday session

R1-2501522 Draft reply LS on LMF-based AI/ML Positioning for Case 2b vivo, Ericsson

Agreement

Send Reply LS R1-2501522 to SA2 about Case 2b with the content of Section 2 of R1-2501521 by revising

> "RAN1 would continue work and may provide additional input in the future to SA2 for Case 2b if any new agreement reached in RAN1."
> To
> "RAN1 may provide additional input in the future to SA2 for Case 2b if any new agreement reached in RAN1."

Final LS is approved in R1-2501523.

R1-2501524 [DRAFT] reply LS on LMF-based AI/ML Positioning for Case 3b Ericsson, vivo

Agreement

Send Reply LS R1-2501524 to SA2 about Case 3b with the content of Section 3 of R1-2501521 by adding all the agreement made in this meeting for Case 3b.
Final LS is approved in R1-2501525.

R1-2500060 AI/ML for Positioning Accuracy Enhancement Ericsson
R1-2500067 Discussion on AI/ML-based positioning enhancement ZTE Corporation, Pengcheng Laboratory
R1-2500090 Discussion on AI/ML for positioning accuracy enhancement Huawei, HiSilicon
R1-2500202 Discussion on AI/ML-based positioning CATT, CICTCI
R1-2500275 Discussion on specification support for positioning accuracy enhancement CMCC
R1-2500338 Specification support for positioning accuracy enhancement vivo
R1-2500405 Specification support for positioning accuracy enhancement Tejas Networks Limited
R1-2500466 On specification for AI/ML-based positioning accuracy enhancements OPPO
R1-2500518 Discussion on specification support for AI-ML based positioning accuracy enhancement Baicells
R1-2500546 AI/ML based PositioningGoogle
R1-2500556 Discussion on AIML positioning TCL
R1-2500606 Discussion on specification support for AIML based positioning accuracy enhancement NEC
R1-2500636 Specification impacts for AI/ML positioning Lenovo
R1-2500643 Specification support for AI/ML for positioning accuracy enhancement Sony
R1-2500670 Discussion on specification support for positioning accuracy enhancement Ruijie Networks Co. Ltd
R1-2500687 Specification support for AI-enabled positioning NVIDIA
R1-2500711 Discussion on AI/ML-based positioning accuracy enhancement Xiaomi
R1-2500746 Discussion on support for AIML positioning InterDigital, Inc.
R1-2500767 Discussion on Specification Support for AI/ML-based positioning Apple
R1-2500835 Discussion for supporting AI/ML based positioning accuracy enhancement Samsung
R1-2500901 Discussion on specification support for positioning accuracy enhancement ETRI
R1-2500926 Discussion on specification support for AIML-based positioning accuracy enhancement Fujitsu
R1-2500971 AI/ML for Positioning Accuracy Enhancement Nokia
R1-2500991 AI/ML positioning accuracy enhancement Fraunhofer IIS, Fraunhofer HHI
R1-2501131 Discussion on specification support for AI/ML based positioning accuracy enhancements Sharp
R1-2501145 Specification support for AI-ML-based positioning accuracy enhancement Qualcomm Incorporated
R1-2501191 Discussion on AI/ML for positioning accuracy enhancement NTT DOCOMO, INC.
R1-2501247 Discussion on Specification Support of AI/ML for Positioning Accuracy Enhancement Indian Institute of Tech (M)
R1-2501259 Design for AI/ML based positioning MediaTek Korea Inc.
R1-2501273 Discussion on specification support for AI/ML Positioning Accuracy enhancement CEWiT
R1-2501283 Discussions on specification support for positioning accuracy enhancement for AI/ML ITL

R1-2501410 Summary #1 of specification support for positioning accuracy enhancement Moderator (Ericsson)

From Monday session

Conclusion

For AI/ML based positioning Case 3a, regarding the time stamp in a measurement report from gNB to LMF,

- Existing IE "Time Stamp" in TS 38.455 can be reused from RAN1 perspective.

R1-2501411 Summary #2 of specification support for positioning accuracy enhancement Moderator (Ericsson)
From Tuesday session
Agreement
For Rel-19 AI/ML based positioning, for Case 3b, "FFS: k" in RAN1#119 agreement is resolved by supporting:
- $k = \{0...5\}$

Agreement
For Rel-19 AI/ML based positioning, for Case 3b, "FFS: N_t' values" in RAN1#119 agreement is resolved by supporting:
- $N_t' = \{8, 16, 24\}$

Agreement
For AI/ML based positioning Case 1, from RAN1 perspective, when the label data of location is generated by LMF and transferred from LMF to UE, label and quality indicator of label can be provided by reusing existing IEs.
- From RAN1 perspective, the existing IE can use one of the geographic shapes defined in TS 23.032. The location estimate uncertainty and confidence (if included with the geographic shapes) can serve as quality indicator of the label.

Agreement
For AI/ML based positioning,
- When channel measurement is to be reported/sent, time stamp of channel measurement refers to the time instance when the channel measurement is performed.
- When label is to be reported/sent, time stamp of label is the time instance for which the label is valid.

R1-2501412 Summary #3 of specification support for positioning accuracy enhancement Moderator (Ericsson)
Presented in Wednesday session

R1-2501413 Summary #4 of specification support for positioning accuracy enhancement Moderator (Ericsson)
From Thursday session
Agreement
For AI/ML based positioning Case 3b, with the existing definition of two channel measurement types (A) or type (B): (A) path-based measurement, i.e., measurement in the existing specifications (up to Rel-18), (B) Rel-19 enhanced measurement (see definition in RAN1#119 agreement for Case 3b).
- Descriptive info of the channel measurement is included together with the channel measurement values, either explicitly or implicitly.
- The descriptive info of the channel measurement includes at least:
  - measurement type (A) or type (B) and
  - measurement parameters.
- Regarding measurement parameters,
  - For (A) path-based measurement, the measurement parameters include: number of additional paths, k.
  - For (B) Rel-19 enhanced measurement, the measurement parameters include at least: N_t', k.
    - FFS: N_t
- Note: Type (A) and Type (B) are used for discuss purpose only.

R1-2501414 Summary #5 of specification support for positioning accuracy enhancement Moderator (Ericsson)
From Friday session
Conclusion
For Rel-19 AI/ML based positioning Case 3b,
- If the gNB cannot report measurements according to measurement type (A) or measurement type (B) requested by LMF, the gNB responds to LMF that: the gNB cannot provide measurement with the requested measurement type.
  - The gNB does not respond to LMF with a measurement report where the measurement type is different from requested.
- It is up to RAN3 to decide the signaling details.

- Note: It is RAN1 understanding that this is the same behavior as in legacy when the NG-RAN node cannot support the requested measurement.

Agreement
RAN1 send an LS to RAN3 and RAN2 to inform them above agreements and conclusions.

Agreement
- For training data collection of AI/ML based positioning 3a/3b, if Part A and Part B are generated by different entities, for pairing between a Part A entry and a Part B entry, the following is needed:
  - The time stamp of Part A (if Part A is transmitted) and the time stamp of Part B (if Part B is transmitted).
- FFS: other information is not precluded (e.g., if Part B is valid for a duration)

Note: Purpose such as "training data collection" will not necessarily be specified in RAN1 specifications.
RAN1 send an LS to RAN3 and RAN2 to inform the above.

R1-2501627 [DRAFT] LS on AI/ML Positioning Case 3b Moderator (Ericsson)
Friday decision: The draft LS doesn't seem to correctly capture the agreements.

[Post-120-AI/ML-01] – Yufei (Ericsson)
- Email approval of Reply LS on AI/ML Positioning Case 3b by February 26 (Wednesday)

R1-2501629 Final summary specification support for positioning accuracy enhancement Moderator (Ericsson)

9.1.3 Specification support for CSI prediction

R1-2500057 AI/ML for CSI prediction Ericsson
R1-2500068 Discussion on specification support for AI CSI prediction ZTE Corporation, Sanechips
R1-2500091 Discussion on AI/ML for CSI prediction Huawei, HiSilicon
R1-2500160 Discussion on AIML for CSI prediction Spreadtrum, UNISOC
R1-2500203 Discussion on AI/ML-based CSI prediction CATT
R1-2500276 Discussion on AI/ML for CSI prediction CMCC
R1-2500319 Discussion on CSI Prediction TCL
R1-2500339 Specification support for CSI prediction vivo
R1-2500406 Specification support for CSI Prediction Tejas Networks Limited
R1-2500467 On specification for AI/ML-based CSI prediction OPPO
R1-2500533 On AI/ML-based CSI prediction InterDigital, Inc.
R1-2500547 AI/ML based CSI Prediction Google
R1-2500566 Discussions on CSI prediction LG Electronics
R1-2500600 Discussion on specification support for CSI prediction NEC
R1-2500637 Specification support for CSI prediction Lenovo
R1-2500644 Specification Support for AI/ML CSI prediction Sony
R1-2500688 Specification support for AI-enabled CSI prediction NVIDIA
R1-2500712 Discussion on AI/ML model based CSI prediction Xiaomi
R1-2500768 Discussion on AI/ML based CSI prediction Apple
R1-2500816 Discussion on AI/ML-based CSI prediction Panasonic
R1-2500836 Views on AI/ML based CSI prediction Samsung
R1-2500902 Discussion on specification support for CSI prediction ETRI
R1-2500927 Discussion on specification support for CSI prediction Fujitsu
R1-2500972 AI/ML for CSI Prediction Nokia
R1-2501014 AI/ML - Specification support for CSI Prediction MediaTek Inc.
R1-2501080 Discussion on AI/ML for CSI prediction AT&T
R1-2501132 Discussion on specification support for AI/ML based CSI prediction Sharp
R1-2501146 Specification support for CSI prediction Qualcomm Incorporated
R1-2501192 Discussion on AI/ML for CSI prediction NTT DOCOMO, INC.

R1-2501528 Summary #1 of CSI prediction Moderator (LG Electronics)
From Wednesday session
Agreement
For CSI prediction using UE-side model, at least for inference, Rel-18 CSI framework is reused.
- For CSI-RS resource type for CMR, periodic, semi-persistent, and aperiodic CSI-RS are supported.
- For inference report,

- for N4>=1, support Rel-18 codebook ('typeII-Doppler-r18')

R1-2501529 Summary #2 of CSI prediction Moderator (LG Electronics)
R1-2501530 Summary #3 of CSI prediction Moderator (LG Electronics)
From Thursday session
Agreement
For CSI prediction using UE-side model, if performance monitoring type 1 or 3 is supported, for calculation of monitoring metric, support
- Based on intermediate KPI
  - Down select between SGCS and NMSE by RAN1#120bis
  - FFS on the definition of SGCS/NMSE and how to calculate monitoring metric

Agreement
For CSI prediction using UE-side model, for CSI processing criteria and timeline, at least for inference further study on
- Whether the CPU should be shared or separately counted between legacy CSI reporting and AI/ML-based CSI reporting
- Whether the Processing Unit should be shared or separately counted among AI/ML related features/functionalities.
- Whether new timeline is needed/updated for inference, and whether a different timeline is needed when functionality switches/activates.
- Whether legacy framework for active CSI-RS resource and port counting can be reused

Note: Strive to study CSI processing criteria considering both BM and CSI case, and take the existing solutions as starting point.

R1-2501610 Summary #4 of CSI prediction Moderator (LG Electronics)
Presented in Friday session
Agreement
For CSI prediction using UE-side model, for data collection for training,
- For resource configuration,
  - For CSI-RS resource type for CMR, periodic and semi-persistent are supported.
    - FFS on support of aperiodic CSI-RS resource
  - FFS on whether to separately configure CSI-RS resource(s) used for measurements for model input and CSI-RS resource(s) used for measurements for ground-truth CSI
  - FFS whether/how to enhance CSI-RS resource configuration to reduce CSI-RS signalling overhead.
- FFS on CSI report configuration

Final summary in R1-2501626.

9.1.4 Study on AI/ML for NR air interface Phase 2

Please refer to RP-243245 for detailed scope of the SI on AI/ML for NR air interface.

9.1.4.1 CSI compression

R1-2500051 Discussion of CSI compression on AI/ML for NR air interface FUTUREWEI
R1-2500058 AI/ML for CSI compression Ericsson
R1-2500069 Discussion on study for AI/ML CSI compression ZTE Corporation, Sanechips
R1-2500151 Discussion on AI/ML for CSI compression Huawei, HiSilicon
R1-2500161 Discussion on AIML for CSI compression Spreadtrum, UNISOC
R1-2500204 Further study on AI/ML-based CSI compression CATT
R1-2500277 Discussion on AI/ML for CSI compression CMCC
R1-2500340 Discussion on CSI compression vivo
R1-2500407 Discussion on AI/ML for CSI Compression Tejas Networks Limited
R1-2500468 Additional study on AI/ML-based CSI compression OPPO
R1-2500534 On AI/ML-based CSI compression InterDigital, Inc.
R1-2500548 AI/ML based CSI Compression Google
R1-2500558 Discussion on AIML CSI compression TCL
R1-2500567 Study on CSI compression LG Electronics
R1-2500601 Discussion CSI compression NEC
R1-2500638 On AI/ML for CSI compression Lenovo
R1-2500689 Additional study on AI-enabled CSI compression NVIDIA

R1-2500713	Views on two-side AI/ML model based CSI compression	Xiaomi
R1-2500769	Discussion on AI based CSI compression	Apple
R1-2500817	Discussion on AI/ML for CSI compression	Panasonic
R1-2500837	Views on additional study for AI/ML based CSI compression	Samsung
R1-2500903	Discussion on AI/ML for CSI compression	ETRI
R1-2500928	Discussion on CSI compression with AI/ML	Fujitsu
R1-2500973	AI/ML for CSI Compression	Nokia
R1-2501015	Additional study on AI/ML for NR air interface - CSI compression	MediaTek Inc.
R1-2501119	AI/ML for CSI feedback enhancement	Mavenir
R1-2501147	Additional study on CSI compression	Qualcomm Incorporated
R1-2501193	Discussion on AI/ML for CSI compression	NTT DOCOMO, INC.
R1-2501274	Discussion on AI/ML for CSI compression	CEWiT
R1-2501354	Discussion on two-sided AI/ML model based CSI compression	IIT Kanpur (rev of R1-2501350)

R1-2501468 Summary#1 of Additional study on AI/ML for NR air interface: CSI compression Moderator (Qualcomm)

From Monday session

Conclusion

Under 10% UCI loss, Case 2 shows small to large performance drop (by Comparing Scenario B and Scenario A). Signaling support to mitigate the UCI loss can improve the performance and helps preserving the gain of Case 2 over Case 0 (by comparing Scenario C and Scenario A and benchmark schemes).

Agreement

Investigate the following approaches for signaling support for mitigating the impact of UCI loss

- NW-signaling to reset of historical CSI information at UE
- NW-triggered CSI retransmission

Conclusion

For direction A, performance impact, if any, due to NW / UE data distribution mismatch with respect to UE side additional condition can be addressed.

Agreement

In Options 3a-1 and 4-1, the exchanged dataset or the model parameters can be associated with an ID for pairing related discussion, then

- The same ID can be used for UE to collect UE-side target CSI for UE-side training
- The same ID can be used for applicability inquiry and reporting
- The same ID can be used for inference configuration
- FFS: whether ID/even same ID is needed for monitoring configuration
- FFS: where the ID is assigned
- Note: whether the purpose for pair will be specified will be discussed separately.

R1-2501469 Summary#2 of Additional study on AI/ML for NR air interface: CSI compression Moderator (Qualcomm)

From Tuesday session

Agreement

For inter-vendor-collaboration Options 3a-1 and 4-1 in Direction A, performance target is confirmed as additional information along with the exchanged dataset or the model parameters.

- FFS: type of performance metric
- FFS: input data for evaluating the performance

Conclusion

- Direction C with standardized model ensures a minimum performance requirement.
  - The so called fully specified reference model discussed in RAN1 for Direction C is assumed to be equivalent to the to-be-specified model, if specified, for performance requirements in RAN4.
- Direction A with standardized model parameter / dataset exchange from NW-side to UE-side may achieve~~s~~ better performance.

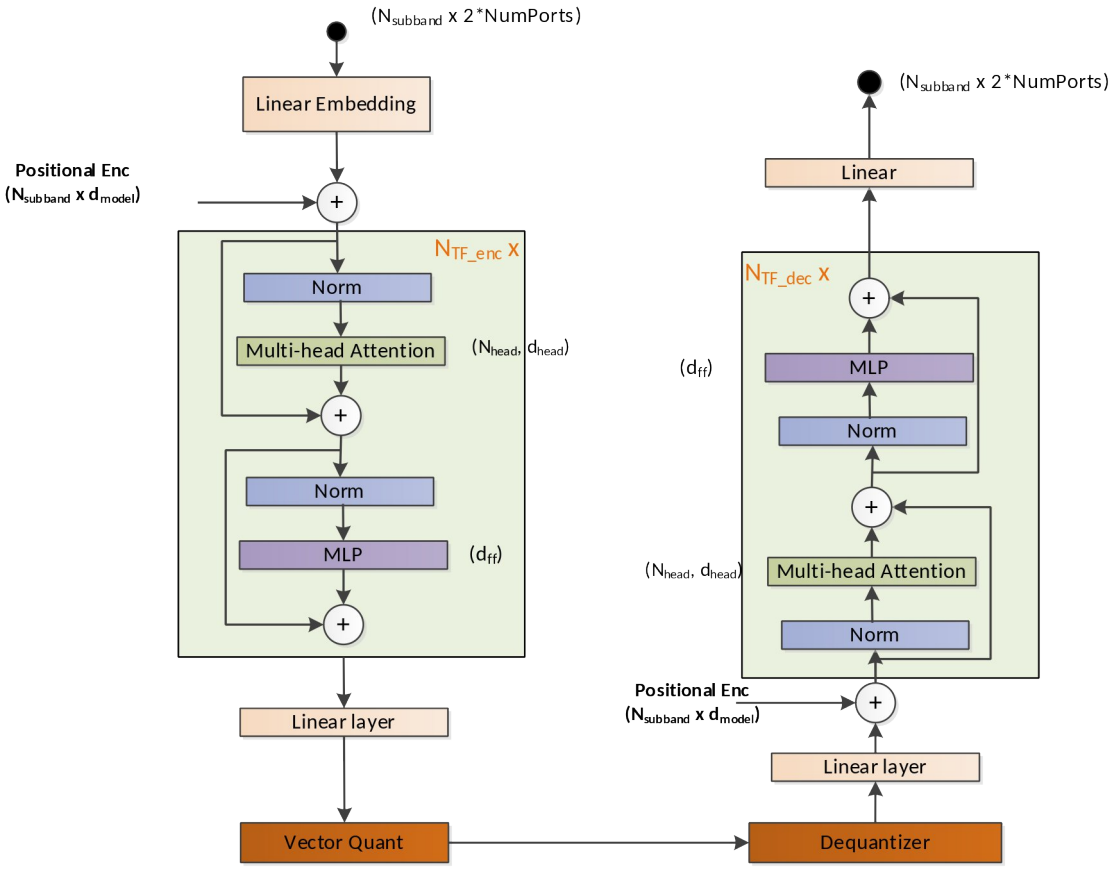
Conclusion

For NW-first training, in inter-vendor training collaboration Direction A and C, identification of root cause for performance issues can be achieved at least by the following

- Data distribution mismatch that is identified by either NW side or UE side

- Via target CSI and CSI feedback pair report, e.g., with the understanding that NW side runs inference using the NW side reference CSI generation part and compare with the inference using UE side CSI generation part.
- Note that this conclusion does not preclude other methods.
- Note that triggering of root cause detection may be after detecting performance degradation

R1-2501470 Summary#3 of Additional study on AI/ML for NR air interface: CSI compression Moderator (Qualcomm)
From Tuesday session
Agreement
For studying the standardized model structure, for temporal domain Case 0, in case of spatial-frequency domain input, adopt the following model structure as one example structure for study purpose,



Encoder description:

Embedding layer (Note: linear embedding in above figure to be revised to Embedding layer)	Number of tokens Feature dimension of each token Output dimension d_{model} per token
Positional encoding	Number of tokens positions Dimension of positional encoding for each token position: d_{model}
Transformer blocks	Number of transformer blocks N_{TF} Dimension of transformer block d_{model} Number of self-attention heads N_{head} Dimension of attention head d_{head} Dimension of latent space inside feedforward module d_{FF} Activation function choice
Output linear layer	Reshape the matrix to a vector Output dimension z_{dim}
Quantization	Scalar quantization: - Number of bits per latent dimension: N_{bit} - Total payload size for Scalar quantization = $N_{bit} * Z_{dim}$ Vector quantization - segment size = N_{seg} - Number of bits per latent dimension: N_{bit}

	- bits per segment = N_{bit} * N_{seg} - Number of segments = Z_{dim} / N_{seg} - Total payload size for Vector quantization = Number of segments * bits per segment

Decoder description:
The decoder has a mirroring design as the encoder. Details are to be discussed.

R1-2501471 Summary#4 of Additional study on AI/ML for NR air interface: CSI compression Moderator (Qualcomm)
From Thursday session
Agreement
For model structure scalability study for temporal domain Case 0,

- For the choice of token dimension and feature dimension,
  - Alt 1: Use subband as the token dimension and Tx port as a feature dimension
    - The number of tokens varies with the number of subbands.
  - Alt 2: Use Tx port as the token dimension and subband as a feature dimension
    - The number of tokens varies with the number of Tx ports.
  - Alt 3: Use a fixed-size sub-block of Tx ports and subbands matrix (e.g., n_Tx_ports*m_Subbands) as a token and represent the input as a sequence of tokens.
    - The number of tokens varies with the number of Tx ports and the number of subbands.
- For scalability over the feature dimension,
  - Alt1: specific embedding layer for each feature size
  - Alt2: a common embedding layer with padding (e.g., zero-padding or other techniques for padding values)
- For scalability over the token dimension,
  - Alt1: positional embedding specific to each token index
    - $N_{token,active}$ tokens out of N_{token} token positions are used as input.
  - Alt2: Padding at the input
- For scalability over payload configurations,
  - Alt1: specific output linear layer for each payload configuration
  - Alt2: truncation/masking of the output linear layer output
  - Alt3: by varying quantization parameters
- Notes
  - Other Alternatives are not precluded.
  - Different Alternatives may be used in combination.
  - Same/similar approach is applied at the decoder side.
- Evaluations to consider:
  - Case 1 (scalable structure): Scalable model structure described above
    - Using model structure as indicated in above diagram with fixed hyperparameters, companies may train a single parameter set or different parameter sets across different {number of Tx ports, CSI feedback payload size, bandwidth} assuming a common model structure.
    - To report whether a single parameter set or different parameter sets were used across different {number of Tx ports, CSI feedback payload size, bandwidth}. (e.g., single parameter set across different payload sizes and bandwidths, different parameter set across different number of Tx ports)
  - Case 2 (dedicated structure): Using model structure as indicated in above diagram with different hyperparameters, where the input and the output related hyperparameters are chosen optimally corresponding to each specific {number of Tx ports, CSI feedback payload size, bandwidth} without scalability considerations.
    - Different parameter sets are trained across different number of Tx ports, CSI feedback payload sizes, and bandwidths.
  - For each scalable model structure choice, to evaluate the SGCS performance of the non-AI/ML benchmark (e.g., Rel-16 eType2), Case 1, and Case 2, for each of {number of Tx ports, CSI feedback payload size, bandwidth}, and report the average gain (%) in SGCS of Case 1 and Case 2 over the non-AI/ML benchmark, as well as the loss (%) in the average gain of Case 1 w.r.t. Case 2. The average is performed by first calculating the SGCS gain (%) for each {number of Tx ports, CSI feedback payload size, bandwidth} and then averaging the SGCS gain (%) values over {number of Tx ports, CSI feedback payload size, bandwidth}.

R1-2501472 Summary#5 of Additional study on AI/ML for NR air interface: CSI compression Moderator (Qualcomm)

From Friday session

Agreement

For model structure scalability study for temporal domain Case 0, for the choice of {number of Tx ports, CSI feedback payload size, bandwidth}, take the following as baseline values.

- Number of Tx ports
  - [N1, N2, P] = [2, 8, 2] (32 CSI-RS ports)
  - [N1, N2, P] = [4,4,2] (32 CSI-RS ports)
  - [N1, N2, P] = [2, 4, 2] (16 CSI-RS ports)
- CSI feedback payload size
  - Multiple values, encouraged to pick at least one value from each of small, medium, and large payload regions X, Y, and Z.
- Bandwidth & subband size
  - All subbands
  - A subset of all the subbands.

Agreement

Capture the following observations into TR:

For the evaluation of performance impact due to UE-side / NW-side data distribution mismatch with respect to UE-side additional condition (issue 4 and 6),

When dataset A include dataset B, for case 2, option 3a-1, with NW-side target CSI sharing

- 1 source [Panasonic] observes minor performance loss (-0.22% ~ -1.09%) relative to case 1 with NW side target CSI sharing.

When dataset A include dataset B, for case 2, option 3a-1, without NW-side target CSI sharing,

- ~~1~~ 2 source [ZTE,vivo] observes minor performance loss (~~-0.67~~ -0.006% ~ -0.97%) related to case 1 without NW side target CSI sharing.
- 1 source [OPPO] observes moderate performance loss (-3.3% ~ -4%) relative to case 1 without NW side target CSI sharing

When dataset A include dataset B, for case 3 (Direction B),

- ~~1~~ 2 source [ZTE,vivo] observes minor performance loss (~~-0.6% ~ -0.9%~~ -0.9% ~ +0.01%) relative to case 1.

When dataset A does not include dataset B, for case 2, option 3a-1, with NW-side target CSI sharing

- 3 sources [QC, Apple, Samsung] observe similar performance (-0.4% ~ +0.24%) as case 1 for Alt1 UE training.
- 3 sources [vivo, Xiaomi, ETRI] observe minor performance loss up to -2.5% relative to case 1 for Alt1 training.
- ~~1~~ 2 source [Ericsson, ZTE] observe moderate performance loss of (-3.8 ~ -8.3%) relative to case 1 for Alt1 training.
- 1 source [OPPO] observe significant performance loss (-51% ~ -62.5%) relative to case 1 for Alt1 UE training.
- 2 sources [CATT, Futurewei] observe minor performance loss (-1.62% ~~-2.78%~~ ~ -3.2%) relative to case 1 for Alt2 UE training
- 1 source [Apple] observe moderate loss of -6.7% relative to case 1 for Alt2 UE training.

When dataset A does not include dataset B, for case 2, option 3a-1, without NW-side target CSI sharing

- ~~4~~ 5 sources [CATT, ZTE, Xiaomi, ETRI, vivo] observe minor to moderate performance loss (-0.4% ~ -3.9%) relative case 1.
- ~~2~~ 3 source [QC, Ericsson, Apple] observe moderate performance loss (-6.7% ~ -8.6%) relative to case 1.
- 1 source [OPPO] observes significant performance loss of -62.1% relative to case 1.

When dataset A does not include dataset B, for case 2, option 4-1,

- 4 sources [QC, Apple, Xiaomi, ETRI] observe zero to minor performance loss (-2.4% ~ 0%) relative to case 1 for Alt1 UE training.
- 1 source [Ericsson] observe minor to moderate performance loss (-2.9% ~ -7.9%) relative to case 1 for Alt1 UE training depending on whether dataset A applies augmentation using various phase normalization methods.

- 4 sources [CATT, Xiaomi, Futurewei, ETRI] observe minor performance loss (-1.41% ~ -3.52%) relative to case 1 for Alt2 UE training.
- 1 source [Apple] observes moderate loss of -7.9% relative to case 1 for Alt2 UE training.
- 1 source [vivo] observes moderate loss of -7.7% relative to case 1 for 3a-1 for Alt1 UE training when backbone are different.

When dataset A does not include dataset B, for case 3, Direction B,

- 4 sources [CATT, vivo, ZTE, ETRI] observe minor loss to positive gain (-3.7% ~ 1%) relative to case 1.
- 1 source [QC] observes significant loss (-20%) relative to case 1.

Agreement

Capture the following observations into TR:

For the evaluation of performance impact due to mismatch between the distribution of the dataset used for reference model(s) training, UE-side data distribution, and NW-side data distribution (Issue 9),

For case 2 (model trained on dataset S but tested on dataset B),

- 2 sources [vivo, QC] observe significant performance loss (-7.2% ~ -17.4%) relative to case 1, where dataset B consists of actual field data.
- 1 source [ETRI] observes significant performance loss of -37.3% relative to case 1, where dataset S and B are different by TxRU mapping.
- 3 resources [ZTE, Panasonic, Ericsson] observe moderate performance loss (-2.12% ~ -4.75%) relative to case 1, where dataset S and B are generated from different scenarios, antenna layout or UE location. E.g., Uma and InH respectively, or dataset S and B are different by antenna layout and indoor/outdoor ratio.

For case 2A (model further finetuned on dataset B),

- If finetune at NW and fix encoder at UE
  - For using actual field data as dataset B
    - When dataset A / B mismatch is considered but dataset B does not contain A, 1 source [vivo] observe moderate loss (-4.63% ~ -7.66%) relative to case 1. The improvement compared to case 2 is minor to moderate (2.58% ~ 9.17%).
    - When dataset A / B mismatch is not considered, ~~2~~ 1 source~~s~~ [~~vivo,~~ QC] observe~~s~~ minor ~~to moderate~~ loss of ~~(~~-3.5% ~~~ -7.66%)~~ loss relative to case 1. The improvement compared to case 2 is moderate to significant (2.6% ~ 13.9%) depending on the specific scenario where the field data is collected.
  - For using synthetic data as dataset B
    - When dataset A / B mismatch is considered but dataset B does not contain A, 1 source [ZTE] observes moderate loss to case 1 (-4.32% ~ -4.75%). The improvement compared to case 2 is negative (-0.68% ~ -1.02%).
    - When dataset A includes or is the same as dataset B, 3 sources [ZTE, Panasonic, Ericsson] observe minor to moderate loss relative to case 1 (-0.97% ~ -4.67%). The improvement compared to case 2 is ~~negative to~~ minor (~~-~~0.6~~1.01~~% ~ 1.61~~49~~%). Dataset S and B are Uma and InH respectively, or dataset S and B are different by antenna layout and indoor/outdoor ratio.
- If finetune at UE and fix decoder at NW
  - For using actual field data as dataset B,
    - 2 sources [vivo, QC] observe moderate to significant loss (-6.8% ~ -12%) relative to case 1. The improvement compared to case 2 is minor moderate (0.4% ~ 7.1%) depending on the specific scenario where the field data is collected.
  - For using synthetic data as dataset B
    - 2 source [Panasonic, Ericsson] observe minor to moderate loss (-1.56% ~ -3.81%) relative to case 1. The improvement compared to case 2 is marginal (0.03% ~ 0.75%). Dataset S and B are different by antenna layout and indoor/outdoor ratio.
    - 1 source [ETRI] observes significant loss (-23.8%) relative to case 1. The improvement compared to case 2 is significant (13.5%). Dataset S and B are different by TxRU mapping
- If finetune at both UE and ~~both~~ NW sides
  - For using actual field data as dataset B
    - When dataset A / B mismatch is considered and dataset A does not contain B, 1 source [vivo] observes significant loss (-8.5% ~ -44.85%) relative to case 1 depending on specific scenarios for data collection. The improvement compared to case 2 is negative (-28% ~ -1.3%). The loss relative to finetune at one side is negative (-1.72% ~ -37.19%).
    - When dataset A / B mismatch is not considered and A is equal to B, 1 ~~2~~ sources [~~vivo,~~ QC] observes moderate ~~to significant~~ loss of ~~(~~-7.3% ~~~ -56.1%)~~ relative to case 1. The improvement compared to case 2 is ~~negative to~~ significant ~~(-39.3% ~~ 10.1%~~)~~. The loss

relative to finetune at one side is ~~minor to significant (-1.7 ~ 48.4%)~~ negative to moderate (-3.82~ 4.67%).

- For using synthetic data as dataset B
  - When dataset A / B mismatch is modelled by different scenarios (i.e., by NW side condition),
    - If the input is in spatial-frequency domain, 1 source [Samsung] observes minor to significant loss (-29.0% ~ -30.5%) relative to case 1 depending on whether data Set S is used in the finetune.
    - If the input is in angle-delay domain, 1 source [Samsung] observes similar performance to case 1 (-0.3% ~-1.3%)
  - When dataset A includes B or is equal to B or has same distribution as B,
    - If the input is in spatial-frequency domain, and dataset A does not contain B, 2 sources [Panasonic, Ericsson] observe minor to moderate loss (-1.1% ~ -3.69%) relative case 1 depending on the modelling of synthetic data. The improvement compared to case 2 is minor (0.57% ~ 1.3%). Compared to finetune at one side, the performance improvement is similar to minor (-0.35% ~ 1%).
    - If the input is in spatial-frequency domain, 1 source [Samsung] observes minor to significant loss (-3.67% ~ -11.18%) relative to case 1 depending on whether data Set S is used in the finetune.
    - If the input is in angle-delay domain, 1 source [Samsung] observes similar performance to case 1 (-0.89% ~1.51%).
  - When dataset A / B mismatch is modelled by Rx antenna spacing and dataset A does not contain B and dataset A has same distribution as S (i.e., synthetic/field data distribution mismatch modeled only at the UE side (via UE-side additional condition) but not modeled at the NW-side), 1 source [Samsung] observes similar performance (-0.89% ~ 0.21%) as case 1.

Agreement
Study performance-complexity trade-off by comparing different AI/ML models, e.g. by optimizing existing designs, and/or by comparing different precoder representation in (spatial-frequency and angle-delay) or (spatial-frequency-time and angle-delay-doppler), by considering the following aspects

- Performance comparison between different AI/ML designs and benchmark schemes
- Complexity numbers (FLOP, calculated/projected latency or power consumption if available, measured latency or power consumption if available) of different AI/ML designs and benchmark schemes

R1-2501474 Final summary of Additional study on AI/ML for NR air interface: CSI compression Moderator (Qualcomm)

9.1.4.2 Other aspects of AI/ML model and data

Including model identification/procedure for two-sided model, collection of UE-sided model training data, and model transfer/delivery

R1-2500052 Discussion on other aspects of AI/ML model and data on AI/ML for NR air-interface FUTUREWEI
R1-2500070 Discussion on other aspects of AI/ML model and data ZTE Corporation, Sanechips
R1-2500152 Discussion on other aspects of the additional study for AI/ML Huawei, HiSilicon
R1-2500162 Discussion on other aspects of AI/ML model and data Spreadtrum, UNISOC
R1-2500205 Further study on AI/ML for other aspects CATT, CICTCI
R1-2500255 Discussion on other aspects of AI ML model and data China Telecom
R1-2500278 Discussion on other aspects of AI/ML model and data CMCC
R1-2500341 Other aspects of AI/ML model and data vivo
R1-2500392 Discussion on other aspects of AI/ML Ericsson
R1-2500408 Other aspects of AI/ML Model and Data Tejas Networks Limited Withdrawn
R1-2500469 Additional study on other aspects of AI/ML model and data OPPO
R1-2500549 AI/ML Model and Data Google
R1-2500559 Discussions on other aspects of AIML In NR air interface TCL
R1-2500568 Discussion on other aspects of AI/ML model and data LG Electronics
R1-2500591 Discussion on other aspects of AI/ML model and data NEC
R1-2500639 Discussion on other aspects of AI/ML model and data Lenovo
R1-2500690 Additional study on other aspects of AI model and data NVIDIA
R1-2500714 Further study on AI/ML model and data Xiaomi

R1-2500770 Discussion on other aspects of AI/ML models and data Apple
R1-2500815 Discussion on the terminology alignment TR in SA Panasonic
R1-2500838 Views on additional study for other aspects of AI/ML model and data Samsung
R1-2500904 Discussion on other aspects of AI/ML model and data ETRI
R1-2500929 Discussion on other aspects of AI/ML model and data Fujitsu
R1-2500974 Other aspects of AI/ML for two-sided model Nokia
R1-2500976 Discussion on other aspects of AI/ML model and data Continental Automotive
R1-2501079 Other Aspects of AI/ML framework AT&T
R1-2501148 Other aspects of AI/ML model and data Qualcomm Incorporated
R1-2501194 Discussion on other aspects of AI/ML model and data NTT DOCOMO, INC.

R1-2501403 Summary #1 for other aspects of AI/ML model and data Moderator (OPPO)
From Wednesday session
Agreement
From RAN1 perspective, for the study of delivery/transfer Case z4, if the known structured model is specified in 3GPP, at least consider the following for the open format.
- Define an open format within 3GPP.

R1-2501404 Summary #2 for other aspects of AI/ML model and data Moderator (OPPO)
From Thursday session
Conclusion
- From RAN1 perspective, the study progress of model identification and model transfer/delivery in Agenda item 9.1.4.2 are sufficient.
  - New work/discussion in AI 9.1.4.2 can be triggered by LS from other working group(s) or other TSG(s), if any
  - The work on pCR (if any) to capture the output of AI 9.1.4.2 to be done in future meeting(s)
  - Other on-demand work (if any)
- From RAN1 perspective, there is no consensus in Rel-19 on the need of standardized solution for model delivery/transfer of one-sided model
- In RAN1, if needed, whether model identification is needed for CSI compression or not is to be decided in Agenda item 9.1.4.1
- In RAN1, if needed, whether standardized solution(s) for model transfer/delivery are needed for CSI compression or not is to be decided in Agenda item 9.1.4.1

Final summary in R1-2501406.

9.2 NR MIMO Phase 5

Please refer to RP-242394 for detailed scope of the WI. Additional RAN guidance on Rel-19 NR MIMO can be found in RP-243265.
Rapporteur to provide initial input on higher layer signalling under agenda item 9.2. For input on higher layer signalling from any other source, please include it as part of your tdoc to relevant sub-agenda items.

[120-R19-MIMO] – Gilwon (Samsung)
Email discussion on Rel-19 MIMO Phase 5
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500844 List of RRC impact for Rel-19 MIMO Ph5 Moderator (Samsung)

R1-2500845 Draft LS to RAN2 on MAC impact for Rel-19 MIMO Ph5 Samsung, MediaTek
From Thursday session
Agreement
LS to RAN2 on MAC impacts for Rel-19 NR MIMO Ph5 is endorsed. Final LS is approved in R1-2500846.

9.2.1 Enhancements for UE-initiated/event-driven beam management

R1-2500056 Enhancements for UE-initiated/event-driven beam management FUTUREWEI
R1-2500097 On UE initiated/event-driven beam management Huawei, HiSilicon
R1-2500104 Enhancements for UE-initiated/event-driven beam management MediaTek Inc.

R1-2500115 Remaining Details on Rel-19 UE/Event-Driven Beam Management InterDigital, Inc.
R1-2500163 Discussion on UE-initiated/event-driven beam management Spreadtrum, UNISOC
R1-2500217 Views on enhancement for Rel-19 UE-initiated/event-driven beam management CATT
R1-2500240 Discussion on UE-initiated/event-driven beam management TCL
R1-2500241 Discussion on enhancements for UE-initiated/event-driven beam management ZTE Corporation, Sanechips
R1-2500279 Discussion on enhancements for UE-initiated/event-driven beam management CMCC
R1-2500342 Remaining issues on UE-initiated/event-driven beam management vivo
R1-2500377 Discussions on UE-initiated/event-driven beam management China Telecom Corporation Ltd.
R1-2500470 Discussions on UE-initiated/event-driven beam management OPPO
R1-2500513 On UE-initiated/event-driven beam management Ofinno
R1-2500569 UE-initiated beam management LG Electronics
R1-2500592 Discussion on enhancements for UE-initiated or event-driven beam management NEC
R1-2500645 Enhancements for UE-initiated/event-driven beam management Sony
R1-2500671 Discussion on enhancements for UE-initiated/event-driven beam management Ruijie Networks Co. Ltd
R1-2500699 Enhancements for UE-initiated/event-driven beam management Lenovo
R1-2500725 Discussion on enhancements for UE-initiated/event-driven beam management Xiaomi
R1-2500771 Remaining issues for UE-initiated beam management Apple
R1-2500839 Views on Rel-19 UE-initiated/event-driven beam management Samsung
R1-2500896 Enhancements to facilitate UE-initiated/event-driven beam management Nokia
R1-2500905 Enhancements for UE-initiated/event-driven beam management ETRI
R1-2500930 Discussion on enhancements for UE-initiated/event-driven beam management Fujitsu
R1-2500963 Enhancements for UE-initiated/event-driven beam management Transsion Holdings
R1-2500977 Enhancements for UE-initiated/Event-Driven Beam Management Panasonic
R1-2501084 Enhancements for UE Initiated Beam Management Meta
R1-2501087 Discussion on enhancements for UE-initiated/event-driven beam management Hyundai Motor Company
R1-2501105 Discussion on the UE-initiated/event-driven beam management HONOR
R1-2501127 Enhancements for UE-initiated/event-driven beam management Sharp
R1-2501149 UE-initiated/event-driven beam management Qualcomm Incorporated
R1-2501195 Discussion on enhancements for UE-initiated/event-driven beam management NTT DOCOMO, INC.
R1-2501230 Discussion on UE-initiated/event-driven beam management ITRI
R1-2501236 Discussion on UE initiated beam report ASUSTeK
R1-2501245 Remaining issues for UE-initiated beam management Ericsson
R1-2501264 Enhancements for UE-initiated/event-driven beam management KT Corp.
R1-2501309 Discussion on enhancements for UE-initiated/event-driven beam management NICT
R1-2501334 Discussion on enhancements for UE-initiated or event-driven beam management Google
R1-2501341 Discussions on enhancements for UE-initiated/event-driven beam management KDDI Corporation

R1-2500942 Offline summary for Rel-19 MIMO UEIBM (9.2.1) pre-RAN1#120 Moderator (ZTE)
R1-2500943 Moderator Summary #1 on UE-initiated/event-driven beam management Moderator (ZTE)
From Monday session

Agreement
On cross-CC beam report transmission procedure for UE-initiated/event-driven beam reporting, regarding Event-2, the following working assumption is confirmed with the following modification.

- The first PUCCH and the second PUSCH associated for UE-initiated/event-driven beam reporting are from different PUCCH groups
  - Subject to separate UE capability
- Note: From RAN1 perspective, the above does NOT introduce any spec impact except for UE capability signaling

Agreement
On cross-CC beam report measurement for UE-initiated/event-driven beam reporting, regarding Event-2, the following is supported

- The current beam RS and new beam RS(s) are in the same CC
  - The above does NOT imply to preclude the cross-CC TCI indication.
- The CC on which the indicated TCI state is applied, and the CC in which new beam RS(s) are can be the same or different.
  - FFS: Whether/how to introduce an RRC parameter to indicate the CC on which the indicated TCI state is applied.

Agreement
On UE-initiated/event-driven beam reporting, for Event 7, the candidate value of RRC parameter *Q* = {1, 2, 3, 4, 5, 6, 7, 8}
- Note: The UE does not expect that the configured Q is greater than the number of the activated DL/joint TCI state(s).

Conclusion
There is no RAN1 consensus on the following proposal:
- On beam report transmission procedure for UE-initiated/event-driven beam reporting, multi-bit indication in the first PUCCH is supported.

Agreement
On beam report transmission procedure for UE-initiated/event-driven beam reporting, regarding Mode-B, the value of X symbols for determining available transmission occasion of the second UL channel is configured by RRC (FFS: subject to a corresponding UE capability).

R1-2500944 Moderator Summary #2 on UE-initiated/event-driven beam management Moderator (ZTE)
From Tuesday session
Agreement
On beam report transmission procedure for UE-initiated/event-driven beam reporting, support the following option of dropping rule for the Case-1: the 1-bit first PUCCH is collided/overlapped with a PUCCH carrying normal SR and/or a PUCCH with normal LRR
- Option-1: LRR > first PUCCH > normal SR

Note: When the 1-bit first PUCCH is collided/overlapped with a PUCCH carrying normal SR and/or a PUCCH with normal LRR, only one of them is transmitted based on the above priority rule

Agreement
On beam report transmission procedure for UE-initiated/event-driven beam reporting, regarding the triggering procedure in Step-2 of Mode-A, the following Option-1 is supported.
- Option-1: A CSI trigger state corresponding to UE-initiated/event-driven beam reporting can NOT be associated with legacy AP-CSI report configuration.

Agreement
Regarding triggering event determination for Event 2, at least Candidate #2 is supported for resetting the counting.
- Candidate#1: RS reconfiguration/update or MAC-CE signaling (if supported) for new beam is received;
  - FFS: whether to reset the counting for all new beams.
  - FFS: whether to maintain the counting whose new beam is NOT updated.
- Candidate#2: [The measured current beam based on] indicated TCI state is updated;
  - In such case, the UE need to reset the counting for all new beams.
- Candidate#3: UEI beam report is transmitted;
  - FFS: Only reset the counting of new beams fulfilling triggering condition and reported by the UEI beam report
- Candidate#4: NW response (e.g., DCI in step-2 of Mode-A) is detected.
- Candidate#5: The time window expires
- Candidate#6: The threshold for event evaluation is re-configured by RRC signaling
- (FFS) Candidate#7: The RRC parameter(s) associated with the CSI report configuration for UEI beam report is reconfigured.
  - FFS: RRC parameter(s)
- FFS: Other candidates

Note: Whether this proposal is captured in RAN1 or RAN2 is a separate discussion point.

R1-2500945 Moderator Summary #3 on UE-initiated/event-driven beam management Moderator (ZTE)
From Wednesday session
Agreement
On UE-initiated/event-driven beam reporting, for Event 1, support Option-2 for RS measurement:
- Option-2: RS resource set for new beam is configured in the CSI reporting configuration, and the following implicit manner for enabling one of either scheme-1 or scheme-2 is used:
  - If the RS(s) for new beam are CSI-RS configured in a CSI-RS resource set configured with *repetition*, Scheme-1 is enabled; otherwise, Scheme-2 is enabled.
  - In such case, the report format for Event-2 is reused unless critical technical issue is identified

- FFS: How to report the RSRP value of the current beam
- FFS: RSRP of the current beam is always reported

Agreement
Regarding triggering event determination for Event 2, on the measurement window for initiating the UE-initiated/event-driven beam reporting procedure, further study the following options for possible down-selection

- Option-1: The measurement window is from T_PUCCH – T_proc – T_window to T_PUCCH – T_proc, where T_PUCCH is a transmission occasion of a first PUCCH, and T_proc is RRC configured.
- Option-2: The measurement window is from T_Instance – T_window to T_Instance, where T_Instance is an evaluation occasion of event instance, and T_proc is RRC configured.
  - The UEI beam report in the second PUSCH is based on the most recent measurement for new/current beam RS(s).
- Option-3: The length, slot offset and periodicity of a measurement window are configured per CSI report configuration by NW.
- Option-4: If an Event-2 instance for a new beam is obtained at the time t, UE (re)starts the timer for the new beam, where the expiry time of the timer is equal to the NW-configured length of the time window (T_window)
- Note: T_window is the agreed time window parameter for measurement.
- Note: Other options are not precluded.

R1-2501556 **Moderator Summary #4 on UE-initiated/event-driven beam management** **Moderator (ZTE)**
From Thursday session
Agreement
On UE-initiated/event-driven beam reporting, Option-1 is supported as an extension on L1-RSRP report format depending on Event-2.

- Option-1: For each of reported CRI/SSBRI, to introduce additional indication of whether the CRI/SSBRI satisfies the condition of Event-2.
  - The presence of this field is enabled by RRC with subjective to UE capability.
  - Note: The presence of this field is only for the case that N > 1 and the time window and M are configured

Above is NOT applicable for event 1.

Agreement
On beam report transmission procedure for UE-initiated/event-driven beam reporting, one PUCCH resource of first PUCCH can be associated with one or multiple CSI report configurations, regarding Event-2.

- Only a single UEI beam report is carried in the second PUSCH.
  - Additional indication of one 'CSI report configuration' is provided in the report format.
    - The CSI report configurations associated with the same PUCCH resource are ordered in ascending order of corresponding *CSI-ReportConfigId*, the number of bits of the additional indication field is ceil($\log_2$(N_CSIconfig)), where the N_CSIconfig denotes the number of CSI report configurations associated with the same PUCCH resource.
  - The payload size of the single UEI beam report is determined according to the max payload size among the associated CSI report configurations.
    - Zero-padding can be appended if the payload size of the UEI beam report is less than the max report payload.
  - The reported UEI beam report should satisfy the triggering condition.
  - If multiple UE initiated beam report procedures occur, down-select one of the following options:
    - Option-1: It is up to UE implementation to select one of configuration.
    - Option-2: The UEI beam report with highest priority is reported
    - Option-3: The report triggered in the latest measurement is reported in PUSCH
  - The multiple CSI report configurations associated with the one first PUCCH resource should be configured in the same CC.
  - Working Assumption: For Mode-A, the multiple CSI report configurations associated with the same PUCCH resource should be associated with a same *CSI-AperiodicTriggerState*.
  - For Mode-B, the multiple CSI report configurations associated with the same PUCCH resource should be associated with the same second configured PUSCH

FFS: Alt-2 Multiple UEI beam reports associated with the same PUCCH resource for first PUCCH can be transmitted in the second PUSCH.

- If RAN1 cannot converge on the support of Alt-2 in RAN1#120bis, this alternative will be dropped from Rel-19

Agreement

On beam report transmission procedure for UE-initiated/event-driven beam reporting, regarding the multiplexing/dropping rule(s) of 1-bit first PUCCH, down-select one of the following rule for the Case-2: the 1-bit first PUCCH is collided/overlapped with a PUSCH

- Option-1: Prioritize first PUCCH over PUSCH, i.e., PUSCH is dropped.
- Option-3: Piggyback 1-bit indication of first PUCCH into the PUSCH.
  - FFS: If the PUSCH should be with UL-SCH or not for UEI beam report
- Option-4: Reuse the SR dropping rules, i.e., the first PUCCH is dropped.
- FFS: whether/how to handle the case of different PHY priorities.

9.2.2 CSI enhancements

Including support for up to 128 CSI-RS ports on FR1 and UE reporting enhancement for CJT and SRS port grouping for TDD low-complexity 6/8RX receiver.

R1-2500098	On 128 CSI-RS ports and UE reporting enhancement	Huawei, HiSilicon
R1-2500105	CSI enhancements	MediaTek Inc.
R1-2500116	Remaining Details on Rel-19 Enhancements of CSI	InterDigital, Inc.
R1-2500164	Discussion on CSI enhancements	Spreadtrum, UNISOC
R1-2500218	Discussion on CSI enhancements for NR MIMO Phase 5	CATT
R1-2500242	Discussion on CSI enhancements	ZTE Corporation, Sanechips
R1-2500280	Discussion on CSI enhancements	CMCC
R1-2500343	Remaining issues on Rel-19 CSI enhancements	vivo
R1-2500471	CSI enhancements for Rel-19 MIMO	OPPO
R1-2500484	Discussion on Rel-19 CSI Enhancements	Tejas Network Limited
R1-2500485	CSI enhancements for Rel. 19 MIMO	Fraunhofer IIS, Fraunhofer HHI
R1-2500550	CSI Enhancement for NR MIMO	Google
R1-2500598	Discussion on CSI enhancements	NEC
R1-2500646	Discussion of CSI enhancements	Sony
R1-2500700	Discussion on CSI enhancements	Lenovo
R1-2500726	Further discussion on Rel-19 MIMO CSI enhancements	Xiaomi
R1-2500772	Views on R19 MIMO CSI enhancement	Apple
R1-2500841	Views on Rel-19 CSI enhancements	Samsung
R1-2500897	CSI enhancement for NR MIMO Phase 5	Nokia
R1-2500906	Discussion on CSI enhancements for NR MIMO Phase 5	ETRI
R1-2500931	Remaining issues on Rel-19 CSI enhancements	Fujitsu
R1-2501073	Discussion on Open Issues of CSI Enhancement	Rakuten Mobile, Inc
R1-2501106	Discussion on CSI enhancements	HONOR
R1-2501128	CSI enhancements	Sharp
R1-2501150	CSI enhancements for >32 ports and UE-assisted CJT	Qualcomm Incorporated
R1-2501196	Discussion on CSI enhancements	NTT DOCOMO, INC., NTT CORPORATION
R1-2501272	Remaining issues on CSI enhancements for large antenna arrays and CJT	Ericsson
R1-2501297	Discussion on CSI enhancements for NR MIMO Phase 5	KDDI Corporation
R1-2501307	Discussion on CSI enhancements	NICT

R1-2500840 Moderator Summary#1 on Rel-19 CSI enhancements: Round 1 Moderator (Samsung)

From Monday session

Conclusion

For the Rel-19 aperiodic standalone CJT calibration (CJTC) reporting, when linking CJTC Dd and Rel-18 eType-II CJT CSI reports is configured with a joint trigger, regarding timeline, the value of d_{relax} will be decided in UE feature session.

- Note: Per previous agreement, only a single non-zero value of $D_{relax}=d_{relax}$ is supported, in addition to the already agreed $D_{relax}=0$.

Agreement

For the Rel-19 aperiodic standalone CJT calibration reporting, when ReportQuantity is ‘cjtc-P’ (DL/UL phase offset), regarding the support of sub-band reporting (<>1),

- The legacy CSI sub-band definition and partitioning rules are reused.

Agreement

For the Rel-19 Type-I SP codebook refinement for P=48, 64, 128 CSI-RS ports, regarding Scheme-B, when the UE is configured to report wideband CSI on PUCCH:

- For PUCCH format 2, one-part CSI is used
- For PUCCH formats 3 and 4, decide, by RAN1#120, between the two alternatives:

- Alt1 (*baseline*). One-part CSI is used
- Alt2. Two-part CSI is used where SD basis selection is reported in CSI-part2
  - CSI fields in CSI-part1 and part2 follows the legacy sub-band CSI

Agreement
For the Rel-19 Type-I SP codebook refinement for P=48, 64, and 128 CSI-RS ports, regarding the CSI reference resource
- For P=128 with Capability 2 timeline, when periodic or semi-persistent CSI reporting is configured, decide (by RAN1#120) between the two alternatives:
  - Alt1. n_{CSI_ref} follows legacy specification.
  - Alt2. scale n_{CSI_ref} from the legacy specification by ceil(P/32).
- For all other cases, n_{CSI_ref} follows legacy specification to determine the CSI reference resource for periodic or semi-persistent CSI reporting.

Conclusion
For the Rel-19 Type-II codebook refinement based on Rel-18 Type-II Doppler for 48, 64, and 128 CSI-RS ports, whether to change the *maxNumberTxPortsPerResource* to *maxNumberTxPortsPerResourceGroup* for Rel-19 codebook triplet capability is to be discussed in UE feature session.

Agreement
For the Rel-19 CRI-based CSI refinement for up to 128 CSI-RS ports, regarding the value of the RRC-configured resource-level slot offset (relative to the resource-set-level slot offset) for aperiodic CSI-RS resource set, at least the following values are supported: {0, 1, 2, 3}
- FFS: Whether to also support {4, 5, …, X-1} where X is either 8, 16, or 32 for only Type-II or all applicable codebooks.
- Note: For the Rel-19 CRI-based CSI refinement for up to 128 CSI-RS ports, the RRC-configured resource-level slot offset (relative to the resource-set-level slot offset) for aperiodic CSI-RS resource set does not apply to CSI-RS for tracking.

R1-2501362 Moderator Summary for 1st offline on Rel-19 CSI enhancements Moderator (Samsung)
R1-2501361 Moderator Summary#2 on Rel-19 CSI enhancements: Round 2 Moderator (Samsung)
From Wednesday session
Agreement
For the Rel-19 Type-I SP codebook refinement for P=48, 64, 128 CSI-RS ports, regarding Scheme-B, when the UE is configured to report wideband CSI on PUCCH formats 3 and 4, one-part CSI is used.

Agreement
For the Rel-19 CRI-based CSI refinement for up to 128 CSI-RS ports, regarding the value of the RRC-configured resource-level slot offset (relative to the resource-set-level slot offset) for aperiodic CSI-RS resource set, in addition to the agreed values of {0, 1, 2, 3}, the following values are supported: $\{4, 5, 6, 7\}$.

Agreement
For the Rel-19 Type-I SP codebook refinement for P=128 CSI-RS ports, with Capability 2 timeline, when periodic or semi-persistent CSI reporting is configured, n_{CSI_ref} is the smallest value greater than or equal to $10 \cdot 2^{\mu}$, such that it corresponds to a valid downlink slot.

R1-2501571 Moderator Summary#3 on Rel-19 CSI enhancements: Round 3 Moderator (Samsung)
From Thursday session
Agreement
Following agreement from RAN1#118bis is updated (in RED):
For the Rel-19 aperiodic standalone CJT calibration reporting, when ReportQuantity is 'cjtc-P' (DL/UL phase offset), regarding the configured associated SRS resource,
- Confirm the following working assumption as agreement: "the configured associated SRS resource can be either periodic, semi-persistent, or aperiodic"
- When periodic or semi-persistent associated SRS resource is configured, the SRS transmission occasion for determining the reference UE antenna port corresponds to the *latest* SRS transmission occasion *before* the occasions of the N_{TRP} CSI-RS resources used for measuring 'cjtc-P' report
  - Note: when periodic or semi-persistent CSI-RS is configured, in case of time-domain Measurement Restriction not configured, the UE can assume that the occasions of the NTRP CSI-RS resources are the latest no later than the CSI reference resource.

9.2.3 Support for 3-antenna-port codebook-based transmissions

Including support for 3T3R and 3T6R SRS antenna switching.

R1-2500099	Discussion on 3-antenna-port UL transmission	Huawei, HiSilicon
R1-2500106	3-antenna-port UL transmissions	MediaTek Inc.
R1-2500117	Remaining Details on Rel-19 CB-based UL for 3TX UE	InterDigital, Inc.
R1-2500219	Views on 3Tx uplink transmissions	CATT
R1-2500243	Discussion on 3-antenna-port codebook-based transmissions	ZTE Corporation, Sanechips
R1-2500281	Discussion on support for 3-antenna-port codebook-based transmissions	CMCC
R1-2500344	Remaining issues on 3-antenna-port codebook-based uplink transmissions	vivo
R1-2500472	Discussion on 3-antenna-port uplink transmissions	OPPO
R1-2500539	Remaining issues for 3 Tx UL transmissions	Ericsson
R1-2500551	Uplink 3 Port Codebook based Transmission	Google
R1-2500701	Support for 3-antenna-port codebook-based transmissions	Lenovo
R1-2500727	Discussion on the support of 3-antenna-port CB based transmissions	Xiaomi
R1-2500773	Remaining issues on R19 MIMO 3Tx transmission	Apple
R1-2500842	Views on Rel-19 3-antenna-port codebook-based transmissions	Samsung
R1-2500879	Discussions on 3-antenna-port codebook-based UL transmission	China Telecom Corporation Ltd.
R1-2500898	On the support for 3-antenna-port codebook-based transmissions	Nokia
R1-2500932	Discussion on uplink enhancement for UE with 3Tx	Fujitsu
R1-2501129	Support for 3-antenna-port transmission	Sharp
R1-2501151	Remaining issues for 3Tx CB based PUSCH	Qualcomm Incorporated
R1-2501197	Discussion on support for 3-antenna-port codebook-based transmissions	NTT DOCOMO, INC.

R1-2500119 FL Summary Support for 3TX CB-based Uplink; First Round Moderator (InterDigital, Inc.)

From Monday session

Agreement

For 3T3R antenna switching, reuse the same mechanism as xTxR for the SRS configuration as follows:

- Up to two SRS resource sets each with one SRS resource can be configured, where the number of SRS ports for each resource is equal to 4 with port 1003 disabled if the UE is not indicating srs-AntennaSwitching2SP-1Periodic.
- Up to two SRS resource sets configured with *resourceType* in *SRS-ResourceSet* set to 'semi-persistent' and one SRS resource set configured with *resourceType* in *SRS-ResourceSet* set to 'periodic' can be configured and the two SRS resource sets configured with 'semi-persistent' are not activated at the same time, or up to two SRS resource sets can be configured, if the UE is indicating srs-AntennaSwitching2SP-1Periodic, where each SRS resource set has one SRS resource, the number of SRS ports for each resource is equal to 4 with port 1003 disabled.
- For 3T=3R, the UE shall not expect to be configured or triggered with more than one SRS resource set with higher layer parameter usage set as '*antennaSwitching*' in the same symbol.

Note: This is not meant to be a text proposal to be captured for the specifications.

Conclusion

To support 3T3R antenna switching for a 3TX UE,

- Reuse the power control parameters configuration principles, i.e., 3T6R, when two aperiodic SRS resource sets are configured,
  - If a UE is provided TCI-State in dl-OrJointTCI-StateList or TCI-UL-State, the UE shall expect that the two SRS resource sets are associated with the same values of the higher layer parameters *p0AlphaSetforSRS* and *pathlossReferenceRS-Id*
  - Otherwise, the UE shall expect that the two SRS resource sets are configured with the same values of the higher layer parameters *alpha, p0, pathlossReferenceRS*, and *srs-PowerControlAdjustmentStates* in *SRS-ResourceSet*.

Conclusion

For a 3TX UE, following agreements for codebook-based transmission can also be extended for non-codebook-based transmission,

Agreement

For codebook-based transmission by a 3TX UE,

- Option- 2: Subject to UE capability, up to 2 PTRS ports may be configured in PTRS-UplinkConfig,
  - FFS whether a single bit or 2 bits are used for PTRS-DMRS association indication.

Above is only for single panel transmission.

Agreement

For codebook-based UL transmission by a 3TX UE, when 1 PTRS port is configured by maxNrofPorts in PTRS-UplinkConfig, PTRS-DMRS association indication is as follows:

- Alt2: 2-bit indication

PTRS-DMRS association when 1 PT-RS port is configured

Value	DMRS port
0	1st scheduled DMRS port
1	2nd scheduled DMRS port

	2	3rd scheduled DMRS port
	3	~~4th scheduled DMRS port~~ Reserved

Agreement

For non-codebook-based UL transmission for a 3TX UE, when 2 PTRS ports are configured by *maxNrofPorts in PTRS-UplinkConfig, PTRS-DMRS association indication for the two DMRS ports that share a PTRS port is as follows:*

Value of MSB	DMRS port	~~Value of LSB~~	~~DMRS port~~
0	1st DMRS port which shares the PTRS port	~~0~~	~~1st DMRS port which shares PTRS port 1~~
1	2nd DMRS port which shares the PTRS port	~~1~~	~~2nd DMRS port which shares PTRS port 1~~

- Number of bits used for the indication
  - 1 bit
- Above is applicable for when a 3TX UE is configured with 3 SRS resources for non-codebook-based UL transmission

Conclusion

For a 3TX UE, following agreements for codebook-based transmission can also be extended for non-codebook-based transmission,

Agreement

For codebook-based M-TRP PUSCH repetition by a 3TX UE, for indication of PTRS-DMRS association,

- When 1 PTRS port is configured by maxNrofPorts in PTRS-UplinkConfig, reuse Rel-17 multi-TRP TDM repetition,
- When 2 PTRS ports are configured by maxNrofPorts in PTRS-UplinkConfig, and maxRank = 2 or 3,
  - A second PTRS-DMRS association field (1 bit) is used to indicate the association between PTRS ports and DMRS ports for the 2nd SRS resource set (i.e.,2nd TRP).

Agreement

For 3T3R and 3T6R antenna switching, support the following separate UE capabilities for SRS configuration,

- *"srs-AntennaSwitching3T6R2SP-1Periodic"* to extend *srs-AntennaSwitching2SP-1Periodic* to the case of 3T6R.
- *"srs-AntennaSwitching3T3R2SP-1Periodic"* to extend *srs-AntennaSwitching2SP-1Periodic* to the case of 3T3R.

Final summary in:

R1-2500121 Summary of Discussion on 3TX CB-based Uplink in RAN1#120 Moderator (InterDigital, Inc.)

9.2.4 Enhancement for asymmetric DL sTRP/UL mTRP scenarios

Including support for 2TA without CoresetPoolIdx association for asymmetric DL sTRP/UL mTRP.

R1-2500100 Enhancements for asymmetric DL sTRP/UL mTRP scenarios Huawei, HiSilicon
R1-2500107 Enhancement for asymmetric DL sTRP/UL mTRP scenarios MediaTek Inc.
R1-2500118 Remaining Details on Rel-19 Asymmetric mTRP Operation InterDigital, Inc.
R1-2500165 Enhancements for asymmetric DL sTRP/UL mTRP scenarios Spreadtrum, UNISOC
R1-2500220 Discussion on enhancement for asymmetric DL sTRP/UL mTRP scenarios CATT
R1-2500239 Discussion on asymmetric DL sTRP/UL mTRP scenariosTCL
R1-2500244 Discussion on enhancements for asymmetric DL sTRP/UL mTRP scenarios ZTE Corporation, Sanechips, China Telecom
R1-2500256 Discussion on enhancements for asymmetric DL sTRP/UL mTRP scenarios China Telecom, ZTE
R1-2500282 Discussion on enhancement for asymmetric DL sTRP/UL mTRP scenarios CMCC
R1-2500345 Remaining issues on asymmetric DL sTRP/UL mTRP scenarios vivo
R1-2500473 Enhancements on asymmetric DL sTRP/UL mTRP scenarios OPPO
R1-2500495 Remaining issues on asymmetric DL sTRP UL mTRP scenarios Ericsson
R1-2500498 Enhancement for asymmetric DL sTRP UL mTRP Tejas Network Limited
R1-2500514 Enhancements on asymmetric DL STRP/UL MTRP scenarios Ofinno
R1-2500570 Discussions on asymmetric DL sTRP/UL mTRP scenarios LG Electronics
R1-2500593 Discussion on enhancements for asymmetric DL sTRP and UL mTRP scenarios NEC
R1-2500647 Enhancement for asymmetric DL sTRP/UL mTRP scenarios Sony
R1-2500702 Enhancement for asymmetric DL sTRP/UL mTRP scenarios Lenovo
R1-2500728 Discussion on enhancement for asymmetric DL sTRP/UL mTRP scenarios Xiaomi
R1-2500774 Enhancements for asymmetric DL sTRP/UL mTRP Apple
R1-2500843 Views on Rel-19 asymmetric DL sTRP/UL mTRP scenarios Samsung
R1-2500899 Enhancement for asymmetric DL sTRP/UL mTRP scenarios Nokia
R1-2500907 Discussion on UL mTRP and DL single TRP operation ETRI
R1-2500933 Discussion on UL-only mTRP operation Fujitsu
R1-2500964 Discussion on enhancements for asymmetric DL sTRP/UL mTRP scenarios Transsion Holdings

R1-2500978 "Enhancement for Asymmetric DL sTRP/UL mTRP Scenarios " Panasonic
R1-2501121 Enhancement for asymmetric DL sTRP/UL mTRP scenarios Sharp
R1-2501152 Enhancement for asymmetric DL sTRP and UL mTRP deployment scenarios Qualcomm Incorporated
R1-2501198 Discussion on enhancement for asymmetric DL sTRP/UL mTRP scenarios NTT DOCOMO, INC.
R1-2501237 Discussion on asymmetric DL sTRP and UL mTRP ASUSTeK
R1-2501335 Discussion on enhancement for asymmetric DL sTRP and UL mTRP scenarios Google

R1-2501358 Summary #1 on Rel-19 asymmetric DL sTRP/UL mTRP Moderator (OPPO)
From Monday session
Agreement
For a UE provided with *SSB-MTC-AddtionalPCI* and not configured with multi-DCI based mTRP, support to reuse the DCI field 'PRACH association indicator' in DCI format 1_0 to indicate PL RS for PDCCH-order PRACH:

- The bit field index 0 of this field indicates the DL RS that DMRS of PDCCH order DCI is QCLed with is used as PL RS for PRACH;
- The bit field index 1 of this field is mapped to the additional PCI associated with the active TCI states and indicates the indicated SSB in this DCI is used as PL RS for PRACH:
  - In this case, the PRACH configuration associated with addition PCI is used.
- This DCI field is present when the corresponding RRC parameter is configured and multi-DCI based mTRP is not configured.

Conclusion
There is no RAN1 consensus to support the following proposal:
Support the UE to report either Type1 PHR or Type 3 PHR in a serving cell configured with one UL carrier and two separate SRS CLPC adjustment states. The UE determines to report Type1 PHR or Type 3 PHR according to:

- UE provides the Type 1 PHR if both the Type 1 PHR and the Type 3 PHR are based on respective actual transmissions or on respective reference transmissions.
- UE provides the PHR that is based on a respective actual transmission if either the Type 1 PHR or the Type 3 PHR is based on a respective reference transmission.

This is subject to UE capability.
This is feature is enabled by one new RRC parameter.

R1-2501359 Summary #2 on Rel-19 asymmetric DL sTRP/UL mTRP Moderator (OPPO)
From Tuesday session
Conclusion
There is no consensus to introduce the restriction that only one of the indicated joint/UL TCI states can be configured with PL offset.

Conclusion
There is no consensus to include PL offset in Type 3 PHR calculation.

Conclusion
In asymmetric DL sTRP/UL mTRP deployment scenario, the PDCCH-order PRACH to UL TRP is CFRA.

Agreement
- The legacy TPC application mechanisms specified in legacy spec are reused for the TPC command for separate SRS CLPC adjustment state(s) indicated in DCI format 1_1 in Rel-19 and the TPC command for separate SRS CLPC adjustment states indicated in DCI format 2_3 in Rel-19.
  - One example specification changes in TS38.213 can be

7.3.1 UE behavior

*** Unchanged parts are omitted ***

- $\delta_{\mathrm{SRS},b,f,c}(m)$ is jointly coded with other TPC commands in a PDCCH with DCI format 2_3, as described in clause 11.4 or provided by TPC command field XXX in a PDCCH with DCI format 1_1

*** Unchanged parts are omitted ***

- $h_{b,f,c}(i) = \delta_{\mathrm{SRS},b,f,c}(i)$ if the UE is not configured for PUSCH transmissions on active UL BWP b of carrier f of serving cell c, or if *srs-PowerControlAdjustmentStates* indicates separate power control adjustment states between SRS transmissions and PUSCH transmissions, and *tpc-Accumulation* is provided, and the UE detects a DCI format 2_3 or DCI format 1_1 carrying DCI field XXX $K_{\mathrm{SRS,min}}$ symbols before a first symbol of SRS transmission occasion i, where absolute values of $\delta_{\mathrm{SRS},b,f,c}$ are provided in Table 7.1.1-1

*** Unchanged parts are omitted ***

  - Note: how to capture this in specification is up to editor.

- In Rel-19, the TPC command for separate SRS CLPC adjustment state(s) indicated in DCI format 1_1 and the TPC command for separate SRS CLPC adjustment states indicated in DCI format 2_3 are applicable to periodic SRS, semi-persistent SRS and aperiodic SRS.

R1-2501360 Summary #3 on Rel-19 asymmetric DL sTRP/UL mTRP Moderator (OPPO)
From Thursday session
Agreement
Study whether/how to reset the CLPC of PUSCH/PUCCH/SRS based on the following alternatives (including if any enhancement is necessary and other alternatives are not precluded)

- Alt1: For a UE configured with two TAGs, when the UE receives a RAR of a PDCCH-order PRACH, according to the new 1-bit DCI field PL offset indication in DCI format 1_0:
  - When there is one indicated joint/UL TCI state in Rel-17 Unified TCI:
    - If the 1-bit DCI field is 0, the UE resets the CLPC that is different from the CLPC associated with the indicated joint/UL TCI state.
    - If the 1-bit DCI field is 1, the UE resets the CLPC associated with the indicated joint/UL TCI state.
  - When there are two indicated joint/UL TCI states in Rel-18 Unified TCI:
    - If the 1-bit DCI field is 0, the UE resets the CLPC associated with the first indicated joint/UL TCI state;
    - If the 1-bit DCI field is 1, the UE resets the CLPC associated with the second indicated joint/UL TCI state
- Alt 2: For a UE configured with two TAGs, one TAG ID is indicated in the RAR
  - The UE resets the closed loop power control adjustment state associated with a joint/UL TCI state associated with the indicated TAG.

9.3 Evolution of NR duplex operation: Sub-band full duplex (SBFD)

Please refer to RP-241614 for detailed scope of the WI.
Rapporteur to provide initial input on higher layer signalling under agenda item 9.3. For input on higher layer signalling from any other source, please include it as part of your tdoc to relevant sub-agenda items.

[120-R19-Duplex] – Xinghua (Huawei)
Email discussion on Rel-19 SBFD

- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500157 Higher layer parameters for Rel-19 SBFD Huawei, HiSilicon

From AI5
R1-2500019 LS on CSI-RS measurement with SBFD operation RAN4, MediaTek
Decision: RAN4 is requesting clarification from RAN1 on CSI-RS measurement with SBFD operation to progress with their RRM requirement. RAN1 response is necessary - Mohamed (MediaTek)
R1-2501536 Summary of Discussion on LS on CSI-RS measurement with SBFD operation Moderator (MediaTek)
From Wednesday session
R1-2501537 Draft Reply LS on CSI-RS measurement with SBFD operation Moderator (MediaTek)
Decision: Response to RAN4 in R1-2501537 is endorsed. Final LS is approved in R1-2501560.

9.3.1 SBFD TX/RX/measurement procedures

Including semi-static indication of SBFD resources.

R1-2500061 Discussion on SBFD TX/RX/measurement procedures TCL
R1-2500093 On subband full duplex design Huawei, HiSilicon
R1-2500144 Discussion on transmission, reception and measurement procedures for SBFD operation ZTE Corporation, Sanechips
R1-2500166 Discussion on SBFD TX/RX/measurement procedures Spreadtrum, UNISOC
R1-2500221 Discussion on SBFD TX/RX/measurement procedures CATT
R1-2500249 Discussion on SBFD TX/RX/measurement procedures LG Electronics
R1-2500257 Discussion on SBFD TX/RX/measurement procedures China Telecom
R1-2500283 Discussion on SBFD TX/RX/measurement procedures CMCC
R1-2500314 Discussion on SBFD TX/RX/measurement procedures MediaTek Inc.

R1-2500346	Discussion on Rel-19 SBFD operation	vivo
R1-2500401	Discussion for SBFD TX/RX/ measurement procedures	Tejas Networks Limited
R1-2500454	Discussion on SBFD Tx/Rx/measurement procedures	OPPO
R1-2500500	Discussion on SBFD Tx/Rx/measurement procedures	Fraunhofer HHI, Fraunhofer IIS
R1-2500515	Discussion on SBFD TX/RX/measurement procedures	Ofinno
R1-2500520	Views on SBFD TX_RX_measurement procedures	InterDigital, Inc.
R1-2500620	Discussion on subband non-overlapping full duplex Tx-Rx and measurement operations	NEC
R1-2500648	SBFD procedures	Sony
R1-2500697	Discussion on SBFD TX/RX/measurement procedures	Panasonic
R1-2500729	Discussion on reception and transmission procedure for SBFD operation	Xiaomi
R1-2500749	SBFD TX/RX/measurement procedures	Ericsson
R1-2500775	Discussion on SBFD TX/RX/measurement procedures	Apple
R1-2500847	SBFD operation and procedures	Samsung
R1-2500880	Discussion on SBFD TX/RX/ measurement procedures	CEWiT
R1-2500908	Discussion on SBFD TX/RX/measurement procedures	ETRI
R1-2500934	Discussion on SBFD Tx/Rx/measurement procedures	Fujitsu
R1-2500965	Discussion on SBFD operation	Transsion Holdings
R1-2500988	Discussion on SBFD TX/RX/measurement procedures	Kookmin University
R1-2501003	On SBFD TX/RX/measurement procedures	Nokia, Nokia Shanghai Bell
R1-2501010	Views on SBFD TX/RX Measurement Procedures	Lenovo
R1-2501052	SBFD Tx/Rx/measurement procedures	Charter Communications, Inc
R1-2501055	SBFD Tx/Rx/measurement aspects	Sharp
R1-2501088	Discussion on SBFD TX/RX/measurement procedures	Hyundai Motor Company
R1-2501107	Discussion on SBFD TX/RX/measurement procedures	HONOR
R1-2501153	SBFD Transmission, Reception and Measurement Procedures	Qualcomm Incorporated
R1-2501199	Discussion on SBFD TX/RX/measurement procedures	NTT DOCOMO, INC.
R1-2501228	Discussion on SBFD TX/RX/measurement procedures	Google Korea LLC
R1-2501231	Discussion on SBFD TX/RX/measurement procedures	ITRI
R1-2501240	Partial PRG handling for SBFD	ASUSTeK
R1-2501285	Discussion on SBFD TX/RX/measurement procedures	WILUS Inc.

R1-2501338 Summary #1 of SBFD TX/RX/measurement procedures Moderator (Xiaomi)

From Tuesday session

Agreement

For aperiodic SRS with available slot counting,

- For *SRS-ResourceSet* configured for SBFD symbol, an available slot is a slot satisfying there are SBFD symbol(s) for the time-domain location(s) for all the SRS resources in the resource set and it satisfies UE capability on the minimum timing requirement between triggering PDCCH and all the SRS resources in the resource set.
- For *SRS-ResourceSet* configured for non-SBFD symbol, an available slot is a slot satisfying there are UL or flexible symbol(s) not configured as SBFD symbols for the time-domain location(s) for all the SRS resources in the resource set and it satisfies UE capability on the minimum timing requirement between triggering PDCCH and all the SRS resources in the resource set.

Agreement

The configuration of Configuration 1 or 2 for DL BWP only applies to PDSCH receptions within the DL BWP.

Agreement

For a CSI report associated with periodic/semi-persistent CSI-RS, the valid symbol type for CSI derivation is configured in *CSI-ReportConfig*.

Agreement

For Configuration 1: The transmissions/receptions are restricted to SBFD symbols only or non-SBFD symbols only,

- The valid symbol type for PUSCH carrying SP-CSI is determined based on the symbol type of the first PUSCH occasion associated with activation DCI.
- For SP-CSI on PUCCH:
  - Option 2: the valid symbol type for PUCCH carrying SP-CSI is configured in *semiPersistentOnPUCCH* in *CSI-ReportConfig*.

Conclusion

- There is no RAN1 consensus to support semi-static link direction indication for SBFD aware UEs.

- For a serving cell configured with SBFD subband time and frequency location, for an SBFD aware UE, *TDD-UL-DL-ConfigDedicated* is only applicable to non-SBFD symbols but not applicable to SBFD symbols

Agreement
For determining starting PRB for PUSCH transmissions in SBFD symbols for Configuration 2,
- For type 1 CG PUSCH, a RB_{offset}^{SBFD} is configured in *rrc-ConfiguredUplinkGrant* in *ConfiguredGrantConfig*.
- For type 2 CG PUSCH and DG PUSCH, a RB_{offset}^{SBFD} is configured per UL BWP in *PUSCH-Config*.
  - The RB_{offset}^{SBFD} is commonly applicable to all DCI formats which schedule PUSCH
- If RB_{offset}^{SBFD} is not configured, it is assumed to be zero

Agreement
When higher layer parameter *ul-FullPowerTransmission* is set to '*fullpowerMode2*', the numbers of SRS ports should be the same for SRS resources with the same corresponding SRI values in the two SRS resource sets associated with SBFD and non-SBFD symbols.

R1-2501339 **Summary #2 of SBFD TX/RX/measurement procedures** **Moderator (Xiaomi)**
From Wednesday session
Agreement
For Configuration 1: The transmissions/receptions are restricted to SBFD symbols only or non-SBFD symbols only,
- The valid symbol type for Type 1 CG PUSCH is configured in *rrc-ConfiguredUplinkGrant* in *ConfiguredGrantConfig*.
- The valid symbol type for PUCCH configured for SR is configured in *SchedulingRequestResourceConfig*.
- The valid symbol type for PUCCH carrying P-CSI is configured in *periodic* in *CSI-ReportConfig*.

Agreement
For a type 1 CG PUSCH with Configuration 2, if FH offset for SBFD symbols is not configured for the type 1 CG PUSCH, frequency hopping is disabled for the type 1 CG PUSCH transmissions in SBFD symbols.
For a type 2 CG PUSCH/PUSCH scheduled by DCI, if FH offset lists for SBFD symbols are not configured for the type 2 CG PUSCH/PUSCH scheduled by DCI, frequency hopping is disabled for the transmission of the type 2 CG PUSCH/PUSCH scheduled by DCI in SBFD symbols.

Agreement:
For SPS PDSCH with Configuration 2, for TBS determination,
- For SPS PDSCH without repetition,
  - The number of PRBs for TBS determination for an SPS PDSCH in SBFD symbols is based on assigned PRBs within DL usable PRBs only.
  - The number of PRBs for TBS determination for an SPS PDSCH in non-SBFD symbols is based on assigned PRBs.
- For SPS PDSCH with repetition, TBS is determined based on the first repetition occasion.
  - If the first repetition occasion of a SPS PDSCH is in SBFD symbols, the number of PRBs for TBS determination for the SPS PDSCH is based on assigned PRBs within DL usable PRBs only.
  - If the first repetition occasion of a SPS PDSCH is in non-SBFD symbols, the number of PRBs for TBS determination for the SPS PDSCH is based on assigned PRBs.

R1-2501340 **Summary #3 of SBFD TX/RX/measurement procedures** **Moderator (Xiaomi)**
From Thursday session
Agreement
If starting PRB is not configured for SBFD symbols for a *PUCCH-Resource*, starting PRB configured for non-SBFD symbols for the *PUCCH-Resource* is used for PUCCH transmissions in SBFD symbols associated with this *pucch-ResourceId*.

Agreement
The one configuration of *intraSlotFrequencyHopping* for a *PUCCH-Resource* is applied to both PUCCH transmissions in SBFD symbols and in non-SBFD symbols associated with this *pucch-ResourceId*.

Agreement
Support SBFD operation on one TDD carrier in multi-carrier scenario.

- Note: If the TDD carrier for SBFD operation is an Scell, support UE dedicated signaling for cell-specific configuration of time and frequency location of SBFD subbands for the Scell.
- FFS whether/how to handle half-duplex CA case

9.3.2 SBFD random access operation

R1-2500062 Discussion on SBFD random access operation TCL
R1-2500094 On subband full duplex random access operation Huawei, HiSilicon
R1-2500145 Discussion on SBFD random access operation ZTE Corporation, Sanechips
R1-2500167 Discussion on SBFD random access operation Spreadtrum, UNISOC
R1-2500222 Discussion on SBFD random access operation CATT
R1-2500250 Discussion on SBFD random access operation LG Electronics
R1-2500258 Discussion on SBFD random access operation China Telecom
R1-2500284 Discussion on SBFD random access operation CMCC
R1-2500315 Discussion on SBFD Random Access Operation MediaTek Inc.
R1-2500347 Discussion on random access for Rel-19 SBFD vivo
R1-2500402 Discussion on SBFD Random Access Operation Tejas Networks Limited
R1-2500455 Discussion on SBFD random access operation OPPO
R1-2500516 Discussion on SBFD for random-access procedure Ofinno
R1-2500521 Discussion on SBFD random access operation InterDigital, Inc.
R1-2500587 Discussion on random access for SBFD NEC
R1-2500649 SBFD RACH operations Sony
R1-2500698 Discussion on SBFD random access operation Panasonic
R1-2500730 Discussion on SBFD random access operation Xiaomi
R1-2500750 SBFD Random access operation Ericsson
R1-2500776 Views on SBFD random access operation Apple
R1-2500848 Random access on SBFD resources Samsung
R1-2500881 Discussion on SBFD random access operation CEWiT
R1-2500883 SBFD random access operation Lenovo
R1-2500909 SBFD random access operation ETRI
R1-2500935 Discussion on SBFD random access operation Fujitsu
R1-2500966 Discussion on SBFD random access operation Transsion Holdings
R1-2500989 Discussion on SBFD random access operation Kookmin University
R1-2501004 SBFD random access operation Nokia, Nokia Shanghai Bell
R1-2501056 SBFD random access aspects Sharp
R1-2501154 SBFD Random Access Operation Qualcomm Incorporated
R1-2501200 Discussion on SBFD random access operation NTT DOCOMO, INC.
R1-2501227 Discussion on SBFD Random Access operation KT Corp.
R1-2501232 Discussion on SBFD random access operation for SBFD aware UEs ITRI
R1-2501286 Discussion on SBFD random access operation WILUS Inc.
R1-2501291 On SBFD random access operation Google Ireland Limited

R1-2501415 Summary#1 on SBFD random access operation Moderator (CMCC)
From Tuesday session
Conclusion
There is no RAN1 consensus on the following proposal:
For RACH configuration Option 1, for support of separate power control parameters for PRACH transmission in additional-ROs and legacy-ROs,
- *also support separate powerRampingStep and powerRampingStepHighPriority for additional-ROs.*

R1-2501416 Summary#2 on SBFD random access operation Moderator (CMCC)
From Wednesday session
Conclusion
For RACH configuration Option 2, for the case that additional ROs overlap with legacy ROs, support Alt-1:
- Alt-1: do not optimize for this case

Conclusion
For RACH configuration Option 2, and for interpretation of the parameter *prach-ConfigurationIndex* provided by the additional RACH configuration:
- For FR1 support Alt-4: no enhancement
- For FR2 support Alt-4: no enhancement

Agreement (confirmed on Thursday with the following changes in red)
For determination of the Msg3 PUSCH transmission power and when separate *preambleReceivedTargetPower* for additional-ROs is configured for RACH configuration Option 1 ~~or Option 2~~
- *preambleReceivedTargetPower* configured for legacy-RO is used if Msg3 PUSCH is transmitted in non-SBFD symbols;
- *preambleReceivedTargetPower* configured for additional-RO is used if Msg3 PUSCH is transmitted in SBFD symbols.
- FFS: Option 2

Companies to check the above proposal for endorsement on Thursday.

Conclusion
For SBFD-aware UEs, there is no restriction on the combination of RO type of the latest PRACH transmission and the symbol type of following Msg3 PUSCH transmission or PUCCH carrying HARQ ACK for Msg4.

Agreement
Consider the following alternatives about power ramping when a SBFD-aware UE switches from additional-ROs to legacy-ROs and from legacy-ROs to additional-ROs (if supported) in PRACH transmission re-attempt in one RACH procedure for RACH configuration Option 1 and Option 2.
- Alt-1: Increase the power ramping counter.
  - FFS whether to introduce a power offset to compensate the power ramping difference for RACH configuration Option 2.
- Alt-2: Reset the power ramping.
- Note: different alternatives can be applied for different RACH configuration options.

Agreement
For an SBFD aware UE, if additional-RO is selected for PRACH transmission, for Msg3 PUSCH repetition Configuration 1,
- In case the valid symbol type is SBFD symbol, a slot is determined as available if the symbols allocated for Msg3 PUSCH are all SBFD symbols and not include a symbol of an SS/PBCH block with index provided by *ssb-PositionsInBurst*.
- In case the valid symbol type is non-SBFD symbol, a slot is determined as available if the symbols allocated for Msg3 PUSCH are all non-SBFD symbols and not include a DL symbol indicated by *tdd-UL-DL-ConfigurationCommon* if provided or a symbol of an SS/PBCH block with index provided by *ssb-PositionsInBurst*.

Agreement
For RACH configuration Option 1, when separate configuration of *rsrp-ThresholdMsg1-RepetitionNum2/4/8* for PRACH transmission with preamble repetitions within additional-ROs is not configured, the *rsrp-ThresholdMsg1-RepetitionNum2/4/8* configured for PRACH transmission with preamble repetitions within legacy-ROs is reused for additional-ROs.

R1-2501417 Summary#3 on SBFD random access operation Moderator (CMCC)
From Thursday session
Conclusion
There is no consensus in RAN1 to support Type-2 random access procedure (2-step RACH) in SBFD symbols for SBFD-aware UEs.

Agreement
For RACH configuration Option 1, down-select from the following two alternatives:
- Alt-1: Shared *ssb-SharedRO-MaskIndex* configuration for Additional-ROs and Legacy-ROs.
- Alt-2: Separate *ssb-SharedRO-MaskIndex* configurations for Additional-ROs and Legacy-ROs.

9.3.3 CLI handling

R1-2500095	On cross-link interference handling for subband full duplex	Huawei, HiSilicon
R1-2500146	Discussion on CLI handling for Rel-19 duplex operation	ZTE Corporation, Sanechips
R1-2500168	Discussion on CLI handling	Spreadtrum, UNISOC
R1-2500223	Discussion on CLI handling for NR duplex evolution	CATT
R1-2500251	Discussion on CLI handling	LG Electronics
R1-2500285	Discussion on CLI handling	CMCC
R1-2500316	Discussion on CLI Handling in SBFD system	MediaTek Inc.
R1-2500348	Discussion on CLI handling for Rel-19 NR duplex	vivo
R1-2500403	Discussion on gNB-gNB CLI handling	Tejas Networks Limited
R1-2500456	CLI handling in NR duplex operation	OPPO
R1-2500522	Discussion on CLI handling for SBFD operation	InterDigital, Inc.
R1-2500588	CLI handling for NR duplex operations	NEC
R1-2500650	SBFD CLI handling	Sony
R1-2500707	Discussion on CLI handling	Panasonic

R1-2500731 Discussion on CLI handling for SBFD operation Xiaomi
R1-2500751 CLI handling Ericsson
R1-2500777 Discussion on CLI handling Apple
R1-2500849 Cross-link interference handling for SBFD Samsung
R1-2500882 Discussion on CLI handling CEWiT
R1-2500910 Discussion on CLI handling for SBFD operation ETRI
R1-2501005 Cross-link interference handling for duplex evolution Nokia, Nokia Shanghai Bell
R1-2501061 Cross-link interference (CLI) handling Sharp
R1-2501155 Enhancements for CLI handling Qualcomm Incorporated
R1-2501201 Discussion on CLI handling NTT DOCOMO, INC.
R1-2501292 On the gNB-to-gNB CLI and the UE-to-UE CLI handling Google Ireland Limited

R1-2501353 Discussion on CLI handling for Rel-19 NR duplex vivo
R1-2501455 Summary #1 of CLI handling Moderator (Huawei)
From Tuesday session
Agreement
For the semi-statically configured time location and frequency location of UL muting symbol(s) for PUSCH,
- Separate configurations of up to two UL muting symbols can be provided per UE for DG PUSCH/Type 2 CG PUSCH and Type 1 CG PUSCH
  - Separate configuration for each of multiple Type 1 CG PUSCH configurations is supported

Agreement
For PUSCH that is scheduled by DCI format 0_0 or Type 2 CG PUSCH activated by DCI format 0_0, UL resource muting is not applied.

Agreement
Confirm the following working assumption
Working assumption
For the frequency location configuration of UL resource muting for a PUSCH, for DFT-S-OFDM
- A comb offset {0, 1} is configured for the up to two UL muting symbols

Agreement
For the configuration of UL resource muting
- Alt.1: A new IE *PUSCH-MutingResources* is introduced to define the time location and frequency location of UL muting resources. The UL muting pattern configures up to two symbols within a slot and a configurable comb offset.

Agreement
For a CSI report with *CSI-ReportConfig* with higher layer parameter *reportQuantity* set to 'cli-RSSI' or 'cli-SRS-RSRP'
- *carrier* in *CSI-ReportConfig* is used to indicate in which serving cell the CLI-RSSI measurement resources or SRS-RSRP measurement resources in *CSI-ResourceConfig* are to be found. If the field is absent, the resources are on the same serving cell as this report configuration.
- *bwp-Id* in the associated *CSI-ResourceConfig* is used to indicate the DL BWP where the CLI-RSSI measurement resources or SRS-RSRP measurement resources are located in.

Note: No RAN1 specification impact.

R1-2501456 Summary #2 of CLI handling Moderator (Huawei)
From Wednesday session
Agreement
A new parameter *ul-MutingIndicator* is introduced to *PUSCH-Allocation-r16* to indicate whether the semi-statically configured UL muting symbol(s) is enabled.

Agreement
The following agreement replaces the earlier agreement from RAN1#119
If transform precoding is enabled for a PUSCH, the following options are provided for the purpose of specification drafting (how UE implementation is done is left to each company).
- **Option 1:** The DFT size is modified as $M_{sc}^{PUSCH}/2$ for symbol(s) with UL resource muting
  - The $M_{sc}^{PUSCH}/2$ modulated symbols are transformed to the frequency domain with a DFT of size $M_{sc}^{PUSCH}/2$, and the result is mapped to the $M_{sc}^{PUSCH}/2$ unmuted scheduled subcarriers.
- **Option 2:** The DFT size is M_{sc}^{PUSCH} for symbol(s) with UL resource muting
  - Option 2A: The block of $M_{sc}^{PUSCH}/2$ modulated symbols is repeated before DFT. After DFT, the $M_{sc}^{PUSCH}/2$ non-zero even-indexed subcarriers are extracted and then mapped to the unmuted scheduled subcarriers.

- Option 2B: The block of $M_{sc}^{PUSCH}/2$ modulated symbols is extended to M_{sc}^{PUSCH} modulated symbols by inserting '0' after every modulated symbol. After DFT, the first $M_{sc}^{PUSCH}/2$ subcarriers are extracted and then mapped to the unmuted scheduled subcarriers.

Note: M_{sc}^{PUSCH} is the total number of subcarriers of the PUSCH.
Note: Above options are mathematically equivalent.
Note: It is up to the spec editor how to capture the above as long as the specification captures the above in a mathematically identical manner.

Agreement
For the PUSCH symbol containing PT-RS with UL resource muting when transform precoding is not enabled, 3 dB power boosting for both PUSCH and PTRS REs if present, regardless of the PTRS density *K* and number of PUSCH layers *L*.

Agreement
- UE determines the subcarrier spacing of the CLI-RSSI measurement resources configured in a *CSI-ResourceConfig* based on the subcarrier spacing of the DL BWP identified by the higher-layer parameter *bwp-Id* in the *CSI-ResourceConfig*.
- UE determines the subcarrier spacing of the SRS-RSRP measurement resources configured in a *CSI-ResourceConfig* based on the subcarrier spacing of the DL BWP identified by the higher-layer parameter *bwp-Id* in the *CSI-ResourceConfig*.
- Note: No RAN1 spec impact is expected.

Agreement
Update the following agreement in RAN1#118 (in red)
For frequency resource allocation of a CLI-RSSI measurement resource, the following are supported:
- Measurement resource type#1: ~~One~~The CLI-RSSI measurement resource is configured within a DL subband or a UL subband.
- Measurement resource type#2: ~~One~~The CLI-RSSI measurement resource is configured across two DL subbands.
- FFS: Number of resources that can be configured.

FFS: UE behavior for measurement.

Working assumption
For L1 based CLI measurement and reporting, the following is adopted
- For L1 SRS-RSRP, a 7-bit value in the quantization range [-140, -44] dBm with 1dB step size, and the differential L1 SRS-RSRP is quantized to a 4-bit value. The differential L1 SRS-RSRP value is computed with 2 dB step size with a reference to the largest measured L1 SRS-RSRP value.
- For L1 CLI-RSSI, a 7-bit value in the quantization range [-100, -25] dBm with 1dB step size, and the differential L1 CLI RSSI is quantized to a 4-bit value. The differential L1 CLI-RSSI value is computed with 2 dB step size with a reference to the largest measured L1 CLI-RSSI value.

Send an LS to RAN4 about the above working assumption.
R1-2501619 **[Draft] LS on L1 based UE-to-UE CLI measurement and reporting Moderator (Huawei)**
Thursday decision: The draft LS is endorsed. Final version is approved in R1-2501620.

Agreement
For discussion purpose, SRS-RSRP measurement resource index is referred to as SRS-RSRP-MRI and CLI-RSSI measurement resource index is referred to as SRS-RSRP-MRI.
The mapping order of CSI fields of one report for SRS-RSRP-MRI/SRS-RSRP or CLI-RSSI-MRI/CLI-RSSI reporting is provided in Table 1 with following modifications based on Table 6.3.1.1.2-8 defined in TS38.212
- CRI and SSBRI is replaced with SRS-RSRP-MRI or CLI-RSSI-MRI.
- Capability index is crossed out in the table.
- RSRP is replaced with L1-SRS-RSRP
- L1-CLI-RSSI is added in the table if the CSI report is associated with CLI-RSSI measurement resource.

Table 1: Mapping order of CSI fields of one report for SRS-RSRP-MRI/SRS-RSRP or CLI-RSSI-MRI/CLI-RSSI

CSI report number	CSI fields
CSI report #n	SRS-RSRP-MRI or CLI-RSSI-MRI #1 as in Table 2, if reported
	SRS-RSRP-MRI or CLI-RSSI-MRI #2 as in Table 2, if reported
	…
	SRS-RSRP-MRI or CLI-RSSI-MRI #4 as in Table 2, if reported
	L1-SRS-RSRP or L1-CLI-RSSI #1 as in Table 2, if reported
	Differential L1-SRS-RSRP or L1-CLI-RSSI #2 as in Table 2, if reported
	…
	Differential L1-SRS-RSRP or L1-CLI-RSSI #4 as in Table 2, if reported

Table 2. Bitwidth for SRS-RSRP-MRI, SRS-RSRP, CLI-RSSI-MRI and CLI-RSSI

Field	Bitwidth
SRS-RSRP-MRI	$\lceil log_2\left(K_s^{SRS-RSRP}\right)\rceil$
CLI-RSSI-MRI	$\lceil log_2\left(K_s^{CLI-RSSI}\right)\rceil$
L1-SRS-RSRP	7
L1-CLI-RSSI	7
Differential RSRP	4
Differential RSSI	4

$K_s^{SRS-RSRP}$: Number of SRS-RSRP measurement resources in the corresponding resource set
$K_s^{CLI-RSSI}$: Number of CLI-RSSI measurement resources in the corresponding resource set

R1-2501457 Summary #3 of CLI handling Moderator (Huawei)
From Thursday session
Agreement
For PUSCH repetition Type A in SBFD symbols with Configuration 2, the same time location of none, one or two UL muting symbol(s) is applied for all PUSCH repetitions in SBFD symbols.
For PUSCH repetition Type B in SBFD symbols with Configuration 2, the same time location of none, one or two UL muting symbol(s) per slot is applied for all the actual PUSCH repetitions in SBFD symbols.
Note: whether UL resource muting is also applied in non-SBFD symbols is discussed separately

Agreement
For UL muting applied for UL TBoMS, the index of the starting coded bit in a slot, except the first slot, within the *Ns* slots allocated for the transmission of TB processing over multiple slots is determined according to Section 6.2.5, TS38.212 as follows

$$k_0 = \left(k_0' + H + \tau\right) mod N_{cb}$$

and *H* is down-selected from one of the following options
- Option 1: *H* is the total number of coded bits available for transmission of the transport block in the previous slot within the *Ns* slots assuming no UCI multiplexing and assuming no muted REs.
- Option 2: *H* is the total number of coded bits available for transmission of the transport block in the previous slot within the *Ns* slots assuming no UCI multiplexing and considering the muted REs

Note: No impact on TBS determination in Option 1 or Option 2.

Agreement
- Periodic L1 CLI-RSSI reporting for L1 based UE-to-UE CLI measurement and reporting is supported
- Semi-persistent L1 CLI-RSSI reporting for L1 based UE-to-UE CLI measurement and reporting is NOT supported
- Periodic/Semi-persistent L1 SRS-RSRP reporting for L1 based UE-to-UE CLI measurement and reporting is NOT supported

Conclusion
For a CSI report with *CSI-ReportConfig* with higher layer parameter *reportQuantity* set to 'cli-RSSI' or 'cli-SRS-RSRP', new reporting criteria considering the least interfered measured L1-CLI-RSSI or L1-SRS-RSRP are not supported.

Agreement
For L1 CLI measurement and reporting, the number of occupied CPU(s) is determined as follows
- $O_{CPU} = 1$ for a CSI report with *CSI-ReportConfig* with higher layer parameter *reportQuantity* set to 'cli-RSSI', 'cli-SRS-RSRP'.

Agreement
If the scheduling offset between the last symbol of the PDCCH carrying the triggering DCI for aperiodic CLI reporting and the first symbol of the aperiodic SRS-RSRP measurement resource or CLI-RSSI measurement resources is smaller than the UE reported threshold *beamSwitchTiming*, down-select from the following alternatives:
- Alt.1: UE does not expect to be configured with a scheduling offset between the last symbol of the PDCCH carrying the triggering DCI for AP CLI and the first symbol of the aperiodic CLI SRS-RSRP or CLI-RSSI resources is smaller than the UE reported threshold *beamSwitchTiming*.
- Alt.2: Same default beam rules as AP CSI-RS resource defined in 38.214 can be applied to AP CLI SRS-RSRP or CLI-RSSI resource.

FFS: whether Alt.1 or Alt.2 has any specification impact.

R1-2501458 Summary #1 of discussion on RRC parameters for 9.3.3 Moderator (Huawei)

9.4 Solutions for Ambient IoT (Internet of Things) in NR

Please refer to RP-243326 for detailed scope of the WI.

R1-2501547 Session notes for 9.4 (Study on solutions for Ambient IoT in NR) Ad-Hoc Chair (Huawei)
Friday decision: The session notes are endorsed and contents reflected below.

[120-R19-A-IoT] – Jingwen (CMCC)
Email discussion on Rel-19 Ambient IoT
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500286 Ambient IoT Work Item work plan CMCC, Huawei, T-Mobile USA

9.4.1 Physical channels design – modulation aspects

Including discussions on CP handling for R2D.

R1-2500033 Modulation aspects for Ambient IoT Ericsson
R1-2500044 Discussion on modulation aspects for A-IoT physical channel FUTUREWEI
R1-2500124 Discussion on Ambient IoT modulation ZTE Corporation, Sanechips
R1-2500153 Physical channels design on modulation Huawei, HiSilicon
R1-2500169 Discussion on modulation aspects of physical channels design for Ambient IoT Spreadtrum, UNISOC
R1-2500224 Ambient IoT physical channel design and modulation CATT
R1-2500259 Discussion on physical channels design about modulation aspects for Ambient IoT China Telecom
R1-2500287 Discussion on modulation aspects of physical channel design CMCC
R1-2500349 Discussion on Modulation Aspects of Physical Channels Design vivo
R1-2500394 AIoT Physical channels design - modulation aspects Nokia
R1-2500424 Discussion on modulation aspects for Ambient IoT physical design TCL
R1-2500450 Discussion on modulation aspects of A-IoT OPPO
R1-2500479 Discussion on modulation aspects for A-IoT physical channel Tejas Network Limited
R1-2500497 Discussion on Physical Channel Modulation Aspects for Ambient-IoT EURECOM
R1-2500583 A-IoT Modulation Aspects Panasonic
R1-2500610 Discussion on modulation aspects of ambient IoT NEC
R1-2500651 Modulation aspects for Ambient IoT Sony
R1-2500732 Discussion on modulation aspects for Ambient IoT Xiaomi
R1-2500778 On modulation aspects for Ambient IoT Apple
R1-2500850 Views on Physical channels design – modulation aspects Samsung
R1-2500936 Modulation for R2D and D2R Fujitsu
R1-2500947 Discussion on D2R modulation schemes for A-IoT IIT Kharagpur
R1-2500949 A-IoT PHY layer design - waveform and modulation aspects LG Electronics
R1-2501016 A-IoT - PHY modulation aspects MediaTek Inc.
R1-2501071 Discussion on modulation aspects for Ambient IoT physical channel design Lenovo
R1-2501092 Discussion on modulation aspects Sharp
R1-2501108 Discussion on Physical channels design for Ambient IoT-modulation aspects HONOR
R1-2501117 Modulation aspects of physical channel design for Ambient IoT InterDigital, Inc.
R1-2501156 Physical channels design – modulation aspects Qualcomm Incorporated
R1-2501202 Discussion on modulation aspects of physical channel design for Ambient IoT NTT DOCOMO, INC.
R1-2501287 Discussion on modulation aspects of physical channels design for Ambient IoT WILUS Inc.
R1-2501310 Discussion on modulation related aspects of AIoT IIT Kanpur
R1-2501351 Discussion on Physical channels design for Ambient IoT Indian Institute of Tech (M)

R1-2501407 FL summary #1 for Ambient IoT: "9.4.1 Physical channels design – modulation aspects Moderator (Huawei)
From Monday session
Agreement
For D2R, only 2SB (Double sideband) modulation is supported.

Agreement
For R2D transmission, only subcarrier spacing of 15 kHz is supported (as per TR 38.769), and no other SCS are further considered.

Agreement
For a D2R transmission, it is up to device implementation to use OOK or BPSK modulation. It is assumed that a device operates consistently with the same D2R modulation in the same inventory/command round.

R1-2501408 FL summary #2 for Ambient IoT: "9.4.1 Physical channels design – modulation aspects Moderator (Huawei)
From Wednesday session
Conclusion
For R2D, $B_{occ,R2D}$ is not specified in RAN1.
- This does not preclude RAN4 to discuss $B_{occ,R2D}$ themselves when defining requirement(s), if needed.

Agreement
RAN1 does not specify the exact bandwidth a reader shall use for $B_{tx,R2D}$.
- For R2D, specify the minimum $B_{tx,R2D}$ # of PRBs associated to each M value.
- For R2D, whether to specify the maximum $B_{tx,R2D}$ # of PRBs is up to RAN4
- Note: Reader can transmit any $B_{tx,R2D}$ subject to this minimum $B_{tx,R2D}$ and the agreed 15 kHz subcarrier spacing, and subject to other limitations that are up to RAN4.

Conclusion
For CW waveform, A-IoT WID RP-243326 and TR 38.769 already give us:
It is left up to specification drafting phase how to capture the following.

F. Carrier wave transmission for waveform 1 only, without hopping, per the following case in TR 38.769:
  - Case 1-4 for D1T1-B

6.8.2 CW characteristics
Candidates for the CW for D2R backscattering are waveforms consisting of:
Waveform 1: A single-tone unmodulated sinusoid, also referred to as 'a single tone'.
Waveform 2: Two single tones.

R1-2501409 FL summary #3 for Ambient IoT: "9.4.1 Physical channels design – modulation aspects Moderator (Huawei)
From Thursday session
Agreement
For R2D, for the OOK-4 modulation for M-chip per OFDM symbol transmission:
- The maximum M value is no less than 16, and to be down-selected from 32, 24 or 16 at RAN1#120bis.

Agreement
For CP handling which retains subcarrier orthogonality, the candidate methods are clarified as follows:
- For Method Type 1 (as per TR 38.769)
  - For purposes of evaluating Method Type 1, it is assumed the device is aware of or determines the boundary of an OFDM symbol using the R2D timing acquisition signal (R-TAS), at least for Alt M1-1
    - Detail is up to R2D preamble design
  - Using Alt M1-1 and/or Alt M1-2 is up to device implementation as per TR 38.769
  - FFS: applicable M values
- For Method Type 2 Alt M2-1 (as per TR 38.769)
  - Detail design of Alt M2-1-1 or Alt M2-1-2 from the followings:
    - Alt M2-1-1
      - Candidate 1:
        - For M less than 6: Use Alt M1-1
        - For M >=6: circular shift of the M chips' samples before CP insertion at the reader side in addition to bit/chip skipping/dropping at the device side after bit decoding
      - Candidate 2:
        - Where a known signal shape in the tail-portion of the R2D signal is copied to the CP.
      - Candidate 3:
        - Insert padding chips at both start and end OOK chips or only at end OOK chips
    - Alt M2-1-2
      - Candidate 4:
        - Change $a = b \neq c$ case to $a = b = c$ case by changing OOK-4 M value to adjacent M +/- 1 for OFDM symbol n.

            - where a, b and c is the last chip of symbol n, first chip of symbol n+1 and last chip of symbol n+1.
        - Candidate 5:
            - Odd number of chips of PRDCH is transmitted in the first OFDM symbols, while M value is kept unchanged during the remaining OFDM symbols.
    - FFS: applicable M values with or without combination of Method Type 1

For CP handling which does not retain subcarrier orthogonality, the candidate methods are clarified as follows:
- For Method Type 2 Alt M2-2 (as per TR 38.769)
    - FFS detail design of Alt M2-2 from the following:
        - Candidate 1:
            - A duration of OFDM symbol with CP has integer number of OOK symbols. CP is not copied from the end of OFDM symbol. Total 14 OFDM symbols fit in a slot.

For further down-selection among candidate methods which retains subcarrier orthogonality, companies are encouraged to provide evaluation (including overhead) among the followings to RAN1#120bis:
- Method Type 1
- Method Type 2 Alt M2-1-1 with or without combination of Method Type 1
- Method Type 2 Alt M2-1-2 with or without combination of Method Type 1
- Note: The evaluation for CP handling at least includes a M value from {2,4,6,8}, a M value from {12, 16}, and a M value from {24, 32}
- Note: In this evaluation, the transmit power of the ON chips remains constant

Agreement
For D2R BPSK modulation
- After line coding or square wave, chip “1” is mapped to the real-valued modulation symbol {1} and chip “0” is mapped to the real-valued modulation symbol {-1} in the baseband.

Agreement
For D2R OOK modulation
- After line coding or square wave, chip “1” is mapped to the real-valued modulation symbol {1} and chip “0” is mapped to the real-valued modulation symbol {0} in baseband.

Agreement
For R2D transmission, from reader perspective, DFT-s-OFDM waveform is supported for OOK-4 modulation. For the details of DFT-s-OFDM waveform generation of OOK-4 modulation:
- Certain specification of DFT-s-OFDM waveform generation for OOK-4 modulation is needed in RAN1.
    - FFS: identify what potential details of 5 steps in section 4.4 of TR 38.769 needs to be specified.
- This does not preclude RAN4 to discuss R2D waveform generation themselves when defining requirement(s), if needed.

R1-2501623 Final summary for Ambient IoT: “9.4.1 Physical channels design – modulation aspects Moderator (Huawei)

9.4.2 Physical channels design – line coding, FEC, CRC, repetition aspects

Including discussions on small frequency shift

R1-2500034 Coding aspects for Ambient IoT Ericsson
R1-2500045 Discussion on coding aspects for A-IoT physical channel FUTUREWEI
R1-2500125 Discussion on Ambient IoTcoding ZTE Corporation, Sanechips
R1-2500154 Physical channel design on channel coding Huawei, HiSilicon
R1-2500170 Discussion on line coding, FEC, CRC, repetition aspects for Ambient IoT Spreadtrum, UNISOC
R1-2500225 Ambient IoT channel coding and small frequency shift CATT
R1-2500260 Discussion on physical channels design about line coding, FEC, CRC, repetition aspects for Ambient IoT China Telecom
R1-2500288 Discussion on coding aspects of physical channel designCMCC
R1-2500350 Discussion on line coding, FEC, CRC and repetition for A-IoT vivo
R1-2500395 AIoT Physical channels design - line coding, FEC, CRC, repetition aspects Nokia
R1-2500425 Discussion on other aspects for Ambient IoT physical design TCL
R1-2500451 Discussion on physical channels design for A-IoT OPPO
R1-2500487 Discussion on Physical Channel Designs for A-IoT Panasonic
R1-2500611 Physical layer design – line coding, FEC, CRC, repetition aspects NEC

R1-2500733 Discussion on line coding, FEC, CRC and repetition aspects for Ambient IoT Xiaomi
R1-2500779 On coding aspects for Ambient IoT Apple
R1-2500851 Views on Physical channels design – line coding, FEC, CRC, repetition aspects Samsung
R1-2500937 Discussion on coding aspects Fujitsu
R1-2500950 A-IoT PHY layer design - line coding, FEC, CRC and repetition aspects LG Electronics
R1-2501017 A-IoT - PHY line coding, FEC, CRC, repetition aspects MediaTek Inc.
R1-2501072 Discussion on the Ambient IoT physical layer design aspects for Line coding, FEC, CRC, Repetition Lenovo
R1-2501093 Discussion on coding aspects Sharp
R1-2501109 Discussion on Physical channels design for Ambient IoT-other aspects HONOR
R1-2501118 Coding aspects of physical channel design for Ambient IoT InterDigital, Inc.
R1-2501124 Discussion on A-IoT Physical channels design ASUSTeK
R1-2501157 Physical channels design – line coding, FEC, CRC, repetition aspects Qualcomm Incorporated
R1-2501203 Discussion on coding and CRC aspects of physical channel design for Ambient IoT NTT DOCOMO, INC.
R1-2501269 Coding aspects for Ambient IoT ITL
R1-2501311 Discussion on other aspects of Phy design for AIoT IIT Kanpur

R1-2501435 Summary #1 for coding aspects of physical channel design Moderator (CMCC)
From Monday session
Agreement
For R2D Manchester line code, the specification is according to TR 38.769 as follows:
- A chip corresponds to one modulated symbol.
- For Manchester encoding, the bit-to-chip mapping is: bit 0→chips{10}, bit 1→chips{01}

Agreement
For D2R transmission with small frequency shift +/- R/T_b Hz, where time duration T_b corresponding to a bit (that is after FEC if applied) and $R = T_b/(2 \times \text{D2R chip length})$:
- the transmitted 2R chips for bit 1 and bit 0 are [0 1 0 1 …] and [1 0 1 0 …], respectively, for OOK modulation
- the transmitted 2R chips for bit 1 and bit 0 are [-1 +1 -1 +1 …] and [+1 -1 +1 -1 …], respectively, for BPSK modulation.
- Note: The case of R=1 is equivalent to using a Manchester line code without repeating each Manchester codeword.

R1-2501436 Summary #2 for coding aspects of physical channel design Moderator (CMCC)
From Tuesday session
Agreement
For D2R repetition, block level repetition from TR38.769 is supported.

Agreement
RAN1 concludes that generator polynomials of LTE CC with constraint length of 7 and coding rate of 1/3 from TS36.212 are reused as the mother code for Ambient IoT.

Agreement
RAN1 concludes that the generator polynomials in TS 38.212 for 6-bit CRC and 16-bit CRC are reused for Ambient IoT.

R1-2501437 Summary #3 for coding aspects of physical channel design Moderator (CMCC)
From Thursday session
Agreement
When CRC is attached to a PRDCH or PDRCH transmission,
- When the number of information bits is ≤ X bits, CRC-6 is used. Otherwise, when the number of information bits is > X bits, CRC-16 is used. Down-selection by RAN1#120bis from the following for X considering the balance of overhead and probability of undetected error:
  - Alt. 1: 24
  - Alt. 2: 56
- FFS impact of segmentation, if any
  - Note: impact may not be in RAN1

R1-2501592 Summary #5 for coding aspects of physical channel design Moderator (CMCC)
From Friday session
Agreement

One or both of the following options are supported to determine when no CRC is used,

- Option 1: A threshold of number of information bits Y. When the number of information bits is ≤ Y bits, no CRC is used. Down-selection from the following for Y:
  - Alt. 1: 16
  - Alt. 2: 8
  - Alt. 3: 6
- Option 2: Specified condition(s), e.g., device transmits PDRCH for Msg 1 upon receiving a PRDCH triggering random access. FFS specified condition(s) and/or how to determine the specified condition(s).

Agreement
For the initial values of the shift register of the CC encoder, down-select by RAN1#120bis from the following two options:

- Option 1: The initial values of the shift register of the CC encoder are set to the values corresponding to the last (K-1) information bits defined in TS 36.212.
- Option 2: The initial values of the shift register of the CC encoder are set to the values corresponding to the first (K-1) information bits, and the first (K-1) bits are appended to the end such that the initial state and ending state of the shift register are the same.
- Note: K is the constraint length.

Final summary in R1-2501634.

9.4.3 Timing acquisition and synchronization

Including discussions on the necessity/functionalities of preamble/midamble/postamble for R2D and D2R, respectively.

R1-2500035 Timing acquisition and synchronization for Ambient IoT Ericsson
R1-2500046 Discussion on timing acquisition and synchronization aspects for A-IoT FUTUREWEI
R1-2500126 Discussion on Ambient IoT signals ZTE Corporation, Sanechips
R1-2500140 Discussion on timing acquisition and synchronization for Ambient IoT Lenovo
R1-2500155 Physical signals design Huawei, HiSilicon
R1-2500171 Discussion on timing acquisition and synchronization for Ambient IoT Spreadtrum, UNISOC
R1-2500226 Ambient IoT Timing and synchronization CATT
R1-2500261 Discussion on timing acquisition and synchronization for Ambient IoT China Telecom
R1-2500289 Discussion on timing acquisition and synchronization CMCC
R1-2500351 Discussion on Timing acquisition and synchronization for AIoT vivo
R1-2500396 AIoT Timing acquisition and synchronization Nokia
R1-2500426 Discussion on timing acquisition and synchronization functionalities for Ambient IoT TCL
R1-2500452 Discussion on timing acquisition and synchronization for A-IoT OPPO
R1-2500480 Discussion on timing acquisition and synchronization aspects Tejas Network Limited
R1-2500493 Discussion on timing acquisition and synchronization InterDigital, Inc.
R1-2500584 A-IoT Timing acquisition and synchronization Panasonic
R1-2500625 Discussion on timing acquisition and synchronization NEC
R1-2500652 Timing acquisition and synchronisation for Ambient IoT Sony
R1-2500734 Discussion on timing acquisition and synchronization for Ambient IoT Xiaomi
R1-2500780 On timing acquisition & synchronization aspects for Ambient IoT Apple
R1-2500852 Views on timing acquisition and synchronization Samsung
R1-2500911 Discussion on timing acquisition and synchronization ETRI
R1-2500938 Discussion on timing acquisition and synchronization Fujitsu
R1-2500951 Timing acquisition and synchronization for A-IoT LG Electronics
R1-2501018 A-IoT - Timing acquisition and synchronization MediaTek Inc.
R1-2501094 Discussion on timing acquisition and synchronization Sharp
R1-2501365 Discussion on timing acquisition and synchronization HONOR (rev of R1-2501110)
R1-2501125 Discussion on end timing acquisition ASUSTeK
R1-2501158 Timing acquisition and synchronization Qualcomm Incorporated
R1-2501204 Discussion on timing acquisition and synchronization for Ambient IoT NTT DOCOMO, INC.
R1-2501275 Discussion on timing acquisition and synchronization aspects for Ambient IoT CEWiT

R1-2500782 FL Summary#1 on timing acquisition & synchronization for Ambient IoT Apple
From Tuesday session
Agreement

For the SIP of R-TAS, for providing the start of the R2D transmission, one single design based on Option 1 is supported and further down-selection to be done among Alt 1 and Alt 2 :

- Option 1: ON-OFF transmission with following alternatives:
  - Alt 1: A single ON-OFF transmission with pre-defined duration for each of the ON-OFF, where ON and OFF may have same or different durations
    - Continue discussion to clarify the duration of each of the ON and OFF
    - Continue discussion to list the different candidate proposals under Alt1
  - Alt 2: A multi-ON-OFF transmission with pre-defined duration for each of the ON(s)-OFF(s), where different ON and different OFF may have same or different durations and different parts may have same or different duration
    - Continue discussion to clarify the duration of each of the ON and OFF
    - Continue discussion to list the different candidate proposals under Alt2
- Only a single fixed value for entire duration of SIP of R-TAS is supported, which is independent of the value of "M" used in CAP and PRDCH
- Note: Specific design and duration for SIP of R-TAS are further discussed, and companies are encouraged to evaluate the designs in terms of target MDR of [10%] for a FAR up to [1%] and at least following assumptions are used:
  - MDR refers to the probability that SIP is not detected when it was actually transmitted
  - FAR probability that the receiver incorrectly detects SIP when SIP was not transmitted
  - Energy/edge detection-based method is the baseline assumption for evaluation purpose
  - Continue discussion on necessary details for simulation assumptions

Agreement
For the CAP of R-TAS, the starting chip has a different voltage level compared to the end of the SIP of R-TAS.

Agreement
R2D transmission does not include a midamble.

R1-2500783 **FL Summary#2 on timing acquisition & synchronization for Ambient IoT** **Apple**
From Wednesday session
Agreement
For the SIP of R-TAS, down-select among the following candidates:

- Alt 1 (Single ON-OFF transmission)
  - Alt 1-1: ON followed by OFF with same duration for both
  - Alt 1-2: ON followed by OFF with a duration ratio of 1:[2,3]
  - Alt 1-3: ON followed by OFF with a duration ratio of [2,3]:1
- Alt 2 (Multi-ON-OFF transmission)
  - Alt 2-1: A number of repetition instances of Alt 1-1 or Alt 1-2 or Alt 1-3
  - Alt 2-2: ON-OFF-ON (duration of ON and OFF can be different)
  - Alt 2-3: OFF-ON-OFF (duration of ON and OFF can be different)
  - Alt 2-4: Combination of single instance of Alt 1-1 and single instance of Alt 1-2
- For the evaluation purpose, for both options, candidate values related to duration are considered:
  - Entire duration of SIP: 1/2 OFDM symbol duration or 1 OFDM symbol duration (including clarifying whether OFDM symbol duration includes CP); additional durations can be considered and reported by companies with justification
  - Companies to report the exact duration(s) for ON or OFF
- Companies are encouraged to report at least the following details for the evaluations:
  - Baseline assumption is that RF transmission is not present; companies can report other consideration
  - For FAR calculation, whether noise and/or PRDCH transmission is considered
  - Details on threshold detection method including whether/how threshold detection training is used based on the proposed design alternative or not
  - BW assumptions for RF-ED and BB-LPF
  - Target MDR of up to 1% for FAR of up to [1%, 10%]

Agreement
For the design of the CAP of R-TAS, at least 2 transition edges in same direction are included, i.e. at least two transitions from "OFF" chip to "ON" chip or two transitions from "ON" chip to "OFF" chip.

R1-2500784 FL Summary#3 on timing acquisition & synchronization for Ambient IoT Apple
From Wednesday session
Agreement
- For D2R preamble design, the functionalities of timing acquisition, SFO estimation/time tracking and channel estimation should be supported
- For D2R midamble design, the functionalities of SFO estimation/time tracking and channel estimation should be supported
  - D2R midamble can be transmitted at the end of the PDRCH transmission. If it is at the end, it is not designed for being used for indicating the end of PDRCH transmission
  - FFS: condition(s) and/or indication where the D2R midamble is present or not

R1-2501616 FL Summary#4 on timing acquisition & synchronization for Ambient IoT Moderator (Apple)
From Friday session
Agreement
For D2R x-ambles:
- Following is considered as the types for base sequence and to be further down-selected:
  - Option 1: M-sequence
  - Option 2: Golay sequence
  - Note: Above doesn't preclude an additional part for preamble, e.g. with ON and/or OFF transmission, if needed/supported
- FFS: Whether/what multiple sequences (using same base sequence type) are supported
  - Note: This in no way implies that there is going to be CDMA between D2R x-ambles
- For evaluation purpose, companies are encouraged to consider following:
  - Performance at least in terms of autocorrelation/cross-correlation property, SFO estimation/Timing accuracy, SNR for target PDRCH BLER of [1%, 10%]
  - Report presence and time-domain resource(s) x-ambles
  - Report sequence type(s) and length(s) for x-ambles
  - Following format can be considered for reporting the evaluation results

X-amble(s) Present	X-ambles time-domain resource(s)	X-ambles Sequence Type(s)	X-ambles Length	PDRCH Duration/Size & data rate	Required SNR for target PDRCH BLER of 1%	Required SNR for target PDRCH BLER of 10%	Performance Metric related to correlation properties, SFO/Timing Accuracy [90%tile]	Other assumptions/metrics can be reported by companies

Agreement
For the CAP of R-TAS:
- Candidate values for maximum duration of CAP to be further down-selected to one value from : 1.5 OFDM symbol duration, 2 OFDM symbol duration, 3 OFDM symbol duration
  - For option 1 for CAP of R-TAS from TR 38.769, maximum duration is applicable to minimum value of M to be supported, and the CAP duration becomes shorter with increasing value of M
    - FFS: whether the number of ON/OFF transmissions in the CAP is fixed or not fixed
  - For option 2 for CAP of R-TAS from TR 38.769, maximum duration is the only (constant) duration that is applicable for all the M values to be supported
- Down-selection between option 1 and option 2 for CAP of R-TAS from TR 38.769 by RAN1#120-bis
- FFS: Values of M to be supported

R1-2501617 Final summary on timing acquisition & synchronization for Ambient IoT Moderator (Apple)

9.4.4 Other aspects incl. multiplexing/multiple access, scheduling information, and timing relationships

Including discussions on L1 control information and any other L1 procedural aspects.

R1-2500036 Other aspects for Ambient IoT Ericsson
R1-2500047 Discussion on multiple access, scheduling and timing aspects for A-IoT FUTUREWEI
R1-2500127 Discussion on Ambient IoT multiple access and timing ZTE Corporation, Sanechips
R1-2500141 Discussion on multiple access, scheduling and timing aspects for Ambient IoT Lenovo
R1-2500156 Multiplexing, scheduling, and other physical-layer procedures Huawei, HiSilicon
R1-2500172 Discussion on other aspects for Ambient IoT Spreadtrum, UNISOC

R1-2500227 Ambient IoT frame structure, system control and resource allocation CATT
R1-2500262 Discussion on other aspects for Ambient IoT China Telecom
R1-2500290 Discussion on access, scheduling and timing relationships CMCC
R1-2500352 Discussion on other aspects for Rel-19 Ambient IoT vivo
R1-2500397 AIoT Other aspects incl. multiplexing/multiple access, scheduling information, and timing relationships Nokia
R1-2500453 Discussions on other aspects of A-IoT communication OPPO
R1-2500481 Discussion on multiple access, scheduling and timing aspects for A-IoT Tejas Network Limited
R1-2500488 Discussion on other aspects of A-IoT Panasonic
R1-2500494 Discussion on other aspects including multiplexing/multiple access, scheduling information, and timing relationships InterDigital, Inc.
R1-2500517 Views on other aspects for AIoT Ofinno
R1-2500563 Discussion on multiple access, scheduling and timing for Ambient IoT Lekha Wireless Solutions
R1-2500612 Discussion on control and other aspects of ambient IoT NEC
R1-2500653 Multiplexing, multiple access and timing relationships for Ambient IoT Sony
R1-2500735 Discussion on other aspects for Ambient IoT Xiaomi
R1-2500781 On multiple Access, scheduling and control for Ambient IoT Apple
R1-2500853 Views on aspects including multiplexing/multiple access, scheduling information, and timing relationships Samsung
R1-2500912 Discussion on other aspects of Ambient IoT ETRI
R1-2500939 Discussion on other aspects for Ambient IoT Fujitsu
R1-2500952 Other aspects for A-IoT LG Electronics
R1-2501019 A-IoT - Other aspects MediaTek Inc.
R1-2501095 Discussion on other aspects Sharp
R1-2501111 Discuss on L1 control information and L1 procedural aspects HONOR
R1-2501126 Discussion on control information and procedural aspects ASUSTeK
R1-2501159 Discussion on other aspects for Rel-19 Ambient IoT Qualcomm Incorporated
R1-2501205 Discussion on other aspects for Ambient IoT NTT DOCOMO, INC.
R1-2501276 Discussion on multiple access, scheduling and timing aspects for Ambient IoT CEWiT
R1-2501288 Discussion on multiplexing/multiple access and timing relationships for Ambient IoT WILUS Inc.
R1-2501298 Discussion on multiple access, scheduling information and timing relationships for Ambient IoT TCL
R1-2501313 Discussion on other aspects of AIoT IIT Kanpur
R1-2501336 Discussion on multiplexing, multiple access, scheduling information, and timing relationships Google
R1-2501347 Discussion on aspects for Ambient IoT Sequans Communications

R1-2501355 FL summary #1 on other aspects for Rel-19 Ambient IoT Moderator (vivo)
From Tuesday session
Agreement
Support FDMA with Y≥1 D2R transmissions for Msg.1 from multiple devices in response to a R2D transmission triggering random access.
- The maximum value of Y >1 is determined in AI 9.4.2, based on feasible values of small frequency shift to be decided in AI 9.4.2.
- Signalling details should be discussed and determined in AI 9.4.4.

Agreement
Support following two options for Msg2 transmission in response to multiple Msg1 transmissions, which are initiated by a R2D transmission triggering random access.
- Option 1: A PRDCH for Msg2 transmission corresponds to a A-IoT Msg1 received from one device
- Option 2: A PRDCH for Msg2 transmission corresponds to multiple A-IoT Msg1 received from different devices
- FFS the maximum number of Msg1 responses that can be carried within a PRDCH for Msg2.

R1-2501356 FL summary #2 on other aspects for Rel-19 Ambient IoT Moderator (vivo)
From Wednesday session
Agreement
Support FDMA of D2R transmissions for Msg3 from multiple devices in response to a PRDCH for Msg2 transmission corresponding to multiple Msg1 during access procedure.

R1-2501357 FL summary #3 on other aspects for Rel-19 Ambient IoT Moderator (vivo)
From Thursday session
Agreement

Support the following option(s) for frequency domain resource for Msg3 transmission:

- Option 2: The frequency domain resource for Msg3 can be determined based on explicit indication in the PRDCH for Msg2 transmission for one or multiples devices.
- FFS: support option 1 as a default if PRDCH does not provide the indication or based on a rule
  - Option 1: the frequency domain resource for Msg3 reuses the frequency domain resource for Msg1 for the same device

Agreement
The following is supported for indicating the D2R chip duration:

- The D2R chip duration is indicated in R2D control information from predefined a set of D2R chip duration values

Agreement
The payload size (i.e. TBS-like) for PDRCH transmission with variable size is explicitly indicated in the corresponding R2D control information.

R1-2501625 Final FL summary on other aspects for Rel-19 Ambient IoT Moderator (vivo)

9.5 Enhancements of network energy savings for NR

Please refer to RP-242354 for detailed scope of the WI. Additional RAN guidance on Rel-19 NES can be found in RP-243297.
Rapporteur to provide initial input on higher layer signalling under agenda item 9.5. For input on higher layer signalling from any other source, please include it as part of your tdoc to relevant sub-agenda items.

[120-R19-NES] – Ajit (Ericsson)
Email discussion on Rel-19 NES enhancement

- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2501248 List of RAN1 agreements for R19 NES WI until RAN1#119 Rapporteur(Ericsson)
R1-2501249 Initial list of RRC parameters for R19 NES WI Rapporteur(Ericsson)

R1-2501647 Updated list of RAN1 agreements for Rel-19 NES WI Rapporteur (Ericsson)

9.5.1 On-demand SSB SCell operation

R1-2500053 Discussion of on-demand SSB Scell operation FUTUREWEI
R1-2500077 On-demand SSB SCell operation for eNES Huawei, HiSilicon
R1-2500128 Discussion on on-demand SSB for NES ZTE Corporation, Sanechips
R1-2500173 Discussion on on-demand SSB SCell operation Spreadtrum, UNISOC
R1-2500228 Discussion on on-demand SSB SCell operation CATT
R1-2500263 Discussion on on-demand SSB operation for SCell China Telecom
R1-2500291 Discussion on on-demand SSB SCell operation CMCC
R1-2500353 Discussion on on-demand SSB Scell operation vivo
R1-2500400 On demand SSB for connected UEs Tejas Networks Limited
R1-2500440 Discussion on the enhancement to support on demand SSB SCell operation OPPO
R1-2500476 On-demand SSB SCell Operation Nokia, Nokia Shanghai Bell
R1-2500489 Discussion on on-demand SSB SCell operation Panasonic
R1-2500525 Discussion on on-demand SSB SCell operation InterDigital, Inc.
R1-2500552 On-demand SSB SCell Operation Google
R1-2500621 Discussion on on-demand SSB for SCell operation NEC
R1-2500682 Discussion on On-demand SSB SCell operation KT Corp.
R1-2500736 Discussion on on-demand SSB SCell operation Xiaomi
R1-2500785 On-demand SSB SCell Operation Apple
R1-2500854 On-demand SSB SCell operation Samsung
R1-2500884 On-demand SSB SCell operation Lenovo
R1-2500913 Discussion on On-demand SSB SCell operation ETRI
R1-2500940 Discussion on on-demand SSB SCell operation Fujitsu
R1-2500953 On-demand SSB SCell operation LG Electronics
R1-2500967 Discussion on On-Demand SSB SCell operation Transsion Holdings
R1-2501020 On-demand SSB SCell operation MediaTek Inc.
R1-2501123 Discussion of On-demand SSB SCell operation Mavenir

R1-2501160 On-demand SSB operation for Scell Qualcomm Incorporated
R1-2501186 Discussion on remaining details of on-demand SSB operation on SCell Sharp
R1-2501206 Discussion on on-demand SSB SCell operation NTT DOCOMO, INC.
R1-2501233 Discussion on on-demand SSB SCell operation ITRI
R1-2501238 Discussion on on-demand SSB Scell operation for NES TCL
R1-2501242 DCI based signaling for on-demand SSB ASUSTeK
R1-2501251 On-demand SSB SCell operation Ericsson
R1-2501277 Discussion on on-demand SSB Scell operation CEWiT

R1-2501493 Summary #1 of on-demand SSB for NES Moderator (LG Electronics)
From Tuesday session
Agreement
Regarding the relation in terms of time location between the always-on SSB and on-demand SSB, at least for the case when the center frequency locations of always-on SSB and on-demand SSB are same, down-select one of the following.

- Alt Time-A: It is not allowed that time location of an SSB within on-demand SSB burst overlaps with time location of any SSB within always-on SSB burst.
  - NOTE: It is assumed that the periodicity of always-on SSB is equal to or larger than that of on-demand SSB.
- Alt Time-B: Time-domain locations of always-on SSB are a subset of time-domain locations of on-demand SSB.
  - NOTE: It is assumed that the periodicity of always-on SSB is equal to or larger than that of on-demand SSB.
- Alt Time-C: The specification has no restriction with regards to overlapping

R1-2501494 Summary #2 of on-demand SSB for NES Moderator (LG Electronics)
From Wednesday session
Agreement

- For a cell supporting on-demand SSB SCell operation, to provide time domain location of on-demand SSB by RRC,
  - For Case #2, adopt Alt A in the previous RAN1 agreement, i.e.,
    - Based on two parameters, where one is to indicate SFN offset from a reference point and the other is to indicate half frame index
      - The reference point is SFN which satisfies (SFN index *10) modulo (OD-SSB periodicity) = 0
      - If SFN offset parameter is NOT configured, UE assumes SFN offset set to 0.
      - If half frame index parameter is NOT configured, UE assumes half frame index set to 0.
      - The value range of SFN offset is 0 to 15 unless longer periodicity for on-demand SSB than 160 ms is introduced.
      - The value range of half frame index is 0 or 1.
    - FFS: Whether/how to indicate time offset between always-on SSB and on-demand SSB (e.g., similar to *ssb-TimeOffset*)
      - By defining reference point as SFN where AO-SSB burst exists

Agreement

- For a cell supporting on-demand SSB SCell operation, introduce NEW higher layer parameter (i.e., *od-ssb-config*) for on-demand SSB configuration, for both Case #1 and Case #2.
- For a cell supporting on-demand SSB SCell operation, introduce NEW higher layer parameters to configure each of the followings, for both Case #1 and Case #2.
  - Frequency of the on-demand SSB (i.e., *od-ssb-absoluteFrequency*)
    - FFS: Whether this parameter can be absent for Case #2
  - SSB positions within an on-demand SSB burst by using signaling similar to *ssb-PositionsInBurst* (i.e., *od-ssb-PositionsInBurst*)
    - FFS: Whether this parameter can be absent for Case #2
  - Periodicity of the on-demand SSB (i.e., *od-ssb-Periodicity*)
  - Sub-carrier spacing of the on-demand SSB (i.e., *od-ssbSubcarrierSpacing*)
    - For Case #2, this parameter is absent and sub-carrier spagcing of on-demand SSB is the same as that of always-on SSB.
  - Physical Cell ID of the on-demand SSB (i.e., *od-ssb-physCellId*)
    - For Case #2, this parameter is absent and physical cell ID of on-demand SSB is the same as that of always-on SSB.
  - Time location of on-demand SSB burst (i.e., *od-ssb-sfn-Offset* and *od-ssb-halfFrameIndex*)

- Downlink transmit power of on-demand SSB (i.e., *od-ss-PBCH-BlockPower*)
  - For Case #2, this parameter is absent and downlink transmit power of on-demand SSB is the same as that of always-on SSB.
- Number N of on-demand SSB bursts to be transmitted after on-demand SSB is indicated (i.e., *od-ssb-nrofTx*)
- FFS: Which of the above parameters are a list

Agreement
For RRC based OD-SSB indication, UE expects on-demand SSB is transmitted from the first on-demand SSB burst after receiving RRC carrying indication of OD-SSB transmission.

Conclusion
The following combination of scenarios and cases for indicating OD-SSB are not supported in Rel-19
- Scenario #3A and Case #1
- Scenario #3A and Case #2

Above does not impact discussion on SSB periodicity adaptation in time domain.

R1-2501495 Summary #3 of on-demand SSB for NES Moderator (LG Electronics)
From Thursday session
Agreement
Regarding the relation in terms of frequency location (i.e., center frequency) between the always-on SSB and on-demand SSB,
- Alt 1: If always-on SSB is CD-SSB on a synchronization raster, the frequency location of on-demand SSB is different from the frequency location of always-on SSB.
  - On-demand SSB is not on sync raster
  - AO-SSB and OD-SSB are located in the same BWP
  - FFS: Additional conditions
  - Subject to separate UE capability
  - Note: UE is not required to measure both AO-SSB and OD-SSB

R1-2501496 Summary #4 of on-demand SSB for NES Moderator (LG Electronics)
From Friday session
Agreement
Regarding the relation in terms of time location between the always-on SSB and on-demand SSB,
- For the case when the center frequency locations of always-on SSB and on-demand SSB are same,
  - Alt Time-C: RAN1 specification has no restriction with regards to overlapping
  - From RAN1 perspective,
    - Alt Time-C1: The case that, during OD-SSB transmission, the union of AO-SSB transmission and OD-SSB transmission has a periodic time domain pattern is supported (the interval between SSB bursts is even and supported in legacy specification).
    - Alt Time-C2: The case that, during OD-SSB transmission, the union of AO-SSB transmission and OD-SSB transmission has a non-periodic time domain pattern is supported.
  - It is up to RAN4 to define requirements, if any, corresponding to both or either of Alt Time-C1 or Alt Time-C2
  - At least the following is supported: PBCH payload for the same SSB index (other than SFN index, half frame index) is the same for AO-SSB and OD-SSB
    - FFS: Whether half frame index is the same or different for AO-SSB and OD-SSB
- For the case when the center frequency locations of always-on SSB and on-demand SSB are different,
  - Alt Time-C: RAN1 specification has no restriction with regards to overlapping
  - UE assumes that frequency resources of always-on SSB are not overlapped with those of on-demand SSB in frequency domain.
  - AO-SSB and OD-SSB are located in the same BWP
  - FFS: PBCH payload for the same SSB index (other than SFN index, half frame index) should be the same for AO-SSB and OD-SSB
- NOTE: AO-SSB periodicity is not adapted

Send an LS to RAN4 to inform them of the above agreement.

R1-2501632 Draft LS on time location of on-demand SSB for Scell LG Electronics
Decision: The draft LS is endorsed. Final version is approved in R1-2501633.

Agreement
For the case where SCell with on demand SSB transmission and cell with signalling transmission have different numerology, when UE determines time instance *A*, the SCS to determine the value of *T* is down selected among the following options

- Option 1: the SCS of the active DL BWP where UE receives MAC CE for on-demand SSB transmission indication
- Option 2: the minimum of "the SCS of the active DL BWP where UE receives MAC CE for on-demand SSB transmission indication" and "the SCS of the active DL BWP where UE receives on-demand SSB"
- Option 3: the SCS of the active UL BWP where the UE transmits ACK corresponding to the MAC-CE for on-demand SSB transmission indication

Final summary in R1-2501635.

9.5.2 On-demand SIB1 for idle/inactive mode UEs

R1-2500054 Discussion of on-demand SIB1 for idle/inactive mode UEs FUTUREWEI
R1-2500078 Discussion on on-demand SIB1 for eNES Huawei, HiSilicon
R1-2500129 Discussion on on-demand SIB1 for NES ZTE Corporation, Sanechips
R1-2500174 Discussion on on-demand SIB1 for idle/inactive mode UEs Spreadtrum, UNISOC
R1-2500229 Discussion on on-demand SIB1 CATT
R1-2500264 Discussion on on-demand SIB1 for idle/inactive mode UEs China Telecom
R1-2500292 Discussion on on-demand SIB1 for UEs in idle/inactive mode CMCC
R1-2500354 Discussion on on-demand SIB1 for idle/inactive mode UEs vivo
R1-2500398 On demand SIB1 for idle/inactive UEs Tejas Networks Limited
R1-2500441 Discussion on the enhancement to support on demand SIB1 for idle/inactive mode UE OPPO
R1-2500477 On-demand SIB1 for Idle/Inactive mode UEs Nokia, Nokia Shanghai Bell
R1-2500490 Discussion on on-demand SIB1 for idle/inactive mode UEs Panasonic
R1-2500526 Discussion on on-demand SIB1 for idle/inactive mode UEs InterDigital, Inc.
R1-2500553 On-demand SIB1 for Idle/Inactive Mode UE Google
R1-2500622 Discussion on on-demand SIB1 for UEs in idle/inactive mode NEC
R1-2500629 Discussion on on-demand SIB1 transmission for network energy savings Fujitsu
R1-2500654 On-demand SIB1 for idle/inactive mode UEs Sony
R1-2500737 Discussion on on-demand SIB1 for idle/inactive mode UEs Xiaomi
R1-2500786 On On-demand SIB1 for IDLE/INACTIVE mode UEs Apple
R1-2500855 On-demand SIB1 for idle/inactive mode UEs Samsung
R1-2500885 On-demand SIB1 for idle/inactive mode UEs Lenovo
R1-2500914 On-demand SIB1 for idle/inactive mode UEs ETRI
R1-2500954 On-demand SIB1 for idle/inactive mode UEs LG Electronics
R1-2500968 Discussion on on-demand SIB1 transmission for idle/inactive mode UEs Transsion Holdings
R1-2501021 On-demand SIB1 for idle or inactive mode UEs MediaTek Inc.
R1-2501049 Views on On-demand SIB1 for idle-inactive UEs Vodafone
R1-2501089 Discussion on on-demand SIB1 for idle/inactive mode UEs DENSO CORPORATION
R1-2501133 Discussion on on-demand SIB1 transmission for idle UEs Sharp
R1-2501161 On-demand SIB1 procedure Qualcomm Incorporated
R1-2501207 Discussion on on-demand SIB1 for idle/inactive mode UEs NTT DOCOMO, INC.
R1-2501252 On-demand SIB1 for UEs in idle/inactive mode for NES Ericsson
R1-2501263 Discussion on on-demand SIB1 in idle/inactive mode CAICT
R1-2501278 Discussion on on-demand SIB1 CEWiT
R1-2501299 On-demand SIB1 for NES Fraunhofer IIS, Fraunhofer HHI

R1-2501378 FL summary 1 for on-demand SIB1 in idle/inactive mode Moderator (MediaTek)
From Tuesday session
Agreement
The reference time point to determine the window starting time for on-demand SIB1 is based on the starting slot of the RAR window for UL WUS wherein UE successfully received a RAR.

Agreement
The time offset between the reference time point and the starting time for the time window of on-demand SIB1 is indicated in the UL WUS configuration.

- Unit of the time offset is in slot

R1-2501379 FL summary 2 for on-demand SIB1 in idle/inactive mode Moderator (MediaTek)

From Wednesday session
Conclusion
On how to use PDCCH-ConfigSIB1 for R19 NES-capable UE in MIB of NES Cell when K_SSB = 30 in FR1 or K_SSB = 14 in FR2 on NES cell if SSB of NES cell is on the sync raster, Option 3 is adopted.

- Option 1: Use it to indicate frequency assistance information to search SSB for Cell A
  - FFS: Details on the range/granularity of the frequency assistance information
  - FFS: Potential impact to RAN3
- Option 2: Use it to indicate *searchSpaceZero* and *controlResourceSetZero* of OD-SIB1
- Option 3: Above feature is not supported.

Agreement
For type 0 PDCCH monitoring occasions of on demand SIB1, searchSpaceZero and controlResourceSetZero for on-demand SIB1 are provided from UL WUS configuration if SSB on NES cell is on sync raster.

Agreement
The UE assumes that, in the OD-SIB1 window, PDCCH for an OD-SIB1 message is transmitted in ~~at least~~ PDCCH monitoring occasions corresponding to at least the SSB associated with the PRACH for UL-WUS if this is indicated via UL WUS configuration

- FFS: Whether UE expect PDCCH on PDCCH monitoring occasion corresponding to SSB other than the one mentioned above
- FFS: Whether gNB can indicate (in UL-WUS configuration) the SSB the UE can expect PDCCH transmission on

Conclusion
There is no RAN1 consensus on the following proposal:
From RAN1 perspective, OD-SIB1 design does not differentiate depending on frequency location of SSB on NES cell

R1-2501380 FL summary 3 for on-demand SIB1 in idle/inactive mode Moderator (MediaTek)
From Thursday session
Conclusion
It's up to RAN2 to discuss whether the value in OD-SIB1 IE can be different from the value of the corresponding IE in the WUS configuration, and how to define the UE behavior if the values are different.

Conclusion (amended as shown in red in Friday session)

- The parameter "ssb-PeriodicityServingCell" is not included in the UL-WUS configuration for FDD system.
- RA-RNTI is not included in the UL-WUS configuration ~~for FDD system~~.
- Cell barring information is not included in the UL-WUS configuration ~~for FDD system~~.
- Cell selection information is not included in the UL-WUS configuration ~~for FDD system~~.
- Accessible UE types is not included in the UL-WUS configuration ~~for FDD system~~.
- CarrierBandwidth (for MPR calculation) is not included in the UL-WUS configuration ~~for FDD system~~.
- InitialUplinkBWP is not included in the UL-WUS configuration ~~for FDD system~~.
- TotalNumberOfRA-Preambles is not included in the UL-WUS configuration ~~for FDD system~~.
- SIB1-RequestResourceRedCap is not included in the UL-WUS configuration ~~for FDD system~~.
- SupplementaryUL is not included in the UL-WUS configuration

Conclusion
There is no RAN1 consensus to support the following in Rel-19:
Support joint PRACH request to trigger both OD-SIB1 and a subset of OD-OSI, e.g. SIB2/SIB3/SIB4.

- FFS: parameters in UL-WUS config to support joint PRACH request.

R1-2501381 FL summary 4 for on-demand SIB1 in idle/inactive mode Moderator (MediaTek)
From Friday session
Agreement
If a UE has SIB1 request configuration of a cell and before transmitting UL WUS,

- If the UE detects a SSB where K_SSB>=24 for FR1 or K_SSB>=13 for FR2, down select from the following:
  - Alt. 1: Only if K_SSB=30 for FR1 and K_SSB=14 for FR2, UE monitors Type 0 PDCCH for SIB1 transmission; Otherwise, UE doesn't monitor Type 0 PDCCH
    - FFS: Whether monitoring time window of Type 0 PDCCH is up to UE implementation or specified.
  - Alt. 2: UE monitors Type 0 PDCCH for SIB1 transmission

        - FFS: Whether monitoring time window of Type 0 PDCCH is up to UE implementation or specified.
    - Alt. 3: It is up to UE implementation on whether to monitor Type 0 PDCCH for SIB1 transmission
    - Alt. 4: UE doesn't monitor Type 0 PDCCH for SIB1 transmission.
    - Alt. 5: UE monitors Type 0 PDCCH for SIB1 based on some repurposed parameters in MIB or PBCH

Note: If the UE detects a SSB where K_SSB<=23 for FR1 or K_SSB<=12 for FR2, UE monitors Type 0 PDCCH for SIB1 transmission as legacy.
Note: From Ericsson point view, Alt 3 and Alt 4 may be inconsistent with existing RAN2 agreements
Note: The above cases are for SSB on the sync raster.

Final summary in R1-2501382.

9.5.3 Adaptation of common signal/channel transmissions

R1-2500055	Discussion of the adaptation of common signal/channel transmissions	FUTUREWEI
R1-2500079	On common channel/signal adaptation for eNES	Huawei, HiSilicon
R1-2500130	Discussion on common signal channel for NES	ZTE Corporation, Sanechips
R1-2500175	Discussion on adaptation of common signal/channel transmissions	Spreadtrum, UNISOC
R1-2500230	Discussion on adaptation of common signal/channel transmissions	CATT
R1-2500265	Discussion on common signal/channel adaptation	China Telecom
R1-2500293	Discussion on adaptation of common signal/channel transmissions	CMCC
R1-2500355	Discussion on adaptation of common signal/channel transmissions	vivo
R1-2500399	Adaptation of common signals/channels	Tejas Networks Limited
R1-2500442	Discussion on adaptation of common signal/channel transmission	OPPO
R1-2500478	Adaptation of common signal/channel transmissions	Nokia, Nokia Shanghai Bell
R1-2500527	Discussion on adaptation of common signal/channel transmissions	InterDigital, Inc.
R1-2500554	Adaptation of Common Signals	Google
R1-2500623	Discussion on adaptation of common signal/channel transmissions	NEC
R1-2500630	Discussion on adaptation of common signal / channel transmissions	Fujitsu
R1-2500655	Adaptation of common signal/channel transmissions	Sony
R1-2500683	Discussion on Adaptation of common signal/channel transmissions	KT Corp.
R1-2500738	Discussion on adaptation of common signal and channel transmissions	Xiaomi
R1-2500787	On adaptation of common signal/channel for NES enhancements	Apple
R1-2500856	Adaptation of common signal/channel transmissions	Samsung
R1-2500915	Adaptation of common signal/channel transmissions	ETRI
R1-2500946	Discussion on adaptation of common signal/channel transmission	Panasonic
R1-2500955	Adaptation of common signal/channel transmissions	LG Electronics
R1-2500969	Discussion on adaptive transmission of common signal or common channel	Transsion Holdings
R1-2501022	Adaptation of common signal/channel transmissions	MediaTek Inc.
R1-2501053	Adaptation of common signals and channels	Lenovo
R1-2501112	Discussion on adaptation of common signal channel transmissions	HONOR
R1-2501134	Discussion on adaptation of common signal/channel transmissions	Sharp
R1-2501162	Adaptation of common channel transmissions	Qualcomm Incorporated
R1-2501208	Discussion on adaptation of common signal/channel transmissions	NTT DOCOMO, INC.
R1-2501253	Adaptation of common signal/channel transmissions for NES	Ericsson
R1-2501279	Discussion on adaptation of common signal and channel transmissions	CEWiT

R1-2501451 Summary#1 of AI 9.5.3 for R19 NES Moderator (Ericsson)
From Tuesday session
Agreement
For adaptation of PRACH in time-domain, for determining the additional PRACH resources in time-domain,

- When an additional RO is overlapped with legacy valid RO in both time and frequency domain, the additional RO is invalid before the SSB-RO mapping
    - Note: the overlapped RO for legacy resource is not impacted
    - FFS: Clarification on configuration of legacy ROs

Conclusion
There is no RAN1 consensus to support the following in Rel-19

- New SSB burst periodicity values other than the legacy values (i.e., 5 ms, 10 ms, 20 ms, 40 ms, 80 ms, or 160 ms).
- New UE trigger
- Adapting the transmitted number of SSBs within a SSB burst

- Adaptation of the time domain positions of SSBs within a burst

Conclusion
There is no RAN1 consensus to support time domain adaptation for CD-SSB in Rel-19 for SCell.

R1-2501452 Summary#2 of AI 9.5.3 for R19 NES Moderator (Ericsson)
From Wednesday session
Agreement
For adaptation of SSB in time-domain, support the following to adapt SSB burst periodicity for an SCell
- UE is configured with SSB burst periodicity using legacy signalling for the SCell
- UE is configured with X additional SSB burst periodicity for the SCell
  - FFS: Value of X
- SSB occasions with larger periodicity are subset of the SSB occasions with shorter periodicity
  - FFS: Whether there is specification impact
  - Note: This does not impact the discussion on OD-SSB
- For switching the periodicity, down-select between
  - (MAC-CE)
    - MAC-CE details for SSB burst periodicity adaptation is up to RAN2.
  - (DCI)
    - Alt 1-3: DCI based signalling is used

Agreement
For adaption of PRACH in time-domain, for a connected mode UE, support a 1-bit field in DCI 1_0 with C-RNTI used to trigger PRACH (i.e. PDCCH order) to indicate whether the additional PRACH resource(s) is available for the triggered PRACH.
- FFS: UE behaviour (e.g. applicable resources for PRACH mask index) when it is indicated of additional PRACH resource(s)
- FFS: Details on how to reuse existing bit for the 1-bit indication

Agreement
For DCI-based adaptation for additional PRACH resources, DCI 1_0 with P-RNTI indicates the availability information for additional PRACH resource from a reference point and for a validity time duration
- FFS: Validity time duration for availability is configured by higher layer signaling or predefined
- FFS: Location of the reference point defined in the specification
- FFS: Value/granularity of the validity time duration.
- FFS: Whether DCI can be used to explicitly deactivate the additional PRACH resources

R1-2501453 Summary#3 of AI 9.5.3 for R19 NES Moderator (Ericsson)
From Thursday session
Agreement
For DCI-based adaptation for additional PRACH resources, support optional semi-static signalling of a single PRACH mask to identify the subset of the additional PRACH resources
- The mask is applicable at unit of
  - Alt 1: PRACH association period
  - Alt 2: PRACH association pattern period
  - Alt 3: SFN level
- The PRACH association period is determined based on valid additional ROs only.
- The mask is applied after valid RO determination and SSB-RO mapping.
- Note: The existing behaviour in TS 38.213 "An association pattern period includes one or more association periods and is determined so that a pattern between PRACH occasions and SS/PBCH block indexes repeats at most every 160 msec." is not impacted due to application of the mask.
- This is applicable at least for adaptation for DCI 1_0 with P-RNTI
- The DCI does not indicate PRACH mask selection
- FFS: how the mask is identified
  - Option 1: The PRACH mask is from a PRACH mask table
    - Pre-defined table with N=[4 or 8 or 16] rows
    - The semi-static signalling indicates a PRACH mask index
  - Option 2: The PRACH mask is based on configuration parameters e.g. bitmap at SFN-level, periodic time domain window, …

Agreement

- Separate configuration of Msg1-FDM for the additional PRACH resources at least for 4-step RACH is supported
  - UE is not expected to be configured such that there are more than 8 FDM-ed valid ROs (legacy + additional ROs)
  - FFS: When there is no configuration of Msg1-FDM
- Separate configuration of number of SSB per RO is supported

R1-2501599 Summary#4 of AI 9.5.3 for R19 NES Moderator (Ericsson)
From Friday session
Agreement
Study the following options for the reference point (for the availability information of additional PRACH resources indicated by DCI 1_0 with P-RNTI in a PF) for RRC idle/inactive mode UE and RRC connected mode UE,
- Option 1: SFN of the first PF from the next I-DRX cycle
- Option 2: SFN of the first PF from the current I-DRX cycle
- Option 3: From the first frame of the first PRACH association period after UE receives the DCI
- Option 4: From the first frame of the current SI modification period
- Option 5: From the first frame of the next SI modification period

Agreement
For adaptation of SSB in time-domain, for adapting SSB burst periodicity for an SCell
- Support group common DCI signalling for switching the SSB burst periodicity using DCI format 2_9 with [*cellDTRX-RNTI]*
- FFS: which scenario(s) is this applicable for (e.g. as defined in 9.5.1)

Note: Above does not prevent RAN2 from designing a MAC CE based on OD-SSB feature and also used for SSB burst adaptation

R1-2501600 Final Summary of AI 9.5.3 for R19 NES Moderator (Ericsson)

9.6 Low-power wake-up signal and receiver for NR (LP-WUS/WUR)

Please refer to RP-241824 for detailed scope of the WI. Additional RAN guidance on Rel-19 LP-WUS/WUR can be found in RP-243266.
Rapporteur to provide initial input on higher layer signalling under agenda item 9.6. For input on higher layer signalling from any other source, please include it as part of your tdoc to relevant sub-agenda items.

[120-R19-LP_WUS] – Xin (vivo)
Email discussion on Rel-19 LP-WUS/WUR
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500356 Rapporteur input on RRC parameters for Rel-19 LP-WUS/WUR vivo, DOCOMO
R1-2501630 Summary of discussion on RRC parameters for Rel-19 LP-WUS/WUR vivo (WI rapporteur)

R1-2501585 Summary of RAN1 agreements on LP-WUS/WUR for NR Rapporteur (vivo)

9.6.1 LP-WUS and LP-SS design

R1-2500048 Discussion on LP-WUS and LP-SS Design FUTUREWEI
R1-2500063 Discussion on LP-WUS and LP-SS Design TCL
R1-2500073 Signal Design of LP-WUS and LP-SS Huawei, HiSilicon
R1-2500131 Discussion on LP-WUS design ZTE Corporation, Sanechips
R1-2500176 Discussion on LP-WUS and LP-SS design Spreadtrum, UNISOC
R1-2500231 Design of LP-WUS and LP-SS CATT
R1-2500294 Discussion on LP-WUS and LP-SS design CMCC
R1-2500357 Discussion on LP-WUS and LP-SS design vivo
R1-2501364 Signal design for LP-WUS and LP-SS OPPO (rev of R1-2500437)
R1-2500491 Discussion on the LP-WUS and LP-SS design Panasonic
R1-2500496 Discussion on LP-WUS and LP-SS Design EURECOM
R1-2500530 Discussion on LP-WUS and LP-SS design InterDigital, Inc.
R1-2500573 LP-WUS and LP-SS design Nokia
R1-2500602 Discussion on LP-WUS and LP-SS design NEC
R1-2500656 LP-WUS and LP-SS design Sony
R1-2500739 Discussion on LP-WUS and LP-SS design Xiaomi

R1-2500788	LP-WUS and LP-SS design	Apple
R1-2500857	Discussion on LP-WUS and LP-SS design	Samsung
R1-2500956	Discussion on LP-WUS and LP-SS design	LG Electronics
R1-2501023	LP-WUS and LP-SS design	MediaTek Inc.
R1-2501096	Discussion on LP-WUS and LP-SS design	Sharp
R1-2501100	Discussion on LP-WUS and LP-SS design	Lenovo
R1-2501113	Discussion on LP-WUS and LP-SS design	HONOR
R1-2501163	LP-WUS and LP-SS design	Qualcomm Incorporated
R1-2501209	Discussion on LP-WUS and LP-SS design	NTT DOCOMO, INC.
R1-2501254	LP-WUS and LP-SS design	Ericsson
R1-2501294	On LP-WUS and LP-SS design	Nordic Semiconductor ASA

R1-2501424 Summary #1 of discussions on LP-WUS and LP-SS design Moderator (vivo)
From Monday session
Agreement
For LP-WUS information carried by OOK, select alt 1 for codepoint to meet FAR and MDR performance.
- Alt 1: Coding with rate matching if any
  - For more than 2 information bits: RM coding as in section 5.3.3.3 of 38.212 (FFS: whether scrambling in the same section is applied or not)
  - 2 information bits (if supported): coding as in section 5.3.3.2 of 38.212 (FFS: whether scrambling in the same section is applied or not)
  - 1 information bit (if supported): repetition as in section 5.3.3.1 of 38.212 (FFS: whether scrambling in the same section is applied or not)
  - FFS: Code block length
  - FFS: Further repetition before or after Manchester coding

Conclusion
For RRC idle/inactive, LP-WUS is not supported for the case where associated CD-SSB and initial DL BWP have different SCS.

R1-2501425 Summary #2 of discussions on LP-WUS and LP-SS design Moderator (vivo)
From Tuesday session
Agreement
At least for M>1, for the overlaid OFDM sequence length, support
- Alt2: $L_{ZC} = 12 * X / M$
- Note: X is the number of RBs of LP-WUS/LP-SS bandwidth (blanked guard RBs are not included)

Agreement
Update the agreements in RAN1 #118bis as below
> Agreement
> Support overlaid OFDM sequence(s) for LP-SS:
> - LP-SS reuses the overlaid OFDM sequence(s) specified for LP-WUS. The design on overlaid OFDM sequence(s) specified for LP-WUS doesn't target for sync and RRM measurement performance based on overlaid OFDM sequence for LP-SS.
> - Applicable to both OOK-1 and OOK-4
> - Whether to transmit LP-SS by using a specified overlaid OFDM sequence is configurable.
> - Applicable ~~at least~~ for OOK-1 only ~~and FFS for OOK-4~~
> - From RAN1 perspective, it is not intended to introduce new RAN4 requirements specific to overlaid sequences

Agreement
For RRC connected, for LP-WUS SCS:
- Alt 1: LP-WUS SCS is same as the active DL BWP

R1-2501519 Summary #3 of discussions on LP-WUS and LP-SS design Moderator (vivo)
From Wednesday session
Working Assumption
Regarding the LP-WUS information to trigger PDCCH monitoring of RRC connected UEs,
- codepoint based
  - The maximum number of codepoints checked per MO by a UE is up to 8
    - Depending on UE capability, a UE may support less than 8

R1-2501520 Summary #4 of discussions on LP-WUS and LP-SS design Moderator (vivo)

From Thursday session

Agreement

For M=2, 4, B_{ZC} is given by the largest prime number such that $B_{ZC} < L_{ZC}$, L_{ZC} is the overlaid OFDM sequence length.

- The base overlaid sequence $s(n)$ is generated by extension of X_q

$$s(n) = X_q\left(n \bmod B_{ZC}\right), n = 0, \ldots, L_{ZC} - 1$$

- With CS(s) applied to the base overlaid OFDM sequence if any: C_v denotes the potential cyclic shift (s)

$$s'(n) = X_q\left((n + C_v) \bmod B_{ZC}\right), n = 0, \ldots, L_{ZC} - 1$$

- Note it doesn't preclude any pulse shaping scheme if any.

Agreement

For RRC connected, LP-WUS frequency resource can be outside of active DL BWP but has to be within the same carrier as the active DL BWP

- Basic capability is LP-WUS, if present, frequency resource within active DL BWP. LP-WUS frequency resource outside of active DL BWP is subject to separate UE capability
- No RAN1 optimization specific to the case where LP-WUS frequency resource is outside of active DL BWP

R1-2501583 **Summary #5 of discussions on LP-WUS and LP-SS design** **Moderator (vivo)**

From Friday session

Agreement

For M=1, L=4 (if supported), the set of LP-SS binary sequences is:
[0 1 0 1]
[0 1 1 0]
[1 0 0 1]
[1 0 1 0]

Agreement

For M=1, L=6 (if supported), the set of LP-SS sequence is:
[1 0 1 0 1 0]
[0 1 0 1 0 1]
[1 0 0 1 0 1]
[1 0 1 0 0 1]

Agreement

For M=1, L=8 (if supported), the set of LP-SS sequence is:
[1 0 1 0 0 1 0 1]
[1 0 1 0 1 0 0 1]
[1 0 0 1 0 1 0 1]
[0 1 0 1 0 1 0 1]

Agreement

For M=2, L=8 (if supported), the set of LP-SS sequence is:
[0 1 0 1 1 0 0 1]
[0 1 1 0 0 1 0 1]
[0 1 1 0 1 0 0 1]
[1 0 0 1 0 1 1 0]

Agreement

For M=2, L=12 (if supported), the set of LP-SS sequence is:
[1 0 0 1 1 0 0 1 1 0 0 1]
[0 1 1 0 1 0 0 1 1 0 0 1]
[0 1 1 0 0 1 1 0 1 0 0 1]
[0 1 1 0 0 1 0 1 1 0 0 1]

Agreement

For M=2, L=16 (if supported), the set of LP-SS sequence is:
[1 0 0 1 0 1 0 1 1 0 0 1 1 0 0 1]
[1 0 0 1 1 0 0 1 0 1 1 0 0 1 0 1]
[1 0 0 1 1 0 1 0 0 1 0 1 1 0 0 1]

[1 0 1 0 0 1 1 0 0 1 1 0 0 1 0 1]

Agreement
For M=4, L=16 (if supported), the set of LP-SS sequence is down-selected between:
- Set 1
  [0 1 1 0 1 0 0 1 1 0 1 0 1 0 1 0]
  [0 1 1 0 1 0 1 0 1 0 0 1 1 0 1 0]
  [1 0 1 0 0 1 1 0 1 0 1 0 1 0 0 1]
  [1 0 1 0 1 0 0 1 1 0 1 0 0 1 1 0]
- Set 2
  [1 0 0 0 1 0 0 0 0 0 0 1 0 0 1 0]
  [1 0 0 0 0 1 0 0 1 0 0 0 0 0 1 0]
  [1 0 0 0 0 1 0 0 0 1 0 0 1 0 0 0]
  [1 0 0 0 0 1 0 0 0 0 0 1 0 0 0 1]

Agreement
For M=4, L=32(if supported), the set of LP-SS sequence is down-selected between:
- Set 1:
  [0 1 0 1 1 0 1 0 1 0 1 0 1 0 0 1 1 0 1 0 0 1 1 0 0 1 1 0 0 1 0 1]
  [0 1 1 0 0 1 0 1 0 1 1 0 0 1 0 1 1 0 0 1 1 0 1 0 1 0 1 0 0 1 0 1]
  [0 1 0 1 0 1 0 1 1 0 1 0 1 0 0 1 1 0 1 0 1 0 0 1 1 0 1 0 0 1 1 0]
  [0 1 0 1 0 1 1 0 0 1 0 1 1 0 1 0 0 1 1 0 0 1 1 0 1 0 1 0 0 1 0 1]
- Set 2
  [0 0 0 1 1 0 0 0 0 0 0 1 0 0 0 1 1 0 0 0 0 0 0 1 0 0 0 1 0 1 0 0]
  [0 0 0 1 0 1 0 0 1 0 0 0 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 0 1 0 0]
  [0 0 1 0 0 0 1 0 1 0 0 0 0 1 0 0 1 0 0 0 0 0 0 1 0 0 1 0 0 0 1 0]
  [0 0 1 0 0 1 0 0 0 0 0 1 1 0 0 0 0 0 1 0 1 0 0 0 0 1 0 0 0 0 1 0]

Agreement
For idle mode, regarding the maximum number of candidates overlaid sequences to carry LP-WUS information per OOK ON chip for one cell:
- support maximum 16 candidates overlaid sequences for M=1
- support maximum 8 candidates overlaid sequences for M=2
- For candidate overlaid sequences across all OOK ON chips of LP-WUS, the number of roots (in specification) is up to [FFS: X], FFS whether the number of roots can be different for different M value.
  - FFS: The number of overlaid sequences applicable for a UE is no more than 2 per OOK ON chip.

Agreement
For WUS information carried by the overlaid OFDM sequence(s), consider at least the following alternatives:
- Alt 1: Raw information bits are mapped to sequence(s)
- Alt 2: Raw information bits are mapped to sequence(s) after channel coding
  - Same channel coding scheme(s) as OOK is applied.
    - FFS same or different rate matching and/or repetition factor as OOK
- Alt 3: Codepoint/Subgroup is mapped to sequence(s)

R1-2501584 Final summary of discussions on LP-WUS and LP-SS design Moderator (vivo)

9.6.2 LP-WUS operation in IDLE/INACTIVE modes

R1-2500049 Discussion on LP-WUS Operation in IDLE/INACTIVE Modes FUTUREWEI
R1-2500064 Discussion on LP-WUS Operation in IDLE/INACTIVE state TCL
R1-2500074 Procedures and functionalities of LP-WUS in IDLE/INACTIVE mode Huawei, HiSilicon
R1-2500132 Discussion on LP-WUS operation in IDLE/INACTIVE mode ZTE Corporation, Sanechips
R1-2500177 Discussion on LP-WUS operation in IDLE/INACTIVE modes Spreadtrum, UNISOC
R1-2500232 System design and procedure of LP-WUS operation for UE in IDLE/Inactive Modes CATT
R1-2500295 Discussion on LP-WUS operation in IDLE/INACTIVE modes CMCC
R1-2500358 Discussion on LP-WUS operation in IDLE/INACTIVE modes vivo
R1-2500438 Further consideration on LP-WUS operation in RRC_IDLE/INACTIVE modes OPPO
R1-2500492 Discussion on LP-WUS operation in IDLE/INACTIVE modes Panasonic
R1-2500531 Discussion on LP-WUS operation in IDLE/INACTIVE modes InterDigital, Inc.
R1-2500574 LP-WUS operation in IDLE/Inactive mode Nokia
R1-2500603 Discussion on LP-WUS operation in RRC IDLE/INACTIVE mode NEC

R1-2500657	LP-WUS operation in IDLE / INACTIVE modes	Sony
R1-2500740	Discussion on LP-WUS operation in Idle/Inactive modes	Xiaomi
R1-2500789	LP-WUS operation in IDLE/INACTIVE modes	Apple
R1-2500858	Discussion on LP-WUS operation in IDLE/INACTIVE modes	Samsung
R1-2500916	Discussion on LP-WUS operation in IDLE/INACTIVE modes	ETRI
R1-2500957	Discussion on LP-WUS operation in IDLE/INACTIVE modes	LG Electronics
R1-2501024	LP-WUS operation in IDLE/INACTIVE modes	MediaTek Inc.
R1-2501097	Discussion on LP-WUS operation in IDLE/INACTIVE modes	Sharp
R1-2501101	Discussion on LP-WUS operation in Idle/Inactive modes	Lenovo
R1-2501164	LP-WUR operation in idle and inactive modes	Qualcomm Incorporated
R1-2501210	Discussion on LP-WUS operation in IDLE/INACTIVE modes	NTT DOCOMO, INC.
R1-2501255	LP-WUS operation in IDLE and INACTIVE modes	Ericsson
R1-2501293	On LP-WUS operation in IDLE/Inactive	Nordic Semiconductor ASA

R1-2500791 Summary #1 on LP-WUS operation in IDLE/INACTIVE mode Moderator (Apple)

From Monday session

Agreement

The EPRE ratio between LP-WUS/LP-SS and SSB can be configured by the gNB.

- FFS whether there is a common configuration or separate configuration for LP-WUS and LP-SS
- Above is not applicable for the case when all the subcarriers for LP-WUS/LP-SS are transmitted with zero power (from baseband perspective)
- Above does not mandate any UE implementation for different receiver types
- FFS: whether/how to support for the case when there are different number of OOK ON symbols in different OFDM symbols (if supported)

Agreement

For the offset value(s) between an LO and a reference PO/PF, at least a frame-level offset is provided.

- The reference point (reference PO/PF) for the frame-level offset is the start of the PF, or the first PF of the PF(s) (if mapping of POs from multiple PFs to one LO is supported), associated with the LO.
- FFS other offset value(s) to determine the MOs of the LO

Agreement

The previous agreement in RAN1#119 is updated as follows:

- Each LP-WUS or LP-SS is QCLed with an SSB with ~~[select one from 'typeA',~~ 'typeC'~~]~~, and when applicable, 'typeD' with the same SSB.

Agreement

Confirm the following working assumption with the additional text in red:

Working Assumption

From RAN1 perspective, for the RRM measurement metrics based on SSS for OFDM-based LP-WUR, use the same definition of SS-RSRP and SS-RSRQ for LP-SSS-RSRP and LP-SSS-RSRQ, respectively.

- Above is applicable for both time-domain processing or frequency-domain processing
- Above does not imply that RAN1 will introduce LP-SSS-RSRP and LP-SSS-RSRQ in the specifications

Existing metrics SS-RSRP and SS-RSRQ are reused for OFDM-based LP-WUR. No separate metrics (LP-SSS-RSRP and LP-SSS-RSRQ) will be introduced in the specifications.

- The metrics SS-RSRP and SS-RSRQ are applicable for OFDM-based LP-WUR for RRC_IDLE and RRC_INACTIVE serving cell measurement.
- FFS: whether to remove or modify the restriction of using SMTC window for LP-SSS-RSRP/RSSI measurement.

R1-2500792 Summary #2 on LP-WUS operation in IDLE/INACTIVE mode Moderator (Apple)

From Tuesday session

Agreement

For the LO to PO mapping from network perspective, support Option 2 (UEs corresponding to different POs monitor the same LO).

- This should not increase the maximum number of codepoints per LO/LP-WUS compared to Option 1.
- FFS conditions/restrictions for mapping multiple POs to one LO
- Down-select between 2 and 4 for max number of POs per LO.

R1-2500793 Summary #3 on LP-WUS operation in IDLE/INACTIVE mode Moderator (Apple)

From Wednesday session

Agreement

For UE capability report on the wake-up delay:

- Alt 1: For the 3 candidate values for the wake-up delay capability report, support {[70ms], [500ms] and [900ms]}.
  - The reported values assume SSB periodicity of 20ms, where [70ms] assumes [3] SSBs needed for synchronization, [500ms] and [900ms] assume [5] SSBs needed for synchronization.
  - FFS: translation of wake-up delay for different SSB periodicities
  - FFS: different sets of 3 candidate value for different SSB periodicities
- Send an LS to RAN4 after the down-selection to confirm the number of SSBs

Agreement
For the offset value(s) between an LO and a reference PO/PF, adopt Option 2B-1.
- gNB can configure 1 or 2 offset values.
  - FFS whether gNB can configure 3 offset values
- If multiple offset values are configured and if the gap between the LO associated with the largest offset value and the corresponding PO is no less than the wake-up delay a UE reports, the UE monitors the LO associated with the smallest offset value that has a gap between the LO and the PO no less than the wake-up delay.
  - Note: if a single offset value is configured, UE behaviour is according to Option 1-1.
- All the UEs supporting LP-WUS for idle/inactive mode supports the configuration of 2 offset values (FFS: 3 values).

R1-2500794 Summary #4 on LP-WUS operation in IDLE/INACTIVE mode Moderator (Apple)
From Thursday session
Agreement
Confirm the following working assumption with the modification:
Working Assumption
The maximum number of subgroups per PO supported in Rel-19 is 31 ~~32~~.

From Friday session
R1-2501615 Draft LS on the wake-up delay for LP-WUS operation in IDLE/INACTIVE mode Moderator (Apple)
Decision: The draft LS is endorsed. Final LS is approved in R1-2501624.

Final summary in R1-2500795.

9.6.3 LP-WUS operation in CONNECTED modes

R1-2500075	Procedures and functionalities of LP-WUS in CONNECTED mode	Huawei, HiSilicon
R1-2500133	Discussion on LP-WUS operation in CONNECTED mode	ZTE Corporation, Sanechips
R1-2500178	Discussion on LP-WUS operation in CONNECTED modes	Spreadtrum, UNISOC
R1-2500233	System design and procedure of LP-WUS operation for UE in CONNECTED Modes	CATT
R1-2500296	Discussion on LP-WUS operation in CONNECTED mode	CMCC
R1-2500359	Discussion on LP-WUS operation in CONNECTED modes	vivo
R1-2500439	Further consideration on LP-WUS operation in connected mode	OPPO
R1-2500505	Discussion on LP-WUS operation in CONNECTED mode	Panasonic
R1-2500532	Discussion on RRC CONNECTED mode LP-WUS monitoring	InterDigital, Inc.
R1-2500575	LP-WUS operation in CONNECTED mode	Nokia
R1-2500604	Discussion on LP-WUS operation in RRC CONNECTED mode	NEC
R1-2500658	LP-WUS operation in CONNECTED mode	Sony
R1-2500741	Discussion on LP-WUS operation in Connected mode	Xiaomi
R1-2500790	LP-WUS operation in CONNECTED modes	Apple
R1-2500859	Discussion on LP-WUS operation in CONNECTED modes	Samsung
R1-2500886	Discussion on LP-WUS operation in CONNECTED modes	Lenovo
R1-2500917	Discussion on LP-WUS operation in CONNECTED modes	ETRI
R1-2500958	Discussion on LP-WUS operation in CONNECTED modes	LG Electronics
R1-2501025	LP-WUS operation in CONNECTED modes	MediaTek Inc.
R1-2501098	Discussion on LP-WUS operation in CONNECTED modes	Sharp
R1-2501165	LP-WUR operation in connected mode	Qualcomm Incorporated
R1-2501211	Discussion on LP-WUS operation in CONNECTED mode	NTT DOCOMO, INC.
R1-2501256	LP-WUS operation in CONNECTED mode	Ericsson
R1-2501296	Discussion on LP-WUS procedures in Connected mode	TCL

R1-2501420 FL summary #1 on LP-WUS operation in CONNECTED mode Moderator (NTT DOCOMO)
From Monday session

Agreement
For RRC connected mode, UE does not monitor LP-WUS during C-DRX active time.
- FFS FAR and MDR impact.

Agreement
LP-WUS can be configured with legacy PDCCH skipping.
- Note: There is no intention to modify legacy PDCCH skipping.

Agreement
For RRC CONNECTED mode, LP-WUS can be configured with Rel-17 SSSG switching.

R1-2501487 **FL summary #2 on LP-WUS operation in CONNECTED mode** **Moderator (NTT DOCOMO)**
From Tuesday session
Agreement
- For the case when UE is configured with CA without dual DRX groups in RRC CONNECTED mode,
  - LP-WUS can be configured on a serving cell
  - LP-WUS indication is applicable to all serving cells
    - Note: There is no impact to PDCCH monitoring behaviour related to SCell dormancy/activation/deactivation
- FFS: Whether the serving cell on which the LP-WUS is configured is only PCell or can be RRC configured cell other than PCell.

Agreement
- For the case when UE is configured with NR-DC with CA without dual DRX groups or without CA in RRC CONNECTED mode,
  - LP-WUS can be configured on a serving cell per cell-group
    - This does not imply that LP-WUS has to be configured on both cell groups
  - LP-WUS indication is applicable to all serving cells in the cell-group
    - Note: There is no impact to PDCCH monitoring behaviour related to SCell dormancy/activation/deactivation
  - FFS whether to support this case as separate UE capability from CA case
  - FFS UE capability for monitoring LP-WUS on one cell group or both cell groups
- FFS: Whether the serving cell on which the LP-WUS is configured is only PCell/PSCell or can be RRC configured cell other than Pcell/PSCell.

R1-2501535 **FL summary #3 on LP-WUS operation in CONNECTED mode** **Moderator (NTT DOCOMO)**
From Wednesday session
Agreement
For RRC CONNECTED mode, LP-WUS can be configured with Cell DTX.
- FFS During Cell DTX inactive time, whether the UE is expected to monitor LP-WUS or not.

Conclusion
From RAN1 perspective, when LP-WUS monitoring is enabled for RRC CONNECTED mode, it is supported that SR, PRACH and CG-PUSCH triggers MR PDCCH monitoring according to the legacy UE behaviour.
- Above applies at least for Option 1-1.

Agreement
For LP-WUS monitoring in RRC CONNECTED mode, select one of the options on how to define/configure/indicate QCL source of LP-WUS
- Opt1: Separate indication/configuration for QCL source of LP-WUS
- Opt2: Association of QCL source between LP-WUS and PDCCH/CORESET

R1-2501561 **FL summary #4 on LP-WUS operation in CONNECTED mode** **Moderator (NTT DOCOMO)**
From Thursday session
Agreement
For LP-WUS MOs in connected mode for Option 1-1, support Approach 1:
- LP-WUS MOs, including periodicity and time offset1, are configured independently from the C-DRX periodicity/offset by new RRC parameter(s).

- Time offset2 indicates a time prior to a slot where the *drx-onDurationTimer* would start
  - UE monitors the first Z LP-WUS MO(s) at or after Time offset2.
    - FFS: How to determine the value of Z
  - Time offset2 is RRC configured
- UE is not required to monitor LP-WUS during the X [slots/symbols] prior to the beginning of a slot where the UE would start the *drx-onDurationTimer*. FFS: How X is determined.

R1-2501586 FL summary #5 on LP-WUS operation in CONNECTED mode Moderator (NTT DOCOMO)
From Friday session
Agreement
For LP-WUS MOs in connected mode for Option 1-2, support Approach 1:
- LP-WUS MOs, including periodicity and time offset3, are configured independently from the C-DRX periodicity/offset by new RRC parameter(s).
  - FFS one or multiple MOs per periodicity
- UE monitors LP-WUS in the LP-WUS MOs and, if triggered to wake up, starts a new timer for PDCCH monitoring triggered by LP-WUS, after a time offset4.
  - The time offset4 configured by the network indicating a time, after which the UE starts PDCCH monitoring via starting the new timer.
  - FFS: Definition of time offset4

Agreement
- For the X = [2, 3] candidate values for the UE capability report on the minimum time gap for each SCS, consider {0.25, 0.5, 1, 2, 3, 6, 10, 20, 30, 40}ms
  - Note: Other values are not precluded
  - Only for reported value of [20ms,] 30ms and 40ms (if supported), the reported values assume SSB periodicity of 20ms and ensures SSB(s) before PDCCH reception and during time gap
    - FFS: translation of minimum time gap for different SSB periodicities including different candidates for different SSB periodicities
  - FFS whether the same set of candidates are used for different SCSs
  - FFS whether different values are selected for OOK-based WUR and OFDM-based WUR
  - FFS whether/how to consider the case of CA
  - FFS: candidate values that can be indicated by UAI

R1-2501587 Final summary on LP-WUS operation in CONNECTED mode Moderator (NTT DOCOMO)

9.7 Study on channel modelling for Integrated Sensing And Communication (ISAC) for NR

Please refer to RP-242348 for detailed scope of the SI.

R1-2501548 Session notes for 9.7 (Study on channel modelling for Integrated Sensing And Communication for NR) Ad-Hoc Chair (Huawei)
Friday decision: The session notes are endorsed and contents reflected below.

[120-R19-ISAC] – Yingyang (xiaomi)
Email discussion on Rel-19 ISAC channel model
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500998 Draft CR for TR 38.901 to introduce channel model for ISAC Xiaomi, AT&T
Revised in
R1-2501640 Draft CR for TR 38.901 to introduce channel model for ISAC Moderator (Xiaomi)
R1-2501641 Summary on discussions on CR to 38.901 on ISAC channel modeling Moderator (Xiaomi)

From Friday session
[Post-120-ISAC-01] – Yingyang (Xiaomi)
Email discussion on values/pattern of A*B1 of RCS for target for monostatic sensing, from March 3-7
- Companies to provide values of the parameters (referring the example table in the agreement) on A*B1 for vehicle, until March 5, targeting agreement on merged values by March 7.
- Collect companies views on the possible extension of the agreement (except the example table) on A*B1 of RCS for vehicle to other target types with angle-dependent RCS until March 7, for potential agreement.

9.7.1 ISAC deployment scenarios

Please provide your inputs on calibrations is to this sub-agenda item.

R1-2500059	Email discussion summary on ISAC CM calibration assumptions	Moderator (AT&T)
R1-2500071	Deployment scenarios for ISAC channel model	Huawei, HiSilicon
R1-2500234	Discussion on ISAC deployment scenarios	CATT, CICTCI
R1-2500266	Discussion on ISAC deployment scenarios	China Telecom
R1-2500297	Discussion on ISAC deployment scenarios	CMCC, China Southern Power Grid
R1-2500312	Discussion on ISAC scenario	CALTTA
R1-2500360	Views on Rel-19 ISAC deployment scenarios	vivo
R1-2500413	Discussion on ISAC deployment scenarios and requirements	EURECOM
R1-2500417	Discussion on ISAC deployment scenarios	Tiami Networks
R1-2500462	Discussion on ISAC deployment scenarios and calibration	OPPO
R1-2500576	Discussion on ISAC deployment scenarios	ZTE Corporation, Sanechips
R1-2500659	Discussion on ISAC deployment scenarios	Sony
R1-2500679	Discussion on ISAC Deployment Scenarios	Nokia, Nokia Shanghai Bell
R1-2500684	Discussion on ISAC deployment scenarios	SK Telecom
R1-2500691	Deployment scenarios for integrated sensing and communication with NR	NVIDIA
R1-2500742	Deployment scenarios and evaluation assumptions for ISAC channel model	Xiaomi
R1-2500747	Discussion on ISAC deployment scenarios	InterDigital, Inc.
R1-2500755	Discussion on ISAC Deployment Scenarios	Ericsson
R1-2500796	Discussion on ISAC deployment scenarios	Apple
R1-2500860	Discussion on ISAC deployment scenarios	Samsung
R1-2500891	Considerations on ISCA deployment scenarios	CAICT
R1-2501011	Discussion on ISAC deployment scenarios for Automotive	Continental Automotive
R1-2501026	Discussion on ISAC deployment scenario	MediaTek Inc.
R1-2501045	Discussion on ISAC deployment scenarios	LG Electronics
R1-2501059	Discussion on ISAC deployment scenarios	Lenovo
R1-2501081	ISAC channel model calibration and scenario parameters	AT&T, FirstNet
R1-2501135	Discussion on ISAC channel calibration	BUPT, CMCC
R1-2501166	Discussion on ISAC deployment scenarios	Qualcomm Incorporated

R1-2501076 FL Summary #1 on ISAC Scenarios and Calibrations Moderator (AT&T)

R1-2501077 FL Summary #2 on ISAC Scenarios and Calibrations Moderator (AT&T)

From Wednesday session

Agreement

For ISAC channel modelling calibration, RAN1 considers both large-scale and full-scale calibration to include parameters and values for at least the following:

- o large scale parameters, where fast fading is not included
- o full-scale calibration parameters, which includes fast fading.

- NOTE0: one part of calibration work does not include additional components and does not include spatial consistency
  - o FFS: whether spatial consistency is specified as an additional component for ISAC CM
- NOTE1: additional calibrations including spatial consistency can also be considered case by case for different scenarios.
- NOTE2: Inclusion of EO in ISAC CM calibrations can also be considered case by case for different scenarios.

Agreement

Calibration of ISAC CM includes separate calibration of the target channel and of the background channel

- FFS: additional calibration for the combined channel (combination of target and background channel).

R1-2501078 FL Summary #3 on ISAC Scenarios and Calibrations Moderator (AT&T)

Presented in Thursday morning session.

R1-2501574 FL Summary #4 on ISAC Scenarios and Calibrations Moderator (AT&T)

From Thursday session

Agreement

For the purposes of large scale calibration for UAV sensing targets, the following calibration parameters are proposed below in Table x.

Table x. Simulation assumptions for large scale calibration for UAV sensing targets

Parameters	Values
Scenario	UMa-AV
Sensing mode	TRP monostatic, TRP-TRP bistatic, TRP-UE bistatic, UE-UE bistatic Note: further down-selection of the sensing modes for UAV sensing is not precluded

Sectorization	3 sectors per cell site: 30, 150 and 270 degrees
Carrier Frequency	FR1: 6 GHz FR2: 30 GHz
BS antenna configurations	Single dual-pol isotropic antenna
BS Tx power	FR1: 56dBm FR2: 41dBm
Bandwidth	FR1: 100MHz FR2: 400MHz
BS noise figure	FR1: 5dB FR2: 7dB
UT antenna configurations	(M,N,P,Mg,Ng;Mp,Np) = (1,1,2,1,1;1,1)
UT noise figure	FR1: 9dB FR2: 10dB
Sensing target distribution	1 target uniformly distributed (across multiple drops) within the center cell. Vertical distribution: Fixed height value of 200 m.
Component A of the RCS for each scattering point	a fixed value of A
Minimum 3D distances between pairs of Tx/Rx and sensing target	10 m
Wrapping Method	No wrapping method is used if interference is not modelled, otherwise geographical distance based wrapping
Metrics	Coupling loss (based on LOS pathloss) • FFS: how to select sensing Tx and Rx FFS: additional metrics, wideband SIR and SINR based on RSRP if interference is modelled.

Final summary in R1-2501607.

9.7.2 ISAC channel modelling

R1-2500072 Channel modelling for ISAC Huawei, HiSilicon
R1-2500179 Discussion on ISAC channel modeling Spreadtrum, UNISOC
R1-2500235 Discussion on ISAC channel modelling CATT, CICTCI
R1-2500267 Discussion on ISAC channel modelling China Telecom
R1-2500298 Discussion on channel modeling methodology for ISAC CMCC, BUPT, SEU, PML
R1-2500313 Discussion on ISAC channel modelling CALTTA
R1-2500361 Views on Rel-19 ISAC channel modelling vivo, BUPT
R1-2500414 Discussion on ISAC channel modeling EURECOM
R1-2500418 Discussion on ISAC Channel Modeling Tiami Networks
R1-2501363 Study on ISAC channel modelling OPPO (rev of R1-2500463)
R1-2500483 Discussion on ISAC channel modelling Tejas Network Limited
R1-2500577 Discussion on channel modelling for ISAC ZTE Corporation, Sanechips
R1-2501369 ISAC Channel Modeling and Measurement Validation BUPT, CMCC, VIVO (rev of R1-2500626)
R1-2500660 Discussion on Channel Modelling for ISAC Sony
R1-2500680 Discussion on ISAC channel modeling Nokia, Nokia Shanghai Bell
R1-2500681 Discussion on ISAC Channel Modeling NIST
R1-2500685 Discussion on ISAC channel modelling SK Telecom
R1-2500692 Channel modeling for integrated sensing and communication with NR NVIDIA
R1-2500743 Discussion on ISAC channel model Xiaomi, BJTU, BUPT
R1-2501368 Discussion on ISAC channel modeling InterDigital, Inc. (rev of R1-2500748 Discussion on ISAC)
R1-2500756 Discussion on ISAC Channel Modelling Ericsson
R1-2500797 Discussion on ISAC channel modelling Apple
R1-2500861 Discussion on ISAC channel modelling Samsung
R1-2500892 Considerations on ISAC channel modelling CAICT
R1-2500979 Discussion on ISAC Channel Modelling Panasonic
R1-2501027 Discussion on ISAC channel modelling MediaTek Inc.
R1-2501046 Discussion on ISAC channel modelling LG Electronics
R1-2501060 Discussion on Channel Modelling for ISAC Lenovo
R1-2501082 Discussions on ISAC Channel Modeling AT&T
R1-2501167 Discussion on ISAC channel modelling Qualcomm Incorporated
R1-2501212 Discussion on ISAC channel modeling NTT DOCOMO, INC.

R1-2500999 Summary #1 on ISAC channel modelling Moderator (Xiaomi)
From Tuesday session
Agreement
For bistatic/monostatic RCS

- RCS values/pattern for a scattering point of a target for bistatic sensing is generated by A*B1*B2 (i.e., Option 3 from the agreement in RAN1 #118bis)
- RCS values/pattern obtained by setting the same incident/scattered angle in the RCS model for bistatic sensing should be aligned with RCS for monostatic sensing

Agreement
RCS model and application in ISAC channel generation
- To define the RCS model (RCS=A*B1*B2) for a scattering point of a target, when the target type is vehicle, large size UAV, human with RCS model 2, AGV
  - The values/pattern of the product of component A and B1, i.e., A*B1 is given per target type, expressed in dBsm scale
  - Component A is expressed in dBsm scale. B1 is dependent on A*B1 and value of component A.
    - A is equal to a single value per target type
      - FFS: this allows different values for the same target type with different size, if needed
      - FFS: this allows different values for monostatic and bistatic sensing, if needed
  - Component B2 follows log-normal distribution. The mean μ_{B2_dB} and variance $\sigma^2_{B2_\mathrm{dB}}$ used to characterize $10lg(B2)$ satisfied a fixed relation $\mu_{B2_\mathrm{dB}} = \frac{-ln(10)}{20}\sigma^2_{B2_\mathrm{dB}}$.
- In the procedure of generating ISAC target channel, B1*B2 is applied after coupling of rays for a STX-SPST link and the corresponding SPST-SRX link before path dropping
- In the procedure of generating ISAC target channel, the following power scaling factor is applied in the last step in target channel generation (i.e., step 14 in the running CR).

$$L_{TX-SPST-RX} = PL_{dB}(d_1) + PL_{dB}(d_2) + 10lg\left(\frac{c^2}{4\pi f^2}\right) - 10lg(\sigma_{RCS,A}) + SF_{dB,1} + SF_{dB,2}$$

  Where,
  - $PL_{dB}(d_1)$ is pathloss between Tx and SPST, where d_1 is the distance between Tx and SPST
  - $PL_{dB}(d_2)$ is pathloss between Rx and SPST, where d_2 is the distance between SPST and Rx
  - $\sigma_{RCS,A}$ is the value of RCS component A
  - $SF_{dB,1}$,$SF_{dB,2}$ are shadow fading respectively generated for the Tx- SPST link and SPST -Rx link referring to step 4 in section 7.5, TR 38.901
  - Note: for monostatic sensing, $PL_{dB}(d_1) = PL_{dB}(d_2)$,$SF_{dB,1} = SF_{dB,2}$

Agreement
RCS upper bound: k equals to 3 is adopted to derive the upper bound of RCS component B2, kσ, where σ is the standard deviation of B2 in dB.

Agreement
For reducing options for reference TRs: for sensing scenario UMi, UMa, RMa, InH, InF, UMi-AV, UMa-AV, and RMa-AV, the reference TR to generate a TRP-TRP channel is:
- TRP-TRP link of scenario UMi, UMa, InH, and InF following the option based on TR 38.901 defined in section A.3 of TR 38.858
  - For InF, h_{UE} is changed to the same height as the BS
- TRP-UE link of scenario RMa defined in section 7 of TR 38.901 by setting h_{UE}=35m
- FFS: whether to add very low power clusters

R1-2501000 Summary #2 on ISAC channel modelling Moderator (Xiaomi)
From Wednesday session
Agreement
For vehicle with single/multiple scattering points:
- For mono-static, the RCS=A*B=A*B1*B2 for a scattering point of a vehicle is generated by
  - The values/pattern A*B1, i.e., $rcs_{dB}(\theta,\varphi)$ is deterministic based on incident/scattered angles

$$rcs_{dB}(\theta,\varphi) = G_{max} - min\left\{-\left(\sigma^V{}_{dB}(\theta) + \sigma^H{}_{dB}(\varphi)\right),\sigma_{max}\right\}$$

Where,

$$\sigma^V{}_{dB}(\theta) = -min\left\{12\left(\frac{\theta-\theta_{center}}{\theta_{3dB}}\right)^2, \sigma_{max}\right\},$$

$$\sigma^{H}_{dB}(\varphi) = -min\left\{12\left(\frac{\varphi-\varphi_{center}}{\varphi_{3dB}}\right)^2, \sigma_{max}\right\},$$

For example, in case of vehicle with multiple scattering points:

	φ_{center}	$\varphi_{3dB,n}$	θ_{center}	$\theta_{3dB,n}$	G_{max}	σ_{max}	Applicable Range of θ	Applicable Range of φ
Left	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Back	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Right	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Front	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Roof	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

- Note: the applicable angular range is 360 degrees per row in horizontal domain in case of vehicle with multiple scattering points, and the applicable angular range is < 360 degrees per row in horizontal domain in case of vehicle with a single scattering point.
  - FFS: angular continuity

Working assumption
For modelling background channel for monostatic sensing:
Solution A: (previous Option 1)
- drop N reference point(s) following certain distribution, and then the background channel is generated based on the channel generated as in existing TR between the real Tx and the reference point for a scenario.
  - The distance between Tx and the reference point, and the height of the reference point follow Gamma distribution,

$$d_{rp_2D_n} \sim \Gamma(\alpha 1, \beta 1) + c1$$
$$h_{rp_n} \sim \Gamma(\alpha 2, \beta 2) + c2$$

  - FFS: value of N (N<4)
- FFS: whether to add additional very low power clusters
- FFS: any update to parameters e.g. angular distribution and delay spread

Agreement
The existing spatial consistency model in TR 38.901 is reused to model correlation of links between one TRP and different STs/UEs.

Agreement
Spatial consistency is not modelled at least for the following links
- Case 1: the links from/to two non-co-located TRPs
  - Link TRP1-X and link TRP2-X, where node X can be a target, a UE or another TRP
  - Link TRP1-X and link TRP2-Y, where node X, Y can be a target, a UE or another TRP
- Case 2: the two links with different LOS/NLOS condition
- Case 3: background channel for monostatic vs. any link (Tx-ST, ST-Rx, background channel) for bistatic

R1-2501001 **Summary #3 on ISAC channel modelling** **Moderator (Xiaomi)**
From Thursday session
Agreement
For sensing scenario UMi, UMa, RMa, InH, InF, UMi-AV, UMa-AV, and RMa-AV, the reference TR to generate a UE-UE channel is
- UE-UE link of scenario UMi, UMa, InH, and InF following the option based on TR 38.901 defined in section A.3 of TR 38.858
- TRP-UE link of scenario RMa defined in section 7 of TR 38.901 by setting h_{BS} =1.5m
- FFS: whether to add very low power clusters

Agreement
The reference TR to generate a TRP-UE channel is

TRP	normal UE	UMi, UMa, RMa, InH, InF, UMi-AV, UMa-AV, and RMa-AV • Option 1: TRP-UE link of scenario UMi, UMa, RMa, InH, and InF in section 7 of TR 38.901 Highway and Urban grid • Option 1: P2B link of scenario Highway and Urban grid in section 6 of TR 37.885 HST • Option 1: TRP-UE link of scenario RMa in section 7 of TR 38.901 for FR1 and TRP-UE link of scenario UMa in section 7 of TR 38.901 for FR2

TRP	vehicle UE	Highway and Urban grid Option 1: V2B link of scenario Highway and Urban grid in section 6 of TR 37.885 UMi, UMa, and RMa • Option 1: TRP-UE link of scenario UMi, UMa, and RMa in section 7 of TR 38.901
TRP	aerial UE	UMa-AV, UMi-AV, and RMa-AV • Option 1: - TRP-aerial UE link of scenario UMa-AV, UMi-AV, and RMa-AV in section Annex A and B of TR 36.777 for FR1 - FFS reuse the channel model of scenario UMa-AV, UMi-AV, and RMa-AV of FR1 for FR2

Agreement
For mono-static, the following values of component A, B2 are agreed for UAV of small size
- Component A: -12.81 dBsm
- Component B1: 0 dB (already agreed in RAN1#118bis)
- Component B2:
  - standard deviation: 3.74 dB

Agreement
For mono-static, the following values of component A, B2 are agreed for RCS model 1 of human
- Component A: -1.37 dBsm
- Component B1: 0 dB (already agreed in RAN1#118bis)
- Component B2:
  - standard deviation: 3.94 dB

Working assumption
Absolute delay model (referring to 7.6.9 in TR 38.901 as starting point) is a mandatory feature for both target channel and background channel for ISAC for UMi, UMa, InH, InF
- Related model referring to $\Delta\tau$ values from 7-24GHz study item

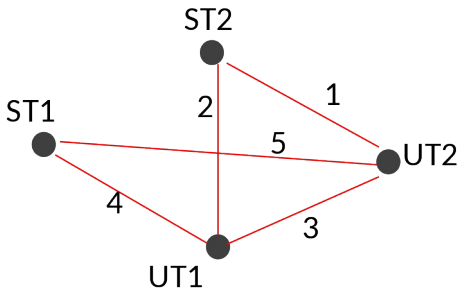
Agreement
When absolute delay model $\Delta\tau$ is configured, it applies to all NLOS clusters in each of Tx-target and target-Rx links and background channel.
- For bistatic sensing: Different values of $\Delta\tau$ are separately generated for the Tx-target link, target-Rx link and the background channel
- For monostatic sensing: the same value of $\Delta\tau$ is used for Tx-target link and target-Rx link, and a different value of $\Delta\tau$ is separately generated for the background channel

Agreement
To generate the LOS ray and NLOS clusters for the multiple scattering points, each scattering point is separately handled as if a different target with single scattering point.

Agreement
- The LOS condition between Tx/Rx and each of the multiple scattering points of a same target are individually generated
- The pathloss between Tx/Rx and each of the multiple scattering points of a same target are individually generated

Agreement
For a target with single/multiple scattering points, the 3D location of each scattering point is defined in the evaluation assumptions.

Agreement
Spatial consistency is needed to model correlation of the following links from ST-UT links and UT-UT links
- Case 5: links between same UT and two nodes X/Y, subjected to correlation distance, i.e., link UT1-X and link UT1-Y, where nodes X/Y can be target or UT
- Case 6: links between same target and two nodes X/Y, subjected to correlation distance, i.e., link target1-X and link target1-Y, where nodes X, Y are different UTs
- Case 7: link X1-Y1 and link X2-Y2, subjected to correlation distance, where X1, X2, Y1, Y2 are 4 different nodes
- FFS: Spatial consistency between multiple scattering points of the same target



Agreement
Correlation type is introduced for large scale parameter, cluster specific parameter and ray specific parameter of ST-UT links and UT-UT links

- Definition of link Correlated: parameters for any two links between STs/UTs are correlated, subjected to correlation distance.

Table 4: Correlation type for links between STs/UTs

Parameters	Correlation type
Delays	link Correlated
Cluster powers	link Correlated
AOA/ZOA/AOD/ZOD offset	link Correlated
AOA/ZOA/AOD/ZOD sign	link Correlated
Random coupling	link Correlated
XPR	link Correlated
Initial random phase	link Correlated
LOS/NLOS states	link Correlated
Blockage (Model A)	All-correlated
O2I penetration loss	All-correlated
Indoor distance	All-correlated
Indoor states	All-correlated

- Note: it is not precluded more parameters for spatial consistency can be discussed and added in the table

Agreement
If a target is modelled with multiple scattering points,

- The number of scattering points of the target is generated in the beginning of the simulation and kept unchanged in the whole simulation
- The number and locations of the scattering points of the target (if it is a vehicle) are common to each pair of sensing Tx/Rx
- RAN1 assumes no ray is scattered from one scattering point to another scattering point of the same target
- RCS values of the multiple scattering points are individually determined

Agreement
To model polarization matrix $CPM_{sp,i} = \begin{bmatrix} \alpha_{i,1} exp\left(j\Phi_{sp,i}^{\theta\theta}\right) & \beta_{i,1} exp\left(j\Phi_{sp,i}^{\theta\phi}\right) \\ \beta_{i,2} exp\left(j\Phi_{sp,i}^{\phi\theta}\right) & \alpha_{i,2} exp\left(j\Phi_{sp,i}^{\phi\phi}\right) \end{bmatrix}$ of a direct/indirect path i of a scattering point of a target

- in Rel-19 study item (e.g., UAV, human, vehicle, AGV), $\alpha_{i,1} = \alpha_{i,2} = 1$, $\beta_{i,1} = \beta_{i,2} = \sqrt{\kappa_{sp,i}^{-1}}$, i.e.,

$$CPM_{sp,i} = \begin{bmatrix} exp\left(j\Phi_{sp,i}^{\theta\theta}\right) & \sqrt{\kappa_{sp,i}^{-1}} exp\left(j\Phi_{sp,i}^{\theta\phi}\right) \\ \sqrt{\kappa_{sp,i}^{-1}} exp\left(j\Phi_{sp,i}^{\phi\theta}\right) & exp\left(j\Phi_{sp,i}^{\phi\phi}\right) \end{bmatrix}$$

  Where,
  - $\kappa_{sp,i}$ is XPR ratio is randomly generated by log-normal distribution per target type
  - The initial random phase $\left\{\Phi_{sp,i}^{\theta\theta}, \Phi_{sp,i}^{\theta\phi}, \Phi_{sp,i}^{\phi\theta}, \Phi_{sp,i}^{\phi\phi}\right\}$ is uniformly distributed within $(-\pi, \pi)$
- FFS: spatial consistency of random phase when a scattering point moves
- FFS: whether the base station rotation procedure in 38.901 can be reused to support rotation of the target

Agreement

- For target channel
  - The power threshold for removing clusters in step 6 in section 7.5, TR 38.901, i.e., -25 dB is reused to generate Tx-target link and target-Rx link
  - The power threshold for path dropping after concatenation is relaxed to [X=-40] dB
- For background channel
  - The power threshold for removing clusters in step 6 in section 7.5, TR 38.901, i.e., [-25 dB] is reused to generated the background channel
    - FFS: whether to add additional very low power clusters
    - FFS: The reference power for removing cluster is the min (max. Tx-target link cluster power, max. target-Rx link cluster power)

Agreement
If EO type-2 is modelled in an indirect path, only specular reflection is modeled for EO type-2
- polarization of the indirect path is product of polarization matrix of the target-Rx link, the target, and the Tx-target link, i.e., $CPM_{tx,sp,rx} = CPM_{sp,rx} \cdot CPM_{sp} \cdot CPM_{tx,sp}$
  - For the specular reflected ray generated by a EO type-2 in the Tx-target link or the target-Rx link (i.e., Tx-EO type-2-target, or target-EO type-2-Rx), $CPM_{tx,sp}$ or $CPM_{sp,rx}$ is the polarization matrix of EO type-2
  - To generate polarization matrix of EO type-2, the procedure in [R1-2409394, R1-2410648] is taken as starting point
    - FFS applicability if the surface of EO type-2 is tilted

Final summary in R1-2501002.

9.8 Study on channel modelling enhancements for 7-24GHz for NR

Please refer to RP-234018 for detailed scope of the WI.

R1-2501549 Session notes for 9.8 (Study on channel modelling enhancements for 7-24GHz for NR) Ad-Hoc Chair (CMCC)
Friday decision: The session notes are endorsed and contents reflected below.

[120-R19-7-24GHz] – Daewon (Intel)
Email discussion on channel modelling for 7-24GHz
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500411 Draft CR for Rel-19 7-24GHz Channel model Intel Corporation, ZTE Corporation

From Friday session
Email discussion on calibration for channel modelling enhancements for 7-24GHz for NR – Daewon (Intel)
- March 2 ~ 5

9.8.1 Channel model validation of TR38.901 for 7-24GHz

R1-2500087 Considerations on the 7-24GHz channel model validation Huawei, HiSilicon
R1-2500122 Discussion on Evaluation of FR3 Channel Modeling InterDigital, Inc.
R1-2500236 Views on channel model validation for 7-24GHz CATT
R1-2500362 Views on channel model validation of TR38.901 for 7-24GHz vivo, BUPT
R1-2500380 Discussion on the channel model validation ZTE Corporation, Sanechips
R1-2500393 Channel model validation of TR38.901 for 7-24 GHz Sharp
R1-2500409 Discussion on channel modeling verification for 7-24 GHz Intel Corporation
R1-2500506 Discussion on validation of channel model Ericsson
R1-2500541 Discussion on channel model validation of TR38.901 for 7-24GHz BUPT, Spark NZ Ltd
R1-2500640 Measurements of the angular spreads in urban and suburban macrocells Vodafone, Ericsson
R1-2500661 Discussion of channel model validation of TR38.901 for 7-24GHz Sony
R1-2500677 Channel model validation for 7-24 GHz Lenovo
R1-2500693 Channel model validation of TR 38901 for 7-24 GHz NVIDIA
R1-2500798 On Validation of Channel Model Apple
R1-2500862 Discussion on channel model validation of TR38.901 for 7 - 24 GHz Samsung
R1-2501048 Intra cluster power profile Keysight Technologies UK Ltd
R1-2501371 Discussion on Validation of the Channel Model in 38901 AT&T (rev of R1-2501083)
R1-2501168 Channel Model Validation of TR38.901 for 7-24 GHz Qualcomm Incorporated
R1-2501213 Discussion on channel model validation for 7-24 GHz NTT DOCOMO, INC.

R1-2501348 Discussion on Channel model validation of TR38.901 for 7-24GHz Nokia

R1-2501427 Summary of issues for Rel-19 7-24 GHz Channel Modeling Validation Moderator (Intel Corporation)

R1-2501428 Summary #1 of discussions for Rel-19 7-24 GHz Channel Modeling Validation Moderator (Intel Corporation)

Conclusion

Based on measurement data provided, RAN1 concludes pathloss parameters in TR38.901 that following scenarios are validated and no updates to TR are made.

- UMi LOS/NLOS

Agreement

Further study whether/how to handle the LOS pathloss convergence regardless of the frequency band beyond the breakpoint distance phenomena.

Conclusion

Based on measurement data provided, RAN1 concludes that following angular spread parameters for 6 – 24 GHz frequency range are validated and no updates to TR are made.

- UMa LOS: ZSD
- UMa NLOS: ZSD

Conclusion

For the following scenarios, there is no consensus to update angular spread models due to lack of consistent and significant observed difference between model and measurements.

- InH LOS : ASD, ASA
- InH NLOS: ASD, ASA
- UMi LOS: ASD
- UMi NLOS: ASD, ZSA

Agreement

- Confirm working assumption (made in RAN1#119) for proposed absolute delay parameters for UMi and UMa scenarios.

Table 7.6.9-1: Parameters for the absolute time of arrival model

Scenarios		UMi	UMa
$lg\Delta\tau = log_{10}(\Delta\tau/1s)$	$\mu_{lg\Delta\tau}$	-7.5	-7.4
	$\sigma_{lg\Delta\tau}$	0.5	0.2
Correlation distance in the horizontal plane [m]		15	50

Agreement

Adopt the following description for suburban scenario:

- In suburban macro-cells base stations are located above the surrounding environment to allow wide area coverage, and mobile stations are outdoors at street level and within commercial and residential buildings. Buildings are typically low residential detached houses with one or two floors, or blocks of apartments/condos or commercial buildings with a few floors. Occasional open areas such as parks or playgrounds between the houses make the environment rather open. Streets do not form urban-like regular strict grid structure.
- FFS: vegetation

Agreement

For suburban scenario, use the BS height and ISD assumption for calibration purposes:

- BS height = 35 m

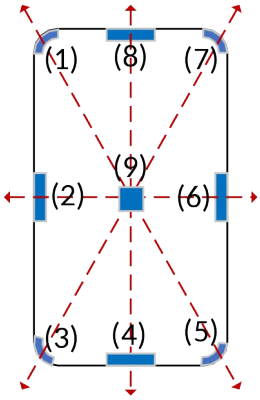
Agreement

For suburban scenario, adopt the following assumption for calibration purposes:

- Minimum BS to UT distance in 2D:
  - 35m

Agreement

Assume enumeration of candidate antenna placement locations as follows:

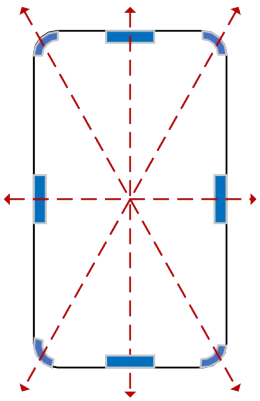


R1-2501429 Summary #2 of discussions for Rel-19 7-24 GHz Channel Modeling Validation Moderator (Intel Corporation)

From Wednesday session

Agreement

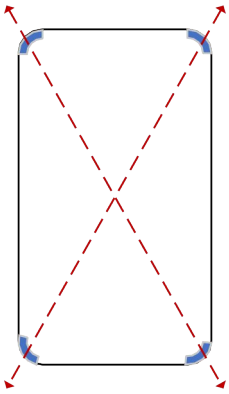
- Amend the previous agreement (from RAN1 #118bis) on handheld UE antenna placement location modeling.
  - The four corners and the centres of each edge are identified as potential locations of the UE antenna. ~~Centre of the device is also identified as potential location of the UE antenna.~~
  - Antennas ~~(except the centre of device antenna)~~ are assumed to be oriented along the direction determined by the vector connecting the centre of the rectangle to the antenna location. ~~centre of device antenna is assumed to be oriented in either forward facing or backward facing the plane of the device.~~



Agreement

For calibration, the antenna placements for 4 elements of handheld devices are defined as follows:

4 Antenna



Agreement

Define candidate antenna locations for CPE devices.

FFS: Device dimensions for CPE antenna modelling.

Agreement

- For cases when a candidate antenna placement location is used for one antenna field pattern (i.e., not used for two distinct antenna polarization field patterns):
  - Reference radiation pattern of the UE antenna model is vertically polarized with all the gain in the theta filed component,
    - $F''_{\theta''}(\theta'',\phi'') = \sqrt{A''(\theta'',\phi'')}$ and $F''_{\phi''}(\theta'',\phi'') = 0$.
- For cases when a candidate antenna placement location is used for two distinct antenna polarization field pattern:
  - Reference radiation pattern of the UE antenna model is FFS,
    - For first antenna field pattern: $F''_{\theta''}(\theta'',\phi'') = FFS$ and $F''_{\phi''}(\theta'',\phi'') = FFS$.
    - For second antenna field pattern: $F''_{\theta''}(\theta'',\phi'') = FFS$ and $F''_{\phi''}(\theta'',\phi'') = FFS$.
- Each polarized field component of the reference radiation pattern $F''_{\theta''}(\theta'',\phi'')$ and $F''_{\phi''}(\theta'',\phi'')$ should be rotated according to the orientation of the each of UE antennae to get $F'_{rx,u,\theta'}(\theta',\phi')$, $F'_{rx,u,\phi'}(\theta',\phi')$ and based on the orientation of the UE in the global coordinate system to get $F_{rx,u,\theta}(\theta,\phi)$ and $F_{rx,u,\phi}(\theta,\phi)$ using the methods already specified in TR 38.901, equation (7.3-3)/(7.1-11).

R1-2501430 Summary #3 of discussions for Rel-19 7-24 GHz Channel Modeling Validation Moderator (Intel Corporation)

From Thursday

Agreement

- Further study and consider update of power angular spectrum (PAS) of the current cluster modeling from equal powered angle offset to unequal power angle offsets.
  - For the update of power angular spectrum (PAS) of the current cluster modeling from equal powered angle offset to unequal power angle offsets. Consider the following options:
    - Option 1) introduce the update as an optional additional modeling component
    - Option 2) update the existing intra-cluster power distribution and angles
  - Consider power adjustments made by "large bandwidth and large antenna array" additional component and review whether measurement observed can be replaced and modeled by large bandwidth and large antenna array" model.
  - The following are examples of changes

Example 1)

Finally add offset angles α_m from Table 7.5-3 to the cluster angles

$$\varphi_{n,m,AOA} = \varphi_{n,AOA} + c_{ASA}\alpha_m, \qquad (7.5\text{-}13)$$

where c_{ASA} is the cluster-wise rms azimuth spread of arrival angles (cluster ASA) in Table 7.5-6.

Table 7.5-3: Ray offset angles within a cluster, given for rms angle spread normalized to 1

Ray number m	~~Basis vector of offset angles $\rightarrow_m$~~	Ray power ratio β_m	Basis vector of offset angles $\rightarrow_m$
1,2	~~± 0.0447~~	0.1193	± 0.1134
3,4	~~± 0.1413~~	0.0933	± 0.3403
5,6	~~± 0.2492~~	0.0729	± 0.5671
7,8	~~± 0.3715~~	0.0570	± 0.7940
9,10	~~± 0.5129~~	0.0446	± 1.0208
11,12	~~± 0.6797~~	0.0348	± 1.2477
13,14	~~± 0.8844~~	0.0272	± 1.4745
15,16	~~± 1.1481~~	0.0213	± 1.7014
17,18	~~± 1.5195~~	0.0166	± 1.9282
19,20	~~± 2.1551~~	0.0130	± 2.1551

<unchanged text omitted>

$$H_{u,s,n}^{NLOS}(t) = \sqrt{\frac{P_n}{M}} \sum_{m=1}^{M} \sqrt{\beta_m} \begin{bmatrix} F_{rx,u,\theta}(\theta_{n,m,ZOA},\phi_{n,m,AOA}) \\ F_{rx,u,\phi}(\theta_{n,m,ZOA},\phi_{n,m,AOA}) \end{bmatrix}^T \begin{bmatrix} \exp(j\Phi_{n,m}^{\theta\theta}) & \sqrt{\kappa_{n,m}^{-1}}\exp(j\Phi_{n,m}^{\theta\phi}) \\ \sqrt{\kappa_{n,m}^{-1}}\exp(j\Phi_{n,m}^{\phi\theta}) & \exp(j\Phi_{n,m}^{\phi\phi}) \end{bmatrix}$$

$$\begin{bmatrix} F_{tx,s,\theta}\left(\theta_{n,m,ZOD},\phi_{n,m,AOD}\right) \\ F_{tx,s,\phi}\left(\theta_{n,m,ZOD},\phi_{n,m,AOD}\right) \end{bmatrix} \exp\left(j2\pi \frac{\hat{r}_{rx,n,m}^{T}\cdot\bar{d}_{rx,u}}{\lambda_0}\right) \exp\left(j2\pi \frac{\hat{r}_{tx,n,m}^{T}\cdot\bar{d}_{tx,s}}{\lambda_0}\right) \exp\left(j2\pi \frac{\hat{r}_{rx,n,m}^{T}\cdot\bar{v}}{\lambda_0} t\right) \quad (7.5\text{-}22)$$

Example 2)

Finally add offset angles α_m from Table 7.5-3 to the cluster angles

$$f_{n,m,AOA} = f_{n,AOA} + c_{ASA} a_m, \quad (7.5\text{-}13)$$

where c_{ASA} is the cluster-wise rms azimuth spread of arrival angles (cluster ASA) in Table 7.5-6.

Table 7.5-3: Ray offset angles within a cluster, given for rms angle spread normalized to 1

Ray number *m*	~~Basis vector of offset angles →$_m$~~	Ray power ratio β_m	Basis vector of offset angles →$_m$
21		$ICP_n/(ICP_n+1)$	0
1,2	~~± 0.0447~~	$0.2029/(ICP_n+1)$	± 0.1134
3,4	~~± 0.1413~~	$0.1679/(ICP_n+1)$	± 0.3403
5,6	~~± 0.2492~~	$0.1390/(ICP_n+1)$	± 0.5671
7,8	~~± 0.3715~~	$0.1151/(ICP_n+1)$	± 0.7940
9,10	~~± 0.5129~~	$0.0952/(ICP_n+1)$	± 1.0208
11,12	~~± 0.6797~~	$0.0788/(ICP_n+1)$	± 1.2477
13,14	~~± 0.8844~~	$0.0653/(ICP_n+1)$	± 1.4745
15,16	~~± 1.1481~~	$0.0540/(ICP_n+1)$	± 1.7014
17,18	~~± 1.5195~~	$0.0447/(ICP_n+1)$	± 1.9282
19,20	~~± 2.1551~~	$0.0370/(ICP_n+1)$	± 2.1551

Value and Distribution of ICPn is FFS

<unchanged text omitted>

$$H_{u,s,n}^{NLOS}(t) = \sqrt{\frac{P_n}{M}} \sum_{m=1}^{M} \sqrt{\beta_{m,n}} \begin{bmatrix} F_{rx,u,\theta}\left(\theta_{n,m,ZOA},\phi_{n,m,AOA}\right) \\ F_{rx,u,\phi}\left(\theta_{n,m,ZOA},\phi_{n,m,AOA}\right) \end{bmatrix}^{T} \begin{bmatrix} \exp\left(j\Phi_{n,m}^{\theta\theta}\right) & \sqrt{\kappa_{n,m}^{-1}}\exp\left(j\Phi_{n,m}^{\theta\phi}\right) \\ \sqrt{\kappa_{n,m}^{-1}}\exp\left(j\Phi_{n,m}^{\phi\theta}\right) & \exp\left(j\Phi_{n,m}^{\phi\phi}\right) \end{bmatrix}$$

$$\begin{bmatrix} F_{tx,s,\theta}\left(\theta_{n,m,ZOD},\phi_{n,m,AOD}\right) \\ F_{tx,s,\phi}\left(\theta_{n,m,ZOD},\phi_{n,m,AOD}\right) \end{bmatrix} \exp\left(j2\pi \frac{\hat{r}_{rx,n,m}^{T}\cdot\bar{d}_{rx,u}}{\lambda_0}\right) \exp\left(j2\pi \frac{\hat{r}_{tx,n,m}^{T}\cdot\bar{d}_{tx,s}}{\lambda_0}\right) \exp\left(j2\pi \frac{\hat{r}_{rx,n,m}^{T}\cdot\bar{v}}{\lambda_0} t\right) \quad (7.5\text{-}22)$$

Example 3)

Finally add offset angles α_m from Table 7.5-3 to the cluster angles

$$f_{n,m,AOA} = f_{n,AOA} + c_{ASA} a_m, \quad (7.5\text{-}13)$$

where c_{ASA} is the cluster-wise rms azimuth spread of arrival angles (cluster ASA) in Table 7.5-6.

Table 7.5-3: Ray offset angles within a cluster, given for rms angle spread normalized to 1

Ray number *m*	~~Basis vector of offset angles →$_m$~~	Ray power ratio $\beta_{m,n}$	Basis vector of offset angles →$_m$
1~~,2~~	~~± 0.0447~~	$ICP_n/(ICP_n+1)$	-0.0901
2		$1/(19(ICP_n+1))$	-0.3611
3,4	~~± 0.1413~~	$1/(19(ICP_n+1))$	±0.2729
5,6	~~± 0.2492~~	$1/(19(ICP_n+1))$	±0.3614
7,8	~~± 0.3715~~	$1/(19(ICP_n+1))$	±0.4626
9,10	~~± 0.5129~~	$1/(19(ICP_n+1))$	±05808
11,12	~~± 0.6797~~	$1/(19(ICP_n+1))$	±0.7227
13,14	~~± 0.8844~~	$1/(19(ICP_n+1))$	±0.9004
15,16	~~± 1.1481~~	$1/(19(ICP_n+1))$	±1.1383
17,18	~~± 1.5195~~	$1/(19(ICP_n+1))$	±1.4995
19,20	~~± 2.1551~~	$1/(19(ICP_n+1))$	±2.2763

$$ICP_n = b \cdot 10^{-a \cdot \frac{\tau_n}{DS}}$$

FFS values of a and b for each deployment scenario

<unchanged text omitted>

$$H_{u,s,n}^{NLOS}(t) = \sqrt{\frac{P_n}{M}} \sum_{m=1}^{M} \sqrt{\beta_{m,n}} \begin{bmatrix} F_{rx,u,\theta}\left(\theta_{n,m,ZOA}, \phi_{n,m,AOA}\right) \\ F_{rx,u,\phi}\left(\theta_{n,m,ZOA}, \phi_{n,m,AOA}\right) \end{bmatrix}^T \begin{bmatrix} \exp\left(j\Phi_{n,m}^{\theta\theta}\right) & \sqrt{\kappa_{n,m}^{-1}} \exp\left(j\Phi_{n,m}^{\theta\phi}\right) \\ \sqrt{\kappa_{n,m}^{-1}} \exp\left(j\Phi_{n,m}^{\phi\theta}\right) & \exp\left(j\Phi_{n,m}^{\phi\phi}\right) \end{bmatrix}$$

$$\begin{bmatrix} F_{tx,s,\theta}\left(\theta_{n,m,ZOD}, \phi_{n,m,AOD}\right) \\ F_{tx,s,\phi}\left(\theta_{n,m,ZOD}, \phi_{n,m,AOD}\right) \end{bmatrix} \exp\left(j2\pi \frac{\hat{r}_{rx,n,m}^T \cdot \bar{d}_{rx,u}}{\lambda_0}\right) \exp\left(j2\pi \frac{\hat{r}_{tx,n,m}^T \cdot \bar{d}_{tx,s}}{\lambda_0}\right) \exp\left(j2\pi \frac{\hat{r}_{rx,n,m}^T \cdot \bar{v}}{\lambda_0} t\right) \quad (7.5\text{-}22)$$

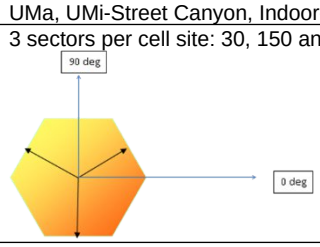
Conclusion

For new scenarios, and if changes are made to existing scenarios, interested companies in RAN1 to perform large scale and full calibration based on the following simulation assumptions. Use the following updates to TR38.901 as starting point for further discussion for calibration.

7.8.1 Large scale calibration

For large scale calibration, fast fading is not modeled. The calibration parameters can be found in Table 7.8-1. The calibration results based on TR 38.900 V14.0.0 can be found in R1-165974.

Table 7.8-1: Simulation assumptions for large scale calibration

Parameter	Values
Scenarios	UMa, UMi-Street Canyon, Indoor-office (open office)
Sectorization	3 sectors per cell site: 30, 150 and 270 degrees 
BS antenna configurations	$M_g = N_g = 1$; (M,N,P) = (10, 1, 1), $d_V = 0.5\lambda$
BS port mapping	The 10 elements are mapped to a single CRS port
BS antenna electrical downtilting	102 degrees for UMa and UMi Street Canyon 110 degrees for indoor
Antenna virtualization	DFT precoding according to TR 36.897 with application of panning and tilting angles
BS Tx power	44 dBm for UMi-Street Canyon, 49 for UMa at 6GHz 35 dBm at 30GHz and 70 GHz for UMa and UMi-Street canyon 24 dBm for Indoor for all carrier frequencies
Bandwidth	20MHz for 6GHz, and 100MHz for 30GHz and 70 GHz
UT antenna configurations	1 element (vertically polarized), Isotropic antenna gain pattern
Handover margin (for calibration)	0dB
UT distribution	Following TR36.873 for UMa and UMi, (3D dropping) uniform dropping for indoor with minimum distance (2D) of 0 m
UT attachment	Based on pathloss considering LOS angle
UT noise figure	9 dB
Fast fading channel	Fast fading channel is not modelled
O2I penetration loss	50% low loss and 50% high loss
Carrier Frequency	6 GHz, 30 GHz, 70GHz
Wrapping method for UMa and UMi	geographical distance based wrapping (mandatory) radio distance (optional)
Metrics	1) Coupling loss – serving cell (based on LOS pathloss)
	2) Geometry (based on LOS pathloss) with and without white noise

Additional calibration parameters can be found in Table 7.8-1A. It is assumed that parameters from Table 7.8-1 is used if unspecified by the additional calibration parameters in Table 7.8-1A.

Table 7.8-1A: Simulation assumptions for large scale calibration

Parameter	Values
Scenarios	UMa, UMi-Street Canyon, SMa
BS antenna electrical downtilting	FFS degrees for SMa
BS antenna configurations	Config 0 for 6GHz: M = 10,N = 1,P=1; $M_g = N_g = 1$; $M_p = 10$, $N_p = 1$; $d_V = 0.5\lambda$ Config 1 for 7GHz : M = 8, N = 16,P = 2; $M_g = 1$, $N_g = 1$; $M_p = 8$, $N_p = 16$; $d_H = d_V = 0.5\lambda$
BS Tx power	FFS dBm for SMa
Bandwidth	20MHz for 6GHz only for UMa and UMi-Street Canyon, 200 MHz for 7 GHz
UT distribution	For SMA, 20% of UT outdoor, 80% of UT indoor. Among indoor UT, 90% of indoor UT are within residential buildings, and 10% of indoor UT in commercial buildings. Indoor UTs are uniformly distributed across all floors for a building type.
O2I penetration loss	For SMa, FFS
Carrier Frequency	6 GHz, 7 GHz

7.8.2 Full calibration

The calibration parameters for full calibration including the fast fading modelling can be found in Table 7.8-2. Unspecified parameters in Table 7.8-2 are the same as those in Table 7.8-1. When P=2, X-pol (+/-45 degree) is used for BS antenna configuration 1 and X-pol (0/+90 degree) is used for UT antenna configuration. The calibration results based on TR 38.900 V14.0.0 can be found in R1-165975.

Table 7.8-2: Simulation assumptions for full calibration

Values
treet Canyon, Indoor-office (open office)
Hz, 60GHz, 70GHz
GHz, and 100MHz for 30GHz, 60 GHz and 70 GHz
UMi-Street Canyon, 49 for UMa at 6GHz 0GHz, 60 GHz and 70 GHz for UMa and UMi-Street canyon ndoor for all carrier frequencies
= 4,N = 4,P = 2, M_g = 1, N_g = 2, d_H = d_V = 0.5λ, $d_{H,g}$ = $d_{V,g}$ = 2.5λ … calibration metrics 1), 2), 3) are calibrated = N_g = 1, M = N = 2, P = 1 … calibration metrics 1), 2), 4) are calibrated
16 elements for each polarization on each panel are mapped to a single CRS port; panning angles of the two subarrays: (0,0) degs; same downtilt a ch antenna element is mapped to one CRS port
M = N = 1, P = 2
R36.873 for UMa and UMi, (3D dropping) pping for indoor with minimum distance (2D) of 0 m
SRP (formula) from CRS port 0
TR36.873
nly distributed on [0,360] degree, $\Omega_{UT,\uparrow}$ = 90 degree, $\Omega_{UT,\leftrightarrow}$ = 0 degree
loss – serving cell
d SIR before receiver without noise
elay Spread and Angle Spread (ASD, ZSD, ASA, ZSA) from the serving cell (according to circular angle spread definition of TR 25.996)
est (1st) PRB singular values (serving cell) at t=0 plotted in 10*log10 scale. llest (2nd) PRB singular values (serving cell) at t=0 plotted in 10*log10 scale. atio between the largest PRB singular value and the smallest PRB singular value (serving cell) at t=0 plotted in 10*log10 scale. RB singular values of a PRB are the eigenvalues of the mean covariance matrix in the PRB.

Additional full calibration parameters can be found in Table 7.8-2A. It is assumed that parameters from Table 7.8-2 is used if unspecified by the additional full calibration parameters in Table 7.8-2A.

Table 7.8-2A: Simulation assumptions for full calibration

Values
-Street Canyon, SMa
GHz
ees for SMa
or 6GHz and only for UMa and UMi: M = 4,N = 4,P = 2, M_g = 1, N_g = 2, d_H = d_V = 0.5λ, $d_{H,g}$ = $d_{V,g}$ = 2.5λ … calibration metrics 1), 2), 3) are calibrated or 6GHz and only for UMa and UMi: M_g = N_g = 1, M = N = 2, P = 1 … calibration metrics 1), 2), 4) are calibrated for 7: M = 8, N = 16,P = 2, M_g = 1, N_g = 1, M_p = 8, N_p = 16,d_H = d_V = 0.5λ … calibration metrics 1), 2), 3), 4) are calibrated for 7: M = 64, N = 16,P = 2, M_g = 1, N_g = 1, M_p = 8, N_p = 16,d_H = d_V = 0.5λ … calibration metrics 1), 2), 3), 4) are calibrated
for SMa
6GHz and only for UMa and UMi-Street Canyon, for 7 GHz
outdoor, 80% of UT indoor. Among indoor UT, 90% of indoor UT are within residential buildings, and 10% of indoor UT in commercial buildings. Inc a building type.
or 6GHz and only for UMa and UMi: M_g = N_g = 1, M = N = 1, P = 2 or 7GHz: reference 4 antenna port with single polarization for calibration based on handheld device antenna model described in Clause 7.3 or 7 GHz: M_g = N_g = 1, M = 2, N = 2, P = 2
or 6GHz: FFS and C for 7 GHz: Based on [directional antenna] for UT described in Clause 7.3.
FFS

7.8.3 Calibration of additional features

The calibration parameters for the calibration of oxygen absorption, large bandwidth and large antenna array, spatial consistency, ~~and~~ blockage, near field channel modeling, and spatial non-stationarity can be respectively found in Table 7.8-3, 7.8-4, 7.8-5, ~~and~~ 7.8-6, 7.8-7, and 7.8-8. Unspecified parameters in these tables are the same as those in Tables 7.8-1 and 7.8-2. When P=2, X-pol (+/-45 degree) is used for BS antenna configuration 1 and X-pol (0/+90 degree) is used for UT antenna configuration. The calibration results based on TR 38.900 V14.0.0 can be found in R1-1700990.

Table 7.8-7: Simulation assumptions for calibration for near field channel modeling

Parameter	Values
Scenarios	UMi-street Canyon, [Indoor-office (open office)]
Carrier Frequency	[6 or 7 GHz]
Bandwidth	20 MHz
BS antenna height	UMi: 10m

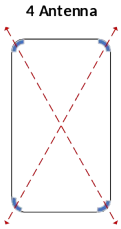
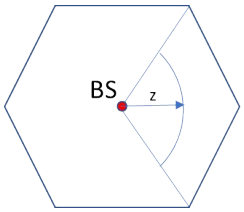
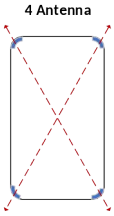
		[Indoor-office: 3m]
BS antenna downtilt		102 degrees for UMi Street Canyon [110 degrees for indoor]
BS Tx power		44 dBm for UMi-Street Canyon [24 dBm for Indoor for all carrier frequencies]
BS antenna configurations		For UMi scenario: 1) Config1: (M, N, P,Mg, Ng,MP,NP) = (24,16,2,1,1, 8,16), dh = 0.5λ,dv = 0.7λ; 2) Config2: (M, N, P,Mg, Ng, MP,NP) = (24,32,2,1,1,8,32), dh = 0.5λ,dv = 0.7λ; [For Indoor-office scenario:FFS]
UE Location	Outdoor/indoor	UMi and UMa: 100% outdoor Indoor-office: 100% indoor
	LOS/NLOS	100% LOS
	UE antenna height	UMi and UMa: 1.5m Indoor-office: 1m
UT antenna configurations		• Config 1: $M_g = N_g = 1$, M = 2, N = 1, P = 2 • Config 2: $M_g = N_g = 1$, M = 2, N = 2, P = 2 • Config 3: 4 antenna ports with single polarization based on the following antenna placements  • Config 4: $M_g = N_g = 1$, M = 4, N = 2, P = 2
Calibration method		Drop multiple users in the multiple cells at the fixed horizontal distance Z from the BS, and collect the metric per distance;  • Note: the value Z can be (1,2,3,4,5,10)* minimum distance of BS and UT o For UMi, the minimum distance is 10 m (ref. Table 6-1: Scenarios in TR36.873) Component of NF channel to be considered: • Phase of Direct path; • Phase of Non-direct path (optionally);
Metric		1) CDF of the ratio between the 2nd,3rd,4th,...,xth singular value and the 1st singular value (serving cell) at t=0 plotted in 10*log10 scale with/without SNS at UE side Note: The value of x are the minimum value of (number of BS antenna ports, number of UE antenna ports) Note: The singular value are calculated in the M PRBs, e.g., M=4;

Table 7.8-7: Simulation assumptions for calibration for spatial non-stationarity

Parameter	Values
Scenarios	UMi-street Canyon
Carrier Frequency	[6 or 7 GHz]
Bandwidth	20 MHz
BS Antenna height	UMi: 10m
BS Tx power	44 dBm for UMi-Street Canyon
BS Ant Config.	M = 4,N = 4,P = 2, $M_g = 1$, $N_g = 2$, $d_H = d_V = 0.5\lambda$, $d_{H,g} = d_{V,g} = 2.5\lambda$
BS port mapping	Config 1: all 16 elements for each polarization on each panel are mapped to a single port; panning angles of the two subarrays: (0,0) degs; same downtilt angles as used for the large-scale calibrations Config 2: each antenna element is mapped to one port
UT Ant Config.	4 antenna ports with single polarization based on the antenna placements for 4 elements of handheld devices 
UT distribution	Following TR36.873 for UMi, (3D dropping)
Metric	1) CDF of coupling loss (serving cell) received per UE port with/without SNS at UE side

		2) CDF of the ratio between the 2nd,3rd,4th,...,xth singular value and the 1st singular value (serving cell) at t=0 plotted in 10*log10 scale with/without SNS at UE side Note: The value of x are the minimum value of (number of BS antenna ports, number of UE antenna ports) Note: The singular value are calculated in the M PRBs, e.g., M=4;	

Note: Email discussion on the simulation assumption for calibration until Mar.5th .

Conclusion

To check and review the following results and measurement data provided in RAN1 #120 for further discussion in next RAN1 meeting. R1-2501426 contains the list of data sources for the results and measurements provided since RAN1 #116-bis until RAN1 #120.

R1-2501426 Data source descriptions for 7 – 24 GHz SI Moderator (Intel Corporation)

Agreement

- Adopt the following absolute delay parameters for InH scenarios.
  - Down-select between option 1, 2 and 3.

Table 7.6.9-1: Parameters for the absolute time of arrival model

Scenarios		InH
$lg\Delta\tau = log_{10}(\Delta\tau/1s)$	$\mu_{lg\Delta\tau}$	Option 1) -7.9 Option 2) -8.6 Option 3) for 40% of the NLOS cluster, Δτ=0; for other 60% of NLOS cluster $\mu_{lg\Delta\tau}$ = -8.9
	$\sigma_{lg\Delta\tau}$	Option 1) 0.3 Option 2) 0.1 Option 3) for 40% of the NLOS cluster with Δτ=0, $\sigma_{lg\Delta\tau}$ is not applicable; for other 60% of the NLOS cluster $\sigma_{lg\Delta\tau}$ = 0.7
Correlation distance in the horizontal plane [m]		10

Working Assumption

- Intermediate angle values of the CDL channel is computed by
  - $\phi_{n,intermediate} = s * \text{WrapTo180}\left(\phi_{n,model} - \mu_{\phi,model}\right)$,
  - where *s* is a scale factor chosen to change the distribution of the angles. Table X shows required scale factor for typical desired angular spread values for AOD, AOD, ZOA, and ZOD.

Table X: Scale factor values for each CDL model

CDL Type	Desired AOD Spread (°)	Scale Factor (AOD)	Desired AOA Spread (°)	Scale Factor (AOA)	Desired ZOA Spread (°)	Scale Factor (ZOA)	Desired ZOD Spread (°)	Scale Factor (ZOD)
CDL-A	5	0.068	30	0.353	5	0.240	1	0.035
	10	0.136	45	0.527	10	0.480	3	0.106
	15	0.204	60	9.525	15	0.723	5	0.176
	25	0.341						
CDL-B	5	0.124	30	0.542	5	0.652	1	0.194
	10	0.248	45	0.808	10	1.302	3	0.582
	15	0.371	60	1.071	15	1.948	5	0.971
	25	0.617						
CDL-C	5	0.128	30	0.431	5	0.648	1	0.364
	10	0.257	45	0.645	10	1.297	3	1.093
	15	0.386	60	7.749	15	1.950	5	1.822
	25	0.651						
CDL-D	5	0.323	30	9.869	5	4.327	1	0.448
	10	0.665	45	N/A	10	8.887	3	1.347
	15	1.059	60	N/A	15	14.034	5	2.258
	25	9.354						
CDL-E	5	0.395	30	9.743	5	6.919	1	0.971
	10	0.801	45	N/A	10	14.838	3	2.918
	15	1.233	60	N/A	15	27.285	5	4.877
	25	7.780						

Note: the values are computed by formula option 2A defined in RAN1#119.

Agreement

- Further study updates to the mean angle calculation in Annex used for calculation of the mean angle values for CDL-D/E channel model
  - Potential update of the power weighted mean angle is

$$\mu_\phi = \arg\left\{ \exp\left(j\phi_{LOS}\right)P_{LOS} + \sum_{n=1}^{N}\sum_{m=1}^{M} \exp\left(j\phi_{n,m}\right)P_{n,m} \right\}$$

Agreement

- Discuss further on the following suggested text changes:

< Unchanged text omitted >

7.7.5.1 CDL extension: Scaling of angles

The angle values of CDL models are fixed, which is not very suitable for MIMO simulations for several reasons; The PMI statistics can become biased, and a fixed precoder may perform better than open-loop and on par with closed-loop or reciprocity beamforming. Furthermore, a CDL only represents a single channel realization. The predefined angle values in the CDL models can be generalized by introducing angular translation and scaling. By translation, mean angle can be changed to $\mu_{\phi,desired}$ and angular spread can be changed by scaling. The translated and scaled ray angles can be obtained according to the following equation:

$$\phi_{n,scaled} = \frac{AS_{desired}}{AS_{model}}\left(\phi_{n,model} - \mu_{\phi,model}\right) + \mu_{\phi,desired} \qquad (7.7\text{-}5)$$

in which:

$\phi_{n,model}$ is the tabulated CDL ~~ray~~ cluster angle

AS_{model} is the rms angular spread of the tabulated CDL ~~including the offset ray angles~~, calculated using the angular spread definition in Annex A

$\mu_{\phi,model}$ is the mean angle of the tabulated CDL, calculated using the definition in Annex A

$\mu_{\phi,desired}$ is the desired mean angle

$AS_{desired}$ is the desired rms angular spread

$\phi_{n,scaled}$ is the resulting scaled ~~ray~~ cluster angle.

The angular scaling is applied on ~~the ray angles including offsets from~~ the tabulated cluster angles. Typical angular spreads for different scenarios can be obtained from the system-level model.

Example scaling values are:

- AOD spread (ASD) for each CDL model: {5, 10, 15, 25} degrees.
- AOA spread (ASA) for each CDL model: {30, 45, 60} degrees.
- ZOA spread (ZSA) for each CDL model: {5, 10, 15} degrees.
- ZOD spread (ZSD) for each CDL model: {1, 3, 5} degrees.

The angular scaling and translation can be applied to some or all of the azimuth and zenith angles of departure and arrival.

Note: The azimuth angles of $\left(\phi_{n,model} - \mu_{\phi,model}\right)$ should be wrapped around to be within [-180, 180] degrees.

Note: The azimuth angles may need to be wrapped around to be within [0, 360] degrees, while the zenith angles may need to be clipped to be within [0, 180] degrees.

Note: The resulting scaled ~~ray~~ cluster angle may or may not achieve the desired angle mean and the desired rms angle spread.

R1-2501431 Summary #4 of discussions for Rel-19 7-24 GHz Channel Modeling Validation Moderator (Intel Corporation)

From Friday session

Agreement

- RAN1 has identified that standard multi-panel glass penetration loss model may require updates at least for 6-24 GHz frequency range. Further discuss changes.
- For standard multi-panel glass penetration loss model, study the following model changes:
  - Option 1) thickness dependent model without resonation effect
    - $L_{glass} = \alpha\cdot(2 + 0.2f)$
    - where $\alpha = \left(\frac{\frac{T_{glass}}{3\text{ cm}} + 0.4}{1.4}\right)$
    - Default reference thickness for standard multi-panel glass is 3 cm
    - If model without resonation effect is selected, add note in TR that while glass material measurement may showcase resonation effect as a function of frequency and thickness, the model abstracts away the effect.
  - Option 2) thickness dependent model with resonation effect
    - $L_{glass} = 0.08\cdot\alpha\cdot f + \sin(1.05\cdot\alpha\cdot f - 7.8)/(\alpha + 0.2) - 0.3\alpha + 1.8$

        - For $\alpha = 3$, $L_{glass} = 0.9 + 0.24 \cdot f + \sin(3.15 \cdot f - 7.8)/3.2$
    - where $\alpha = \left(\frac{T_{glass}}{1\ cm}\right)$
    - Default reference thickness for standard multi-panel glass is 3 cm
  - Option 3) thickness dependent model without resonation effect and with air interface loss
    - $L_{glass} = a + b \cdot \alpha \cdot f$, $\alpha = \left(\frac{T_{glass}}{X\ cm}\right)$
    - Where X is the default thickness of the glass.
    - Parameters a, b, and X are FFS.
  - Note: for all options, consider the impact to overall building exterior penetration loss from the potential changes to glass penetration loss.

Agreement

- RAN1 has identified that IRR glass penetration loss model may require updates at least for 6-24 GHz frequency range. Further discuss changes For IRR glass penetration loss model, study the following model changes:
  - Option 1) coating dependent model without resonation effect
    - $L_{IRRglass} = 0.11f - 0.9\alpha^2 + 8.3\alpha + 18$
      - For $\alpha = 1$, $L_{IRRglass} = 25.4 + 0.11f$
    - where $\alpha = 1$ for single silver-coating; $\alpha = 2$ for double silver-coating; $\alpha = 3$ for triple silver-coating.
    - Default reference coating for standard multi-panel glass is single coating, i.e. $\alpha = 1$
    - Default reference thickness for IRR glass is 3 cm
    - If model without resonation effect is selected, add note in TR that while glass material measurement may showcase resonation effect as a function of frequency and thickness, the model abstracts away the effect.
  - Option 2) coating dependent model with resonation effect
    - $L_{IRRglass} = 0.11f - 3\sin(0.5f + 1.36) - 0.9\alpha^2 + 8.3\alpha + 18$
      - For $\alpha = 1$, $L_{IRRglass} = 25.4 + 0.11f - 3\sin(0.5f + 1.36)$
    - where $\alpha = 1$ for single silver-coating; $\alpha = 2$ for double silver-coating; $\alpha = 3$ for triple silver-coating.
    - Default reference coating for IRR glass is single coating, i.e. $\alpha = 1$
  - Option 3) thickness dependent model without resonation effect and with air interface loss
    - $L_{IRRglass} = a + b \cdot \alpha \cdot f$, $\alpha = \left(\frac{T_{IRRglass}}{X\ cm}\right)$
    - Where X is the default thickness of the glass.
    - Parameters a, b, and X are FFS.
  - Note: for all options, consider the impact to overall building exterior penetration loss from the potential changes to glass penetration loss.
  - FFS: whether and how to incorporate thickness dependent affect for the IRR model. In this context, incorporation of thickness impact is considered a minor refinement of option 1 or 2 to better match the measurement data.

Agreement

- Discuss further on potential update of concrete penetration model based on measurement provided by companies.
  - For updates of the concrete penetration loss model, further study between the following model changes:
    - Option 1) model with no separate air to interface insertion loss
      - $L_{concrete} = \alpha(5 + 4f)$
      - $\alpha = \left(\frac{T_{concrete}}{23\ cm}\right)$
      - Default reference thickness for concrete is 23 cm
    - Option 2) model with separate air to interface insertion loss
      - $L_{concrete} = 1.4 + 4.6\alpha f$
      - $\alpha = \left(\frac{T_{concrete}}{23cm}\right)$
      - Default reference thickness for concrete is 23 cm

    Note: for all options, consider the impact to overall building exterior penetration loss from the potential changes to concrete penetration loss.

Agreement

- Discuss further on potential introduction of new material mode "cinder block" penetration model based on measurement provided by companies for concrete based walls.

Agreement

- Discuss further on potential update of wood penetration loss model based on measurement provided by companies. The following is an example of a potential update:
  - Option 1) thickness dependent model without air interface loss
    - $L_{wood} = \alpha(4.85 + 0.12f)$
    - $\alpha = \left(\frac{T_{wood}}{6cm}\right)$
    - Default reference thickness for wood is 6 cm
  - Option 2) thickness dependent model with air interface loss
    - $L_{wood} = a + b \cdot \alpha \cdot f$
    - $\alpha = \left(\frac{T_{wood}}{X\ cm}\right)$
    - Where X is the default thickness of the wood.
    - Parameters a, b, and X are FFS.

Agreement

Discuss further on whether to introduce additional model to handle LOS probability impact from vegetation for SMa and details of the additional modeling component.

For suburban scenario, the down-select among the following LOS probability

- Option 1)

$$Pr_{LOS} = exp(-\frac{d_{2D-out} - 10}{exp(0.02044h_{BS} + 5.746)})$$

- Option 2)

$$P_{\mathrm{LOS}} = \begin{cases} 1, & d_{2\mathrm{D}} \le 10 \\ \exp\left(-\frac{d_{2\mathrm{D}}}{k_{\mathrm{commercial}}}\right)\exp\left(-\frac{d_{2\mathrm{D}}}{k_{\mathrm{residential}}}\right)\exp\left(-\frac{d_{2\mathrm{D}}}{k_{\mathrm{vegetation}}}\right), & d_{2\mathrm{D}} > 10 \end{cases}$$

$$\mathrm{k}_{\mathrm{commercial}} = -\frac{d_{\mathrm{commercial}}}{\ln\left(1 - r_{\mathrm{commericial}}\right)} \cdot \frac{h_{\mathrm{BS}} - h_{\mathrm{UT}}}{h_{\mathrm{commercial}} - h_{\mathrm{UT}}}$$

$$k_{\mathrm{residential}} = -\frac{d_{\mathrm{residential}}}{\ln\left(1 - r_{\mathrm{residential}}\right)} \cdot \frac{h_{\mathrm{BS}} - h_{\mathrm{UT}}}{h_{\mathrm{residential}} - h_{\mathrm{UT}}}$$

$$k_{\mathrm{vegetation}} = -\frac{d_{\mathrm{residential}}}{\ln\left(1 - r_{\mathrm{residential}}\right)} \cdot \frac{h_{\mathrm{BS}} - h_{\mathrm{UT}}}{h_{\mathrm{residential}} - h_{\mathrm{UT}}}$$

$$d_{\mathrm{commercial,residential,vegetation}} = \text{clutter bin size}$$

$$r_{\mathrm{commercial,residential,vegetation}} = \text{clutter density}$$

$$h_{\mathrm{commercial,residential,vegetation}} = \text{clutter height}$$

Parameter	
$d_{\mathrm{commercial,residential,vegetation}}$	30 m
$h_{\mathrm{commercial}}$	20 m
$h_{\mathrm{residential}}$	8 m
$h_{\mathrm{vegetation}}$	15 m
$r_{\mathrm{commercial}}$	2%
$r_{\mathrm{residential}}$	18%
$r_{\mathrm{vegetation}}$	0% (no vegetation), 10% (sparse vegetation), or 20% (dense vegetation)

- Option 3)

$$\mathrm{P}_{LOS} = \begin{cases} 1, & d_{2D-out} \le 10 \\ \exp\left(-\frac{d_{2D-out} - 10}{200}\right), & d_{2D-out} > 10 \end{cases}$$

Working Assumption

- For parameters with multiple options, choose from one of the options in meeting RAN1#120-bis.
- For information: O2I are barrowed from TR38.901 UMa O2I, LOS and NLOS parameters are barrowed from ITU M.2135 SMa if values are available. For coloured values listed, values are based on measurement data from companies.

Scenarios	SMa	*Reference – ITU M.2135 SMa*	*TR38.901 UMa*

		LOS	NLOS	O2I	LOS	NLOS	O2I	LOS	NLOS	O2I
Delay spread (DS) lgDS=$\log_{10}$(DS/1s)	μ_{lgDS}	–7.23	Option 1) -7.12 Option 2) -7.2 Option 3) -6.98	-6.62	–7.23	–7.12	N/A	-6.955 - 0.0963 $\log_{10}(f_c)$	-6.28 - 0.204 $\log_{10}(f_c)$	-6.62
	σ_{lgDS}	0.38	Option 1) 0.33 Option 2) 0.58 Option 3) 0.74	0.32	0.38	0.33	N/A	0.66	0.39	0.32
AOD spread (ASD) lgASD=$\log_{10}$(ASD/1)	μ_{lgASD}	0.55	0.55	0.55	0.78	0.90	N/A	1.06 + 0.1114 $\log_{10}(f_c)$	1.5 - 0.1144 $\log_{10}(f_c)$	1.25
	σ_{lgASD}	0.25	0.25	0.25	0.12	0.36	N/A	0.28	0.28	0.42
AOA spread (ASA) lgASA=$\log_{10}$(ASA/1)	μ_{lgASA}	Option 1) 1.48 Option 2) 1.75	Option1) 1.65 Option 2) 1.74 Option 3) 1.27	1.76	1.48	1.65	N/A	1.81	2.08 - 0.27 $\log_{10}(f_c)$	1.76
	σ_{lgASA}	Option 1) 0.20 Option 2) 0.04	Option 1) 0.25 Option 2) 0.24 Option 3) 0.86	0.16	0.20	0.25	N/A	0.20	0.11	0.16
ZOA spread (ZSA) lgZSA=$\log_{10}$(ZSA/1)	μ_{lgZSA}	1.31	Option 1) 1.32 Option 2) -0.388	1.01	N/A	N/A	N/A	0.95	-0.3236 $\log_{10}(f_c)$ + 1.512	1.01
	σ_{lgZSA}	0.02	Option 1) 0.08 Option 2) 1.17	0.43	N/A	N/A	N/A	0.16	0.16	0.43
Shadow fading (*SF*) [dB]	σ_{SF}	4	8	7	4	8	N/A	See Table 7.4.1-1	See Table 7.4.1-1	7
K-factor (*K*) [dB]	μ_K	9	N/A	N/A	9	N/A	N/A	9	N/A	N/A
	σ_K	7	N/A	N/A	7	N/A	N/A	3.5	N/A	N/A
Cross-Correlations	*ASD* vs *DS*	0	0	0.4	0	0	N/A	0.4	0.4	0.4
	ASA vs *DS*	0.8	Option 1) 0.7 Option 2) 0.5	0.4	0.8	0.7	N/A	0.8	0.6	0.4
	ASA vs *SF*	–0.5	Option 1) 0 Option 2) -0.25	0	–0.5	0	N/A	-0.5	0	0
	ASD vs *SF*	–0.5	-0.4	0.2	–0.5	–0.4	N/A	-0.5	-0.6	0.2
	DS vs *SF*	–0.6	Option 1) -0.4 Option 2) -0.13	-0.5	–0.6	–0.4	N/A	-0.4	-0.4	-0.5
	ASD vs *ASA*	0	0	0	0	0	N/A	0	0.4	0
	ASD vs K	0	N/A	N/A	0	N/A	N/A	0	N/A	N/A
	ASA vs K	0	N/A	N/A	0	N/A	N/A	-0.2	N/A	N/A
	DS vs K	0	N/A	N/A	0	N/A	N/A	-0.4	N/A	N/A
	SF vs K	0	N/A	N/A	0	N/A	N/A	0	N/A	N/A
Cross-Correlations [1)]	*ZSD* vs *SF*	FFS	FFS	0	N/A	N/A	N/A	0	0	0
	ZSA vs *SF*	FFS	-0.47	0	N/A	N/A	N/A	-0.8	-0.4	0
	ZSD vs *K*	FFS	N/A	N/A	N/A	N/A	N/A	0	N/A	N/A
	ZSA vs *K*	FFS	N/A	N/A	N/A	N/A	N/A	0	N/A	N/A
	ZSD vs *DS*	FFS	-0.06	-0.6	N/A	N/A	N/A	-0.2	-0.5	-0.6
	ZSA vs *DS*	FFS	FFS	-0.2	N/A	N/A	N/A	0	0	-0.2
	ZSD vs *ASD*	0.5	0.5	0.5	N/A	N/A	N/A	0.5	0.5	-0.2
	ZSA vs *ASD*	FFS	FFS	0	N/A	N/A	N/A	0	-0.1	0
	ZSD vs *ASA*	FFS	FFS	0	N/A	N/A	N/A	-0.3	0	0
	ZSA vs *ASA*	FFS	0.36	0.5	N/A	N/A	N/A	0.4	0	0.5
	ZSD vs *ZSA*	FFS	FFS	0.5	N/A	N/A	N/A	0	0	0.5
Delay scaling parameter r_τ		2.4	1.5	2.2	2.4	1.5	N/A	2.5	2.3	2.2
XPR [dB]	μ_{XPR}	8	4	9	8	4	N/A	8	7	9
	σ_{XPR}	N/A	N/A	5	N/A	N/A	N/A	4	3	5
Number of clusters N		15	14	12	15	14	N/A	12	20	12
Number of rays per cluster M		20	20	20	20	20	N/A	20	20	20
Cluster *DS* (c_{DS}) in [ns]		N/A	N/A	11	N/A	N/A	N/A	max(0.25, 6.5622 -3.4084 $\log_{10}(f_c)$)	max(0.25, 6.5622 -3.4084 $\log_{10}(f_c)$)	11
Cluster *ASD* (c_{ASD}) in [deg]		Option 1) 5 Option 2) 1.5	Option 1) 2 Option 2) 1.5	1.5	5	2	N/A	5	2	5
Cluster *ASA* (c_{ASA}) in [deg]		5	10	8	5	10	N/A	11	15	8
Cluster *ZSA* (c_{ZSA}) in [deg]		N/A	N/A	3	N/A	N/A	N/A	7	7	3
Per cluster shadowing std ξ [dB]		3	3	4	3	3	N/A	3	3	4
Correlation distance in the horizontal plane [m]	*DS*	6	40	10	6	40	N/A	30	40	10
	ASD	15	30	11	15	30	N/A	18	50	11
	ASA	20	30	17	20	30	N/A	15	50	17
	SF	40	50	7	40	50	N/A	37	50	7
	K	10	N/A	N/A	10	N/A	N/A	12	N/A	N/A
	ZSA	N/A	N/A	25	N/A	N/A	N/A	15	50	25
	ZSD	N/A	N/A	25	N/A	N/A	N/A	15	50	25

Scenarios		SMa LOS	SMa NLOS	SMa O2I	For reference - UMa LOS	For reference - UMa NLOS	For reference – RMa O2I
ZOD spread (ZSD) lgZSD=$\log_{10}$(ZSD/1°)	μ_{lgZSD}	Option 1) 0.14 Option 2) 0.55	Option 1) 0.14 Option 2) 0.55	0.55	max[-0.5, -2.1(d_{2D}/1000) -0.01 (h_{UT} - 1.5)+0.75]	max[-0.5, -2.1(d_{2D}/1000) -0.01(h_{UT} - 1.5)+0.9]	max[-1, -0.19(d_{2D}/1000) -0.01(h_{UT} - 1.5) + 0.28]

	$\sigma_{\lg ZSD}$	Option 1) 0.16 Option 2) 0.25	Option 1) 0.16 Option 2) 0.25	0.25	0.40	0.49	0.30
ZOD offset	$\mu_{offset,ZOD}$	FFS	FFS	FFS	0	$e(f_c)$-10^{$a(f_c)$ $\log_{10}$(max($b(f_c)$, d_{2D})) +$c(f_c)$ -0.07(h_{UT}-1.5)}	arctan((35 - 3.5)/d_{2D}) -arctan((35 - 1.5)/d_{2D})
Note: For NLOS ZOD offset: $a(f_c)$ = FFS; $b(f_c)$ = FFS; $c(f_c)$ = FFS; $e(f_c)$ = FFS.					Note: For NLOS ZOD offset: $a(f_c) = 0.208\log_{10}(f_c) - 0.782$; $b(f_c) = 25$; $c(f_c) = -0.13\log_{10}(f_c)+2.03$; $e(f_c) = 7.66\log_{10}(f_c)-5.96$.		

Agreement

Further study on introduction of a loss outdoor-to-indoor (O2I) building penetration loss model for SMa:

- $PL_{tw} = 5 - 10\log_{10}\left(0.3 \cdot 10^{\frac{-L_{glass}}{10}} + 0.7 \cdot 10^{\frac{-L_{wood}}{10}}\right)$
- Consider the impact to overall building exterior penetration loss

Working Assumption

Scenario	LOS /NLOS	Pathloss [dB], f_c is in GHz and d is in meters	Shadow fading std [dB]	default values	Applicability range
SMa	LOS	$PL_1 = 20\log_{10}(40\pi d f_c/3) + \min(0.03h^{1.72}, 10)\log_{10}(d) + \min(0.044h^{1.72}, 14.77) + 0.002\log_{10}(h)d$, h = 10 m,	$\sigma = 4$ $\sigma = 6$	10 m < d< d_{BP} d_{BP} < d < 5000 m	$0.8 \le f_c \le 37$ GHz 25 m < h_{BS} < 35 m 1 m < h_{UT} < 14 m
	NLOS	$PL = 161.04 - 7.1\log_{10}(W) + 7.5\log_{10}(h) - (18.7 - 3.7\log_{10}(h_{BS}))\log_{10}(h_{UT}) + (43.42 - 3.1\log_{10}(h_{BS}))(\log_{10}(d) - 3) + 20\log_{10}(f_c) - 3.2(\log_{10}(11.75h_{UT}))^2 - 4.97$, h = 10 m, W = 10 m FFS potential changes: $PL = 161.04 - 7.1\log_{10}(W) + 7.5\log_{10}(h) - (18.7 - 3.7\log_{10}(h_{BS}))\log_{10}(h_{UT}) + (43.42 - 3.1\log_{10}(h_{BS}))(\log_{10}(d) - 3) + 20\log_{10}(f_c) + 10 \cdot \log_{10}(\min(d, 10)) - 10 \cdot \log_{10}(h_{UT}) - 3.2(\log_{10}(11.75h_{UT}))^2 - 4.97$, h = 10 m, W = 10 m	$\sigma = 8$	10 m < d_{BP} < 5000 m	

Corresponding Working Assumption (made in RAN1#119) is dropped.

Agreement

For UE antenna modeling of handheld devices, at least support directional antenna radiation pattern for calibration purposes.

- Further study the following directional radiation pattern parameters

For directional radiation pattern		
Vertical Radiation Pattern	**Horizontal**	**Max Gain**
$\theta_{3dB} = 90°$ SLA_V = 25 dB	$\phi_{3dB} = 120°$~$180°$ A_m = 25 dB	
$\theta_{3dB} = 115°$ SLA_V = 25 dB	$\phi_{3dB} = 85°$ A_m = 25 dB	
$\theta_{3dB} = 131°$ SLA_V = 20 dB	$\phi_{3dB} = 115°$ A_m = 20	4.5 ~ 6.5 dBi
$\theta_{3dB} = 90°$ SLA_V = 25 dB	$\phi_{3dB} = 90°$ A_m = 25	≤ 5 dBi
$\theta_{3dB} = 90$~$120°$ SLA_V = 15~25 dB	$\phi_{3dB} = 90°$~$120°$ A_m = 15~25 dB	0 ~ 5 dBi
$\theta_{3dB} = 131°$ SLA_V = 15 dB	$\phi_{3dB} = 131°$ A_m = 15 dB	5 dBi

Agreement

RAN1 has identified that angular spread for following scenarios require necessary updates at least for 6-24 GHz frequency range. Further discuss necessary changes. Note: For UMa ASD, weak dependency to frequency was observed and the potential necessary changes may include changes to values across wide frequency ranges.

- UMi LOS ASA
- UMi NLOS ASA
- UMa LOS ASD

- UMa LOS ASA
- UMa LOS ASD
- UMa NLOS ASD
- UMa NLOS ASA
- UMa NLOS ZSA

Agreement

Conclude in RAN1 #120bis among the following:

- Alt 1)
  - RAN1 has identified that delay spread for UMi LOS/NLOS and UMa LOS/NLOS scenario is smaller compared to delay spread in the TR at least for 6-24 GHz frequency range. Further discuss, necessary changes for UMi LOS/NLOS and UMa LOS/NLOS
  - Proponent companies to provide detailed information on potentially necessary changes to DS for scenarios in question.
    - Data samples for discussion on necessary changes for UMi LOS

Delay spread For UMi LOS	TR38.901 @ 6.5 GHz	Measured @ 6.5 GHz	TR38.901 @ 10 GHz	Measured @ 10 GHz	TR38.901 @ 13 GHz	Measured @ 13 GHz	TR38.901 @ 13.5 GHz	Measured @ 13.5 GHz
μ_{DS} (log_10 〖DS/1s〗)	-7.35	-8.16	-7.39	-7.47	-7.42	-7.85	-7.41	-8.19
σ_{DS}	0.38	0.83	0.38	0.28	0.38	0.54	0.38	0.91

- Alt 2)
  - For the following scenarios, there is no consensus to update delay spread models due to lack of consistent and significant observed difference between model and measurements.
    - UMi LOS/NLOS and UMa LOS/NLOS

Agreement

Conclude in RAN1 #120bis among the following:

- Alt 1)
  - RAN1 has identified that number of clusters spread for [InH LOS/NLOS, UMi LOS/NLOS, and UMa LOS/NLOS] scenario is smaller compared to delay spread in the TR at least for 6-24 GHz frequency range. Further discuss, necessary changes for [InH LOS/NLOS, UMi LOS/NLOS, and UMa LOS/NLOS].
    - RAN1 to determine which scenarios among [InH LOS/NLOS, UMi LOS/NLOS, and UMa LOS/NLOS] requires necessary changes.
    - Data samples for discussion on necessary changes

Scenario	InH @10 GHz				Umi @10 GHz				UMa @6.5 GHz				UMa @13 GHz				UMa @15GHz			
	TR 38.901		Measureme		TR 38.901		Measureme		TR 38.901		Measureme		TR 38.901		Measureme		TR 38.901		Measureme	
	LOS	NLOS	LOS	NLOS	LOS	NLOS	LOS	NLOS	LOS	NLOS	LOS	NLOS	LOS	NLOS	LOS	NLOS	LOS	NLOS	LOS	NLOS
Number of clusters	15	19	10	11	12	19	5	7	12	20	13	17	12	20	11	15	12	20	7	13

  - Proponent companies to provide detailed information on potentially necessary changes to number of clusters for scenarios in question.
- Alt 2)
  - For the following scenarios, while sources have observed smaller number of clusters for various deployment scenario from measurement taken for 6 – 24 GHz frequency range, due to impact on frequency continuity outside 6 – 24 GHz frequency range, RAN1 concludes that there is no census to update number of clusters for all existing scenarios.

Agreement

Conclude in RAN1 #120bis to either (1) introduce an optional modeling component for polarization variability for each cluster for NLOS component of the channel or (2) no consensus to introduce polarization variability for each cluster for NLOS component of the channel.

- Distribution and normalization of $\eta_{n,\theta\theta,\theta\phi,\phi\theta,\phi\phi}$ is FFS
- $$H_{u,s,n}^{\mathrm{NLOS}}(t) = \sqrt{\frac{P_n}{M}} \sum_{m=1}^{M} F_{rx,u,\theta}(\theta_{n,m,ZOA}, \phi_{n,m,AOA}) \begin{bmatrix} \sqrt{\eta_{n,\theta\theta}} \exp(j\Phi_{n,m}^{\theta\theta}) & \sqrt{\eta_{n,\theta\phi}\kappa_{n,m}^{-1}} \exp(j\Phi_{n,m}^{\theta\phi}) \\ \sqrt{\eta_{n,\phi\theta}\kappa_{n,m}^{-1}} \exp(j\Phi_{n,m}^{\phi\theta}) & \sqrt{\eta_{n,\phi\phi}} \exp(j\Phi_{n,m}^{\phi\phi}) \end{bmatrix} \begin{bmatrix} F_{rx,u,\phi}(\theta_{n,m,ZOA}, \phi_{n,m,AOA}) \end{bmatrix}^{T} \begin{bmatrix} F_{tx,s,\theta}(\theta_{n,m,ZOD}, \phi_{n,m,AOD}) \\ F_{tx,s,\phi}(\theta_{n,m,ZOD}, \phi_{n,m,AOD}) \end{bmatrix} \exp\left(j2\pi \frac{\hat{r}_{rx,n,m}^{T} \cdot d_{rx,u}(m)}{\lambda_0}\right) \exp\left(j2\pi \frac{\hat{r}_{tx,n,m}^{T} \cdot d_{tx,s}(m)}{\lambda_0}\right) \exp\left(j2\pi \frac{r_{rx,n,m}^{T} \cdot v(m)}{\lambda_0} t\right)$$

Final summary in R1-2501638.

9.8.2 Channel model adaptation/extension of TR38.901 for 7-24GHz

Including near-field propagation and spatial non-stationarity

R1-2500088 Considerations on the 7-24GHz channel model extension Huawei, HiSilicon
R1-2500123 Discussion on Extension of FR3 Channel Modeling InterDigital, Inc.
R1-2500237 Views on channel model adaptation/extension for 7-24GHz CATT
R1-2500252 Discussion on channel modelling adaptation/extension for 7-24GHz LG Electronics
R1-2500363 Views on channel model adaptation/extension of TR38.901 for 7-24GHz vivo, BUPT
R1-2500381 Discussion on the channel model adaptation and extension ZTE Corporation, Sanechips
R1-2500410 Discussion on channel model adaptation/extension Intel Corporation
R1-2500464 Channel model adaptation and extension for 7-24GHz OPPO
R1-2500507 Discussion on adaptation and extension of channel model Ericsson
R1-2500542 Discussion on modeling near-field propagation and spatial non-stationarity in TR38.901 for 7-24GHz BUPT, CMCC
R1-2500561 Discussions on FR3 Channel Modelling Lekha Wireless Solutions
R1-2500628 Near-field channel modelling for FR3 VIAVI Solutions (rev of R1-2500627)
R1-2500678 Channel model extension for 7-24 GHz Lenovo
R1-2500694 Channel model adaptation of TR 38901 for 7-24 GHz NVIDIA
R1-2500799 On Channel Model Adaptation and Extension of TR38.901 for 7-24 GHz Apple
R1-2500863 Discussion on channel model adaptation/extension of TR38.901 for 7 - 24 GHz Samsung
R1-2501028 Discussion on channel modelling enhancements for 7-24GHz for NR MediaTek Inc.
R1-2501169 Channel Model Adaptation/Extension of TR38.901 for 7-24GHz Qualcomm Incorporated
R1-2501281 Discussion of channel model adaptation/extension of TR38.901 for 7-24GHz Sony
R1-2501349 Discussion on Channel model adaptation/extension of TR38.901 for 7-24GHz Nokia

R1-2501482 Summary#1 of channel model adaptation and extension Moderator (ZTE)

From Tuesday session

Agreement

For the modelling of spatial non-stationarity at UE side mainly due to close proximity, if blocker-based approach is adopted, the Option-3 is supported with following details:

- Three typical cases are introduced: one hand grip, dual hand grip, head and one hand grip.
  - probability of each case in the simulation is defined with the following table:

Usage scenario	one hand grip	dual hand grip	head and one hand grip
Proportion	58%	29%	13%

- For each case, to model the attenuation for each of antenna in the candidate antenna location defined in the UE antenna model for handheld device with Alt 1:
  - Alt 1: Introduce a fixed value for each candidate antenna position per case.
- The percentage of the UE without impact from the SNS at UE side is 10%.

Related Working Assumption made in RAN1#119 doesn't need to be confirmed.

R1-2501483 Summary#2 of channel model adaptation and extension Moderator (ZTE)

From Wednesday session

Agreement

For the modelling of spatial non-stationarity at UE side, the fixed values of attenuation for candidate antenna locations for handheld device will be defined.

Note:
The fixed values of attenuation for candidate antenna locations for handheld device can be defined as:

Antenna index	*Power attenuation (dB)*											
	One hand grip				*Dual hand grip*				*Head and one hand grip*			
	Set-1@6GHz	*Set-2@[0.7-8.4] GHz*	*[Set-3@15GHz]*		*Set-1@6GHz*	*Set-2@[0.7-8.4] GHz*	*[Set-3@15GHz]*		*Set-1@6GHz*	*Set-2@[0.7-8.4] GHz*	*[Set-3@15GHz]*	
			Alt-1	*Alt-2*			*Alt-1*	*Alt-2*			*Alt-1*	*Alt-2*
1	*0*	*1*	*1.0*	*0*	*7.31*	*9*	*3.8*	*9.68*	*10.57*	*1*	*4.0*	*14.16*
2	*0*	*FFS*	*FFS*	*0*	*0*	*FFS*	*FFS*	*0*	*7.08*	*FFS*	*FFS*	*9.87*
3	*0*	*3*	*3.8*	*0*	*7.31*	*9*	*3.8*	*10.6*	*0*	*3*	*3.3*	*0*
4	*6.15*	*FFS*	*3.6*	*9.13*	*6.15*	*FFS*	*FFS*	*9.12*	*0*	*FFS*	*3.3*	*9.13*
5	*7.31*	*12*	*3.8*	*10.6*	*0*	*5*	*1.0*	*0*	*6.55*	*12*	*3.3*	*10.6*
6	*0*	*FFS*	*FFS*	*0*	*0*	*FFS*	*FFS*	*0*	*10.12*	*FFS*	*FFS*	*9.70*
7	*0*	*1*	*1.0*	*0*	*0*	*5*	*1.0*	*0*	*10.57*	*3*	*4.0*	*14.16*
8	*0*	*FFS*	*1.0*	*0*	*6.15*	*FFS*	*FFS*	*10.62*	*10.67*	*FFS*	*4.0*	*14.27*

- Note1: The selection of number of antenna and antenna location is determined as part of assumption in the calibration and simulation.
- Note2: For both one hand grip and Head and one hand grip cases, the handheld is assumed to be held by right hand as example.
- The standardized CTIA hand grip can be considered as reference.

Conclusion
For near-field channel, no changes are expected on following parameter for both direct and non-direct path between TRP and UE:
- Delay

Agreement
For near-field channel, for the non-direct paths, the distance d_1 between the TRP and the 1st point associated with cluster is generated by Option-2:
- The distance d_1 is generated by a scaling factor ($s_{TRP,n}$) multiplied by the absolute distance corresponding to the absolute delay of cluster, where the scaling factor ($s_{TRP,n}$) is generated by:
  - For UMa scenarios, a Beta distribution with [$\alpha = 1.8, \beta = 1.22$]:
  - For UMi scenarios, a Beta distribution with [$\alpha = 2.35, \beta = 1.45$]:
  - For Indoor Office scenarios, a Beta distribution with [$\alpha = 1.13, \beta = 1.42$]
- For the value of k_1 and k_2
  - For UMa, the value of k_1 is 2.
  - For UMi, the value of k_1 is 2.
  - For InH, the value k_1 is [75%, 10.9%] of the total number of clusters for InH
  - Note: the k_1 clusters are the strongest of the total number of clusters in the channel
  - Note: the value of k_2 is equal to total number of clusters minus k_1.

R1-2501484 Summary#3 of channel model adaptation and extension Moderator (ZTE)
From Thursday session
Agreement
Confirm the following working assumption.
Working Assumption (made in RAN1#119)
For near-field channel, for the non-direct paths, the impact of spherical wavefront is optionally considered at UE side.

Agreement
For near-field channel, for number of k_2 non-direct paths, the distance d_2 between the UE and the 2nd point associated with cluster is generated by:
- a scaling factor ($s_{UE,n}$) multiplied by the absolute distance corresponding to the absolute delay of cluster.

Agreement
- For the modelling of spatial non-stationarity at BS side, if physical blocker-based approach is adopted, the nearest K blockers from BS are selected.
- For the modelling of spatial non-stationarity at BS side, if physical blocker-based approach is adopted, at least for blockage model B, the following parameters to define the building edge are introduced in the Table 7.6.4.2-5 in TR 38.901:

	Typical set of blockers	Blocker dimensions	Mobility pattern
Outdoor	Building edge	Cartesian: w=50 m; h=20 m	Stationary

- In the blockage model B, the following equation is used to calculate the attenuation caused by the building edge:
  - $L_{dB} = -20\log_{10}(0.5 - F)$, where F represents one of F_{h1}, F_{h2}, F_{w1} and F_{w2} defined in equation 7.6-29 in section 7.6.4.2 of TR 38.901.

Note: Corresponding Working Assumption made in RAN1#119 does not need to be confirmed.

Agreement
For near-field channel, the following formula is adopted to model the phase parameters of non-direct path between TRP and UE as antenna element-wise channel parameter:
- For the TRP side, the element-wise phase is calculated as:

$$\exp\left(j2\pi\frac{d_{1,n} - \left\|d_{1,n}\cdot\hat{r}_{tx,n,m} - \bar{d}_{tx,s}\right\|}{\lambda_0}\right)$$

where $d_{1,n}$ is the generated distance of the cluster n. $\hat{r}_{tx,n,m}$ is the spherical unit vector with azimuth departure angle and elevation departure angle for ray m of cluster n. $\bar{d}_{tx,s}$ is the vector pointing from reference point to transmit antenna element s, wherein the reference point is the physical center of the antenna array/center at Tx side.

- For the UE side, the element-wise phase is calculated as:

$$\exp\left(j2\pi\frac{d_{2,n}-\left\|d_{2,n}\cdot\hat{r}_{rx,n,m}-\bar{d}_{rx,u}\right\|}{\lambda_0}\right)$$

where $d_{2,n}$ is the generated distance of the cluster n. $\hat{r}_{rx,n,m}$ is the spherical unit vector with azimuth arrival angle and elevation arrival angle for ray m of cluster n. $\bar{d}_{rx,u}$ is the vector pointing from reference point to receive antenna element u, wherein the reference point is the physical center of the antenna array/center at Rx side.

Final summary in R1-2501485.

9.9 NR mobility enhancements Phase 4

Please refer to RP-242356 for detailed scope of the WI.

[120-R19-Mobility] – Yosuke (Fujitsu)
Email discussion on channel modelling for NR mobility enhancement

- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

9.9.1 Measurements related enhancements for LTM

R1-2500081	Measurements related enhancements for LTM	Huawei, HiSilicon
R1-2500180	Discussion on measurements related enhancements for LTM	Spreadtrum, UNISOC
R1-2500238	Views on measurements related enhancements for LTM	CATT
R1-2500245	Discussion on measurements related enhancements for LTM	ZTE Corporation, Sanechips
R1-2500253	Discussion on measurements related enhancements for LTM	LG Electronics
R1-2500299	Discussion on measurements related enhancements for LTM	CMCC
R1-2500318	Measurements enhancements for LTM	TCL
R1-2500364	Discussion on measurements related enhancements for LTM	vivo
R1-2500415	Measurement related enhancements for LTM	Nokia
R1-2500475	Discussions on measurement enhancement for LTM	OPPO
R1-2500562	Discussion on measurement related enhancements for LTM	Lekha Wireless Solutions
R1-2500624	Discussion on measurements related enhancements for LTM	NEC
R1-2500631	Discussion on measurement related enhancements for LTM	Fujitsu
R1-2500641	Measurements related enhancements for LTM	InterDigital, Inc.
R1-2500662	Measurements related enhancements for LTM	Sony
R1-2500703	Measurements related enhancements for LTM	Lenovo
R1-2500744	Discussion on measurements related enhancements for LTM	Xiaomi
R1-2500803	Measurement related enhancements for LTM	Apple
R1-2500864	Views on Rel-19 measurement related enhancements for LTM	Samsung
R1-2500918	Discussion on measurements related enhancements for LTM	ETRI
R1-2500992	Measurement related enhancements for LTM	Ericsson
R1-2501086	Measurement Enhancements for LTM	Meta
R1-2501091	Discussion on measurements related enhancements for LTM	KDDI Corporation
R1-2501170	Measurements related enhancement for LTM	Qualcomm Incorporated
R1-2501214	Discussion on measurement related enhancements for LTM	NTT DOCOMO, INC.
R1-2501244	Discussion on measurements related enhancements for LTM	Sharp
R1-2501315	LTM measurements related enhancements	MediaTek
R1-2501337	Discussion on measurements related enhancements for LTM	Google

R1-2500419 FL plan for mobility enhancements in RAN1#120 Moderator (Fujitsu)
R1-2500420 FL summary 1 of Measurements related enhancements for LTM Moderator (Fujitsu)
From Monday session
Agreement

- Confirm the following working assumption made in RAN1#118bis

Working Assumption(RAN1#118bis)

In addition to periodic CSI-RS, semi-persistent CSI-RS is supported for candidate cell L1-RSRP measurement for gNB scheduled reporting from RAN1 perspective
Send an LS to RAN3 (CC RAN2) to ask for the feasibility of specifying the signalling for coordination between serving cell and candidate cell(s) on the transmission of semi-persistent CSI-RS(s) and any other potential issues (e.g. RAN3 workload).
Support of semi-persistent CSI-RS is subject to UE capability.

- MAC CE is used to activate/deactivate the semi-persistent CSI-RS resource similarly to the legacy mechanism for a serving cell which will be specified in RAN2

Send an LS to RAN2 to inform about this agreement.

R1-2501499 [Draft] LS on activation/deactivation of semi-persistent CSI-RS resource for LTM candidate cells Moderator (Fujitsu)

Thursday decision: The draft LS is endorsed. Final LS is approved in R1-2501500.

Conclusion
There is no RAN1 consensus to support SP CSI-RS for event triggered reporting.

Conclusion
There is no consensus on the support of aperiodic CSI-RS resource for gNB scheduled reporting or event triggered reporting.

Agreement
The parameters defined in legacy *NZP-CSI-RS-resource* can be different for CSI-RS resources associated with the resource set.

Agreement
An LTM report configuration for L1-RSRP is associated with a single resource config that includes:

- Alt.A: a single resource set containing CSI-RS resources corresponding to multiple candidate cells. i.e., the same design as that of SSB in Rel-18 LTM.
- FFS: how to associate between the measurement CSI-RS resources and candidate cells
  - explicit or implicit signaling of candidate cells

R1-2500421 FL summary 2 of Measurements related enhancements for LTM Moderator (Fujitsu)

From Thursday session

Conclusion
There is no consensus in RAN1 to support specified filtering of L1-RSRP for LTM.
Send an LS to RAN2 to inform that L1 filtering is up to UE implementation.

R1-2501576 [Draft] LS on L1filtering for LTM event triggered reporting Moderator (Fujitsu)

Thursday decision: The draft LS is endorsed. Final LS is approved in R1-2501577.

Agreement
For target cell CSI acquisition,

- A UE is provided with RRC configurations for periodic CSI-RS resource(s) and CSI report(s) for one or more candidate cell
  - For a candidate cell,
    - down-select from the following alternatives:
      - Alt 1: A single CSI report configuration is configured
      - Alt 2: Multiple CSI report configurations can be configured
    - down-select from the following alternatives:
      - Alt X: A single CSI-RS resource for CMR is associated with a CSI report configuration
      - Alt Y: Multiple CSI-RS resources for CMR can be associated with a CSI report configuration
  - FFS: Semi-persistent CSI-RS resource
- After the RRC configuration and before the reception of CSC, the UE may measure CSI based on the configured CSI-RS resource(s), which is subject to UE capability
  - FFS: whether or how to select a subset of CSI-RS resources to measure
  - FFS: when the UE may start measuring the configured CSI-RS resources
- UE determines the CSI report configuration based on the CSC
- After the reception of cell switch command, the UE may measure (depending on the timeline) CSI-RS resource(s) associated with determined CSI report configuration

- The latest available measured CSI on target cell resource(s) is conveyed at least by a single report, and the report is sent to the target cell
  - Option 1: to use UCI
  - Option 2: to use MAC CE

Note: with this agreement, the working assumption made in RAN1#119 is automatically confirmed.

Conclusion
The following is up to RAN2: Coexistence of LTM event triggered reporting and mTRP operation at serving cells.

For future meetings:
Companies are encouraged to check with their RAN2 colleagues if there are any issues for RS type determination for event LTM2. Come back in RAN1#120bis if necessary.

Final FL summary in R1-2500423.

9.10 XR (eXtended Reality) for NR Phase 3

Please refer to RP-243318 for detailed scope of the WI.
Rapporteur to provide initial input on higher layer signalling under agenda item 9.10. For input on higher layer signalling from any other source, please include it as part of your tdoc to relevant sub-agenda items.

[120-R19-XR] – Jorma (Nokia)
Email discussion on Rel-19 XR Phase 3
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2501171 Initial higher layer parameters for Rel-19 XR for NR Phase3 Qualcomm, Nokia (WI rapporteurs)

9.10.1 Enabling TX/RX for XR during RRM measurements

R1-2500102 Discussions on scheduling enhancements considering RRM measurements for XR Huawei, HiSilicon
R1-2500134 Discussion on measurement gap for XR ZTE Corporation, Sanechips
R1-2500181 Discussion on enabling TXRX for XR during RRM measurements Spreadtrum, UNISOC
R1-2500206 Signaling control of scheduling restriction during measurement gap in support of XR services CATT
R1-2500300 Discussion on enabling TX/RX for XR during RRM measurements CMCC
R1-2500365 Remaining issues on enabling data transmissions for XR during RRM measurements vivo
R1-2500459 Enhancements to enable TX/RX for XR during RRM measurements OPPO
R1-2500486 Discussion on enabling TX/RX for XR during RRM measurements Panasonic
R1-2500528 Discussion on enabling TX/RX for XR during RRM measurements InterDigital, Inc.
R1-2500605 Discussion on enabling TX/RX for XR during RRM measurements NEC
R1-2500663 Discussion on enabling TX/RX for XR during RRM measurements Sony
R1-2500715 Discussion on enabling TX/RX for XR during RRM measurements Xiaomi
R1-2500804 Enabling TX/RX for XR during RRM measurements Apple
R1-2500865 Remaining topics on enabling TX/RX during RRM measurements for XR Samsung
R1-2500959 Discussion on XR during RRM measurements LG Electronics
R1-2501008 Enabling TX/RX for XR during RRM measurements Nokia
R1-2501172 Enabling Tx/Rx for XR during RRM measurements Qualcomm Incorporated
R1-2501215 Discussion on Enabling TX/RX for XR during RRM NTT DOCOMO, INC.
R1-2501289 Discussion on Enabling TX/RX for XR During RRM Measurements Meta
R1-2501290 On enabling Tx/Rx for XR during RRM measurements Google Ireland Limited
R1-2501343 Enabling TX/RX for XR traffic during RRM measurement gaps Ericsson

R1-2501009 Moderator summary #1 - Enabling TX/RX for XR during RRM measurements Moderator (Nokia)
From Monday session
Agreement
Confirm the Working assumption in RAN1#118 for supporting Alt. 1-1:

Working Assumption
For solutions based on triggering/enabling by network signaling to enable Tx/Rx in gaps/restrictions that are caused by RRM measurements select the following option:
- Alt. 1: Dynamic indication to enable Tx/Rx in particular gap/restriction that are caused by RRM measurements.
  - **Alt 1-1**: Explicit indication by DCI to skip a particular gap/restriction;

- Indication is included as part of scheduling DCI:
  - Bit-field size is one bit;
    - The bit in the DCI is used to indicate whether to skip the first gap/restriction occasion after a minimum time offset required between the last symbol of the PDCCH carrying the DCI format and the start of corresponding skipped gap/restriction occasion indicated by the DCI.

Conclusion
It is RAN1 understanding that if measurement gap/restriction is indicated to be skipped by explicit indication by DCI, the TX/RX is enabled on all the serving cells in the applicable range of the skipped measurement gap/restriction.
- Note: Applicable range of measurement gap/restriction is based on RAN4 specifications.

R1-2501531 Moderator summary #2 - Enabling TX/RX for XR during RRM measurements Moderator (Nokia)
From Wednesday session
Agreement
Agreement from RAN1#119 is updated (red part)
For explicit indication by DCI to skip a particular gap/restriction, in addition to DCI formats 0_1/1_1 and 0_2/1_2, one bit indication can be included as a part of DCI formats 0_3/1_3.
- Explicit indication can be configured for each DCI format individually by higher layer.
- The skipping indication corresponds to the first gap/restriction occasion among co-scheduled cells by the DCI (Type-1C field)

Above applies only for intra-band CA cases (same SCS for cells in the set of co-scheduled cells).
- Alt1: If skipping field is configured to DCI format 0_3/1_3, only cells that are intra-band (and have same SCS) can be co-scheduled. In other words, if co-scheduled cells could be inter-band, skipping indication field cannot be configured.

Agreement
When measurement gap/restriction is indicated to be skipped by explicit indication by DCI format 1_1/1_2 that does not schedule PDSCH and
- if CIF field is present, the scheduled cell for the skipping indication is determined by CIF field.
- if CIF field is not present, the scheduled cell for the skipping indication is the scheduling cell.

9.11 Non-Terrestrial Networks (NTN) for NR Phase 3, Internet of Things (IoT) Phase 3, and IoT-NTN TDD mode

Please refer to RP-243300 for detailed scope of the WI for NR-NTN Phase 3. Please refer to RP-243278 for detailed scope of the WI for IoT-NTN Phase 3. Please refer to RP-243293 for detailed scope of the WI for Introduction of IoT-NTN TDD mode.
Rapporteurs to provide initial input on higher layer signalling under agenda item 9.11 for NR-NTN and IoT-NTN. For input on higher layer signalling from any other source, please include it as part of your tdoc to relevant sub-agenda items.

R1-2501550 Session notes for 9.11 (Non-Terrestrial Networks for NR Phase 3 and Internet of Things Phase 3) Ad-Hoc Chair (Huawei)
Friday decision: The session notes are endorsed and contents reflected below.

[119-R20-NTN] – Mohamed (Thales)
Email discussion on Rel-19 NTN enhancement
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500043 Work plan for Rel-19 NR_NTN_Ph3 THALES
R1-2500523 Work plan WID: introduction of IoT-NTN TDD mode Iridium Satellite LLC

From AI 5
R1-2500018 LS on Ku band numerology RAN4, Eutelsat
Decision: RAN4 is requesting RAN1 input on whether there is any issue if different numerologies are used for different Ku bands. RAN1 response is necessary - Keith (Eutelsat).
From Thursday session
Agreement
Endorse the following response from RAN1 to RAN4:
- RAN1 does not see any reason why two sets of bands, one using a FR1-NTN numerology (30 kHz) and the other using a FR2-NTN numerology (120 kHz) cannot be defined.

- At this time, RAN1 has not identified a need for RAN1 specification update. RAN1 will consider whether changes to the RAN1 specifications will be needed, based on RAN4 progress.

Comeback for draft LS (Keith, Eutelsat).
R1-2501608 Draft Reply LS on Ku band numerology Eutelsat Group
Friday decision: The draft LS reply to RAN4 is endorsed in R1-2501608, changing "kindly' to "respectfully". Final LS is approved in R1-2501609.

9.11.1 NR-NTN downlink coverage enhancement

R1-2500037	NR NTN Downlink coverage enhancements	THALES
R1-2500082	Discussion on downlink coverage enhancements for NR NTN	Huawei, HiSilicon
R1-2500109	Discussions on downlink coverage enhancement for NR NTN	Fraunhofer IIS, Fraunhofer HHI
R1-2500182	Discussion on NR-NTN downlink coverage enhancement	Spreadtrum, UNISOC
R1-2500207	Discussion on downlink coverage enhancement for NR NTN	CATT
R1-2500268	Discussion on NR NTN downlink coverage enhancement	China Telecom
R1-2500301	Discussion on NR-NTN DL coverage enhancement	CMCC
R1-2500366	Discussion on NR-NTN downlink coverage enhancement	vivo
R1-2500382	Discussion on DL coverage enhancement for NR NTN	ZTE Corporation, Sanechips
R1-2500443	Discussion on NR-NTN downlink coverage enhancement	OPPO
R1-2500501	NR-NTN downlink coverage enhancement	InterDigital, Inc.
R1-2500519	Discussion on downlink coverage enhancement for NR NTN	Baicells
R1-2500607	NR-NTN downlink coverage enhancement	NEC
R1-2500632	Discussion on downlink coverage enhancements	Fujitsu
R1-2500672	On NR-NTN downlink coverage enhancement	Ericsson
R1-2500716	Discussion on NR-NTN downlink coverage enhancement	Xiaomi
R1-2500800	NR-NTN Downlink Coverage Enhancement	Apple
R1-2500866	Discussion on downlink coverage enhancement for NR-NTN	Samsung
R1-2500878	Discussion on downlink coverage enhancements for NR NTN	CCU
R1-2500887	Discussion on downlink coverage enhancement for NR NTN	Lenovo
R1-2500919	Discussion on NR-NTN downlink coverage enhancement	ETRI
R1-2500924	NR-NTN Downlink Coverage Enhancement	Panasonic
R1-2500983	Discussion on NR-NTN downlink coverage enhancement	TCL
R1-2500993	Discussions on downlink coverage enhancements	Nokia, Nokia Shanghai Bell
R1-2501029	NR-NTN downlink coverage enhancement	MediaTek Inc.
R1-2501040	Discussion on NR-NTN downlink coverage enhancement	LG Electronics
R1-2501099	Discussion on DL coverage enhancements for NR-NTN	NICT
R1-2501114	Discussion on NR-NTN downlink coverage enhancement	HONOR
R1-2501122	Discussion on downlink coverage enhancement for NR-NTN	CSCN
R1-2501173	Downlink coverage enhancement for NR NTN	Qualcomm Incorporated
R1-2501216	Discussion on DL coverage enhancement for NR-NTN	NTT DOCOMO, INC.
R1-2501229	Downlink Coverage Enhancement for NR NTN	Google Korea LLC
R1-2501280	Discussion on Downlink Coverage Enhancements for NR NTN	CEWiT

R1-2500039 FL Summary #1: NR-NTN downlink coverage enhancements THALES
From Monday session
Agreement
For NR NTN, support extended periodicity of the half frames with SS/PBCH blocks assumed by UE during initial access.
- The additional default value assumed by UE during initial access (apart from the existing 20ms value) is 160 ms.

Agreement
Support only inter-slot repetition for Type0 PDCCH CSS.

Agreement
For Msg4 PDSCH repetition support, RAN1 to consider:
- Option 1: UE specific repetition indication via DCI
- Option 2: Msg4 repetition is configured by SIB1
- Option 3: Msg4 PDSCH repetition is implicitly determined by SIB1 PDSCH repetition

R1-2500040 FL Summary #2: NR-NTN downlink coverage enhancements THALES
From Wednesday session
Agreement

At least for enabling PDCCH repetition for Type0 PDCCH CSS of searchSpaceZero configured within MIB pdcch-ConfigSIB1, RAN1 to consider the following options

- Option 1: Using the spare 1 bit in MIB
- Option 2: Using reserved bit(s) in PBCH payload
- Option 3: Using codepoint(s) in PBCH payload
- Option 4: UE blind decoding without signaling from the network during initial access

Agreement
For PDCCH repetition for Type0 PDCCH CSS of searchSpaceZero configured within MIB pdcch-ConfigSIB1, consider the following:

- Option 1: Support repeated PDCCH candidates in the two consecutive slots n_0 and $n_0 + 1$ associated with the same SSB index (n_0 as defined in section 13 of TS 38.213)
  - Repeated PDCCH candidates share the same aggregation level (AL), coded bits and same candidate index
  - FFS: Details including how the two PDCCH candidates are counted toward the BD limits
  - Note: with option 1, if the network repeats the Type 0 PDCCH across two consecutive slots, a legacy UE might decode the PDCCH and associated PDSCH in one slot and skip PDCCH monitoring in the other slot.
  - FFS: whether/how option 1 can be applicable for M=1 and M= ½
- Option 2: Support repeated PDCCH candidates in the two slots n_0 and $n_0 + X$ [or $n_0 + 1$ and $n_0 + 1 + X$] associated with the same SSB index (n_0 as defined in section 13 of TS 38.213)
  - Value of X>1, predefined or configured
    - FFS: Value of X
  - Repeated PDCCH candidates share the same aggregation level (AL), coded bits and same candidate index
  - FFS: Backward compatibility to legacy UE
- Option 3:
  - The PDCCH candidates in slots n0 associated respectively with different SSB indexes are repetitions of each other and share the same aggregation level, coded bits and same candidate index
    - For M=1/2 and M=1, the repeated PDCCH candidates in two consecutive slots associated with different SSB indexes;
    - For M=2, the PDCCH candidates in slots n_0+1 associated respectively with different SSB indexes are repetitions of each other and share the same aggregation level, coded bits and same candidate index
- Option 4: Option 2 with cross SSB beam repetition support
  - The PDCCH candidates in slots n0 associated respectively with different SSB indexes are repetitions of each other and share the same aggregation level, coded bits and same candidate index

R1-2500041 FL Summary #3: NR-NTN downlink coverage enhancements THALES
From Friday session
Agreement
For SIB1 link level enhancement, RAN1 to consider the following options:

- Option 1: PDSCH repetition of SIB1 is transmitted within the same slot as the type0-CSS PDCCH repetition.
  - UE supporting SIB1 PDSCH coverage enhancement assumes that the PDCCH and associated PDSCH to be repeated in both slots where the corresponding PDCCHs are transmitted.
  - Each PDSCH SIB1 repetition is within the same slot of each PDCCH candidate for scheduling DCI
  - The two associated PDSCHs have the same RV
- Option 2: Option 1 and an additional PDSCH with SIB1 repetition can occur after the slot of type0-PDCCH CSS repetition.
  - FFS: How to schedule the SIB1 repetition
- Option 3: The repetition of PDSCH with SIB1 is indicated by the scheduling PDCCH
  - PDSCH is repeated in two slots

Note: Backward compatibility should be maintained.

Final summary in R1-2500042.

9.11.2 Support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands

R1-2500083 Discussion on HD-FDD RedCap UEs and eRedCap UEs for FR1-NTN Huawei, HiSilicon
R1-2500183 Discussion on support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands Spreadtrum, UNISOC
R1-2500208 Discussion on the enhancement of RedCap and eRedCap UEs In NTN CATT

R1-2500269 Discussion on Support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands China Telecom
R1-2500302 Discussion on the collision issues of HD-FDD Redcap UE in FR1-NTN CMCC
R1-2500367 Discussion on support of RedCap and eRedCap UEs with NR-NTN vivo
R1-2500383 Discussion on support of RedCap/eRedCap UEs for NR NTN ZTE Corporation, Sanechips
R1-2500444 Discussion on supporting of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands OPPO
R1-2500502 Discussion on half-duplex RedCap issues for NTN FR1 operation InterDigital, Inc.
R1-2500673 On HD-FDD Redcap UEs for NTN Ericsson
R1-2500717 Discussion on the support of Redcap and eRedcap UEs in NR NTN Xiaomi
R1-2500805 Discussion on support of RedCap UEs with NR NTN operation Apple
R1-2500867 Discussion on support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands Samsung
R1-2500893 Discussion on support of RedCap/eRedCap UEs in NTN CAICT
R1-2500920 Discussion on HD UEs with NR NTN ETRI
R1-2500984 Discussion on HD-FDD Redcap UEs and eRedcap UEs for FR1-NTN TCL
R1-2500994 Discussion of support for RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands Nokia, Nokia Shanghai Bell
R1-2501030 Support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands MediaTek Inc.
R1-2501041 Discussion on support of (e)RedCap UEs with NR-NTN operating in FR1-NTN bands LG Electronics
R1-2501062 Support of (e)RedCap UEs with NR NTN Sharp
R1-2501115 Discussion on support of (e)RedCap UEs in NR NTN HONOR
R1-2501174 Support of Redcap and eRedcap UEs in NR NTN Qualcomm Incorporated
R1-2501217 Discussion on support of RedCap and eRedCap UEs in FR1-NTN NTT DOCOMO, INC.
R1-2501261 Support of RedCap and eRedCap UEs in NR NTN Nordic Semiconductor ASA

R1-2501465 Summary #1 for 9.11.2 Support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands Moderator (CATT)
From Monday session
Agreement
When network indicates UL overriding DL in RRC-Connected mode for collision case 3, one UE-specific RRC parameter is used to indicate overriding for all the applicable other use cases except for collisions with Type-0/0A/1/2-PDCCH CSS.

Conclusion
For Rel-19 HD-FDD (e)Redcap UE with CG-SDT procedure ongoing in RRC-inactive mode, for collision case 3,
- Handling of collision with PDCCH CSS in RRC-inactive mode is left to UE implementation whether to prioritize UL or prioritize DL with the constraint in the following note.

Note: UE shall comply to the following existing procedure in 38.331:
- UEs in RRC_INACTIVE while SDT procedure is not ongoing shall monitor for SI change indication in its own paging occasion(s) that the UE monitors as specified in TS 38.304 [20].
- UEs in RRC_INACTIVE while SDT procedure is ongoing shall monitor for SI change indication in any paging occasion at least once per modification period, if the initial downlink BWP on which the SDT procedure is ongoing is associated with a CD-SSB.
- ETWS or CMAS capable UEs in RRC_INACTIVE while SDT procedure is not ongoing shall monitor for indications about PWS notification in its own paging occasion every DRX cycle.
- ETWS or CMAS capable UEs in RRC_INACTIVE while SDT procedure is ongoing shall monitor for indication about PWS notification in any paging occasion at least once every defaultPagingCycle.

R1-2501466 Summary #2 for Support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands Moderator (CATT)
Presented in Wednesday session.

R1-2501467 Summary #3 for Support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands Moderator (CATT)
From Friday session
Working assumption
For Rel-19 NTN HD-FDD (e)Redcap UE in RRC connected mode, the following handling rule for collision case 4 is supported:

- Handling of collision with PDSCH (at least for system information) scheduled by Type-0/0A[/1][/2]-PDCCH CSS in RRC-Connected mode is left to UE implementation whether to prioritize UL or prioritize DL with the constraint in the following note.
- FFS: handling of PDCCH ordered PRACH transmission
- For other use cases, default priority rule for collision case 4 in RRC-Connected mode is that [DL or UL] is prioritized.
    - Network is allowed to indicate [UL or DL] overriding [DL or UL] for all cases
        - This is signaled by one UE specific RRC parameter
    - Note: if DL is prioritized, the DL prioritization applies only if the UL cancellation timeline can be satisfied, otherwise UL is prioritized.

Note: UE shall comply to the following existing procedure in 38.331:

- UEs in RRC_CONNECTED shall monitor for SI change indication in any paging occasion at least once per modification period if the UE is provided with common search space, including *pagingSearchSpace*, *searchSpaceSIB1* and *searchSpaceOtherSystemInformation*, on the active BWP to monitor paging, as specified in TS 38.213 [13], clause 13.

R1-2501605 Final Summary for Support of RedCap and eRedCap UEs with NR NTN operating in FR1-NTN bands Moderator (CATT)

9.11.3 NR-NTN uplink capacity/throughput enhancement

R1-2500084 Discussion on uplink capacity/throughput enhancement for FR1-NTN Huawei, HiSilicon
R1-2500184 Discussion on NR-NTN uplink capacity/throughput enhancement Spreadtrum, UNISOC
R1-2500209 Discussion on UL capacity enhancement for NR NTN CATT
R1-2500270 Discussion on NR-NTN uplink enhancement China Telecom
R1-2500303 Discussion on the NR-NTN uplink capacity/throughput enhancements CMCC
R1-2500368 Discussion on NR-NTN uplink capacity enhancement vivo
R1-2500384 Discussion on UL capacity enhancement for NR NTN ZTE Corporation, Sanechips
R1-2500445 Discussion on NR-NTN uplink capacity/throughput enhancement OPPO
R1-2500503 NR-NTN uplink capacity/throughput enhancement InterDigital, Inc.
R1-2500608 NR-NTN uplink capacity/throughput enhancement NEC
R1-2500633 Discussion on uplink capacity/cell throughput enhancement for FR1-NTN Fujitsu
R1-2500718 Discussion on NR-NTN PUSCH capacity enhancement Xiaomi
R1-2500801 NR-NTN Uplink Capacity Enhancement Apple
R1-2500868 Discussion on uplink capacity/throughput enhancement for NR-NTN Samsung
R1-2500890 Uplink capacity/throughput enhancement for NR-NTN Panasonic
R1-2500921 Discussion on NR-NTN uplink capacity/throughput enhancement ETRI
R1-2500980 Discussion on the NR-NTN uplink capacity/throughput enhancements TCL
R1-2500987 On uplink capacity/cell throughput enhancement for NR NTN Ericsson
R1-2500995 Discussion of NR-NTN uplink capacity enhancements Nokia, Nokia Shanghai Bell
R1-2501031 NR-NTN uplink capacity and throughput enhancements MediaTek Inc.
R1-2501042 Discussion on NR-NTN uplink capacity/throughput enhancement LG Electronics
R1-2501051 Discussion on NR-NTN uplink capacity/throughput enhancement Lenovo
R1-2501090 Discussion on NR-NTN uplink capacity/throughput enhancement Hyundai Motor Company
R1-2501116 Discussion on NR-NTN UL capacity/throughput enhancement HONOR
R1-2501175 NR-NTN uplink capacity / throughput enhancement Qualcomm Incorporated
R1-2501218 Discussion on NR-NTN uplink capacity/throughput enhancement NTT DOCOMO, INC.

R1-2501400 Feature lead summary #1 of AI 9.11.3 on NR-NTN uplink capacity and throughput enhancements Moderator (MediaTek)

From Tuesday session

Conclusion

For OCC time synchronization / alignment of multiplexed UEs to maintain orthogonality of the codes used for OCC, OCC group alignment is handled by network scheduling without specification impact.

Conclusion

OCC with Msg3 PUSCH is not in scope of Release 19 NR NTN Ph3.

R1-2501401 Feature lead summary #2 of AI 9.11.3 on NR-NTN uplink capacity and throughput enhancements Moderator (MediaTek)

From Thursday session
Agreement
For RV cycling for OCC with DG-PUSCH, support Option 1
- Option 1: RV cycling is used across OCC groups.
  - Note: when option 1 is applied, RV cycling is applied when the number of repetitions is greater than the OCC length

No optimization in Rel-19 for pairing UEs with OCC2 and UEs with OCC4.

Final summary in R1-2501402.

9.11.4 IoT-NTN uplink capacity/throughput enhancement

From AI 5
R1-2500009 LS on PWS support in RRC_CONNECTED for NB-IoT NTN RAN2, InterDigital
Decision: RAN2 is requesting RAN1 input on the support for reception of PWS (ETWS and CMAS) notifications in RRC_CONNECTED for NB-IoT from RAN1 perspective. RAN1 response is necessary - Umer (InterDigital).

R1-2501481 Moderator Summary for response to LS on PWS Support for NB-IoT NTN UEs Moderator (InterDigital, Inc.)
From Tuesday session
R1-2501517 Draft reply LS on PWS support in RRC_CONNECTED for NB-IoT NTN Moderator (InterDigital)
Proposal
Response to RAN2 LS:
From RAN1 perspective, it is not feasible to introduce support of NB-IoT UEs in RRC connected mode monitor PWS notifications in Rel-19, due to increased implementation complexity, power consumption at UE side and the amount of RAN1 workload for specifying this considering the Rel-19 TU allocation.

R1-2501555 Moderator Summary#2 for response to LS on PWS Support for NB-IoT NTN UEs Moderator (InterDigital, Inc.)
From Thursday session
R1-2501518 Reply LS on PWS support in RRC_CONNECTED for NB-IoT NTN RAN1, InterDigital
Agreement
Response to RAN2 LS, to be sent to RAN plenary as well:
From RAN1 perspective, it may be possible to find solutions with specification impact for supporting NB-IoT UEs in RRC connected mode monitor PWS notifications, while some companies expressed concerns due to increased implementation complexity, power consumption at UE side and timely delivery of PWS notifications in NTN context. However, the amount of RAN1 workload for specifying this may not fit in the Rel-19 TU allocation for NTN. RAN1 would like to ask RAN plenary to decide whether RAN1 should work on specifying support of NB-IoT UEs in RRC connected mode monitor PWS notifications in Rel-19.

R1-2501612 Draft reply LS on PWS support in RRC_CONNECTED for NB-IoT NTN Moderator (InterDigital)
Friday decision: The draft LS in R1-2501612 is endorsed. Final LS is approved in R1-2501613.

R1-2500085 Discussion on UL capacity enhancements for IoT NTN Huawei, HiSilicon
R1-2500185 Discussion on IoT-NTN uplink capacity/throughput enhancement Spreadtrum, UNISOC
R1-2500210 Discussion on UL capacity enhancement for IoT NTN CATT
R1-2500304 Discussion on the IoT-NTN uplink capacity/throughput enhancements CMCC
R1-2500369 Discussion on IoT-NTN uplink capacity enhancement vivo
R1-2500385 Discussion on UL capacity enhancement for IoT NTN ZTE Corporation, Sanechips
R1-2500446 Discussion on IoT-NTN uplink capacity/throughput enhancement OPPO
R1-2500504 IoT-NTN uplink capacity/throughput enhancement InterDigital, Inc.
R1-2500508 IoT-NTN uplink capacity enhancement Nokia, Nokia Shanghai Bell
R1-2500609 IoT-NTN uplink capacity/throughput enhancement NEC
R1-2500664 On IoT-NTN uplink capacity enhancements Sony
R1-2500674 On uplink capacity enhancements for IoT-NTN Ericsson
R1-2500719 Discussion on IoT-NTN uplink capacity enhancement Xiaomi
R1-2500806 Discussion on IoT-NTN Uplink Capacity Enhancement Apple
R1-2500869 Discussion on uplink capacity/throughput enhancement for IoT-NTN Samsung
R1-2500888 Discussion on uplink capacity enhancement for IoT NTN Lenovo
R1-2500922 Discussion on uplink capacity/throughput enhancement for IoT NTN ETRI

R1-2500981 Discussion on the IoT-NTN uplink capacity/throughput enhancements TCL
R1-2501032 IoT-NTN uplink capacity and throughput enhancement MediaTek Inc.
R1-2501043 Discussion on IoT-NTN uplink capacity/throughput enhancement LG Electronics
R1-2501074 IoT NTN OCC methods and signaling for NPUSCH and NPRACH Sharp
R1-2501176 IOT-NTN uplink capacity/throughput enhancement Qualcomm Incorporated
R1-2501260 IoT-NTN uplink capacity/throughput enhancement Nordic Semiconductor ASA

R1-2500665 FL Summary #1 for IoT-NTN Moderator (Sony)
From Tuesday session
Agreement
For the support of OCC length 2 for NPUSCH Format 1 single-tone with 3.75 kHz SCS and 15 kHz SCS, the orthogonal sequences are [1 1; 1 -1].

Working assumption
For 3.75kHz SCS OCC for NPUSCH format 1, support TDM DMRS over 4 slots where DMRS are transmitted in the first 2 slots and DMRS REs are blanked in the next 2 slots, or vice-versa, where the DMRS REs are as in legacy NB-IoT and the guard period within the slot is as in legacy NB-IoT.
Send LS to RAN4 asking whether there would be any issue (e.g. phase continuity) for supporting such TDM DMRS for IoT NTN.

R1-2500666 FL Summary #2 for IoT-NTN Moderator (Sony)
From Thursday session
Agreement
Send the following LS to RAN4:

Overall description
RAN1 has made the following working assumption on the DMRS pattern for OCC for 3.75kHz SCS OCC for NPUSCH format 1:

For 3.75kHz SCS OCC for NPUSCH format 1, support TDM DMRS over 4 slots where DMRS are transmitted in the first 2 slots and DMRS REs are blanked in the next 2 slots, or vice-versa, where the DMRS REs are as in legacy NB-IoT and the guard period within the slot is as in legacy NB-IoT.

Question:
1. From a RAN4 perspective, is it feasible to introduce support of TDM DMRS, as per the description above, in Rel-19?

Comeback for final LS on Friday.
R1-2501621 [Draft] LS on TDM DMRS for 3.75kHz SCS OCC Moderator (Sony)
Friday decision: The draft LS to RAN4 in R1-2501621 is endorsed. Final LS is approved in R1-2501622.

Agreement
For CONNECTED mode, UE-specific RRC signalling is used for enabling of the OCC feature.

From Friday session
Conclusion
RAN1 did not reach consensus on whether OCC for NPRACH is beneficial or not. RAN1 will not specify support for OCC for NPRACH in Rel-19 IoT NTN.

R1-2500667 Final FL Summary for IoT-NTN Moderator (Sony)

9.11.5 IoT-NTN TDD mode

R1-2500038 Discussion on IoT-NTN TDD mode THALES
R1-2500086 Discussion on IoT-NTN TDD mode Huawei, HiSilicon
R1-2500186 Discussion on IoT-NTN TDD mode Spreadtrum, UNISOC
R1-2500211 Physical layer design on IoT-NTN TDD mode CATT
R1-2500305 Discussion on the new NB-IoT NTN TDD mode CMCC
R1-2500370 Discussion on IoT-NTN TDD mode vivo
R1-2500386 Discussion on the IoT-NTN TDD mode ZTE Corporation, Sanechips
R1-2500447 Discussion on IoT-NTN TDD mode OPPO
R1-2500509 Discussion on IoT-NTN TDD mode Nokia, Nokia Shanghai Bell
R1-2500524 IoT-NTN TDD mode scheduling, timing and channel decoding performance Iridium Satellite LLC
R1-2500720 Discussion on IoT-NTN TDD mode Xiaomi
R1-2500807 Discussion on IoT-NTN TDD mode Apple

R1-2500870	Discussion on IoT-NTN TDD mode	Samsung
R1-2500889	Discussion on IoT-NTN TDD mode	Lenovo
R1-2500982	Discussion on the IoT-NTN TDD mode	TCL
R1-2501044	Discussion on IoT-NTN TDD mode	LG Electronics
R1-2501075	IoT NTN TDD frame structure and synchronization signal mapping	Sharp
R1-2501177	IOT-NTN TDD mode	Qualcomm Incorporated
R1-2501219	Discussion on IoT-NTN TDD mode	NTT DOCOMO, INC.
R1-2501295	On R19 TDD IoT-NTN	Nordic Semiconductor ASA

R1-2501366 Feature lead summary #1 on IoT-NTN TDD mode Moderator (Qualcomm Incorporated)
From Tuesday session
Agreement
The guard period between the end of the downlink subframes and the beginning of the uplink subframes is fixed in specifications to be [48…50] ms at the ULSRP.
- NOTE: This corresponds to the largest separation between DL and UL Iridium slot pairs. Adjustments to align with different UL/DL pair can be achieved by adjusting the common TA.

Agreement
The downlink subframes within D=8 are fixed to:
- Option 1: [3 4 5 6 7 8 9 0] (across two consecutive radio frames).

Agreement
NPSS/NSSS/NPBCH transmissions are dropped in non-D NB-IoT subframes
- NOTE: “non-D NB-IoT subframes” are subframes not contained within the set of D=8 usable consecutive downlink subframes in the TDD structure.
- FFS: handling for SIB1-NB.

Agreement
The non-U NB-IoT subframes are not considered by the UE as “NB-IoT UL subframes” from the specifications
- NPUSCH is mapped as per current specifications.
- NOTE: “non-U NB-IoT subframes” are subframes not contained within the set of U=8 usable consecutive uplink subframes in the TDD structure.

Agreement
The non-D NB-IoT subframes are not considered by the UE as “NB-IoT DL subframes” from the specifications
- NPDCCH / NPDSCH (not containing SI) are mapped as per current specifications.
- FFS: additional NB-IoT TDD-specific rules for handling non “NB-IoT DL subframes”
- FFS: Introduction of new periodicities for NPDCCH monitoring (e.g. by the introduction of scaling factor)
- FFS: whether all the configuration values for NPDCCH are valid for NB-IoT NTN TDD.
- FFS: handling of DL transmission gap
- NOTE: “non-D NB-IoT subframes” are subframes not contained within the set of D=8 usable consecutive downlink subframes in the TDD structure.

R1-2501367 Feature lead summary #2 on IoT-NTN TDD mode Moderator (Qualcomm Incorporated)
From Thursday session
Conclusion
For determining the offset between the 90ms TDD structure and the 10240ms H-SFN: the UE may derive the 90ms TDD structure based on observations of downlink signals.
No specification impact.

Agreement
For NPRACH, further consider the two options until RAN1#120bis:
- Option 1: New NPRACH periodicities (multiple of 90ms) are introduced for NB-IoT NTN TDD.
- Option 2: NPRACH transmissions are postponed in non-U NB-IoT subframes until the next U NB-IoT subframe(s)
  - FFS: The unit of postponement (e.g. PRU)

Agreement
NPRACH format 2 is not supported in NB-IoT NTN TDD.

Working assumption

Uplink transmission gaps do not apply in NB-IoT NTN TDD.

Conclusion
RAN1 does not further discuss the issue of Kmac extension unless triggered by RAN2 LS.

R1-2501614 Feature lead summary #3 on IoT-NTN TDD mode Moderator (Qualcomm Incorporated)
From Friday session
Conclusion
It is RAN1's understanding that the issue of scheduling and transmitting SI messages in the presence of non-D subframes will be solved mainly (or completely) by RAN2 specifications. RAN1 may discuss potential impact on RAN1 specifications, if any.

Agreement
For segmented pre-compensation, RAN1 to further discuss at least the following options:
- Option 1: Dropped/postponed NPUSCH / NPRACH are counted for segment determination
- Option 2: A new pre-compensation segment is started at the beginning of each uplink burst
- Option 3: UL segment should only be defined in consecutive U subframes
- Option 4: Segmented pre-compensation does not apply for NB-IoT NTN TDD.
- FFS: How to handle the UL segment gap.

9.12 Multi-Carrier Enhancements

R1-2501551 Session notes for 9.12 (Multi-Carrier Enhancements for NR Phase 2) Ad-Hoc Chair (CMCC)
Friday decision: The session notes are endorsed and contents reflected below.

9.12.1 Multi-carrier enhancements for NR phase 3

Please refer to RP-242904 for detailed scope of the WI for Multi-Carrier Enhancements for NR Phase 3.

[120-R19-MCE] – Haipeng (Lenovo)
Email discussion on Rel-19 multi-carrier enhancement
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2500065 On Rel-19 Multi-carrier enhancements Nokia
R1-2500092 Discussion on Rel-19 Multi-carrier enhancements Huawei, HiSilicon
R1-2500139 Discussion on multi-cell scheduling with a single DCI Lenovo
R1-2500147 Discussion on multi-cell PUSCH/PDSCH scheduling with a single DCI ZTE Corporation, Sanechips
R1-2500187 Discussion on multi-cell PUSCH/PDSCH scheduling with a single DCI Spreadtrum, UNISOC
R1-2500212 Discussion on multi-cell PUSCH/PDSCH scheduling with a single DCI CATT
R1-2500271 Discussion on multi-carrier enhancements for NR phase 3 China Telecom
R1-2500306 Discussion on multi-cell PUSCH/PDSCH scheduling with a single DCI CMCC
R1-2500371 Discussion on enhancement of multi-cell PUSCH/PDSCH scheduling with a single DCI vivo
R1-2500461 Discussion of multi-cell scheduling with a single DCI OPPO
R1-2500808 On multi-cell PUSCH/PDSCH scheduling with single DCI Apple
R1-2500818 Discussion on multi-carrier enhancements for NR Phase 3 Panasonic
R1-2500871 Enhancements for multi-cell PUSCH/PDSCH scheduling Samsung
R1-2500923 Discussion on multi-cell PUSCH/PDSCH scheduling with a single DCI ETRI
R1-2500960 Discussion on single DCI based multi-cell scheduling for Rel-19 LG Electronics
R1-2500985 Discussion on Multi-cell PUSCH/PDSCH scheduling with a single DCI TCL
R1-2501033 Multi-cell PUSCH/PDSCH scheduling with a single DCI MediaTek Inc.
R1-2501178 Multi-cell PUSCH/PDSCH scheduling with a single DCI Qualcomm Incorporated
R1-2501220 Discussion on multi-cell PUSCH/PDSCH scheduling with a single DCI NTT DOCOMO, INC.
R1-2501345 Multi-cell PxSCH scheduling with a single DCI Ericsson

R1-2501374 Feature lead summary#1 on multi-cell scheduling with a single DCI Moderator (Lenovo)
From Monday session
Agreement
- For determining the timing of a PUCCH carrying HARQ-ACK information corresponding to a set of co-scheduled PDSCHs by a DCI format 1_3, follow Rel-18 operation.
  - If the UE is not provided *subslotLengthForPUCCH*, the DL slot n_D is the DL slot ending last, amongst the DL slots where the more than one PDSCH are scheduled by the DCI format 1_3.

        - FFS: RAN1 spec impact till RAN1#120-bis
    - If the UE is provided *subslotLengthForPUCCH*, no spec change is necessary.
- Note: Specification of this feature shall not impact the existing UE processing PDSCH timeline requirement for any individual PDSCH, as specified in 5.3.1 of TS38.214.

Agreement
For the second sub-codebook, the number of HARQ-ACK information bits for each DCI format 1_3 is equal to M.
- M is the maximum number of HARQ-ACK information bits which can be generated for a DCI format 1_3 across all the configured cell set(s) in the PUCCH group for the UE. M is implicitly derived based on RRC configuration.

Agreement
For NDI indication in DCI format 0_3/1_3 for a TB,
- Option 1 is adopted.

Agreement
For RV indication in DCI format 0_3/1_3 for a TB,
- Option 1 is adopted.

Agreement
For multi-PUSCH/PDSCH scheduling using a DCI format 0_3/1_3, 1 bit RV indication is determined according to Table 7.3.1.2.3-1 of TS 38.212.
- Note: This is aligned with Rel-18 DCI format 0_3/1_3 for cells configured with 1 bit RV by *numberOfBitsForRV-DCI-0-3/1-3*.

Agreement
For the second sub-codebook, the HARQ-ACK information bits for a DCI format 1_3 are ordered firstly according to same ordering as in Rel-17 multi-PDSCHs scheduling for PDSCH receptions on a same serving cell, then according to ascending order of associated serving cell indexes.

R1-2501375 Feature lead summary#2 on multi-cell scheduling with a single DCI Moderator (Lenovo)
From Tuesday session
Agreement
For Type-1C fields in DCI format 0_3, when the TDRA field indicates more than one scheduled PUSCHs on the scheduled cell with the smallest cell index:
- The CSI request field applies to the PUSCH determined based on Rel-17 multi-PUSCH scheduling on same serving cell.
- Note for background: When the TDRA field of DCI format 0_3 indicates only one scheduled PUSCH on the scheduled cell with the smallest cell index, DCI interpretation and UE procedure is same as in Rel-18.

Working Assumption
For a DCI format 1_3 with fields repurposed for SCell dormancy indication and scheduling one or more PDSCHs, if TDRA field indicates multiple SLIVs for the serving cell with smallest serving cell index with invalid FDRA, the HARQ-ACK information bit for SCell dormancy indication is ACK for the first SLIV and followed by NACK bits for the remaining SLIVs.

R1-2501376 Feature lead summary#3 on multi-cell scheduling with a single DCI Moderator (Lenovo)
From Thursday session
Agreement
- Regarding presence of UL-SCH field,
    - 1 bit UL-SCH field is always present in DCI format 0_3.
- UL-SCH field and CSI request field in a DCI format 0_3 are applied to the same PUSCH.

Agreement
- For NDI/RV indication in DCI format 0_3/1_3 for a TB,
    - Within each block of NDI field, the NDI bits are placed in the LSBs based on the SLIV position in the indicated TDRA row. Padding bits, if any, are placed in the MSBs within the block.
    - Within each block of RV field, the RV bits are placed in the LSBs based on the SLIV position in the indicated TDRA row. Padding bits, if any, are placed in the MSBs within the block.

Final summary in R1-2501377.

9.12.2 Low band carrier aggregation via switching

Placeholder only. RAN1 discussions on 9.12.2 will start in 2025.Q2. Please refer to RP-243317 for detailed scope of the WI for low band carrier aggregation via switching.

9.13 LTE-based 5G Broadcast Phase 2

Please refer to RP-243290 for detailed scope of the WI for LTE-based 5G Broadcast Phase 2.

R1-2501552 Session notes for 9.13 (LTE-based 5G Broadcast Phase 2) Ad-Hoc Chair (Huawei)
Friday decision: The session notes are endorsed and contents reflected below.

[120-R19-Broadcast] – Jordi (EBU)
Email discussion on Rel-19 LTE-bad 5G Broadcast Phase 2
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2501057 Introduction of WI "Time and Frequency Interleaving for LTE-based 5G Terrestrial Broadcast" EBU

9.13.1 Time-frequency interleaver design for MBMS dedicated cells

R1-2500148 Discussion on time-frequency interleaver design for LTE-based 5G Broadcast ZTE Corporation, Sanechips
R1-2500150 On time-frequency interleavers for LTE-based 5G Broadcast Huawei, HiSilicon
R1-2500582 Design of Time and Frequency Interleaver for LTE-based 5G Broadcast Shanghai Jiao Tong University
R1-2500872 Time-frequency interleaver design for MBMS dedicated cells Samsung
R1-2501058 Details of Time and Frequency Interleaving for LTE-based 5G Terrestrial Broadcast EBU
R1-2501179 Time and Frequency Interleaving for 5G Broadcast Qualcomm Incorporated

R1-2501432 FL summary#1 on Rel-19 LTE-based 5G Broadcast Phase 2 Moderator (EBU)
From Tuesday session
Agreement
For TBS scaling by N, down-select one of the two options below:
- Option 1: Calculate the total number of PRBs N_{RB}^{total} as $N_{RB}^{total} = N_{RB}^{DL} \times N$, wherein, N_{RB}^{DL} is system bandwidth;
  - If $N_{RB}^{total} \leq 110$, reuse the transport block size table in TS 36.213 and according to the obtained I_{TBS} and N_{RB}^{total}, obtain the corresponding TBS;
  - Else, reuse the transport block size table in TS 36.213 and according to the obtained I_{TBS} and N_{RB}^{DL}, obtain the corresponding TBS'; and calculate the final TBS as TBS = round{TBS′ × N}
  - The operation "round{x}" is mapping x to the closest TBS from all the existing TBS values in section 7.1.7.2 TS 36.213.
- Option 2: Obtain the initial TBS (*TBS′*) as per current specifications, and calculate the final TBS as TBS = round{TBS′ × N}
  - The operation "round{x}" is mapping x to the closest TBS from all the existing TBS values in section 7.1.7.2 TS 36.213.

Agreement
At least values of N={1,4,8} are supported, where N=1 indicates time interleaving is disabled.
- FFS: other values

Agreement
LBRM is applied for MCH (excluding MCCH and MSI) with time interleaving.
- FFS spec impact

Agreement
For PMCH with time interleaving, the maximum TBS a UE supports for the scaled TB is derived based on DL-SCH as per TS 36.306 set by UE category.
- FFS when the UE category has multiple maximum TBS

Agreement
The maximum number of transport block bits received within a TTI defined by the UE category should include the number of bits for unicast and broadcast, where the number of bits for broadcast is the one after being scaled up by N.

Agreement
The basic transmission pattern of PMCH with time interleaving consists of a set of (M x N) consecutive MBSFN subframes, excluding MCCH and MSI, where two transmissions of the same TB are separated by (M-1) MBSFN subframes, excluding MCCH and MSI.

Agreement
Specify time interleaving per MBMS session/MTCH without supporting interleaving across different MBMS sessions/MTCH.

Agreement
Regarding configuration for time interleaving transmission, the following options can be considered:
- Option 1: Per MBSFN area/MCCH configuration, e.g., via mcch-Config
- Option 2: Per PMCH configuration, e.g., via pmch-Config
- Option 3: Per MBMS session, e.g., via MBMS-SessionInfo or via MAC CE

Agreement
RAN1 to further study the potential issue of mismatch between MSI periodicity and the basic transmission pattern of (M x N) subframes.

Agreement
The equation of k0 for time interleaving is modified to avoid puncturing of systematic bits.
- Further study the exact equation for k0 for the n-th subframe of a transmission belonging to a same TB k_0^n, $n \in \{0, 1, 2, ..., N-1\}$), including
  - Option 1: for each CB of the TB, $k_0^n = 2R_{subblock}^{TC} + rv_n E_{min}$, where $E_{min} = N_L Q_m \lfloor G'/C \rfloor$ (as per TS 36.212), and $rv_n \in \{0...N-1\}$ is the redundancy version of the n-th subframe.
  - Option 2: For each CB of the TB, follow the principle of NR TBoMS to determine k_0^n
  - Option 1 and Option 2 assumes all CBs of the TB have the same RV index for all CBs of the TB, indexed by k_0^n
- FFS whether different CBs of the TB mapped to each subframe may have different RV indices (e.g., indexed by different k_0^n per CB)

Agreement
The modulation symbol vector (prior to frequency interleaving) $y = [y(0), y(1), ..., y(Z-1)]$ is interleaved, according to the principles of subclause 5.1.4.1.1 of TS 36.212 as follows
- Set the number of columns of the frequency interleaver, $C = X$, set the number of rows R to be the minimum integer satisfying $Z \le (R \times C)$, where Z is the number of data REs in one OFDM symbol
  - FFS: How to determine X
- Append y with $< NULL >$ elements, as required, to obtain y_{app}
- Write the elements of y_{app} into the $(R \times C)$-matrix column-wise, with $y_{app}(0)$ in column 0 of row 0.
- FFS whether the rows and elements in each row will be permuted, with details to be further studied, e.g., intra-row cyclic shifts, inter-row permutations
- Obtain the vector y_Π by reading the elements of the above $(R \times C)$-matrix row-wise, excluding the $< NULL > s$.
- The vector y_Π shall then be mapped to resource elements, according to clause 6.3.5 of TS 36.211.

FFS: whether to apply a cyclic shift offset, and the details of the cyclic shift offset for the concatenated frequency interleaver outputs of all OFDM symbols before the mapping to REs of a subframe, where the cyclic shift offset is dependent on the RV index.

Agreement
The granularity of the frequency interleaver is 1 RE.

R1-2501433 **FL summary#2 on Rel-19 LTE-based 5G Broadcast Phase 2** **Moderator (EBU)**
From Thursday session
Agreement
For TBS scaling by N, the following is clarified for the rounding operation in option 1 and option 2 in previous agreement:
- In case two TBSs are at the same distance of $TBS' \times N$, the larger TBS is chosen.

Agreement
Regarding configuration for time interleaving transmission, the following options can be considered:
- ~~*Option 1: Per MBSFN area/MCCH configuration, e.g., via mcch-Config*~~
- *Option 2: Per PMCH configuration, e.g., via pmch-Config*
- *Option 3: Per MBMS session, e.g., via MBMS-SessionInfo or via MAC CE*

Agreement
Values of M = 16 and M = 8 are supported.
- FFS other values
- Note: some UE categories and depending on factors such as (value of N, bandwidth, MCS) may not be able to receive an interleaved MTCH with M=16.

Agreement
The previous agreement is modified as follows:
The modulation symbol vector (prior to frequency interleaving) $y = [y(0), y(1), ..., y(Z-1)]$ is interleaved, according to the principles of subclause 5.1.4.1.1 of TS 36.212 as follows
- Set the number of columns of the frequency interleaver, $C = X$, set the number of rows R to be the minimum integer satisfying $Z \leq (R \times C)$, where Z is the number of data REs in one OFDM symbol
  - FFS: How to determine X
    - Option 1: X is equal to the number of codeblocks that are mapped to the OFDM symbol
    - Option 2: X=32
    - Other options are not precluded.
- Append y with $< NULL >$ elements, as required, to obtain y_{app}
- Write the elements of y_{app} into the $(R \times C)$-matrix column-wise, with $y_{app}(0)$ in column 0 of row 0.
- FFS whether the rows and elements in each row will be permuted, with details to be further studied, e.g., intra-row cyclic shifts, inter-row permutations
- Obtain the vector y_{Π} by reading the elements of the above $(R \times C)$-matrix row-wise, excluding the $< NULL >$ s.
- The vector y_{Π} shall then be mapped to resource elements, according to clause 6.3.5 of TS 36.211.

FFS: whether to apply a cyclic shift offset, and the details of the cyclic shift offset for the concatenated frequency interleaver outputs of all OFDM symbols before the mapping to REs of a subframe, where the cyclic shift offset is dependent on the RV index.

Guidance for the next meeting:
To evaluate frequency interleavers, the "Two Path Ensemble" channel profile defined in Table A.1.1, Annex A of A325-2024-04" (with delay to be selected by proponents but less than 0.95*CP) can be used by companies.

9.14 TEI

R1-2501553 Session notes for 9.14 (TEI19) Ad-Hoc Chair (CMCC)
Friday decision: The session notes are endorsed and contents reflected below.

[120-R19-TEI] – Haruhi (NTT DOCOMO)
Email discussion on Rel-19 TEI proposals
- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

R1-2501418 Enhancement of CDM DMRS port allocation combinations for enhanced uplink and downlink performance Tejas Network Limited (rev of R1-2500499)
R1-2500579 Rel-19 TEI on UE frequency hopping for positioning ZTE Corporation, Sanechips
R1-2500615 On TEI-19 Ericsson
R1-2500809 Rel-19 TEI proposal on remaining ambiguities with SRS carrier switching Apple
R1-2501034 Proposals for TEI19 MediaTek Inc.
R1-2501180 TEI proposal on SR triggered SSSG fallback Qualcomm Incorporated
R1-2501284 TEI19: Counting of active NZP-CSI-RS resources Nokia, Apple, Ericsson, MediaTek, NTT DOCOMO, Spreadtrum
R1-2501323 Enhancement of SRS beamforming for FR2 positioning Huawei, HiSilicon, China Unicom, ZTE, Sanechips, CATT

R1-2501464 FL Summary #1 on Rel-19 TEIs Moderator (NTT DOCOMO, INC.)
From Tuesday session
Agreement
- Extend Rel-18's UL frequency hopping UL SRS for positioning transmission to non-RedCap UEs in a single carrier
- UE capability for non-RedCap UEs for UL SRS frequency hopping for positioning transmission

Send LS to RAN2 to inform this agreement, whether new parameter is needed is up to RAN2 discussion.

R1-2501534 FL Summary #2 on Rel-19 TEIs Moderator (NTT DOCOMO, INC.)
From Wednesday session
Agreement
- If a UE is instructed to monitor PDCCH according to search space sets with a group index other than a-designated index, the UE stops PDCCH monitoring according to search space sets with the group index and start PDCCH monitoring according to search space sets with the designated group index from the first slot that is at least P_switch symbols after the last symbol of a PUCCH carrying an SR.
- Introduce corresponding UE capability and RRC parameters to enable/disable the above feature and indicate the designated SSSG index.
- Send LS to RAN2 to inform above agreement and ask for the support of the UE capability and the corresponding RRC parameters.

R1-2501562 FL Summary #3 on Rel-19 TEIs Moderator (NTT DOCOMO, INC.)
From Thursday session
R1-2501539 Draft LS on non-RedCap UE UL SRS frequency hopping for positioning ZTE Corporation, Sanechips
Decision: The draft LS to RAN2 is endorsed with following information:

In RAN1#120 meeting TEI agenda, RAN1 has made the following agreement for enabling non-RedCap UE performing UL SRS frequency hopping for positioning:

> Agreement
> Extend Rel-18's UL frequency hopping UL SRS for positioning transmission to non-RedCap UEs in a single carrier
> UE capability for non-RedCap UEs for UL SRS frequency hopping for positioning transmission
> Send LS to RAN2 to inform this agreement, whether new parameter is needed is up to RAN2 discussion.

As this agreement may relate to RAN2 specification, this liaison informs RAN2 about this agreement and whether new parameter for non-RedCap UE UL frequency hopping is needed is up to RAN2 discussion.

To RAN2

ACTION: RAN1 respectfully asks RAN2 to take the agreement into account in their Rel-19 specification, and feedback if there is any spec impact.

Final LS is approved in R1-2501573.

Agreement
To resolve ambiguities with concurrent configuration of SRS-CS and ulTxswitch in a case where a UE configured with SRS CS on target CC and its "switch-from" CC on CC2 and configured with UL Tx switching operation for UL CC2 and at least one UL CC1
- Confirm that the prioritization rules in 38.214 Sec. 6.2.1.3 are applied between target and CC1, regardless of SRS-AS antenna port configuration on target CC, if UE indicates based on srs-SwitchingAffectedBandsListNR-r17 that SRS-CS on target impacts CC1, where CC1 is one of the CC(s) which may share Tx chains with source CC. No spec change is needed.
  - Note: if UE does not indicate srs-SwitchingAffectedBandsListNR-r17, UE can only perform simultaneous transmission when the total number of involved Tx chains for simultaneous transmission on all the bands/carriers is not greater than number of Tx chains supported by the UE for simultaneous UL transmission. No spec change is needed.
- If the UE is under the operation state in which all Tx chains are available at the source CC, the required switching time before the beginning of SRS-CS transmission on target CC is *SRS-SwitchingTimeNR*. Otherwise, the required switching time before the beginning of SRS-CS transmission on target CC is indicated by a UE capability. Details about UE capability will be discussed in UE feature session.
- The existing scheduling restriction of maximum one switching per reference slot for UL Tx switching is also taking into account the triggered SRS CS.
- After SRS transmissions within an SRS resource set is done, if UE is indicated to transmit on CC1, the required switching time between the end of SRS-CS transmission on target CC and start of UL transmission on CC1 is indicated by the UE capability defined above. Otherwise, it is assumed for the determination of any future switching time that all Tx chains are returned to the source CC, and the RF tuning time to switch from target to source will be *SRS-SwitchingTimeNR*.

Final summary in R1-2501598.

9.15 Rel-19 UE Features

[120-R19-UE_features] – Ralf (AT&T), Naoya (NTT DOCOMO)
Email discussion on Rel-19 UE features

- To be used for sharing updates on online/offline schedule, details on what is to be discussed in online/offline sessions, tdoc number of the moderator summary for online session, etc

From Friday session
R1-2501397 Initial RAN1 UE features list for Rel-19 NR after RAN1 #120 Moderators (AT&T, NTT DOCOMO, INC.)
Decision: The updated list after RAN1#120 in R1-2501397 is agreed.

R1-2501398 Draft LS on Rel-19 RAN1 UE features lists for NR after RAN1#120 Moderators (AT&T, NTT DOCOMO, INC.)
Decision: The draft LS to RAN2 (cc: RAN4) on the updated RAN1 UE features list for NR after RAN1#120 is endorsed. Final LS is approved in R1-2501399.

9.15.1 UE features for AI/ML for NR Air Interface

Placeholder only. No contributions in RAN1#120.

9.15.2 UE features for NR MIMO Phase 5

R1-2501602 Session Notes of AI 9.15.2 Ad-Hoc Chair (AT&T)
Friday decision: The session notes are endorsed.

R1-2500101 UE features for NR MIMO Phase 5 Huawei, HiSilicon
R1-2500108 UE features for NR MIMO Phase 5 MediaTek Inc.
R1-2500110 Views on NR MIMO Phase 5 UE features Nokia
R1-2500213 Discussion on UE features for NR MIMO Phase 5 CATT
R1-2500246 Discussion on UE features for NR MIMO Phase 5 ZTE Corporation, Sanechips
R1-2500307 Discussion on UE features for NR MIMO Phase 5 CMCC
R1-2500372 Discussion on UE features for NR MIMO Phase 5 vivo
R1-2500474 UE features for NR MIMO Phase 5 OPPO
R1-2500721 Discussion on UE features for NR MIMO Phase 5 Xiaomi
R1-2500802 Views on UE features for NR MIMO Phase 5 Apple
R1-2501120 UE features for NR MIMO Phase 5 BEIJING SAMSUNG TELECOM R&D
R1-2501181 UE features for NR MIMO phase 5 Qualcomm Incorporated
R1-2501221 Discussion on UE features for NR MIMO Phase 5 NTT DOCOMO, INC.
R1-2501271 UE features for NR MIMO Phase 5 Ericsson

R1-2501601 Summary of UE features for NR MIMO Phase 5 Moderator (AT&T)
Summary of email discussion [120-R19-UE_features] during RAN1 #120. The agreements reached during RAN1 #120 are summarized in R1-2411602.

9.15.3 UE features for evolution of NR duplex operation: SBFD

For RAN1#120, in addition to individual company/organization/university tdocs, the rapporteurs are to provide initial input.

R1-2501392 Session Notes of AI 9.15.3 Ad-Hoc Chair (NTT DOCOMO, INC.)
Friday decision: The session notes are endorsed.

R1-2500096 UE features for evolution of NR duplex operation Huawei, HiSilicon
R1-2500111 Views on SBFD UE features Nokia
R1-2500149 Discussion on UE features for Rel-19 SBFD ZTE Corporation, Sanechips
R1-2500188 Discussion on UE features for evolution of NR duplex operation: SBFD Spreadtrum, UNISOC
R1-2500214 UE features for SBFD CATT
R1-2500308 Discussion on UE features for evolution of NR duplex operation: SBFD CMCC
R1-2500373 Discussion on UE features for evolution of NR duplex operation vivo
R1-2500457 Discussion on UE features for Rel-19 NR Duplex OPPO
R1-2500722 UE features for SBFD Xiaomi
R1-2500754 UE-features for SBFD Ericsson
R1-2500810 Views on UE features for Rel-19 SBFD Apple
R1-2500874 UE features for Rel-19 Duplex Evolution for NR Samsung
R1-2501182 UE features for evolution of NR duplex operation Qualcomm Incorporated
R1-2501222 Discussion on UE features for Rel-19 duplex evolution enhancements NTT DOCOMO, INC.

R1-2501391 Summary of discussion on Rel-19 NR UE features for SBFD Moderator (NTT DOCOMO, INC.)
R1-2501508 Summary#2 of discussion on Rel-19 NR UE features for SBFD Moderator (NTT DOCOMO, INC.)
R1-2501588 Summary#3 of discussion on Rel-19 NR UE features for SBFD Moderator (NTT DOCOMO, INC.)

9.15.4 UE features for enhancements of network energy savings for NR

For RAN1#120, in addition to individual company/organization/university tdocs, the rapporteurs are to provide initial input.

R1-2501604 Session Notes of AI 9.15.4 Ad-Hoc Chair (AT&T)
Friday decision: The session notes are endorsed.

R1-2500080 UE features for Rel-19 NES Huawei, HiSilicon
R1-2500112 Views on Network Energy Saving Enhancement UE features Nokia
R1-2500135 Discussion on NES features ZTE Corporation, Sanechips
R1-2500310 Discussion on UE features for enhancements of network energy savings for NR CMCC
R1-2500374 Discussions on UE features for enhancements of network energy savings for NR vivo
R1-2500427 Discussion of Rel-19 NES UE Features FUTUREWEI
R1-2500723 UE features for NES enhancements Xiaomi
R1-2500811 Views on UE features for Rel-19 NES Apple
R1-2500873 UE features for enhancements of network energy savings for NR Samsung
R1-2500961 Discussion on UE features for enhancements of NES LG Electronics
R1-2501183 UE features for Rel-19 NES Qualcomm Incorporated
R1-2501223 Discussion on UE features for eNES NTT DOCOMO, INC.
R1-2501250 Rapporteur initial input for R19 NES WI UE features Rapporteur(Ericsson)
R1-2501257 UE features for R19 NES Ericsson

R1-2501603 Summary of UE features for enhancements of network energy savings for NR Moderator (AT&T)
Summary of email discussion [120-R19-UE_features] during RAN1 #120. The agreements reached during RAN1 #120 are summarized in R1-2411604.

9.15.5 UE features for LP-WUS/WUR for NR

For RAN1#120, in addition to individual company/organization/university tdocs, the rapporteurs are to provide initial input.

R1-2501394 Session Notes of AI 9.15.5 Ad-Hoc Chair (NTT DOCOMO, INC.)
Friday decision: The session notes are endorsed.

R1-2500076 UE features for Rel-19 LP-WUS Huawei, HiSilicon
R1-2500113 Views on LP-WUS/WUR UE features Nokia
R1-2500136 Discussion on WUR features ZTE Corporation, Sanechips
R1-2500215 LP-WUS/WUR UE feature CATT
R1-2500311 Discussion on UE features for low-power wake-up signal and receiver for NR CMCC
R1-2500375 Rapporteur input on UE features for Rel-19 LP-WUS/WUR vivo, DOCOMO
R1-2500724 UE features for NR LP-WUS/WUR Xiaomi
R1-2500876 UE features for LP-WUS/LP-WUR Samsung
R1-2501039 UE features for LP-WUS/WUR for NR MediaTek Inc.
R1-2501050 UE Features for LP-WUS Vodafone
R1-2501184 UE features for LP-WUS/WUR for NR Qualcomm Incorporated
R1-2501258 Initial view on UE features for Rel-19 LP-WUS/WUR WI Ericsson

R1-2501393 Summary of discussion on Rel-19 NR UE features for LP-WUS/WUR Moderator (NTT DOCOMO, INC.)
R1-2501570 Summary#2 of discussion on Rel-19 NR UE features for LP-WUS/WUR Moderator (NTT DOCOMO, INC.)
R1-2501589 Summary#3 of discussion on Rel-19 NR UE features for LP-WUS/WUR Moderator (NTT DOCOMO, INC.)

9.15.6 UE features for NR mobility enhancements Phase 4

Placeholder only. No contributions in RAN1#120.

9.15.7 UE features for XR for NR Phase 3

For RAN1#120, in addition to individual company/organization/university tdocs, the rapporteurs are to provide initial input.

R1-2501396 Session Notes of AI 9.15.7 Ad-Hoc Chair (NTT DOCOMO, INC.)
Friday decision: The session notes are endorsed.

R1-2500103 UE feature for R19 XR Huawei, HiSilicon
R1-2500114 Views on Rel-19 XR UE features Nokia
R1-2500137 Discussion on XR features ZTE Corporation, Sanechips
R1-2500189 Discussion on UE features for XR for NR Phase 3 Spreadtrum, UNISOC
R1-2500216 XR UE feature CATT
R1-2500309 Discussion on UE features for XR for NR Phase 3 CMCC
R1-2500376 Discussion on Rel-19 UE features for XR for NR Phase 3 vivo
R1-2500460 Discussion on UE features for Rel-19 XR OPPO
R1-2500812 Views on UE features for XR Apple
R1-2500875 UE features for XR for NR Phase 3 Samsung
R1-2501007 Initial UE feature list for Rel-19 XR for NR Phase3 Nokia, Qualcomm (WI rapporteurs)
R1-2501185 UE features for Rel-19 XR for NR Phase3 Qualcomm Incorporated
R1-2501224 Discussion on UE features for Rel-19 XR enhancements NTT DOCOMO, INC.
R1-2501344 UE features for XR enhancements Phase 3 Ericsson

R1-2501395 Summary of discussion on Rel-19 NR UE features for XR Phase 3 Moderator (NTT DOCOMO, INC.)
R1-2501509 Summary#2 of discussion on Rel-19 NR UE features for XR Phase 3 Moderator (NTT DOCOMO, INC.)
R1-2501590 Summary#3 of discussion on Rel-19 NR UE features for XR Phase 3 Moderator (NTT DOCOMO, INC.)

9.15.8 UE features for NTN for NR Phase 3

Placeholder only. No contributions in RAN1#120.

9.15.9 UE features for NTN for Internet of Things (IoT) Phase 3

Placeholder only. No contributions in RAN1#120.

9.15.10 UE features for IoT-NTN TDD mode

Placeholder only. No contributions in RAN1#120.

9.15.11 UE features for MCE for NR Phase 3

Placeholder only. No contributions in RAN1#120.

9.15.12 UE features for low band CA via switching

Placeholder only. No contributions in RAN1#120.

9.15.13 UE features for LTE based 5G broadcast Phase 2

Placeholder only. No contributions in RAN1#120.

9.15.14 Others

For discussions on UE features for Rel-19 TEI and other relevant issues. Placeholder only. No contributions in RAN1#120.

10 Closing of the meeting

Meeting was closed on February 21st 12:59 UTC (excluding post email discussions - see Annex F).

Annex A-1: List of Tdocs at RAN1 #120

Please see excel file attached to this report

3GPP TSG RAN WG1 Meeting #120bis
Wuhan, China, April 07th – 11th, 2025

R1-250xxxx

RAN1 Main | RAN1 Brk 1 | RAN1 Brk 2

Annex A-2: RAN1#120 Online Session Schedule

	Monday			Tuesday			Wednesday			Thursday			Friday		
08:30 ~ 10:30 (120 min)	RAN1#120 commences at 09:00 on Monday Agenda items 1, 2, 3, 4, 5 Agenda items 6, 7, 8.1			R19 Duplex (120)	R19 7–24GHz (90) AI 8.1 (30)	R19 A-IoT (120)	R19 MIMO (90) R19 XR (30)	R19 UE Features (120)	R19 A-IoT (120)	AI 7 (20) R19 LP-WUS (100)	R19 AI/ML (120)	ISAC (45) R19 UE Features (75)	R19 LP-WUS (120)	R19 AI/ML (120)	R19 NTN (90) R19 A-IoT (30)
Morning coffee break: 10:30 ~ 11:00															
11:00 ~ 13:00 (120 min)	Agenda item 8.1 (45) R19 MIMO (75)			R19 NES (120)	R19 TEI (60) R19 AI/ML (60)	ISAC (90) AI 8.1 (30)	R19 LP-WUS (120)	R19 AI/ML (75) R19 TEI (45)	R19 ISAC (120)	R19 MIMO (75) R19 Mobility (45)	R19 UE Features (90) R19 TEI (30)	R19 A-IoT (120)	R19 NES (115) R19 XR (5)	R19 UE Features (60) R18 UE Features (30) R19 7–24GHz (30-60)	R19 A-IoT (90)
Lunch break: 13:00 ~ 14:30															
14:30 ~ 16:30 (120 min)	R19 MIMO (45) R19 Mobility (45) R19 XR (30)	AI 8.1 (40) R19 MCE (50) R18 UE Features (30)	R19 A-IoT (120)	R19 MIMO (90) AI 8.1 (30)	R19 UE Features (90) R19 MCE (30)	R19 A-IoT (60) R19 Broadcast (60)	R19 Duplex (120)	R19 7–24GHz (90) AI 8.1 (30)	AI 8.1(20) R19 A-IoT (100)	R19 NES (120)	R19 AI/ML (120)	R19 Broadcast (30) R19 ISAC (90)	R19 Mobility (30) Any other issues Session reports RAN1#120 expected to close at 17:00		
Afternoon coffee break: 16:30 ~ 17:00															
17:00 ~ 19:30 (150 min)	R19 LP-WUS (150)	R19 AI/ML (150)	AI 8.1 (40) R19 NTN (110)	AI 6, 7 R19 LP-WUS (120)	R19 AI/ML (150)	R19 NTN (100) R19 UE Features (50)	R19 NES (150)	R19 AI/ML (150)	R19 NTN (100) R19 UE Features (50)	R19 Duplex (90) AI 8.1, 7 (30) TBD (30)	R19 7–24GHz (105) R19 MCE (45)	R19 A-IoT (60) AI 8.1 NTN (3) R19 NTN (87)			
All sessions end at 19:45 – no exceptions															

Annex B: List of CRs agreed at RAN1 #120

TD#	Title	Source to WG	Rel	TS	Vsn	Work Item	CR	Rev	Cat
R1-2501501	Correction on transition time of UL BWP change triggered by DCI format 0_1 scheduling multiple PUSCHs	vivo, ASUSTeK	Rel-18	38.213	18.5.0	NR_ext_to_71GHz-Core, TEI18	0693		F
R1-2501502	CR on PSCCH DMRS sequence generation in a dedicated SL PRS resource pool	Moderator (CATT), CATT, CICTCI	Rel-18	38.211	18.5.0	NR_pos_enh2-Core	0148		F
R1-2501503	CR on power control of PSFCH in TS 38.213	Moderator (ZTE), ZTE Corporation, Sanechips, Samsung, vivo, OPPO	Rel-18	38.213	18.5.0	NR_SL_enh2-Core	0694		F
R1-2501559	Correction on the zero padding for CSI part 1 on NES	Moderator (Samsung)	Rel-18	38.214	18.5.0	Netw_Energy_NR-Core	0654		F
R1-2501564	CR on timeline for STxMP	Moderator (Samsung), ZTE, Huawei, HiSilicon	Rel-18	38.214	18.5.0	NR_MIMO_evo_DL_UL-Core	0655		F
R1-2501572	Corrections on determination of DCI format 0_3 in case of dormancy	Huawei, HiSilicon, Ericsson, Fujitsu, Nokia, Samsung, ZTE	Rel-18	38.212	18.5.0	NR_MC_enh-Core	0210		F
R1-2501575	CR on LTM PRACH and serving cell UL transmission in a same band	Moderator (MediaTek), Apple, NTT DOCOMO, Ericsson, Nokia, ZTE, Huawei, Vivo, Google, Fujitsu, Samsung	Rel-18	38.213	18.5.0	NR_Mob_enh2-Core	0695		F
R1-2501579	CR on Slot aggregation configuration with multi-PUSCH	Moderator (Nokia), Ericsson	Rel-17	38.214	17.12.0	NR_ext_to_71GHz-Core	0656		F
R1-2501580	CR on Slot aggregation configuration with multi-PUSCH	Moderator (Nokia), Ericsson	Rel-18	38.214	18.5.0	NR_ext_to_71GHz-Core	0657		A
R1-2501581	Correction on LBRM TBS determination for UE indicating 1024QAM rank restricted capability	Moderator (Nokia), Ericsson, xiaomi	Rel-17	38.212	17.10.0	NR_DL1024QAM_FR1-Core	0211		F
R1-2501582	Correction on LBRM TBS determination for UE indicating 1024QAM rank restricted capability	Moderator (Nokia), Ericsson, xiaomi	Rel-18	38.212	18.5.0	NR_DL1024QAM_FR1-Core	0212		A
R1-2501596	Correction on RACH-less handover for NTN	Moderator(ASUSTeK), Huawei, Hisilicon	Rel-18	38.213	18.5.0	NR_NTN_enh-Core	0696		F
R1-2501637	CR on the PRACH retransmission indicator field included in the PDCCH order	Moderator (vivo), Nokia	Rel-18	38.212	18.5.0	NR_Mob_enh2-Core	0213		F
R1-2501648	Alignment of parameter names	Ericsson	Rel-18	38.211	18.5.0	NR_MIMO_evo_DL_UL-Core	0149		F
R1-2501649	Rel-18 editorial corrections for TS 38.213	Samsung	Rel-18	38.213	18.5.0	NR_MIMO_evo_DL_UL-Core, NR_cov_enh2-Core, NR_SL_enh2-Core	0697		F
R1-2501650	Rel-18 editorial corrections for TS 38.212	Huawei	Rel-18	38.212	18.5.0	NR_MIMO_evo_DL_UL-Core	0214		F
R1-2501651	Rel-18 editorial corrections for TS 38.214	Nokia	Rel-18	38.214	18.5.0	NR_MIMO_evo_DL_UL-Core, NR_SL_enh2-Core	0658		F
R1-2501652	Rel-17 editorial corrections for TS 38.214	Nokia	Rel-17	38.214	17.12.0	NR_FeMIMO-Core, NR_cov_enh-Core	0659		F
R1-2501653	Rel-17 editorial corrections for TS 38.214 (mirrored to Rel-18)	Nokia	Rel-18	38.214	18.5.0	NR_FeMIMO-Core, NR_cov_enh-Core	0660		A

Annex C-1: List of Outgoing LSs from RAN1 #120

TDoc #	Title	Source	Release	Related WIs	Reply to	To	Cc	Original LS	Reply in
R1-2500846	LS to RAN2 on MAC impact for Rel-19 MIMO Ph5	RAN1, Samsung	Rel-19	NR_MIMO_Ph5-Core		RAN2			
R1-2501390	LS on Rel-18 RAN1 UE features lists for NR after RAN1#120	RAN1, AT&T, NTT DOCOMO, INC.	Rel-18	NR_MIMO_evo_DL_UL, NR_pos_enh2, Netw_Energy_NR, NR_netcon_repeater, NR_NTN_enh, NR_Mob_enh2, NR_SL_enh2, NR_redcap_enh, NR_MC_enh, NR_XR_enh, NR_FR1_lessthan_5MHz_BW, NR_DSS_enh, NR_BWP_wor, NR_cov_enh2, TEI18		RAN2	RAN4		
R1-2501399	LS on Rel-19 RAN1 UE features lists for NR after RAN1#120	RAN1, AT&T, NTT DOCOMO, INC.	Rel-19	NR_MIMO_Ph5-Core, NR_duplex_evo-Core, Netw_Energy_NR_enh-Core, NR_LPWUS-Core, NR_XR_Ph3-Core		RAN2	RAN4		
R1-2501479	Reply LS on co-existence of SL-CA and SL PRS transmission/reception in Dedicated SL-PRS resource pool	RAN1, vivo	Rel-18	NR_pos_enh2-Core	R2-2411251	RAN2			
R1-2501500	LS on activation/deactivation of semi-persistent CSI-RS resource for LTM candidate cells	RAN1, Fujitsu	Rel-19	NR_Mob_Ph4-Core		RAN2	RAN3		
R1-2501512	LS on measurement quantities for IoT NTN	RAN1, vivo	Rel-18	IoT_NTN_enh-Core		RAN4			
R1-2501523	Reply LS on LMF-based AI/ML Positioning for Case 2b	RAN1, vivo	Rel-19	AIML_CN	S2-2501133	SA2	RAN2, RAN3		
R1-2501525	Reply LS on LMF-based AI/ML Positioning for Case 3b	RAN1, Ericsson	Rel-19	AIML_CN	S2-2501341	SA2	RAN2, RAN3		
R1-2501540	LS on RRC parameter for PRACH transmission in 2TA	RAN1, CATT	Rel-18	NR_MIMO_evo_DL_UL-Core		RAN2			
R1-2501560	Reply LS on CSI-RS measurement with SBFD operation	RAN1, MediaTek	Rel-19	NR_duplex_evo-Core	R4-2420165	RAN4	RAN2		
R1-2501573	LS on non-RedCap UE UL SRS frequency hopping for positioning	RAN1, ZTE Corporation	Rel-19	TEI19		RAN2			
R1-2501577	LS on L1filtering for LTM event triggered reporting	RAN1, Fujitsu	Rel-19	NR_Mob_Ph4-Core		RAN2			
R1-2501593	LS on Slot aggregation configuration with multi-PUSCH	RAN1, Nokia	Rel-17	NR_ext_to_71GHz-Core		RAN2			
R1-2501606	Reply LS on soft satellite switch with re-sync	RAN1, Huawei	Rel-18	NR_NTN_enh-Core	R2-2410978	RAN2			
R1-2501609	Reply LS on Ku band numerology	RAN1, Eutelsat Group	Rel-19	NR_NTN_Ku_bands-Core	R4-2419902	RAN4	RAN, RAN2		
R1-2501611	Reply LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP	RAN1, CATT	Rel-18	NR_MIMO_evo_DL_UL-Core	R2-2411244	RAN2			
R1-2501613	Reply LS on PWS support in RRC_CONNECTED for NB-IoT NTN	RAN1, InterDigital	Rel-19	IoT_NTN_Ph3-Core	R2-2411194	RAN2, RAN			
R1-2501620	LS on L1 based UE-to-UE CLI measurement and reporting	RAN1, Huawei	Rel-19	NR_duplex_evo-Core		RAN4			
R1-2501622	LS on TDM DMRS for 3.75kHz SCS OCC	RAN1, Sony	Rel-19	IoT_NTN_Ph3-Core		RAN4			
R1-2501624	LS on the wake-up delay for LP-WUS	RAN1, Apple	Rel-19	NR_LPWUS-Core		RAN4			

TDoc #	Title	Source	Release	Related WIs	Reply to	To	Cc	Original LS	Reply in
	operation in IDLE/INACTIVE mode								
R1-2501628	LS on AI/ML Positioning Case 3b	RAN1, Ericsson	**Rel-19**	**NR_AIML_air-Core**		RAN3, RAN2			
R1-2501633	LS on time location of on-demand SSB for Scell	RAN1, LG Electronics	**Rel-19**	**Netw_Energy_NR_enh-Core**		RAN4	RAN2		
R1-2501636	Reply LS on UL 8Tx	RAN1, Samsung	**Rel-18**	**NR_MIMO_evo_DL_UL-Core**	R2-2411220	RAN2			
R1-2501644	LS on Rel-19 higher layers parameters list Post RAN1#120	RAN1, Ericsson	**Rel-19**	NR_MIMO_Ph5, NR_duplex_evo, Netw_Energy_NR_enh, NR_LPWUS, NR_XR_Ph3		RAN2, RAN3	RAN4		

Annex C-2: List of Incoming LSs from RAN1 #120

TDoc #	Title	Source	Release	Related WIs	Reply to	To	Cc	Original LS	Reply in
R1-2500005	Reply LS to R2-2407921/ R1-2407559	RAN2, Vodafone	Rel-19	NR_LPWUS-Core	R1-2407559	RAN1, RAN4		R2-2410957	
R1-2500006	LS on soft satellite switch with re-sync	RAN2, Huawei	Rel-18	NR_NTN_enh-Core		RAN4, RAN1		R2-2410978	R1-2501606
R1-2500007	LS response on waveform determination for PUSCH scheduled by fallbackRAR UL grant and MsgA PUSCH	RAN2, CATT	Rel-17	NR_redcap-Core, NR_SmallData_INACTIVE-Core, NR_cov_enh-Core, NR_slice-Core	R1-2409301	RAN1		R2-2411031	
R1-2500008	Reply LS on the time period for R18 preamble repetition	RAN2, Huawei	Rel-18	NR_cov_enh2-Core	R1-2409262	RAN1		R2-2411168	
R1-2500009	LS on PWS support in RRC_CONNECTED for NB-IoT NTN	RAN2, InterDigital	Rel-19	IoT_NTN_Ph3-Core		RAN1		R2-2411194	R1-2501613
R1-2500010	LS on LTM UE capabilities for cross-band operation	RAN2, Qualcomm	Rel-18	NR_Mob_enh2-Core		RAN1		R2-2411219	R1-2501390
R1-2500011	RAN2 agreements on Inter-CU LTM and Conditional LTM	RAN2, Lenovo	Rel-19	NR_Mob_Ph4-Core		RAN1, RAN3, RAN4		R2-2411134	
R1-2500012	LS on UL 8Tx	RAN2, Samsung	Rel-18	NR_MIMO_evo_DL_UL-Core		RAN1		R2-2411220	R1-2501636
R1-2500013	LS on differentiation of sDCI mTRP, mDCI mTRP and sTRP	RAN2, CATT	Rel-18	NR_MIMO_evo_DL_UL-Core		RAN1		R2-2411244	R1-2501611
R1-2500014	LS on co-existence of SL-CA and SL PRS transmission/reception for Dedicated SL-PRS resource pool	RAN2, vivo	Rel-18	NR_pos_enh2-Core		RAN1		R2-2411251	R1-2501479
R1-2500015	Reply LS to SA5 on AIML data collection	RAN2, Nokia	Rel-19	FS_AIML_MGT_Ph2	S5-246299	SA5	RAN, RAN1, RAN3, SA, SA2, SA3	R2-2411114	
R1-2500016	Reply LS to SA2 on AIML data collection	RAN2, InterDigital	Rel-19	NR_AIML_air-Core	S2-2411191	SA2	RAN, RAN1, RAN3, SA, SA5, SA3	R2-2411152	
R1-2500017	LS on inter-gNB RACH-less HO in NTN	RAN2, Nokia	Rel-18	NR_NTN_enh-Core		RAN3	RAN1	R2-2411191	
R1-2500018	LS on Ku band numerology	RAN4, Eutelsat	Rel-19	NR_NTN_Ku_bands-Core		RAN1, RAN2	RAN	R4-2419902	R1-2501609
R1-2500019	LS on CSI-RS measurement with SBFD operation	RAN4, MediaTek	Rel-19	NR_duplex_evo-Core		RAN1	RAN2	R4-2420165	R1-2501560
R1-2500020	Reply LS on timeline for On-demand SSB operation on SCell	RAN4, Ericsson	Rel-19	Netw_Energy_NR_enh-Core	R1-2407565	RAN1		R4-2420192	
R1-2500021	Response LS on gaps/restrictions that are caused by RRM measurements	RAN4, Qualcomm	Rel-19	NR_XR_Ph3-Core	R1-2407561	RAN1	RAN2	R4-2420198	
R1-2500022	Reply LS on carrierBandwidth configuration for less-than-5MHz carriers	RAN4, ZTE Corporation	Rel-18	NR_FR1_lessthan_5MHz_BW-Core	R2-2409409	RAN2	RAN1	R4-2420382	
R1-2500023	Reply LS to RAN1 and RAN2 on FR2-NTN inclusion to specifications	RAN4, vivo	Rel-18	NR_NTN_enh-Core	R1-2407406	RAN1, RAN2		R4-2420476	
R1-2500024	LS on AdditionalSpectrumEmission in NR SL Pre-configuration	RAN4, LG Electronics	Rel-18	NR_SL_enh2-Core		RAN2	RAN1	R4-2418075	
R1-2500025	LS on simultaneous operation between	RAN4, Huawei	Rel-19	NR_NTN_enh-Core		RAN2	RAN1	R4-2419898	

TDoc #	Title	Source	Release	Related WIs	Reply to	To	Cc	Original LS	Reply in
	GNSS and NR NTN								
R1-2500026	LS on UE capability for FDD-FDD inter-band CA simultaneous Rx/Tx	RAN4, CATT	Rel-19	NR_CADC_SUL_R19		RAN2	RAN1	R4-2420410	
R1-2500027	Reply LS for LS on the support of semi-persistent CSI-RS resource for LTM candidate cells	RAN3, CATT	Rel-19	NR_Mob_Ph4-Core	R1-2409283	RAN1	RAN2	R3-247911	
R1-2500028	Reply LS on RAN3 vs RAN2 Basketball Match	RAN3, ZTE	Rel-19	TEI19	R2-2409711	RAN2	RAN1, RAN4, RAN5	R3-247925	
R1-2500029	Reply on AIML data collection	SA5, Nokia	Rel-19	NR_AIML_air-Core	RP-242389	RAN	SA, SA2, SA3, RAN1, RAN2, RAN3	S5-247219	
R1-2500030	Reply LS on AIML data collection	SA5, Samsung	Rel-19	FS_AIML_MGT_Ph2	R2-2411114	RAN2	SA, SA2, SA3, RAN1, RAN, RAN3	S5-247336	
R1-2500031	Reply LS on AIML data collection	SA2, InterDigital	Rel-19	NR_AIML_air-Core	RP-242389	RAN, SA	RAN1, RAN2, RAN3, SA3, SA5	S2-2412726	
R1-2500032	LS on SSB position restrictions for less-than-5MHz Scells	RAN4, Qualcomm	Rel-19	NR_FR1_lessthan_5MHz_BW_Ph2-Core		RAN2	RAN1	R4-2420383	
R1-2500388	LS on LMF-based AI/ML Positioning for Case 2b	SA2, vivo	Rel-19	AIML_CN		RAN1, RAN2	RAN3	S2-2501133	R1-2501523
R1-2500389	LS on LMF-based AI/ML Positioning for case 3b	SA2, Ericsson	Rel-19	AIML_CN		RAN3, RAN1	RAN2	S2-2501341	R1-2501525
R1-2501370	Availability of Group Reports from ETSI ISAC ISG on use cases and channel models	ETSI ISG ISAC, Mitsubishi, ZTE	Rel-19	FS_Sensing_NR		RAN, RAN1		ISC(25)000032r3	
R1-2501569	Reply LS on the STx2P UE Features	RAN2, ZTE	Rel-18	NR_MIMO_evo_DL_UL-Core	R1-2410665	RAN1		R2-2501516	R1-2501390

Annex D: List of Approved updated WIDs

None

Annex E: List of draft TSs/TRs agreed at RAN1 #120

Tdoc Number	Title	Source	Conclusion/Decision

Annex F: List of actions

- **Outgoing LS**

[Post-120-AI/ML-01] – Yufei (Ericsson)
- Email approval of Reply LS on AI/ML Positioning Case 3b by February 26 (Wednesday)

R1-2501639 [DRAFT] LS on AI/ML Positioning Case 3b Moderator (Ericsson)
Decision: As per email decision posted on Feb 26th, the draft LS is endorsed in principle. Final LS is approved in R1-2501628.

[Post-120-Rel-19-RRC] – Sorour (Ericsson)
LS to RAN2 on RRC parameter list post RAN1#120
- Approval by February 26 (Wednesday)

R1-2501643 DRAFT LS on Rel-19 higher layers parameters list Post RAN1#120 Moderator (Ericsson)
Decision: As per email decision posted on Feb 26th, the draft LS is endorsed and includes the agreed RRC parameters (in R1-2501645) for Rel-19 WIs on MIMO, SBFD, NES, LP WUS/WUR and XR. Final LS is approved in R1-2501644.

- **CR approval**

[Post-120-Rel19-AAA-BBB] – Editors
Email discussion on endorsement of initial Rel-19 draft CRs
- Editors to prepare draft CRs by March 17 (Monday)
- Endorsements by March 20 (Thursday)
- Only for MIMO and XR
- Replace AAA with specification number, replace BBB with WI code

Note that the draft CRs will endorsed for internal RAN1 work only. Final CR submission to RAN will be made in June.

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[Post-120-RelXX-YYY-Alignment] – Editors
Email discussion on endorsement of alignment CRs
- Editors to prepare draft CRs by February 25 (Tuesday)
- Endorsements by February 27 (Thursday)
- Replace XX with Release # and YYY with specification number

Rel-17
R1-2501652 Rel-17 editorial corrections for TS 38.214 Nokia
(Rel-17, 38.214, NR_FeMIMO-Core, NR_cov_enh-Core, CR0659, Cat F) is agreed.
R1-2501653 Rel-17 editorial corrections for TS 38.214 (mirrored to Rel-18) Nokia

(Rel-18, 38.214, NR_FeMIMO-Core, NR_cov_enh-Core, CR0660, Cat A) is agreed.

Rel-18
R1-2501648 Alignment of parameter names Ericsson
(Rel-18, 38.211, NR_MIMO_evo_DL_UL-Core, CR0149, Cat F) is agreed.
R1-2501649 Rel-18 editorial corrections for TS 38.213 Samsung
(Rel-18, 38.213, NR_MIMO_evo_DL_UL-Core, NR_cov_enh2-Core, NR_SL_enh2-Core, CR0697, Cat F) is agreed.
R1-2501650 Rel-18 editorial corrections for TS 38.212 Huawei
(Rel-18, 38.212, NR_MIMO_evo_DL_UL-Core, CR0214, Cat F) is agreed.
R1-2501651 Rel-18 editorial corrections for TS 38.214 Nokia
(Rel-18,38.214, NR_MIMO_evo_DL_UL-Core, NR_SL_enh2-Core, CR0658, Cat F) is agreed.

- **TR updates**

None.

- **Miscellaneous**

[Post-120-Rel-19-RRC] – Sorour (Ericsson)
Approval of RRC parameter list and LS to RAN2
- Approval by February 26 (Wednesday)

R1-2501642 Email discussion summary on Rel-19 higher layers parameters list Post RAN1#120 Moderator (Ericsson)
R1-2501645 Consolidated Rel-19 higher layers parameters list Post RAN1#120 Moderator (Ericsson)
Agreement
- RRC parameters (in R1-2501645) for Rel-19 WIs on MIMO, SBFD, NES, LP WUS/WUR and XR are agreed.
- Send an LS to RAN2 and RAN3 (CC: RAN4) to share the agreed RRC parameters (in R1-2501645).

For information
R1-2501646 Collection of Rel-19 higher layers parameters list Post RAN1#120 Moderator (Ericsson)

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[Post-120-ISAC-01] – Yingyang (Xiaomi)
Email discussion on values/pattern of A*B1 of RCS for target for monostatic sensing, from March 3-7
- Companies to provide values of the parameters (referring the example table in the agreement) on A*B1 for vehicle, until March 5, targeting agreement on merged values by March 7.
- Collect companies views on the possible extension of the agreement (except the example table) on A*B1 of RCS for vehicle to other target types with angle-dependent RCS until March 7, for potential agreement.

R1-2501654 Email summary on values/pattern of A*B1 of monostatic RCS for target Moderator (Xiaomi)

Decision: As per email decision posted on March 7th, the following is agreed with the understanding that additional details will be collected at the next RAN1 meeting.

Agreement

On the monostatic RCS of vehicle with single scattering point,

- The values/pattern of component A*B1 are generated by the following parameters

	φ_{center}	$\varphi_{3dB,n}$	θ_{center}	$\theta_{3dB,n}$	G_{max}	σ_{max}	*Applicable Range of* θ	*Applicable Range of* φ
Left	90	26.90	79.70	44.42	20.75	13.68	[30°,180°]	(45°,135°]
Back	180	36.32	79.65	36.73	14.56	7.50	[30°,180°]	(135°,225°]
Right	270	26.90	79.70	44.42	20.75	13.68	[30°,180°]	(225°,315°]
Front	0	40.54	71.75	29.13	15.52	8.45	[30°,180°]	(-45°, 45°]
Roof	-	-	0.00	18.13	21.26	14.19	[0°,30°)	[0°,360°)

  - When θ is in the range [0°,30°), $\sigma^{H}_{\mathrm{dB}}(\varphi) = 0$
- The standard deviation of component B2 is 3.41 dB

Agreement

On the monostatic RCS for each scattering point of vehicle with multiple scattering points,

- The values/pattern of component A*B1 are generated by the following parameters

	φ_{center}	$\varphi_{3dB,n}$	θ_{center}	$\theta_{3dB,n}$	G_{max}	σ_{max}	*Applicable Range of* θ	*Applicable Range of* φ
Left	90	26.90	79.70	44.42	20.60	20.52	[0°,180°]	[0°,360°]
Back	180	36.32	79.65	36.73	13.90	13.82	[0°,180°]	[0°,360°]
Right	270	26.90	79.70	44.42	20.60	20.52	[0°,180°]	[0°,360°]
Front	0	40.54	71.75	29.13	14.99	14.91	[0°,180°]	[0°,360°]
Roof	-	-	0.00	18.13	21.12	21.05	[0°,180°]	[0°,360°]

  - For the scattering point associated with roof of the vehicle, $\sigma^{H}_{\mathrm{dB}}(\varphi) = 0$
- The standard deviation of component B2 is 3.41 dB

Agreement

On the monostatic RCS for UAV with large size and AGV

- The monostatic RCS for a scattering point of the target is generated by
  - The values/pattern A*B1, i.e., $rcs_{\mathrm{dB}}(\theta,\phi)$ is deterministic based on incident/scattered angles

$$rcs_{\mathrm{dB}}(\theta,\varphi) = G_{max} - min\left\{-\left(\sigma^{V}_{\mathrm{dB}}(\theta) + \sigma^{H}_{\mathrm{dB}}(\varphi)\right), \sigma_{max}\right\}$$

where,

$$\sigma^{V}{}_{\mathrm{dB}}(\theta) = -\,min\left\{12\left(\frac{\theta-\theta_{center}}{\theta_{3dB}}\right)^{2}, \sigma_{max}\right\},$$

$$\sigma^{H}{}_{\mathrm{dB}}(\varphi) = -\,min\left\{12\left(\frac{\varphi-\varphi_{center}}{\varphi_{3dB}}\right)^{2}, \sigma_{max}\right\},$$

- o FFS how many rows of the values/pattern A*B1 are defined for the target
  - Note: each row has a defined applicable range of θ and φ
- FFS human RCS model 2

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[Post-120-7-24GHz-01] – Daewon (Intel)
Email discussion on calibration for channel modelling enhancements for 7-24GHz for NR

- March 3 ~ 7

R1-2501655 Summary of email discussions for calibration assumptions for 7-24 GHz channel modeling Moderator (Intel Corporation)
Decision: As per email decision posted on March 13th,
Agreement
Proposal #1D, Proposal #2E, Proposal #3D, Proposal #4C, and Proposal #5C in R1-2501655 are endorsed.

Annex G: List of participants at RAN1 #120

Please see excel file attached to this report

Annex H: TSG RAN WG1 meetings in 2025 – 2026

TITLE	TYPE	DATES	LOCATION	CTRY
3GPPRAN1#120bis	WG	07 – 11 Apr 2025	Wuhan	China
3GPPRAN1#121	WG	19 – 23 May 2025	St Julian's	Malta
3GPPRAN1#122	WG	25 – 29 Aug 2025	TBD	India
3GPPRAN1#122bis	WG	13 – 17 Oct 2025	Prague	CZ
3GPPRAN1#123	WG	17 – 21 Nov 2025	Dallas	US
3GPPRAN1#124	WG	09 – 13 Feb 2026	TBD	EU
3GPPRAN1#124bis	WG	13 – 17 Apr 2026	TBD	EU
3GPPRAN1#125	WG	18 – 22 May 2026	TBD	China
3GPPRAN1#126	WG	24 – 28 Aug 2026	TBD	EU
3GPPRAN1#126bis	WG	12 – 16 Oct 2026	TBD	Korea
3GPPRAN1#127	WG	16 – 20 Nov 2026	TBD	US

MEETING TYPES		
AH = Ad Hoc	ST = Startup Meeting	RG = Rapporteurs Group
JM = Joint	WG = Working Group	SG = Steering Group
PM = Preparatory Meeting	CM = Chairmen's meeting	TG = Task Group
RM = Resolution Meeting	OR = Ordinary	XO = Extraordinary

End of document